

ACADEMIC REGULATIONS AND COURSE STRUCTURE

CHOICE BASED CREDIT SYSTEM

AR22

**Computer Science & Engineering
(Artificial Intelligence and Machine Learning)**

Bachelor of Technology (B.Tech)

B. Tech. - Regular Four Year Degree Programme

(For batches admitted from the academic year 2022-2023)

&

**(For batches admitted Lateral Entry Scheme from the
academic year 2023-2024)**

MLR Institute of Technology

(Autonomous)

Laxman Reddy Avenue, Dundigal (V), Quthbullapur (M),
Hyderabad – 500043, Telangana State

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COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

VISSION

To develop competent and socially responsible engineers in the domain of AI&ML architect for a strong society.

MISSION

- To impart knowledge in Artificial Intelligence technologies to meet industrial standards.
- To transform professionals into technically competent through innovation and socially responsible.
- To provide necessary strengths and develop self-learning ability among the student to improve innovative Thinking and become entrepreneurs.

PROGRAM EDUCATIONAL OBJECTIVES (PEOS):

PEO 1: To enable the graduates of the program to have a globally competent professional career in Artificial Intelligence and Machine Learning domain.

PEO 2: To prepare the students to excel in Artificial Intelligence and Machine Learning with the technical skills and competency to carry out research and address the basic needs of the society.

PEO 3: To empower the graduates of the program to have entrepreneurial skills with a lifelong learning attitude in order to support the growth of the economy of a country.

PROGRAM OUTCOMES

PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

B.Tech – CSE(AIML) : PROGRAM SPECIFIC OUTCOMES

PSO 1: Develop an in-depth knowledge and skill set in human cognition, Artificial Intelligence, Machine Learning and data engineering for designing intelligent systems to address modern computing challenges.

PSO 2: Evaluate, analyse and synthesize solutions for real time problems in the Artificial Intelligence and Machine Learning domain to conduct research in a wider theoretical and practical context.

FOREWORD

The autonomy is conferred on MLR Institute of Technology by UGC, based on its performance as well as future commitment and competency to impart quality education. It is a mark of its ability to function independently in accordance with the set norms of the monitoring bodies like UGC and AICTE. It reflects the confidence of the UGC in the autonomous institution to uphold and maintain standards it expects to deliver on its own behalf and thus awards degrees on behalf of the college. Thus, an autonomous institution is given the freedom to have its own **curriculum, examination system and monitoring mechanism**, independent of the affiliating University but under its observance.

MLR Institute of Technology is proud to win the credence of all the above bodies monitoring the quality in education and has gladly accepted the responsibility of sustaining, if not improving upon the standards and ethics for which it has been striving for more than a decade in reaching its present standing in the arena of contemporary technical education. As a follow up, statutory bodies like Academic Council and Boards of Studies are constituted with the guidance of the Governing Body of the College and recommendations of the JNTU Hyderabad to frame the regulations, course structure and syllabi under autonomous status.

The autonomous regulations, course structure and syllabi have been prepared after prolonged and detailed interaction with several expertise solicited from academics, industry and research, in accordance with the vision and mission of the college in order to produce quality engineering graduates to the society.

All the faculty, parents and students are requested to go through all the rules and regulations carefully. Any clarifications, if needed, are to be sought, at appropriate time with principal of the college, without presumptions, to avoid unwanted subsequent inconveniences and embarrassments. The Cooperation of all the stake holders is sought for the successful implementation of the autonomous system in the larger interests of the college and brighter prospects of engineering graduates.

PRINCIPAL

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ACADEMIC REGULATIONS (AR22)
B. Tech. - Regular Four Year Degree Programme
(For batches admitted from the academic year 2022-23)

1.0 For pursuing four year Under Graduate Degree Programme of study in Engineering & Technology (UGP in E&T) offered by MLR Institute of Technology under Autonomous status is herein referred to as MLRIT (Autonomous):

All the rules specified herein approved by the Academic Council will be in force and applicable to students admitted from the Academic Year 2022-23 onwards. Any reference to “Institute” or “College” in these rules and regulations shall stand for M L R Institute of Technology (Autonomous).

All the rules and regulations, specified hereafter shall be read as a whole for the purpose of interpretation as and when a doubt arises, the interpretation of the Chairman, Academic Council is final. As per the requirements of statutory bodies, the Principal, M L R Institute of Technology shall be the chairman Academic Council.

2.0 Eligibility for Admission

2.1 Admission to the under graduate(UG) programme shall be made either on the basis of the merit rank obtained by the qualified student in entrance test conducted by the Telangana State Government(EAMCET)or the College or on the basis of any other order of merit approved by the University/TSHE, subject to reservations as prescribed by the government from time to time.

2.2 The medium of instructions for the entire undergraduate programme in Engineering & Technology will be **English** only.

3.0 B.Tech. Programme Structure

3.1 A student after securing admission shall complete the B.Tech. programme in a minimum period of **four** academic years (8 semesters), and a maximum period of **eight** academic years (16 semesters) starting from the date of commencement of first year first semester, failing which student shall forfeit seat in B.Tech course. Each student shall secure 160 credits (with CGPA ≥ 5) required for the completion of the undergraduate programme and award of the B.Tech. degree.

3.2 **UGC/AICTE** specified definitions/descriptions spreadeagled appropriately for various terms and abbreviations used in these academic regulations/ norms, which are listed below.

3.2.1 Semester Scheme

Each undergraduate programme is of 4 academic years (8 semesters) with the academic year divided into two semesters of 22 weeks (≥ 90 instructional days) each and in each semester - ‘Continuous Internal Evaluation (CIE)’ and ‘Semester End Examination (SEE)’ under Choice Based Credit System (CBCS) and Credit Based Semester System (CBSS) indicated by UGC, and curriculum/course structure suggested by AICTE are followed.

3.2.2 Credit Courses

All subjects/courses are to be registered by the student in a semester to earn credits which shall be assigned to each subject/course in an L:T:P:C(lecture periods: tutorial periods: practical periods: credits) structure based on the following general pattern.

- One credit for one hour/ week/ semester for Theory/ Lecture (L) courses or Tutorials.

- One credit for two hours/ week/ semester for Laboratory/ Practical (P)courses.

Courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab are mandatory courses. These courses will not carry any credits.

3.2.3 Subject Course Classification

All subjects/ courses offered for the undergraduate programme in E&T (B.Tech. degree programmes) are broadly classified as follows. The College has followed almost all the guidelines issued by AICTE/UGC.

S. No.	Broad Course Classification	Course Group/ Category	Course Description
1	Foundation Courses (FnC)	BS–Basic Sciences	Includes Mathematics, Physics and Chemistry Subjects.
2		ES–Engineering Sciences	Includes Fundamental Engineering Subjects.
3		HS–Humanities and Social Sciences	Includes subjects related to Humanities, Social Sciences and Management.
4	Core Courses (CoC)	PC–Professional Core.	Includes core subjects related to the parent discipline/ department/ branch of Engineering.
5	Elective Courses (ElC)	PE–Professional Electives	Includes elective subjects related to the parent discipline/ department/ branch of Engineering.
6		OE–Open Electives	Elective subjects which include inter-disciplinary subjects or subjects in an area outside the parent discipline/ department/ branch of Engineering.
7	Core Courses	Project Work	B.Tech. Project or UG Project or UG Major Project or Project Stage I & II.
8		Industry Training/ Internship / Industry Oriented Mini- project/ Mini- Project/ Skill Development Courses.	Industry Training/ Internship/ Industry Oriented Mini-Project/ Mini-Project/ Skill Development Courses.
9		Seminar	Seminar/ Colloquium based on core contents related to parent discipline/ department/ branch of Engineering.
10	Minor Courses	-	1 or 2 Credit Courses (subset of HS)
11	Mandatory Courses (MC)	-	Mandatory Courses (non-credit)

4.0 Course Registration

- 4.1** A ‘faculty advisor or counselor’ shall be assigned to a group of 20 students, who will advise the students about the undergraduate programme, its course structure and curriculum, choice/option for subjects/courses, based on their competence, progress, pre- requisites and interest.

- 4.2 The academic section of the college invites ‘registration forms’ from students before the beginning of the semester through ‘on-line registration’, ensuring ‘date and time stamping’. The on-line registration requests for any ‘current semester’ shall be **completed before the commencement of SEEs (Semester End Examinations) of the ‘preceding semester’**.
- 4.3 A student can apply for **on-line** registration, **only after** obtaining the ‘**written approval**’ from faculty advisor/counselor, which should be submitted to the college academic section through the Head of the Department. A copy of it shall be retained with the Head of the Department, Faculty Advisor/ Counselor and the student.
- 4.4 A student may be permitted to register for all the subjects/ courses in a semester as specified in the course structure with maximum additional subject(s)/course(s) limited to 6 Credits(any 2 elective subjects),based on **progress** and SGPA/CGPA, and completion of the ‘**pre-requisites**’ as indicated for various subjects/courses, in the department course structure and syllabus contents.
- 4.5 Choice for ‘**additional subjects/ courses**’, not more than any 2 elective subjects in any Semester, must be clearly indicated, which needs the specific approval and signature of the Faculty Advisor/Mentor/HOD.
- 4.6 If the student submits ambiguous choices or multiple options or erroneous entries during **on-line** registration for the subject(s) / course(s) under a given/ specified course group/ category as listed in the course structure, only the first mentioned subject/ course in that category will be taken into consideration.
- 4.7 Subject/ course options exercised through **on-line** registration are final and **cannot** be changed or inter -changed; further, alternate choices also will not be considered. However, if the subject/ course that has already been listed for registration by the Head of the Department in a semester could not be offered due to any inevitable or unexpected reasons,thenthestudentshallbeallowedtohavealternatethechoiceeitherforanewsubject (subject to offering of such a subject), or for another existing subject (subject to availability of seats). Such alternate arrangements will be made by the Head of the Department, with due notification and time-framed schedule, within **a week** after the commencement of class-work for that semester.
- 4.8 Dropping of subjects/courses may be permitted, only after obtaining prior approval from the faculty advisor/ counselor ‘within a period of 15 days’ from the beginning of the current semester.
- 4.9 **Open Electives:** The students have to choose three Open Electives (OE-I,II&III)from the list of Open Electives given by other departments. However, the student can opt for an Open Elective subject offered by his own (parent) department, if the student has not registered and not studied that subject under any category (Professional Core, Professional Electives, Mandatory Courses etc.) offered by parent department in any semester. Open Elective subjects already studied should not repeat/should not match with any category (Professional Core, Professional Electives, Mandatory Courses etc.) of subjects even in the forthcoming semesters.
- 4.10 **Professional Electives:** The students have to choose six Professional Electives (PE-I to VI) from the list of professional electives given.
- 5.0 **Subjects/ courses to be offered**
- 5.1 A subject/ course may be offered to the students, **only if** a minimum of 15 students opt for it.
- 5.2 More than **one faculty member** may offer the **same subject** (lab/ practical may be included

with the corresponding theory subject in the same semester) in any semester. However, selection of choice for students will be based on-‘**first come first serve** basis and CGPA criterion’ (i.e. the first focus shall be on early **on-line entry** from the student for registration in that semester, and the second focus, if needed, will be on CGPA of the student).

- 5.3 If more entries for registration of a subject come into picture, then the Head of the Department concerned shall decide whether or not to offer such a subject/course for **two (or multiple) sections**.
- 5.4 In case of options coming from students of other departments/ branches/ disciplines (not considering **open electives**), first **priority** shall be given to the student of the ‘**parent department**’.

6.0 Attendance requirements:

- 6.1 A student shall be eligible to appear for the semester end examinations, if the student acquires a minimum of 75% of attendance in aggregate of all the subjects/ courses (including attendance in mandatory courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab) for that semester. **Two periods** of attendance for each theory subject shall be considered, if the student appears for the mid-term examination of that subject.
- 6.2 Shortage of attendance in aggregate up to 10% (65% and above, and below 75%) in each semester may be condoned by the college academic committee on genuine and valid grounds, based on the student’s representation with supporting evidence.
- 6.3 A stipulated fee shall be payable for condonation of shortage of attendance.
- 6.4 Shortage of attendance below 65% in aggregate shall in **NO** case be condoned.
- 6.5 **Students whose shortage of attendance is not condoned in any semester are not eligible to take their end examinations of that semester. They get detained and their registration forth at semesters hall stand cancelled**, including all academic credentials (internal marks etc.) of that semester. **They will not be promoted to the next semester.** They may seek re-registration for all those subjects registered in that semester in which the student is detained, by seeking readmission into that semester as and when offered; if there are any professional electives and/ or open electives, the same may also be re-registered if offered. However, if those electives are not offered in later semesters, then alternate electives may be chosen from the **same** set of elective subjects offered under that category.
- 6.6 A student fulfilling the attendance requirement in the present semester shall not be eligible for readmission into the same class.

7.0 Academic Requirements

The following academic requirements have to be satisfied, in addition to the attendance requirements mentioned Item.6.

- 7.1 A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course, if student secures not less than 35% (21 marks out of 60 marks) in the semester end examination, and a minimum of 40%(40 marks out of 100 marks) in the sum total of the CIE(Continuous Internal Evaluation) and SEE(Semester End Examination) taken together; in terms of letter grades, this implies securing ‘**C**’ grade or above in that subject/course.

- 7.2** A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship (or) Seminar, if the student secures not less than 40% marks (i.e. 40 out of 100 allotted marks) in each of them. The student is deemed to have failed, if he (i) does not submit a report on Industry Oriented Mini Project/Internship, or (ii) not make a presentation of the same before the evaluation committee as per schedule, or (iii) secures less than 40% marks in Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship evaluations.

A student may reappear once for each of the above evaluations, when they are scheduled again; if the student fails in such ‘one reappearance’ evaluation also, the student has to reappear for the same in the next subsequent semester, as and when it is scheduled.

7.3 Promotion Rules

S. No.	Promotion	Conditions to be fulfilled
1	First year first semester to first year second semester	Regular course of study of first year first semester.
2	First year second semester to Second year first semester	(i) Regular course of study of first year second semester. (ii) Must have secured at least 20 credits out of 40 credits i.e., 50% credits up to first year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3.	Second year first semester to Second year second semester	Regular course of study of second year first semester.
4	Second year second semester to Third year first semester	(i) Regular course of study of second year second semester. (ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Third year first semester to Third year second semester	Regular course of study of third year first semester.
6	Third year second semester to Fourth year first semester	(i) Regular course of study of third year second semester. (ii) Must have secured at least 72 credits out of 120 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
7	Fourth year first semester to Fourth year second semester	Regular course of study of fourth year first semester.

- 7.4 A student(i) shall register for all courses/subjects covering 160 credits as specified and listed in the course structure, (ii) fulfills all the attendance and academic requirements for 160 credits, (iii) earn all 160 credits by securing $SGPA \geq 5.0$ (in each semester), and $CGPA \geq 5$ (at the end of 8 semesters), (iv) passes all the mandatory courses, to successfully complete the undergraduate programme. The performance of the student in these 160 credits shall be considered for the calculation of the final CGPA (at the end of undergraduate programme), and shall be indicated in the grade card / marks memo of IV-year II semester.
- 7.5 If a student registers for 'extra subjects' (in the parent department to other departments/branches of Engg.) other than those listed subjects totaling to 160 credits as specified in the course structure of his department, the performances in those 'extra subjects' (although evaluated and graded using the same procedure as that of the required 160 credits) will not be considered while calculating the SGPA and CGPA. For such 'extra subjects' registered, percentage of marks and letter grade alone will be indicated in the grade card / marks memo as a performance measure, subject to completion of the attendance and academic requirements as stated in regulations Items 6 and 7.1–7.4 above.
- 7.6 A student eligible to appear in the semester end examination for any subject/ course, but absent from it or failed (thereby failing to secure C grade or above) may reappear for that subject/ course in the supplementary examination as and when conducted. In such cases, internal marks (CIE) assessed earlier for that subject/course will be carried over, and added to the marks to be obtained in the SEE supplementary examination for evaluating performance in that subject.
- 7.7 A student **detained in a semester due to shortage of attendance may be readmitted in the same semester in the next academic year for fulfillment to face academic requirements.** The academic regulations under which a student has been re-admitted shall be applicable. Further, no grade allotments or SGPA/ CGPA calculations will be done for the entire semester in which the student has been detained.
- 7.8 A student detained **due to lack of credits, shall be promoted to the next academic year only after acquiring the required number of academic credits.** The academic regulations under which the student has been readmitted shall be applicable to him.
- 8.0 **Evaluation - Distribution and Weight age of Marks**
- 8.1 The performance of a student in every subject/course (including practicals and Project Stage – I & II) will be evaluated for 100 marks each, with 40 marks allotted for CIE (Continuous Internal Evaluation) and 60 marks for SEE (Semester End-Examination).
- 8.2 In CIE, for theory subjects, during a semester, there shall be two mid-term examinations. Each Mid-Term examination consists of two parts i) **Part – A** for 10 marks, ii) **Part–B** for 20 marks with a total duration of 2 hours as follows:
1. Mid Term Examination for 30 marks:
 - a. Part-A: Objective/quiz paper for 10 marks.
 - b. Part-B: Descriptive Paper For 20 Marks.

The objective/quiz paper is set with multiple choice, fill-in the blanks and match the following type of questions for a total of 10 marks. The descriptive paper shall contain 6

full questions out of which, the student has to answer 4 questions, each carrying 5 marks. The **average of the two Mid Term Examinations** shall be taken as the final marks for Mid Term Examination (for 30 marks).

The remaining 10 marks of Continuous Internal Evaluation are distributed as:

2. Assignment for 5 marks. (**Average of 2 Assignments** each for 5 marks)
3. Subject Viva-Voce/PPT/Poster Presentation/Case Study on a topic in the concerned subject for 5 marks.

While the first mid-term examination shall be conducted on 50% of the syllabus, the second mid-term examination shall be conducted on the remaining 50% of the syllabus.

Five (5) marks are allocated for assignments (as specified by the subject teacher concerned). The first assignment should be submitted before the conduct of the first mid-term examination, and the second assignment should be submitted before the conduct of the second mid-term examination. The average of the two assignments shall be taken as the final marks for assignment (for 5 marks).

Subject Viva-Voce/PPT/Poster Presentation/ Case Study on a topic in the subject concerned for 5 marks before II Mid-Term Examination.

- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Over all 40% of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

The student is eligible to write Semester End Examination of the concerned subject, if the student scores $\geq 35\%$ (14 marks) of 40 Continuous Internal Examination (CIE) marks.

In case, the student appears for Semester End Examination (SEE) of the concerned subject but has not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled in spite of appearing the SEE

There is NO Computer Based Test (CBT) for R22 Regulations.

The details of the end semester question paper pattern are as follows:

- 8.2.1** The semester end examinations (SEE), for theory subjects, will be conducted for 60 marks consisting of two parts viz. i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

- ☐ Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- ☐ Part-B consists of five questions (numbered from 2 to 6) carrying 10 marks each. Each of these questions is from each unit and may contain sub-questions. For each question there will be an “either” “or” choice, which means that there will be two questions from each unit and the student should answer either of the two questions.
- ☐ The duration of the Semester End Examination is 3 hours.

- 8.3** For practical subjects there shall be a Continuous Internal Evaluation (CIE) during the semester for 40 marks and 60 marks for semester end examination. Out of the 40 marks for internal evaluation:

1. A write-up on day-to-day experiment in the laboratory (in terms of a

- components/procedure, expected outcome) which shall be evaluated for 10 marks
2. **10 marks for viva-voce** (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.
 3. Internal practical examination conducted by the laboratory teacher concerned shall be evaluated for 10 marks.
 4. The remaining 10 marks are for Laboratory Project, which consists of the Design(or)Software/Hardware Model Presentation(or)App Development(or)Proto type Presentation submission which shall be evaluated after completion of laboratory course and before semester end practical examination.

The Semester End Examination shall be conducted with an external examiner and the laboratory teacher. The external examiner shall be appointed from the other colleges which will be decided by the principal of the College.

In the Semester End Examination held for 3 hours, total 60 marks are divided and allocated as shown below:

1. 10 marks for write-up
 2. 15 for experiment/program
 3. 15 for evaluation of results
 4. 10 marks for presentation on another experiment/program in them same laboratory course and
 5. 10 marks for viva-voice on a concerned laboratory course.
- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks)in CIE,35% of marks (i.e.21 marks out of 60)in SEE and Overall40%of marks(i.e.40 marks out of100 marks)both CIE and SEE marks put together.

The student is eligible to write Semester End Examination of the concerned subject, if the student scores $\geq 35\%$ (14 marks) of 40 Continuous Internal Examination (CIE) marks.

In case, the student appears for Semester End Examination (SEE) of the concerned subject but has not scored minimum 35% of CIE marks (14 marks out of 40 internal marks),his performance in that subject in SEE shall stand cancelled in spite of appearing the SEE.

8.4 The evaluation of courses having ONLY internal marks in I Year I Semester and II Year II Semester is as follows:

1. I Year I,II Semester course (ex., Elements of CE/ME/EEE/ECE/CSE/Seminar & etc): The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two MID-Term examinations is the final for 50marks.Student shall have to earn 40%,i.e 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than40% marks in this course. Seminar evaluation conducted through day-to-day evaluation of the students in presentation/ abstract.
2. II Year II Semester Real-Time(or)Field-based Research Project course:The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two Mid-Term examinations is the final

for 50marks. Student shall have to earn 40%, i.e. 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the internal committee as per schedule, or (iii) secures less than 40% marks in this course.

- 8.5** There shall be an Industry training (or) Internship (or) Industry oriented Mini-project (or) Skill Development Courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project in collaboration with an industry of their specialization. Students shall register for this immediately after II-Year II Semester Examinations and pursue it during summer vacation/semester break & during III Year without affecting regular coursework. Internship at reputed organization (or) Skill development courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in III-year II semester before end semester examination. It shall be evaluated for 100 external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project (or) Internship etc, Internal Supervisor and a Senior Faculty Member of the Department. There shall be NO internal marks for Industry Training (or) Internship (or) Mini-Project (or) Skill Development Courses (or) Paper Presentation in reputed journal (or) Industry Oriented Mini Project.
- 8.6** The UG project shall be initiated at the end of the IV Year I Semester and the duration of the project work is one semester. The student must present Project Stage – I during IV Year I Semester before II Mid examinations, in consultation with his Supervisor, the title, objective and plan of action of This Project work to the departmental committee for approval before commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can start this project work.
- 8.7** UG project work shall be carried out in two stages: Project Stage – I for approval of project before Mid-II examinations in IV Year I Semester and Project Stage–II during IV Year II Semester. Student has to submit project work report at the end of IV Year II Semester. The project shall be evaluated for 100 marks before commencement of SEE Theory examinations.
- 8.8** For Project Stage–I, the departmental committee consisting of the Head of the Department, project supervisor and a senior faculty member shall approve the project work to begin before II Mid-Term examination of IV Year I Semester. The student is deemed to be not eligible to register for the Project work, if he does not submit a portion Project Stage - I or does not make a presentation of the same before the evaluation committee as per schedule.
- A student who has failed may reappear once for the above evaluation, when it is scheduled again; if he fails in such ‘one reappearance’ evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.
- 8.9** For Project Stage – II, the external examiner shall evaluate the project work for 60 marks and the internal project committee shall evaluate it for 40 marks. Out of 40 internal marks, the departmental committee consisting of Head of the Department, Project Supervisor and a Senior Faculty Member shall evaluate the project work for 20 marks and Project Supervisor shall evaluate for 20 marks. The topics for Industry Oriented Mini

Project/Internship/SDC etc. and the main Project shall be different from the topic already taken. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together.

For conducting viva-voce of the project, Principal/COE selects an external examiner from the list of experts in the relevant branch submitted by the Principal of the College.

A student who has failed, may reappear once for the above evaluation, when it is scheduled again; if student fails in such 'one reappearance' evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

8.10 A student shall be given only one time chance to re-register for a maximum of two subjects in a semester:

- If the internal marks secured by a student in the Continuous Internal Evaluation marks for 40 (Sum of average of two mid-term examinations consisting of Objective & descriptive parts, Average of two Assignments & Subject Viva-voce/PPT/Poster presentation/Case Study on a topic in the concerned subject) are less than 35% and failed in those subjects.

A student must re-register for the failed subject(s) for 40 marks within four weeks of commencement of the class work in next academic year. In the event to the student taking this chance, his Continuous Internal Evaluation marks for 40 and Semester End Examination marks for 60 obtained in the previous attempt stand cancelled.

8.11 For mandatory non-credit Audit courses, the internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two MID-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e. 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than 40% marks in this course.

8.12 For Mandatory Non-Credit Courses offered in a Semester, after securing ≥ 65% attendance and has secured not less than 40% of marks in the sum Total. Then the student is **PASS** and will be qualified for the award of the degree. No marks or Letter Grade shall be allotted for these courses/activities. However, for non-credit courses 'Satisfactory' or 'Unsatisfactory' shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA.

9.0 Grading Procedure

9.1 Grades will be awarded to indicate the performance of students in each Theory Subject, Laboratory/Practicals/ Industry-Oriented Mini Project/Internship/SDC and Project Stage. Based on the percentage of marks obtained (Continuous Internal Evaluation plus Semester End Examination, both taken together) as specified in item 8 above, a corresponding letter grade shall be given.

9.2 A same measure of the performance of a student, a 10-point absolute grading system using the following letter grades (as per UGC/AICTE guidelines) and corresponding percentage of marks shall be followed:

% of Marks Secured in a Subject/Course (Class Intervals)	Letter Grade (UGC Guidelines)	GradePoints
Greater than or equal to 90%	O (Outstanding)	10
80 and less than 90%	A+ (Excellent)	9
70 and less than 80%	A (Very Good)	8
60 and less than 70%	B+ (Good)	7
50 and less than 60%	B (Average)	6
40 and less than 50%	C (Pass)	5
Below 40%	F (FAIL)	0
Absent	Ab	0

- 9.3 A student who has obtained an ‘F’ grade in any subject shall be deemed to have ‘**failed**’ and is required to reappear as a ‘supplementary student’ in the semester end examination, as and when offered. In such cases, internal marks in those subjects will remain the same as those obtained earlier.

To a student who has not appeared for an examination in any subject, ‘**Ab**’ grade will be allocated in that subject, and he is deemed to have ‘**Failed**’. A student will be required to reappear as a ‘supplementary student’ in the semester end

- 9.4 To a student who has not appeared for an examination in any subject, ‘**Ab**’ grade will be allocated in that subject, and he is deemed to have ‘**Failed**’. A student will be required to reappear as a ‘supplementary student’ in the semester end examination, as and when offered next. In this case also, the internal marks in those subjects will remain the same as those obtained earlier.

- 9.5 A letter grade does not indicate any specific percentage of marks secured by the student, but it indicates only the range of percentage of marks.

- 9.6 A student earns Grade Point (GP) in each subject/course, on the basis of the letter grade secured in that subject/ course. The corresponding ‘Credit Points’ (CP) are computed by multiplying the grade point with credits for that particular subject/course.

Credit Points (CP) = Grade Point (GP) x Credits For a course

- 9.7 A student passes the subject/ course only when **GP $\geq$ 5** (‘C’ grade or above)

- 9.8 The Semester Grade Point Average (SGPA) is calculated by dividing the sum of credit points (Σ CP) secured from all subjects/ courses registered in a semester, by the total number of credits registered during that semester. SGPA is rounded off to **two** decimal places. SGPA is thus computed as

$$\text{SGPA} = \{\sum_{i=1}^N C_i G_i\} / \{\sum_{i=1}^N C_i\} \dots \text{For each Semester,}$$

where ‘i’ is the subject indicator index (considering all subjects in a semester), ‘N’ is the no. of subjects ‘**registered**’ for the semester (as specifically required and listed under the course structure of the parent department), C_i is the no. of credits allotted to the i^{th} subject, and G_i represents the grade points (GP) corresponding to the letter grade awarded for that i^{th} subject.

- 9.9 The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student in all semesters considered for registration. The CGPA is the ratio of the total credit points secured by a student in **all** registered courses (of 160) in **all**

semesters, and the total number of credits registered in **all** the semesters. CGPA is rounded off to **two** decimal places. CGPA is thus computed from the I year II semester onwards at the end of each semester as per the formula

$$\text{CGPA} = \{ \sum_{j=1}^M C_j G_j \} / \{ \sum_{j=1}^M C_j \} \dots \text{for all S Semesters registered}$$

(i.e., up to and inclusive of S Semesters, $S \in \mathbb{N}$),

where 'M' is the TOTAL no. of Subjects (as specifically required and listed under the Course Structure of the parent Department) the Student has 'REGISTERED' from the 1st Semester onwards up to and inclusive of the Semester S (obviously $M > N$), 'j' is the Subject indicator index (takes into account all Subjects from 1 to S Semesters), C_j is the no. of Credits allotted to the jth Subject, and G_j represents the Grade Points (GP) corresponding to the Letter Grade awarded for that jth Subject. After registration and completion of I Year I Semester however, the SGPA of that Semester itself may be taken as the CGPA, as there are no cumulative effects.

Illustration of calculation of SGPA:

Course/Subject	Credits	Letter Grade	Grade Points	Credit Points
Course 1	4	A	8	4 x 8 = 32
Course 2	4	O	10	4 x 10 = 40
Course 3	4	C	5	4 x 5 = 20
Course 4	3	B	6	3 x 6 = 18
Course 5	3	A+	9	3 x 9 = 27
Course 6	3	C	5	3 x 5 = 15
	21			152

$$\text{SGPA} = 152/21 = 7.24$$

Illustration of Calculation of CGPA up to 3rd Semester:

Semester	Course/Subject Title	Credits Allotted	Letter Grade Secured	Corresponding Grade Point (GP)	Credit Points (CP)
I	Course 1	3	A	8	24
I	Course 2	3	O	10	30
I	Course 3	3	B	6	18
I	Course 4	4	A	8	32
I	Course 5	3	A+	9	27
I	Course 6	4	C	5	20
II	Course 7	4	B	6	24
II	Course 8	4	A	8	32
II	Course 9	3	C	5	15
II	Course 10	3	O	10	30
II	Course 11	3	B+	7	21
II	Course 12	4	B	6	24
II	Course 13	4	A	8	32
II	Course 14	3	O	10	30
III	Course 15	2	A	8	16

III	Course 16	1	C	5	5
III	Course 17	4	O	10	40
III	Course 18	3	B+	7	21
III	Course 19	4	B	6	24
III	Course 20	4	A	8	32
III	Course 21	3	B+	7	21
	Total Credits	69		Total Credit Points	518

$$\text{CGPA} = 518/69 = 7.51$$

The calculation process of CGPA illustrated above will be followed for each subsequent semester until 8th semester. The CGPA obtained at the end of 8th semester will become the final CGPA secured for entire B.Tech. programme.

- 9.10** For merit ranking or comparison purposes or any other listing, **only** the ‘**rounded off**’ values of the CGPAs will be used.
- 9.11** SGPA and CGPA of a semester will be mentioned in the semester Memorandum of Grades if all subjects of that semester are passed in the first attempt. Otherwise the SGPA and CGPA shall be mentioned only on the Memorandum of Grades in which sitting he passed his last exam in that semester. However, mandatory courses will not be taken into consideration.

10.0 Passing Standards

- 10.1** A student shall be declared successful or ‘passed’ in a semester, if he secures a $GP \geq 5$ (‘C’ grade or above) in every subject/course in that semester (i.e. when the student gets an SGPA ≥ 5.0 at the end of that particular semester); and he shall be declared successful or ‘passed’ in the entire undergraduate programme, only when gets a CGPA ≥ 5.00 (‘C’ grade or above) for the award of the degree as required.
- 10.2** After the completion of each semester, a grade card or grade sheet shall be issued to all the registered students of that semester, indicating the letter grades and credits earned. It will show the details of the courses registered (course code, title, no. of credits, grade earned, etc.) and credits earned. **There is NO exemption of credits in any case.**

11.0 Declaration of results

- 11.1** Computation of SGPA and CGPA are done using the procedure listed in 9.6 to 9.9.
- 11.2** For the final percentage of marks equivalent to the computed final CGPA, the following formula may be used.

$$\% \text{ of Marks} = (\text{final CGPA} - 0.5) \times 10$$

12.0 Award of Degree

- 12.1** A student who registers for all the specified subjects/ courses as listed in the course structure and secures the required number of 160 credits (with CGPA ≥ 5.0), within 8 academic years from the date of commencement of the first academic year, shall be declared to have ‘**qualified**’ for the award of B.Tech. degree in the branch of Engineering selected at the time of admission.

- 12.2 A student who qualifies for the award of the degree as listed in item 12.1 shall be placed in the following classes.
- 12.3 A student with final CGPA (at the end of the undergraduate programme) > 8.00 , and fulfilling the following conditions-shall be placed in '**First Class with Distinction**'. However, he/she
- (i) Should have passed all the subjects/courses in '**First Appearance**' within the first 4 academic years (or 8 sequential semesters) from the date of commencement of first year first semester.
 - (ii) Should not have been detained or prevented from writing the semester end examinations in any semester due to shortage of attendance or any other reason.
- A student not fulfilling any of the above conditions with final CGPA > 8 shall be placed in '**First Class**'.
- 12.4 Students with final CGPA (at the end of the undergraduate programme) ≥ 7.0 but < 8.00 shall be placed in '**First Class**'.
- 12.5 Students with final CGPA (at the end of the undergraduate programme) ≥ 6.00 but < 7.00 , shall be placed in '**Second Class**'.
- 12.6 All other students who qualify for the award of the degree (as per item 12.1), with final CGPA (at the end of the undergraduate programme) ≥ 5.00 but < 6 , shall be placed in '**pass class**'.
- 12.7 A student with final CGPA (at the end of the undergraduate programme) < 5.00 will not be eligible for the award of the degree.
- 12.8 Students fulfilling the conditions listed under item 12.3 alone will be eligible for award of '**Gold Medal**'.
- 12.9 **Award of 2-Year B.Tech. Diploma Certificate**
- 1. A student is awarded a 2-Year UG Diploma Certificate in the concerned engineering branch on completion of all the academic requirements and earned all the 80 credits (within 4 years from the date of admission) up to B. Tech. – II Year – II Semester, if the student wants to exit the 4-Year B.Tech. program. The student **once opted and awarded for 2-Year UG Diploma Certificate, the student will not be permitted to join** in B. Tech. III Year – I Semester and continue for completion of remaining years of study for 4-Year B. Tech. Degree.
 - 2. A student may be permitted to take one year break after completion of II Year – II Semester or B. Tech. – III Year – II Semester (with Principal permission through the HOD of the department well in advance) and can re-enter the course in **next Academic Year in the same college** and complete the course on fulfilling all the academic credentials within a stipulated duration i.e. double the duration of the course (Ex. within 8 Years for 4-Year program).
- 13.0 **Withholding of results**
- 13.1 If the student has not paid the fees to the College at any stage, or has dues pending due to any reason whatsoever, or if any case of indiscipline is pending, the result of the student may be withheld, and the student will not be allowed to go into the next higher semester. The award or issue of the degree may also be withheld in such cases.

14.0 Transitory Regulations

A. For students detained due to shortage of attendance:

1. A Student who has been detained in I year of R18 Regulations due to lack of attendance, shall be permitted to join I year I Semester of AR22 Regulations and he is required to complete the study of B.Tech./B. Pharmacy programme within the stipulated period of eight academic years from the date of first admission in I Year.
2. A student who has been detained in any semester of II, III and IV years of R18 regulations for want of attendance, shall be permitted to join the corresponding semester of AR22 Regulations and is required to complete the study of B.Tech./B. Pharmacy within the stipulated period of eight academic years from the date of first admission in I Year. The AR22 Academic Regulations under which a student has been readmitted shall be applicable to that student from that semester. See rule (C) for further Transitory Regulations.
3. For students detained due to shortage of credits: A student of MLR18 Regulations who has been detained due to lack of credits, shall be promoted to the next semester of AR22 Regulations only after acquiring the required number of credits as per the corresponding regulations of his/her first admission. The total credits required are 160 including both MLR18 & AR22 regulations. The student is required to complete the study of B.Tech. within the stipulated period of eight academic years from the year of first admission. The AR22 Academic Regulations are applicable to a student from the year of readmission. See rule (C) for further Transitory Regulations.

B. For readmitted students in AR 22 Regulations:

4. A student who has failed in any subject under any regulation has to pass those subjects in the same regulations.
5. The maximum credits that a student acquires for the award of degree, shall be the sum of the total number of credits secured in all the regulations of his/her study including AR22 Regulations. **There is NO exemption of credits in any case.**
6. If a student is readmitted to AR22 Regulations and has any subject with 80% of syllabus common with his/her previous regulations, that particular subject in AR22 Regulations will be substituted by another subject to be suggested by the Concerned BOS.

15.0 Student Transfers

- 15.1 There shall be no branch transfers after the completion of the admission process.
- 15.2 There shall be no transfers from one college/stream to another within the constituent colleges and units of Jawaharlal Nehru Technological University Hyderabad.
- 15.3 The students seeking transfer to colleges affiliated to JNTUH from various other Universities/institutions have to pass the failed subjects which are equivalent to the subjects of JNTUH, and also pass the subjects of JNTUH which the students have not studied at the earlier institution. Further, though the students have passed some of the subjects at the earlier institutions, if the same subjects are prescribed in different semesters of JNTUH, the students have to study those subjects in JNTUH in spite of the fact that those subjects are repeated.
- 15.4 The transferred students from other Universities/Institutions to College who are on rolls are to be provided one chance to write the CBT (for internal marks) in the **equivalent subject(s)** as per the clearance letter used by the University.

- 15.5** The college has to provide one chance to write the internal examinations in the **equivalent subject(s)** to the students transferred from other universities/institutions to JNTUH autonomous affiliated colleges who are on rolls, as per the clearance (equivalence) letter issued by the University.

16.0 Malpractice Prevention Committee

A malpractice prevention committee shall be constituted to examine and punish the students who involve in malpractice / indiscipline in examinations. The committee shall consist of:

- a) Controller of examinations - Chairman
- b) Addl. Controller of examinations.- Member Convener
- c) Subject expert - member
- d) Head of the department of which the student belongs to. - Member
- e) The invigilator concerned - member

The committee shall conduct the meeting after taking explanation of the student and punishment will be awarded by following the malpractice rules meticulously.

Any action on the part of a candidate at the examination like trying to get undue advantage in the performance at examinations or trying to help another, or derive the same through unfair means is punishable according to the provisions contained hereunder. The involvement of the Staff who are in charge of conducting examinations, valuing examination papers and preparing / keeping records of documents relating to the examinations, in such acts (inclusive of providing incorrect or misleading information) that infringe upon the course of natural justice to one and all concerned at the examination shall be viewed seriously and will be recommended for appropriate punishment after thorough enquiry.

17.0 Scope

- 17.1** The academic regulations should be read as a whole, for the purpose of any interpretation.
- 17.2** In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Chairman of Academic Council is final.
- 17.3** The College may change or amend the academic regulations, course structure or syllabi at any time, and the changes or amendments made shall be applicable to all students with effect from the dates notified by the college authorities.
- 17.4** Where the words “he”, “him”, “his”, occur in the regulations, they include “she”, “her”, “hers”.

ACADEMIC REGULATIONS FOR B.TECH (LATERAL ENTRY SCHEME)
FROM THE AY 2023-24

1. Eligibility for the award of B.Tech Degree(LES)

The LES students after securing admission shall pursue a course of study for not less than three academic years and not more than six academic years.

2. The students shall register for 120 credits and secure 120 credits with CGPA ≥ 5 from II year to IV-year B.Tech. programme (LES) for the award of B.Tech. degree.

3. The students, who fail to fulfil the requirement for the award of the degree in six academic years from the year of admission, shall forfeit their seat in B.Tech.

4. The attendance requirements of B. Tech. (Regular) shall be applicable to B.Tech. (LES).

5. Promotion rule

S. No	Promotion	Conditions to be fulfilled
1	Second year first semester to second year second semester	Regular course of study of second year first semester.
2	Second year second semester to third year first semester	(i) Regular course of study of second year second semester. (ii) Must have secured at least 24 credits out of 40 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3	Third year first semester to third year second semester	Regular course of study of third year first semester.
4	Third year second semester to fourth year first semester	i) Regular course of study of third year second semester. ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Fourth year first semester to fourth year second semester	Regular course of study of fourth year first semester.

6. All the other regulations as applicable to B. Tech. 4-year degree course (Regular) will hold good for B. Tech. (Lateral Entry Scheme).

7. LES students are not eligible for 2-Year B. Tech. Diploma Certificate.

Malpractices Rules

Disciplinary Action For / Improper Conduct in Examinations

	Nature of Malpractices/ Improper conduct	Punishment
	If the student:	
1. (a)	Possesses or keeps accessible in examination hall, any paper, notebook, programmable calculators, cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which student is appearing but has not made use of (material shall include any marks on the body of the student which can be used as an aid in the subject of the examination).	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives It from any other student or all yor by any other body language methods or communicates through cell phones with any student or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the students involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2.	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the student is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The hall ticket of the student is to be cancelled
3.	Impersonates any other student in connection with the examination.	The student who has impersonated shall be expelled from the examination hall. The student is also debarred and forfeits the seat. The performance of the original student who has been impersonated, shall be cancelled in all the subjects of the examination(including practicals and project work)already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.

4.	Smuggles in the answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.
5.	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject.
6.	Refuses to obey the orders of the chief superintendent/COE/ACoE/any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the college campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct has the tendency to disrupt the orderly conduct of the examination.	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the student(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The students also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.
7.	Leaves the exam hall taking away answer script or intentionally tears off the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.

8.	Possesses any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat.
9.	If a student of the college, who is not a student for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat. Person(s) who do not belong to the college will be handed over to the police and a police case will be registered against them.
10.	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared for including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year.
11.	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject and all other subjects the student has appeared for including practical examinations and project work of that semester/year examinations.
12	If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the Principal for further action to award a suitable punishment.	

Malpractices identified by squad or special invigilators

1. Punishments to the students as per the above guidelines.
2. Punishment for staff: (if the squad reports that the staff is also involved in encouraging malpractices)
 - a. A show-cause notice shall be issued to the staff.
 - b. Impose a suitable fine on the staff.

* * * * *

COURSE STRUCTURE

Course Structure

B. TECH – Computer Science and Engineering (AIML)

Regulations: R22

I B.Tech.- I Semester									
Induction program for one weeks									
Code	Course Title	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6BS01	Linear Algebra and Calculus	BSC	3	1	-	4	40	60	100
A6BS07	Applied Physics	BSC	3	1	-	4	40	60	100
A6CS02	Programming for Problem Solving	ESC	3	-	-	3	40	60	100
A6ME02	Engineering Drawing	ESC	1	-	3	2.5	40	60	100
A6CS03	Programming for Problem Solving Lab	ESC	-	-	3	1.5	40	60	100
A6BS08	Applied Physics Lab	BSC	-	-	3	1.5	40	60	100
A6ME04	Engineering workshop	ESC	-	-	3	1.5	40	60	100
A6AI01	Elements of Artificial Intelligence	ESC	1	-	2	2	50	-	50
TOTAL			11	2	14	20	330	420	750
I B.Tech.- II Semester									
Code	Course Title	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6BS02	Numerical Methods and Integral Transforms	BSC	3	1	-	4	40	60	100
A6HS01	English for Skill Enhancement	HSMC	3	-	-	3	40	60	100
A6BS09	Engineering Chemistry	BSC	3	1	-	4	40	60	100
A6EE60	Basic Electrical and Electronics Engineering	BSC	3	-	-	3	40	60	100
A6EC03	Electronic Devices and Circuits	ESC	2	-	-	2	40	60	100
A6HS02	English Language and Communication Skills Lab	HSMC	-	-	3	1.5	40	60	100
A6CS04	Python Programming Lab	ESC	-	-	3	1.5	40	60	100
A6HS04	Seminar	HSMC	-	-	2	1	50	-	50
A6BS11	Environmental Science	MC	3	-	-	0	50	-	50
TOTAL			17	2	8	20	380	420	800

II B.Tech.- I Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6CS08	Discrete Mathematics	BSC	3	-	-	3	40	60	100
A6CS07	Software Engineering	PCC	3	-	-	3	40	60	100
A6CS28	Digital Electronics Computer Organization	ESC	3	-	-	3	40	60	100
A6IT02	Object Oriented Programming through JAVA	ESC	3	-	-	3	40	60	100
A6CS05	Data structures	PCC	3	-	-	3	40	60	100
A6CS06	Data Structures Lab	PCC	-	-	3	1.5	40	60	100
A6IT03	Object Oriented Programming through JAVA Lab	ESC	-	-	3	1.5	40	60	100
A6AI04	Skill Development (Android)	PCC	-	-	4	2	-	100	100
Total			15	00	10	20	280	520	800
Mandatory Course (Non-Credit)									
A6HS06	Constitution of India	MC	2	-	-	-	50	-	50

II B.Tech.- II Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6BS03	Computer Oriented Statistical Methods	BSC	3	1	-	4	40	60	100
A6HS08	Business Economics and Financial Analysis	HSC	3	-	-	3	40	60	100
A6AI02	Artificial Intelligence	PCC	2	-	-	2	40	60	100
A6CS09	Database Management Systems	PCC	3	-	-	3	40	60	100
A6CS15	Design and Analysis of Algorithms	PCC	3	-	-	3	40	60	100
A6AI03	Artificial Intelligence Lab	PCC	-	-	3	1.5	40	60	100
A6CS10	Database Management Systems Lab	PCC	-	-	3	1.5	40	60	100
A6AI05	Real-time Research Project / Societal Related Project	PWC	-	-	4	2	50	-	50
Total			14	1	10	20	330	420	750
Mandatory Course (Non-Credit)									
A6HS05	Gender Sensitization	MC	-	-	2	-	50	-	50

III B.Tech.- I Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6IT11	Automata and Compiler Design	PCC	3	-	-	3	40	60	100
A6AI06	Machine Learning	PCC	3	-	-	3	40	60	100
A6AI07	Web Programming	PCC	3	-	-	3	40	60	100
	Professional Elective-I	PEC	3	-	-	3	40	60	100
A6CS11	Operating Systems	PCC	3	-	-	3	40	60	100
A6AI08	Web Programming Lab	PCC	-	-	3	1.5	40	60	100
A6AI09	Machine Learning Lab	PCC	-	-	3	1.5	40	60	100
A6IT15	Linux Programming Lab	PCC	-	-	2	1	40	60	100
A6AI16	Independent Study/Moocs/Certification	PWC	-	-	2	1	-	100	100
Total			15	00	08	20	280	520	900
Mandatory Course (Non-Credit)									
A6HS10	Human Values and Professional Ethics	MC	2	-	-	-	50	-	50
III B.Tech.- II Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6CS18	Computer Networks	PCC	3	-	-	3	40	60	100
A6AI14	Natural Language Processing	PCC	3	-	-	3	40	60	100
	Professional Elective-II	PEC	3	-	-	3	40	60	100
	Professional Elective-III	PEC	3	-	-	3	40	60	100
	Open Elective-I	OEC	3	-	-	3	40	60	100
	Professional Elective - II Lab	PEC	-	-	3	1.5	40	60	100
A6AI15	Natural Language Processing Lab	PCC	-	-	3	1.5	40	60	100
A6AI10	Mini Project	PCC	-	-	4	2	-	100	100
Total			15	00	10	20	280	520	800
Mandatory Course (Non-Credit)									
A6BS11	Environmental Science	MC	3	0	0	0	50	-	50

Note: Environmental Science (A6BS11) will be applicable for Lateral Entry students

IV B.Tech.- I Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6AI17	Deep Learning	PCC	3	-	-	3	40	60	100
A6AI18	Knowledge Representation and Reasoning	PCC	3	-	-	3	40	60	100
	Professional Elective-IV	PEC	3	-	-	3	40	60	100
	Professional Elective-V	PEC	3	-	-	3	40	60	100
	Open Elective-II	OEC	3	-	-	3	40	60	100
A6AI19	Deep Learning Lab	PCC	-	-	2	1	40	60	100
	Research Project Phase - 1	PCC	-	-	8	4	100	-	100
Total			15	00	10	20	340	360	700

IV B.Tech. - II Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6HS15	Organizational Behavior	HSMC	3	-	-	3	40	60	100
	Professional Elective-VI	PEC	3	-	-	3	40	60	100
	Open Elective - III	OEC	3	-	-	3	40	60	100
	Research Project Phase - 2	PWC	-	-	22	11	40	60	100
Total			09	00	22	20	160	240	400

SEM	I	II	III	IV	V	VI	VII	VIII	Total
Credits	20	20	20	20	20	20	20	20	160

*Environmental Studies have to be taken by LE in 3-2

**MC: Satisfactory / Unsatisfactory

PROFESSIONAL ELECTIVES			
PE - I		PE – II	
A6DS29	Cloud Computing	A6DS02	Data Analytics Using R
A6AI11	Data Mining Techniques	A6DS25	Object Oriented Analysis and Design
A6AI12	Image Processing	A6DS23	Internet of Things
A6AI13	Software Testing Methodologies	A6IT10	Full Stack Development
PE - III		PE - IV	
A6CT05	Scripting Languages	A6CY22	Mobile Computing
A6AI28	Pattern Recognition	A6CT06	Robotic Process Automation
A6AI26	Neural Networks	A6AI21	Cognitive Computing
A6IT26	Information Retrieval Systems	A6AI22	Augmented Reality and Virtual Reality
PE - V		PE - VI	
A6AI23	Computer Vision	A6CY10	Ethical hacking
A6AI24	Prompt Engineering	A6DS22	Business Intelligence & Analysis
A6AI25	Quantum Computing	A6IT33	e-Commerce
A6CY06	Cryptography and Network Security	A6CY25	Block Chain Technology
PE – II LAB			
A6DS03	Data Analytics USING R Lab		
A6DS26	Object Oriented Analysis and Design Lab		
A6DS24	Internet of things Lab		
A6IT14	Full Stack Development Lab		

OPEN ELECTIVES

OPEN ELECTIVE – I			
S. No.	Course Code	Course Name	Offering Department
1	A6AE19	Fundamentals of Avionics	Aeronautical Engineering
2	A6AE20	Introduction to Aerospace Technology	
3	A6CS48	Software Project Management Fundamentals	Computer Science and Engineering
4	A6CS49	Data Analytics	
5	A6EC55	Microprocessors and Interfacing	Electronics & Communication Engineering
6	A6EC56	Principles of Communications	
7	A6EE54	Electrical Wiring and Safety Measures	Electrical & Electronics Engineering
8	A6EE55	Electrical Materials	
9	A6IT28	Ad hoc & Sensor Networks	Information Technology
10	A6IT35	Object Oriented Programming	
11	A6ME55	Basic Mechanical Engineering	Mechanical Engineering
12	A6ME60	Fundamentals of Engineering Materials	
13	A6HS08	Business Economics and Financial Analysis	HSM
14	A6HS09	Basics of Entrepreneurship	

OPEN ELECTIVE – II			
S. No.	Course Code	Course Name	Offering Department
1	A6AE25	Introduction to Jets and Rockets	Aeronautical Engineering
2	A6AE26	Non-Destructive Testing Methods	
3	A6CS09	Database Management System	Computer Science and Engineering
4	A6CS50	Data Visualization Techniques	
5	A6EC57	Micro controllers and Applications	Electronics & Communication Engineering
6	A6EC58	Fundamentals of Image processing	
7	A6EE56	Neural Networks and Fuzzy Logic	Electrical & Electronics Engineering
8	A6EE57	Energy Storage Systems	
9	A6IT30	Security Testing	Information Technology
10	A6IT36	Human Computer Interaction	
11	A6ME65	Fundamentals of Operations Research	Mechanical Engineering
12	A6ME70	Industrial Engineering & Management	
13	A6HS11	Advanced Entrepreneurship	HSM

OPEN ELECTIVE - III			
S. No.	Course Code	Course Name	Offering Department
1	A6AE40	Introduction to Aircraft Industry	Aeronautical Engineering
2	A6AE41	Unmanned Aerial Vehicles	
3	A6CS51	Artificial Intelligence and Applications	Computer Science and Engineering
4	A6CS36	Cyber Security & Cyber laws	
5	A6EC59	Introduction to Sensors and Actuators	Electronics & Communication Engineering
6	A6EC60	Introduction to Computer Vision	
7	A6EE58	Solar Energy and Applications	Electrical & Electronics Engineering
8	A6EE59	Instrumentation and Control	
9	A6IT37	Introduction to Computer Networks	Information Technology
10	A6IT31	High Performance Computing	
11	A6ME75	Renewable Energy Sources	Mechanical Engineering
12	A6ME80	Elements of Electric and Hybrid vehicles	
13	A6HS12	Management Science	HSM

I B.TECH I SEMESTER SYLLABUS

LINEAR ALGEBRA & CALCULUS

I B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS01	BSC	L	T	P	C	CIA	SEE	Total
		3	1	0	4	40	60	100
Contact Classes: 44	Tutorial Classes: 08	Practical Classes: Nil			Total Classes: 52			

Course Objectives

The course should enable the students to:

1. Concept of Rank of a matrix, Consistency and solving system of linear equations.
2. Concept of eigenvalues, eigenvectors and diagonalization of the matrix.
3. The concept of differential equations and solve them using appropriate methods.
4. Evaluate multiple integrals and improper integrals
5. The partial derivatives of several variable functions.

Course Outcomes

At the end of the course, student will be able to:

1. Solve the system of linear equations using rank of the matrices.
2. Find the Eigenvalues and Eigenvectors of a matrix
3. Identify the different types of differential equations and solve them using appropriate methods.
4. Evaluate the improper integrals using beta and gamma functions.
5. Find the Maxima and Minima of several variable functions.

UNIT-I	MATRICES AND THEIR APPLICATIONS	Classes: 08
Real matrices: Symmetric-skew-symmetric and orthogonal matrices –Complex matrices: Hermitian, Skew –Hermitian and Unitary matrices –Elementary row and column transformations –Elementary matrix-Finding rank of a matrix by reducing to Echelon form and Normal form-Finding the inverse of a matrix using elementary row/column transformations (Gauss-Jordan method)-Consistency of system of linear equations (homogeneous and non-homogeneous) using the rank of a matrix –Solving m n and n n linear system of equations by Gauss Elimination, Gauss seidel Method		
UNIT-II	EIGEN VALUES, EIGEN VECTORS	Classes: 08
Eigenvalues and Eigenvectors and its properties(without proof), Cayley-Hamilton theorem (Statement and verification)- Finding inverse and powers of a matrix by Cayley-Hamilton theorem, Diagonalization of matrices. The n-dimensional vector space(only definition and examples)-linear dependence of vectors-Basis and dimension		
UNIT-III	ORDINARY DIFFERENTIAL EQUATIONS AND THEIR APPLICATIONS	Classes: 10
Introduction- Exact and reducible to Exact differential equations-Newton's Law of cooling-Law of Growth and Decay. Linear differential equations of second and higher order with constant coefficients - Non-Homogeneous term of the type $Q(x) = e^{ax}$, $\sin ax$, $\cos Ax$, $e^{ax}v(x)$, $x^n v(x)$ - Method of variation of parameters L-C-R Circuits.		
UNIT-IV	MULTIPLE INTEGRALS, BETA AND GAMMA FUNCTIONS	Classes: 10
Double and triple integrals (Cartesian and polar), Change of order of integration in double integrals, Change of variables (Cartesian to polar) in double integrals. Finding the area and volume of a region using double and triple integrals. Beta- Gamma Functions and their Properties-Relation between them- Evaluation of improper integrals using Gamma and Beta functions.		

UNIT-V	CALCULUS OF SEVERAL VARIABLES	Classes: 08
Limit, Continuity - Partial derivative- Partial derivatives of higher order -Total derivative - Chain rule, Jacobians-functional dependence & independence. Applications: Maxima and Minima of functions of two variables without constraints and Lagrange's method (with constraints).		
Text Books:		
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.		
Reference Books:		
G.B.Thomas, calculus and analytic geometry,9th Edition, Pearson Reprint 2006. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008. E.L.Ince, Ordinary Differential Equations,Dover publications,1958.		
Web references:		
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/		
E -Text Books:		
1. https://www.e-booksdirectory.com/details.php?ebook=10166		
MOOCS Course:		
1. https://swayam.gov.in/ 2. https://onlinecourses.nptel.ac.in/		

APPLIED PHYSICS

I B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS07	BSC	L	T	P	C	CIA	SEE	Total
		3	1	0	4	40	60	100
Contact Classes: 50	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 50			

COURSE OBJECTIVES

The course should enable the students to:

1. Learn the basic principles of quantum physics and its applications
2. Understand the formation of energy bands and atomic structure in solids for material classification
3. Understand the underlying mechanism involved in construction and working properties of different types of semiconductor devices
4. Learn the basic principles of laser and optical fiber in information technology
5. Identify the importance of nano scale and various fabrication and characterization techniques and quantum computations in engineering applications

COURSE OUTCOMES

At the end of the course students will be able to:

1. Analyze the microscopic properties of materials using principles of quantum physics for engineering applications
2. Explain the behavior of different electronic materials based on the concepts of band theory
3. Apply the knowledge of Solar PV cells for choice of materials in efficient alternate energy generation
4. Gain the knowledge of production of laser and usage of fibers in fiber optic communication technology
5. Comprehend the knowledge of quantum physics in quantum computation for secure information technology

UNIT-I	Quantum Physics	Classes: 10
Black body radiation, Stefan-Boltzmann's law, Planck's radiation law (Qualitative treatment), Photoelectric effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation, Physical Significance of the wave Function, Particle in One Dimensional Potential Box.		
UNIT-II	Electronic properties of Materials & Band theory of solids	Classes: 10
Electronic properties of Materials: Classical free electron theory and Quantum free electron theories of metals, success and drawbacks, Bloch theorem, Kronig-Penny model (Qualitative treatment), E-k diagram, effective mass of electron. Band theory of solids: Origin of energy band formation in solids, Fermi energy level, Fermi-Dirac distribution law, Classification of materials as conductors, insulators and semiconductors.		
UNIT-III	Semi -conductors & Semiconductor Devices	Classes: 08
Semiconductors: Intrinsic and Extrinsic Semiconductors, formation of PN junction diode and its V-I characteristics, Direct and Indirect band gap semiconductors, Hall effect and its applications. Semiconductor Devices: Construction, working and V-I characteristics of PIN Diode, LED, Solar cell and their applications.		

UNIT-IV	Laser & Fiber Optics	Classes: 12
Laser: Characteristics of Laser, Absorption, Spontaneous and Stimulated emission of radiations. Lasing actions- Pumping mechanism, Meta stable state and Population inversion, Nd-YAG laser, CO2 laser, Applications of lasers in different fields. Fiber Optics: Structure of fibers, Total Internal Reflection, Acceptance angle – Numerical Aperture, Types of fibers- SI and GI fibers, Single and Multimode fibers - SMSI, MMSI, MMGI, Fiber Optic Communication system, Signal Degradation - Attenuation mechanism, Dispersion, Applications of fibers in different fields.		
UNIT-V	Physics of Quantum computing & Quantum gates	Classes: 10
Physics of Quantum computing: Idea of classical bits and qubits, advantages with qubits over classical bits, Bloch vector representation of state of qubits. Quantum gates-Single qubit logic gates- Pauli X, Y, Z and Hadmard gate in matrix form, Two level gates-CNOT and SWAP gates and representation in matrix form, Comments on No cloning theorem, Entanglement, Quantum Teleportation – Basic Idea, Quantum Key distribution protocol - BB84 protocol.		
Text Books:		
1. P.K Palanisamy, Engineering Physics, Sitech Publications, 2013, IVth Ed. 2. Nielsen M. A.I. L Chung, Quantum Computation & Quantum Information, Cambridge Univ. Press. 3. M. N. Avadhanulu, P.G. Kshirsagar & TVS Arun Murthy" A Textbook of Engineering Physics"- S. Chand Publications, 11th Edition 2019.		
Reference Books:		
1. Quantum Physics, H.C. Verma, TBS Publication, 2nd Edition 2012 2. Fundamentals of Physics – Halliday, Resnick and Walker, John Wiley & Sons, 11th Edition, 2018. 3. Engineering Physics by Shatendra Sharma and Jyotsna Sharma, Pearson Publication, 2019 4. B.K Pandey and Chaturvedi, Engineering Physics, Cengage Learning, 2nd Edition, 2022		

PROGRAMMING FOR PROBLEM SOLVING

I B. TECH- I SEMESTER

I B. TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS02	ESC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 64	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 64			
Course Objectives The course should enable the students to:								
1. Familiarize with the syntax and semantics of C programming language.								
2. Learn the usage of structured programming approaches in solving problems.								
3. Use arrays, pointers, strings and structures in solving problems.								
4. Understand how to solve problems related to matrices, Searching and sorting.								
5. Understand how to use files to perform read and write operations.								
Course Outcomes At the end of the course students will be able to:								
1. Apply algorithmic thinking to understand, define and solve problems								
2. Develop computer programs using programming constructs and control structures and to use arrays to develop C programs								
3. Decompose a problem into functions to develop modular reusable code and to use pointers to solve complex problems.								
4. Use Strings and structures to formulate algorithms and programs.								
5. Use FILE to perform read and write operations.								
UNIT-I	INTRODUCTION - PROBLEM SOLVING AND ALGORITHMIC THINKING & INTRODUCTION TO C LANGUAGE					Classes: 12		
Algorithm -Definition, Characteristics of Algorithm. Constituents of algorithms: - Sequence, Selection and Repetition. Algorithm with Example: Roots Of a Quadratic Equations, Minimum And Maximum Numbers of a Given Set, Given number is prime number or not, given integer is palindrome or not, etc. Flowchart/PseudoCode with examples. Introduction To C Language: Structure of C Program, Data Types, data input and output statements, Operators, P precedence and Associativity of operators, Evaluation of Expressions, Type Conversions In Expressions.								
UNIT-II	CONTROL STRUCTURES AND ARRAYS					Classes: 15		

Control structures: Decision statements: if and switch statement; Loop control statements: while, for and do while loops, Jump statements: break, continue, goto statements. Arrays: Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi dimensional arrays		
UNIT-III	FUNCTIONS AND POINTERS	Classes: 17
Functions: Function definition, Types of Functions: User defined and built-in Functions, Advantages of User Defined Functions. Parameter passing in functions: Call by value, Call by reference, Passing arrays to functions, Recursion as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Towers of Hanoi etc., Storage classes. Pointers: Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, Functions returning pointers, Dynamic memory allocation.		
UNIT-IV	STRINGS AND USER DEFINED DATA TYPES	Classes: 10
Strings: Arrays of characters, variable length character strings, inputting character strings, character library functions, String Handling Functions, Arrays Of Strings Structures and Unions: Structure definition, initialization, accessing structures, nested structures, arrays of structures, Structures and functions, Self-referential structures, unions, type def, enumerations.		
UNIT-V	FILE HANDLING , SEARCHING AND SORTING	Classes: 10
File Handling: Command Line Arguments, File Modes, Basic File Operations Read, Write And Append, Example Programs. Random Access Using fseek, ftell and rewind Functions. Basic Searching And Sorting Algorithms: Linear and Binary Search, Bubble Sort, Insertion Sort, Quick Sort.		

Text Books:
1. B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition) 2. Byron Gottfried, "Programming with C", Schaum's Outlines Series, McGraw Hill Education, 3rd edition, 2017. 3. Programming in C E. Balagurusamy Edition 3 Publisher Tata McGraw-Hill Publishing, 1990
Reference Books:
1. W. Kernighan Brian, Dennis M. Ritchie, "The C Programming Language", PHI Learning, 2nd Edition, 1988. 2. Yashavant Kanetkar, "Exploring C", BPB Publishers, 2nd Edition, 2003. 3. Schildt Herbert, "C: The Complete Reference", Tata McGraw Hill Education, 4th Edition, 2014.4. R. S. Bichkar, "Programming with C", Universities Press, 2nd Edition, 2012. 4. Dey Pradeep, Manas Ghosh, "Computer Fundamentals and Programming in C", Oxford University Press, 2nd Edition, 2006. 5. Stephen G. Kochan, "Programming in C", Addison-Wesley Professional, 4th Edition, 2014.

Web References:

1. https://en.wikipedia.org/wiki/Computational_thinking
2. <https://nptel.ac.in/courses/106/104/106104128/>
3. <https://en.cppreference.com/w/c/language>
4. <https://www.learn-c.org/>

ENGINEERING DRAWING

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME02	ESC	L	T	P	C	CIA	SEE	Total
		1	0	3	2.5	40	60	100

COURSE OUTCOMES:

At the end of the course the student should be able to:

1. Understand various commands and create drawings in Autocad.
2. Construct various engineering curves and know their importance.
3. Prepare orthographic projections of objects by visualizing them in different positions.
4. Solve the problem of projections of planes and solids in different positions.
5. Construct the isometric view into orthographic views and vice versa.

UNIT-I

INTRODUCTION

Introduction to Engineering Drawing: Principles and their significance.

INTRODUCTION TO COMPUTER AIDED DRAFTING: Initial Setup Commands, Draw Commands, modify commands, 2D Drawings-Simple Exercises.

UNIT-II

ENGINEERING CURVES

Engineering Curves: Ellipse, Parabola, and Hyperbola (General Method only), Involute.

UNIT-III

ORTHOGRAPHIC PROJECTION

Principles of Orthographic Projections –Projections of points.

Projection of lines inclined to both planes (First angle projection only).

UNIT-IV

PROJECTION OF PLANES AND SOLIDS

PROJECTIONS OF PLANES: Projections of regular planes inclined to one plane.

PROJECTION OF SOLIDS: Solids inclined to one plane (Regular solids).

UNIT-V

ISOMETRIC DRAWINGS

Isometric view: Drawing Isometric circles, Dimensioning Isometric Objects.

Conversion of Isometric view to Orthographic views and Orthographic to isometric view.

Text Books:

1. Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House.
2. Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication.

Reference Books:

- 1.D.M. Kulkarni, A.P.Rastogi, A.K. Sarka "Engineering Graphics with AutoCAD" PHI publications,2013.
- 2.Narayana, K.L. & P Kanniah (2008), Text book on Engineering Drawing, Scitech Publishers.
- 3.Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education.

PROGRAMMING FOR PROBLEM SOLVING LAB								
I B.TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS03	ESC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100
Contact Classes: Nil	Tutorial Classes: Nil	Practical Classes: 36			Total Classes: 36			
COURSE OBJECTIVES The course should enable the students to:								
1. To be familiarize with flowgorithm to solve simple problems 2. To develop program to solve basic problems by understanding basic concepts in C like operators, control statements etc. 3. To develop modular, reusable and readable C Programs using the concepts like functions, arrays, strings, pointers and structures.								
COURSE OUTCOMES At the end of the course student will be able to								
1. Solves simple mathematical problems using Flowgorithm. 2. Corrects syntax errors as reported by the compilers and logical errors encountered at runtime Develop program by using decision making and looping constructs. 3. Implement real time applications using the concept of array, pointers, functions and structures. 4. Solve real world problems using matrices, searching and sorting.								
LIST OF EXPERIMENTS								
WEEK- 1	INTRODUCTION TO FLOWGORITHM							
1. Installation and working of Flowgorithm Software. 2. Write and implement basic arithmetic operations using Flowgorithm sum, average, product, difference, quotient and remainder of given numbers etc. 3. Draw a flow chart to calculate area of Shapes (Square, Rectangle, Circle and Triangle). 4. Draw a flow chart to find the sum of individual digits of a 3 digit number 5. Draw a flow chart to read input name, marks of 5 subjects of a student and display the name of the student, the total marks scored, percentage scored. 6. Draw a flow chart to find roots of a quadratic equation. 7. Draw a flowchart to find the large stand smallest among three entered numbers and also display whether the identified largest/smallest number is even or odd 8. Draw a flow chart to check whether the triangle is equilateral, is isosceles scalene triangle 9. Draw a flow chart to check whether a given number is palindrome or not.								
WEEK- 2	BASIC DATA TYPES							
1. Write a C program to find division of 1 integer 1 float numbers. 2. Write a C program to find division of 2 integer numbers. 3. Write a C program to find the average of (1 int, 1 float) numbers. 4. Write a C Program to Swap Numbers without Using Temporary Variable								

5.	Write a C Program to Swap two Numbers Using Temporary Variable
6.	Write a C program to read the values of x, y and z and print the results of the following expressions in one line. a) $(x+y+z)/(x-y-z)$ b) $(x+y+z)/3$ c) $(x+y)*(x-y)*(y-z)$
WEEK- 3	OPERATORS
1.	Write a C program to convert temperature from Fahrenheit to Celsius and vice versa ($c=(f-32)/1.8$)
2.	Write a C program to find area and perimeter of a circle
3.	Write a C program to calculate the area and perimeter of a right-angled triangle. a. $(Area=1/2*b*perimeter=w+h+sqrt(w*w+h*h))$
4.	Find the sum of natural numbers 1 to n. (read n as input) (Use formula $sum=n(n+1)/2$)
5.	Write a C program to calculate Simple interest ($SI=PTR/100$)
6.	Write a C program to calculate area and perimeter of a rectangle. $Area=l*b$ $Perimeter=2*(l+b)$
7.	Write a C program to calculate the value of the third angle of a triangle if two angles are given as input. ($a+b+c=180$)
8.	Write a C program to read the consumer number and number of units consumed and the cost per unit and print the amount to be paid. ($Amt=number\ of\ units*cost$)
9.	Write a C Program to calculate the area and perimeter of a triangle. b. $Perimeter=(a+b+c)$ c. $s=(a+b+c)/2$ d. $Area=sqrt(s*(s-a)*(s-b)*(s-c))$
10.	Write a C program to read five Subject marks and find average.
11.	Write a C program to Calculate Compound Interest $CI=p(1+r/100)^n$
WEEK- 4	CONDITIONAL STATEMENTS
1.	Write a C program to find the largest and smallest of given numbers.
2.	Write a C program which takes two integer operands and one operator from the user (+, -, *, /, %) use switch)
3.	Write a program to compute the grade of students using an if else ladder. The grades are assigned as followed: marks < 50 F $50 \leq marks < 60$ C $60 \leq marks < 70$ B $70 \leq marks < 80$ B+ $80 \leq marks < 90$ A $90 \leq marks \leq 100$ A+
4.	Write a C program to determine whether given year is leap year or not.
5.	Write a C program to find whether a given triangle is scalene or isosceles or equilateral.
WEEK- 5	LOOPING STATEMENTS
1.	Write a C program to find Sum of individual digits of given integer
2.	Write a C program to generate first Fibonacci series
3.	Write a C Program to find the Sum of Series $SUM=1-x^2/2!+x^4/4!-x^6/6!+x^8/8!-x^{10}/10!$
4.	Write a C program to print the Fibonacci sequence up to given value of n.
5.	Write a C program to print the multiplication table to the given value of n.
6.	Write a C program to check whether given number is palindrome or not
7.	Write a C program to check whether given number is perfect or not
WEEK- 6	NESTED LOOPING STATEMENTS

Write a C program to generate prime numbers between 1 and n

1. Write a C program to generate Pascal's triangle.
2. Write a C program to generate the following pyramid of numbers.

```

      1
    1 3 1
  1 3 5 3 1
  
```

3. Write a C program to generate the pattern

```

*****
***
**
*

```

(area= πr)

WEEK- 7 ARRAYS

1. Write a C Program to implement following searching methods
 - i. Binary Search
 - ii. Linear Search
2. Write a C program to find largest and smallest number in a list of integers
3. Write a C program
 - iii. To add two matrices
 - iv. To multiply two matrices
4. Write a C program to find Transpose of a given matrix

WEEK- 8 FUNCTIONS

1. Write a C program to find the factorial of a given integer using non recursive functions
2. Write a C program to find GCD of given integers using non recursive functions
3. Write a C Program to find the power of a given number using non recursive functions
4. Write a C program to find the sum of natural numbers using a non recursive function.
5. Write a C program to reverse a given integer number using non recursive functions
6. Write a C Program to find binary equivalent of a given decimal number using recursive functions.
7. Write a C Program to print Fibonacci Sequence Using Recursive Functions.
8. Write a C Program to find LCM of 3 given numbers using recursive functions
9. Write a C program to find the factorial of a given integer using recursive functions
10. Write a C program to print fibonacci series till n terms using recursion.

WEEK- 9 STRINGS

1. Write a C program using to Insert a substring into a given main string from a given position 2. Write a C program using to Delete n characters from a given position in a string 3. Write a C program to determine if given string is palindrome or not 4. Write C Programs to demonstrate the following string handling functions. a. strcat() b. strcmp() c. strrev() d. strcpy() e. strlen() f. strstr() g. strncpy()	
h. Strncat () i. Strncmp ()	
WEEK-10	POINTERS
1. Write a C program to read the elements of 1-darray using pointers and print temin reverse order using pointers. 2. Write a C Program to read two elements dynamically using malloc() function and interchange the two numbers using call by reference. 3. Write a C Program to read and print the elements of 1-D array using calloc() memory allocation function and reallocate memory for the array by increasing the size of the array,read and print the elements of reallocate array. 4. Write a C Program to print 2-D array using pointers.	
WEEK-11	STRUCTURES
1. Write a C Program using functions to a) Reading A Complex Number b) Writing A Complex Number c) Add Two Complex Numbers d) Multiply two complex numbers Note:represent complex number using structure 2. Write a C program to read employee details employee number,employee name,basic salary,hra and da of n employees using structures and print employee number,employee name and gross salary of employees.	
WEEK-12	FILES
1. Write a C program to read and print the content of a file. 2. Write a C program copy the content of one file to another file 3. Write a C program to merge two file into thirdfile. 4. Write a C Program to find the number of lines in a textfile	
TEXTBOOKS	
1. RileyDD,Hunt.A.Computational Thinking for the Modern ProblemSolver.CRCpress,2014Mar27. 2. B.A.Forouzan andR.F.Gilberg Programming DataStructures,CengageLearning,(3rdEdition) Edition,2012.	

REFERENCE BOOKS

1. FerraginaP,Lucci.Computational Thinking:FirstAlgorithms,ThenCode.Springer;2018.
2. LindenPeterV,stEdition,1994.

WEB REFERENCES

1. <http://www.flowgorithm.org/documentation/>
2. <http://www.sanfoundry.com/c-programming-examples>
3. <http://www.geeksforgeeks.org/c>
4. <http://www.cprogramming.com/tutorial/c>

APPLIED PHYSICS LAB

I B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS08	BSC	L	T	P	C	CIA	SEE	Total
		0	0	3	1.5	40	60	100
Contact Classes: Nil	Tutorial Classes: Nil	Practical Classes: 36			Total Classes: 36			

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the temperature and other dependent properties of semiconductor devices and their usages in different applications
2. To know the Laser and Fibre optic technologies and its applications in real time scenario
3. Understand the electromagnetic properties using experimental knowledge
4. Understand the method of least squares fitting

COURSE OUTCOMES:

By the end of the course students will be able to:

1. **Analyze** the electric properties of semiconductor materials by determining energy gap of semiconductors, charge carrier concentration in Semiconductors using Hall effect and threshold voltage of LEDs, photo current in Photodiodes, solar cell, and temperature effect on resistance using a thermistor.
2. **Identify** the optical properties of light such as diffraction phenomenon using grating material for calculation of the wavelength of Laser and acceptance angle, NA of optical fiber using OFC and determine the value of Plank's constant using a light source and interference by using Newton's rings
3. **Analyze** the electromagnetic properties of a current carrying coil by using Stewart Gee's experiment
4. **Analyze** the least squares fitting method for data analysis using experimental data of Tensi

LIST OF EXPERIMENTS

Experiment-1	ENERGY GAP OF P-N JUNCTION DIODE: To determine the energy gap of a given semiconductor diode
Experiment-2	AR CELL: To study the V-I and V-P characteristics and determine the fill factor of solar cell
Experiment-3	LIGHT EMITTING DIODE: To study the characteristics of LED by plotting V-I graph and determine the threshold value of given LEDs
Experiment-4	HALL EFFECT: To determine Hall coefficient and charge carrier concentration of a given semi conductor
Experiment-5	PIN PHOTO DIODE: To study the V-I Characteristics of Photo Diode with respect to intensity of light
Experiment-6	OPTICAL FIBRE: To determine the numerical aperture and acceptance angle of an optical fiber
Experiment-7	LASER: To determine the wavelength of a given laser source by using diffraction grating method
Experiment-8	NEWTON'S RINGS: To determine the radius of curvature of a given Plano convex lens by forming Newton's rings

Experiment-9	THERMISTOR: To study the variation of resistance with respect to temperature using thermistor
Experiment-10	Understanding the method of least squares - Torsional pendulum as an example
Experiment-11	PLANCK'S CONSTANT: To determine value of planck's constant by measuring radiation in fixed spectral range
Experiment-12	STEWART GEE'S EXPERIMENT: To study the variation of magnetic field along the axis of a circular coil and calculation of magnetic flux
Note: Students have to perform any 8 experiments	
Reference Books:	
1. "Applied Physics Lab Manual"- Dr. Radhika Devi, Mr. A V Laxman Rao, N. Noel 2. S. Balasubramanian, M.N Srinivasan "A Text book of practical Physics" – S Chand Publishers, 2017.	

ENGINEERING WORKSHOP

I B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME04	ESC	L	T	P	C	CIA	SEE	Total
		0	0	3	1.5	40	60	100
Contact Classes: Nil	Tutorial Classes: Nil	Practical Classes: 36				Total Classes:36		

COURSE OBJECTIVES:

The course should enable the students to:

1. Study of different hand operated tools, uses and their demonstration.
2. Gain a good basic working knowledge required for the production of various engineering products.
3. provide hands on experience about use of different engineering materials, tools, equipment and processes that are common in the engineering field.
4. It explains the construction, function, use and application of different working tools, equipment and machines.
5. Study commonly used carpentry joints
6. Develop a right attitude, team working, precision and safety at work place

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Assemble different components
2. Identify and apply suitable tools for different trades of engineering processes.
3. Practice on manufacturing of components using workshop trades including Soldering, Carpentry, Fitting and Tin smithy & Fabricate Components with their own hands.
4. Apply basic electrical engineering knowledge for house wiring practice.
5. Learn the safety precautions for various operations in basic trade operations.

WEEKS	TRADES FOR EXERCISES
	Fitting
Week 1	Filing Four Sides of MS Work piece
Week 2	L Fit
	Carpentry
Week 3	Half Lap Joint
Week 4	Dove Tail Joint
	Tin Smithy
Week 5	Prepare a Rectangular Tray
Week 6	Prepare A Square Tin
	Electrical
Week 7	Series and Parallel Connection One Way Switch
Week 8	Two Way Switch Connection Stair Case Wiring

	Electronics
Week 9	Soldering - Series Connection & Parallel Connection
Week10	Desoldering & Construction of Wheatstone bridge
	TRADES FOR DEMONSTRATION & EXPOSURE
Week 11	Introduction to Black smithy , Introduction to Plumbing
TEXT BOOKS	
1. Workshop Manual - P. Kannaiah / K.L. Narayana/ Scitech. 2. Workshop Manual / K. Venugopal / Anuradha.	
REFERENCE BOOKS:	
1. Hajra Choudhury S.K., Hajra Choudhury A.K.and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010. 2. Workshop Manual / Venkat Reddy/ BSP	

ELEMENTS OF ARTIFICIAL INTELLIGENCE

I B. TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AI01	ESC	L	T	P	C	CIA	SEE	Total
		1	0	2	2	50	0	50
Contact Classes: 50	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes:50			
COURSE OBJECTIVES The course should enable the students to: 1. understand the applications of AI: namely Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing 2. Ability to comprehend the concept of supervised and unsupervised learning techniques 3. Understand the biological neural network and to model equivalent neuron models. 4. Understand the architecture, learning algorithm and issues of various feed forward and feedback neural networks. 5. Understand the context of neural networks and deep learning								
COURSE OUTCOMES At the end of the course, student will be able to: 1.Implement the concepts of AI in Healthcare, Robotics and Agriculture 2.Recognize the characteristics of machine learning that makes it useful to solve real-world problems. 3.Explain different neural networks of various architectures both feed forward and feed backward. 4.Analyse impact of various neural networks concepts for various pattern recognition applications 5.Analyze the limitation of Single layer Perceptron and Develop MLP with 2 hidden layers								
UNIT-I	INTRODUCTION TO ARTIFICIAL INTELLIGENCE					CLASSES:10		
Introduction, history, intelligent systems, foundations of AI, applications, tic-tac-toe game playing, current trends in AI.								
UNIT-II	MACHINE LEARNING					CLASSES:10		
Introduction, Well- Posed Learning Problems, Perspectives and Issues in Machine Learning, Supervised learning, unsupervised learning,								
UNIT-III	ARTIFICIAL NEURAL NETWORKS					CLASSES:10		
Introduction, neural network representation, models of a neuron, network architecture ,multi layer networks								
UNIT-IV	AGENTS					CLASSES:10		
Agent Definition, Agent, Agent Vs Object, Aglet, Mobile Agents, Agent Frameworks, Agent Reasoning.								

UNIT-V	EXPERT SYSTEM AND APPLICATIONS	CLASSES:10
Introduction, phases in building expert systems, expert system versus traditional systems, application of expert systems.		
Text Books:		
1 Saroj Kaushik. Artificial Intelligence. Cengage Learning. 2011 2 Russell, Norvig: Artificial intelligence, A Modern Approach, Pearson Education, Second Edition. 2004		
Reference Books:		
1. Rich, Knight, Nair: Artificial intelligence, Tata McGraw Hill, Third Edition 2009. 2. Introduction to Artificial Intelligence by Eugene Charniak, Pearson. 3. Introduction to Artificial Intelligence and expert systems Dan W.Patterson. PHI. 4 Artificial Intelligence by George Fluger pearson fifth edition.		

I B.TECH II SEMESTER SYLLABUS

NUMERICAL METHODS AND INTEGRAL TRANSFORMS

I B. TECH- II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6BS02	BSC	L	T	P	C	CIE	SEE	Total
		3	1	0	4	40	60	100
Contact Classes: 46	Tutorial Classes: 08	Practical Classes: Nil			Total Classes: 54			
COURSE OBJECTIVES The course should enable the students to: 1. Curve fitting and Interpolation techniques. 2. Numerical techniques. 3. Fourier series for periodic function 4. Laplace transforms 5. Concept and application of Fourier Transforms and Vector differentiation.								
COURSE OUTCOMES: At the end of the course, student will be able to: 1. Apply Curve fitting and Interpolation techniques. 2. Apply various numerical techniques 3. Find the Fourier series of the periodic functions. 4. Obtain the Laplace transforms of functions 5. Find Fourier transforms and apply vector differentiation techniques.								
UNIT-I	INTERPOLATION AND CURVE FITTING					Classes: 08		
INTERPOLATION: Finite differences: Forward, Backward and Central differences - Other difference operators and relations between them - Difference of a polynomial – Missing terms - Newton's forward interpolation, Newton's backward interpolation, Gauss's forward and backward interpolation formula. Interpolation with unequal intervals – Lagrange's interpolation. CURVE FITTING: Method of least squares - Fitting a straight line, second degree parabola and non-linear curves of the form $y = a e^{bx}$, $y = a x^b$, $y = a b^x$ by the method of least squares.								
UNIT-II	NUMERICAL TECHNIQUES					Classes: 08		
ROOT FINDING TECHNIQUES: Bisection method, Regula Falsi method, Iteration method and Newton Raphson method. NUMERICAL INTEGRATION : Trapezoidal rule - Simpson's one-third rule - Simpson's three-eighth rule. NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS: Taylor's series method – Euler's - modified Euler's Method – Runge-Kutta method.								
UNIT-III	FOURIER SERIES					Classes:10		
Periodic function-Determination of Fourier Coefficients-Fourier Series-Even and Odd functions-Fourier series in arbitrary interval-Half range Fourier sine and cosine expansions.								
UNIT-IV	LAPLACE TRANSFORMS					Classes: 10		
Laplace transforms of elementary functions- First shifting theorem - Change of scale property – Multiplication by t^n - Division by t – Laplace transforms of derivatives and integrals – Unit step function – Second shifting theorem – Periodic function – Evaluation of integrals by Laplace transforms – Inverse Laplace transforms- Method of partial fractions – Other methods of finding inverse transforms – Convolution theorem – Applications of Laplace transforms to ordinary differential equations.								

UNIT-V	FOURIER TRANSFORMS AND VECTOR DIFFERENTIATION	Classes: 10
Fourier Integral theorem (Statement only)-Fourier Sine and Cosine Integrals, Fourier Transforms, Cosine and Sine transforms, properties, Inverse transforms. Vector functions, vector differentiation, gradient, directional derivative, divergence, curl and scalar potential.		
Text Books:		
Erv in Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.		
Reference Books:		
1. G.B.Thomas, calculus and analytic geometry,9th Edition, Pearson Reprint 2006. 2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008. 3. E.L.Ince, Ordinary Differential Equations,Dover publications,1958.		
Web references:		
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/		
E -Text Books:		
1.https://www.e-booksdirectory.com/details.php?ebook=10166		
MOOCS Course:		
1. https://swayam.gov.in/ 2. https://onlinecourses.nptel.ac.in/		

ENGLISH FOR SKILL ENHANCEMENT

I B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS01	HSMC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 64	Tutorial Classes: Nil	Practical Classes: Nil				Total Classes: 64		

COURSE OBJECTIVES

The course should enable the students to:

1. Develop language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills.
2. Apply the theoretical and practical components of English syllabus to study academic subjects more effectively and critically.
3. Analyze a variety of texts and interpret them to demonstrate in writing or speech.
4. Write/ compose clearly and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject.
5. Develop language components to communicate effectively in formal and informal situations.

COURSE OUTCOMES:

By the end of this course, students will be able to:

1. Understand the importance of vocabulary and sentence structures.
2. Choose appropriate vocabulary and sentence structures for their oral and written communication.
3. Demonstrate their understanding of the rules of functional grammar.
4. Develop Comprehension Skills From the known and unknown passages.
5. Take An Active Part in drafting paragraphs, letters, essays, abstracts, precis and reports various contexts.Acquire basic proficiency in reading and writing modules of English.

UNIT-I	'Toasted English' by R. K. Narayan	Classes: 12
Vocabulary	: The Concept of Word Formation -The Use of Prefixes and Suffixes - Acquaintance with Prefixes and Suffixes from Foreign Languages to form Derivatives - Synonyms and Antonyms	
Grammar	: Identifying Common Errors in Writing with Reference to Articles and Prepositions	
Reading:	: Reading and Its Importance- Techniques for Effective Reading	
Writing	: Sentence Structures – Use of Phrases and Clauses in Sentences – Importance of Proper Punctuation – Techniques for Writing precisely – Paragraph Writing – Types, Structures and Features of a Paragraph – Creating Coherence – Organizing Principles of Paragraphs in Documents.	

UNIT-II	‘Appro JRD’ by Sudha Murthy	Classes: 14
Vocabulary	: Words Often Misspelled–Homophones, Homonyms and Homographs	
Grammar	: Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject - verb Agreement	
Reading	: Sub-Skills of Reading –Skimming and Scanning– Exercises for Practice	
Writing	: Nature and Style of Writing-Defining/Describing People,Objects,Places and Events– Classifying-Providing Examples or Evidence.	
UNIT-III	‘Lessons from Online Learning’ by F.Haider Alvi, Deborah Hurst et al	Classes: 14
Vocabulary	: Words Often Confused– Words from Foreign Languages and Their Use of English.	
Grammar	: Identifying Common Errors in Writing with Reference to Misplaced Modifiers and Tenses	
Reading	: Sub -Skills of Reading–Intensive Reading and Extensive Reading – Exercises for Practice	
Writing	: Format of a Formal Letter-Writing Formal Letters E.g.,Letter of Complaint, Letter of Requisition, Email Etiquette, Job Application with CV/Resume	
UNIT-IV	Art and Literature	Classes: 12
Vocabulary	: Standard Abbreviations in English	
Grammar	: Redundancies and Clichés in Oral and Written Communication	
Reading	: Survey,Question,Read,Recite and Review(SQ3RMethod) - Exercises for Practice	
Writing	: Writing Practices- Essay Writing-Writing Introduction and Conclusion-PrécisWriting	
UNIT-V	Go, Kiss the World’ by Subroto Bagchi	Classes: 12
Vocabulary	: Technical Vocabulary and their Usage	
Grammar	: Common Errors in English (Covering all the other aspects of grammar which were not covered in the previous units)	
Reading	: Reading Comprehension-Exercises For Practice	
Writing	: Technical Reports- Introduction – Characteristics of a Report – Categories of ReportsFormats- Structure of Reports (Manuscript Format) -Types of Reports - Writing aReport	
Text Books:		
“English: Language, Context and Culture” by Orient Black Swan Pvt. Ltd, Hyderabad.2022.Print.		
Reference Books:		

1. Effective Academic Writing by Liss and Davis (OUP)
2. Richards, Jack C. (2022) Interchange Series. Introduction, 1, 2, 3. Cambridge University Press
3. Wood, F. T. (2007). Remedial English Grammar. Macmillan.
4. Chaudhuri, Santanu Sinha. (2018). Learn English: A Fun Book of Functional Language, Grammar and Vocabulary. (2nd ed.). Sage Publications India Pvt. Ltd.
5. (2019). Technical Communication. Wiley India Pvt. Ltd.
6. Vishwamohan, Aysha. (2013). English for Technical Communication for Engineering Students. McGraw-Hill Education India Pvt. Ltd.
7. Swan, Michael. (2016). Practical English Usage. Oxford University Press. Fourth Edition

Web References:

1. http://www.bbc.co.uk/learning_english
2. <http://learnenglish.britishcouncil.org>
3. <https://www.cambridgeenglish.org/learning-english/>
4. <https://study.com/academy/subj/english.html>

E-Text Books:

<https://www.pdfdrive.com/advanced-english-books.html>

MOOC Courses:

1. <http://nptel.ac.in/courses/109/106/109106067>
2. <https://www.britishcouncil.org/tr/en/english/mooc>

ENGINEERING CHEMISTRY

I B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6BS09	BSC	L	T	P	C	CIE	SEE	Total
		3	1	0	4	40	60	100
Contact Classes:50	Tutorial Classes:nil	Practical Classes: Nil			Total Classes: 50			

COURSE OUTCOMES:

At the end of the course student will be able to:

1. Illustrate the types of hard and soft water, treatment of hard water.
2. Demonstrate the basic principles of electrochemistry in electrochemical cells.
3. Impart knowledge on the basic concepts of battery and fuels cells.
4. Apply the methods of metal finishing in solving corrosion related problems
5. Identify the significance of polymers, sensors and smart materials.

UNIT-I	WATER AND ITS TREATMENT	Classes: 10
Introduction to hardness of water – Estimation of hardness of water by complexometric method and related numerical problems. Potable water and its specifications - Steps involved in the treatment of potable water - Disinfection of potable water by chlorination and breakpoint chlorination. Defluoridation - Determination of F⁻ ion by ion-selective electrode method. Boiler troubles: Sludges, Scales and Caustic embrittlement. Internal treatment of Boiler feed water Calgon conditioning Phosphate conditioning Colloidal conditioning, External treatment methods - Softening of water by ion-exchange processes. Desalination of water – Reverse osmosis.		
UNIT-II	ELECTROCHEMISTRY AND ITS APPLICATIONS	Classes: 10
Introduction - Electrochemical cells -Electrode potential - Standard electrode potential - Nernst Equation -Types of electrodes: SHE, Calomel, Quinhydrone, Glass Electrode-Electrochemical series, and its applications-Numerical Problems-Potentiometric titrations: Acid-base, Redox titration.		
UNIT-III	BATTERY CHEMISTRY AND FUEL CELLS	Classes: 10
Introduction - Classification of batteries- primary, secondary and reserve batteries with examples Basic requirements for commercial batteries. Construction, working and applications of: Zn-air, Lead acid and Lithium ion battery- Applications of Li-ion battery to electrical vehicles. Fuel cells- Introduction-Differences between fuel cells and batteries - Construction, working and applications of methanol-Oxygen fuel cells-Molten carbonate fuel cells-solid oxide fuel cell		
UNIT-IV	CORROSION CONTROL	Classes: 10
Introduction-causes and effects -Mechanism of chemical and electrochemical Corrosion-Types of corrosion - Galvanic, water-line and pitting corrosion - Factors affecting rate of Corrosion-Corrosion control methods-Cathodic protection (Sacrificial anodic, Impressed current cathodic) Protective Coatings Metallic Coatings-Hot Dipping Methods-Galvanization-Tinning-Cementation(sherardizing)-Electroplating of Cu-Electroless plating of Ni-Organic Coatings-Paints (constituents and functions).		

UNIT-V	ENGINEERING MATERIALS	Classes: 10
Definition Classification of polymers with examples Types of polymerization addition (free radical addition) and condensation polymerization with examples Nylon 6:6, Terylene Plastics: Definition and characteristics- thermoplastic and thermo setting plastics, Preparation, Properties and engineering applications of PVC and Bakelite, Teflon, Fiber reinforced plastics (FRP). Rubbers: Natural rubber and its vulcanization. Elastomers: Characteristics –preparation – properties and applications of Buna-S, Butyl and Thiokol rubber. Sensors- Introduction Classification (active and passive) -Types (Light, Temperature, Pressure, Position, Sound and Touch). Biosensors Working Principle-Applications and advantages. Smart materials - Introduction Types of smart materials Principle and applications of piezoelectric, magnetostrictive, Shape memory- PH sensitive and Chromatic.		
Text Books:		
1. Text Book of Engineering Chemistry by Dr. Bharathi Kumari.Y, Dr. Jyothsna, V.G.S Book Links. 2. Text Book of Engineering Chemistry by C. Parameshwaramurthy, C.V. Agarwal, B.S. Publications. 3. A text book of Engineering Chemistry by M. Thirumala Chary, E. Laxminarayana and K. Shashikala, Pearson Publications, 2021. 4. Textbook of Engineering Chemistry by Jaya Shree Anireddy, Wiley Publications.		
Reference Books:		
1. Shashi Chawla, A Textbook of Engineering Chemistry, Dhanapath Rai and sons, 2003. 2. Engineering Chemistry (NPTEL Web-book), 11th edition by B.L. Tembe, Kamaluddin and M.S.Krishnan. 3. V. Raghavan, A Material Science and Engineering, Prentice-Hall India Ltd, 2004 4. K. Sesha Maheshwaramma and Mridula Chugh, Engineering Chemistry, Pearson India Edn services, 2016.		

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

I B. TECH- II SEMESTER

Course Code	Category	Hours/ Week			Credis	Maximum Marks		
A6EE60	BSC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
ContactClasses:58	Tutorial Classes:Nil	Practical Classes:Nil			Contact Classes:58			

COURSE OBJECTIVES

The course should enable the students to:

1. Develop fundamentals, including Ohm's law, Kirchhoff's laws and be able to solve for currents, voltages and power in electrical circuits.
2. Develop EMF equation and analyze the operation of DC Machines.
3. Analyze the working principle of Transformer.
4. Discuss the operation of AC Machines.
5. Analyze the operation of PN junction diodes and rectifiers.
6. Discuss the operation and characteristics of Transistors.

COURSE OUTCOMES:

At the end of the course students will be able to:

1. Analyze and solve for current values in resistive circuits with independent sources.
2. Analyze the working of DC machines and solve the numerical problems..
3. Analyze the working of AC electrical machines and solve the numerical problems.
4. Analyze the V-I characteristics of PN – junction diode and describe the operation of rectifiers.
5. Analyze the different configurations of Transistors and obtain its characteristics.

UNIT - I	ELECTRICAL CIRCUITS	CLASSES: 12
Basic definitions-Ohm's Law, types of elements, types of sources, Kirchhoff's Laws – simple problems., series & parallel resistive networks with DC excitation, star to delta and delta to star transformations.		
UNIT - II	DC MACHINES	CLASSES: 12
Principle of Operation of DC Motor, types of DC motor, Torque equation & Losses and problems. DC Generator construction and working Principle, EMF Equation types of generators and problems.		
UNIT - III	AC MACHINES	CLASSES:12
Working principle and Construction of transformer, Emf Equation & problems, Principle operation of 3-phase induction motor, slip and torque Equation, Torque –slip characteristics & problems, principle operation of 3-phase Alternator, Emf Equation of Alternator & problems.		
UNIT - IV	DIODE AND ITS CHARACTERISTICS	CLASSES: 12

PN JUNCTION DIODE: Operation of PN junction Diode: forward bias and reverse bias, Characteristics of PN Junction Diode – Zener Effect – Zener Diode and its Characteristics. Rectifiers, Half wave, Full wave and bridge Rectifiers –capacitor filters, inductor filters		
UNIT - V	TRANSISTORS	CLASSES: 10
Bipolar Junction Transistor - NPN & PNP Transistor, CB, CE, CC Configurations and Characteristics – Transistor Amplifier.		
TEXT BOOKS:		
1. Basic Electrical Engineering by <i>M.S.Naidu and S.Kamakshaiah</i>TMH 2. Electronic Devices and circuits by <i>J.Millman, C.C.Halkias and Satyabrata Jit</i>2ed.,		
REFERENCE BOOKS:		
1. Muthusubramanian R,SalivahananS and Muraleedharan K,“Basic Electrical,Electronics and Computer Engineering”,Tata McGraw Hill, Second Edition,(2006). 2. Nagsarkar T K and Sukhija M S, “Basics of Electrical Engineering”, Oxford press(2005). 3. Mehta V K, “Principles of Electronics”, S.Chand & Company Ltd,(1994). 4. MahmoodNahviandJosephA.Edminister,“Electric Circuits”,Schaum’OutlineSeries,McGraw Hill,(2002).		

ELECTRONIC DEVICES AND CIRCUITS

I B. TECH- II SEMESTER

Course Code	Category	Hours /Week			Credits	Maximum Marks		
A6EC03	ESC	L	T	P	C	CIE	SEE	Total
		2	0	0	2	40	60	100
Contact Classes:50	Tutorial Classes:Nil	Practical Classes:Nil			Total Classes:50			

Course Objectives:

The course should enable the students to:

- 1.To introduce components such as diodes,BJT s and FETs.
- 2.To know the applications of devices.
- 3.To know the switching characteristics of devices.

Course Outcomes:

Upon completion of the Course, the students will be able to:

1. Acquire knowledge of various electronic devices and their use on real life.
2. Know the applications of various devices.
3. Acquire the knowledge about the role of special purpose devices and their applications.

UNIT-I	DIODES	Classes:10
Static and Dynamic resistances, Equivalent circuit, Diffusion and Transition Capacitances, VI Characteristics, Diode as a switch- switching times.		
UNIT-II	DIODE APPLICATION	Classes:10
Rectifier - Half Wave Rectifier, Full Wave Rectifier, Bridge Rectifier, Rectifiers with Capacitive and Inductive Filters, Clippers-Clipping at two independent levels, Types of Clampers.		
UNIT-III	BIPOLAR JUNCTION TRANSISTOR (BJT)	Classes:10
Principle of Operation, Common Emitter, Common Base and Common Collector Configurations, Transistor as a switch, Need for Biasing, BJT as Amplifier.		
UNIT-IV	JUNCTION FIELD EFFECT TRANSISTOR (BJT)	Classes:10
struction, Principle of Operation, Pinch-Off Voltage, Volt- Ampere Characteristic, Comparison of BJT and FET, FET as Voltage Variable Resistor, FET as amplifier.		
UNIT-V	SPECIAL PURPOSE DEVICES	Classes:10

Zener Diode - Characteristics, Zener diode as Voltage Regulator, Operation of -SCR, UJT, Photo diode, Solar cell, LED.

Text Books:

1. Jacob Millman - Electronic Devices and Circuits, McGraw Hill Education
2. Robert L. Boylestead, Louis Nashelsky- Electronic Devices and Circuits theory, 11th Edition, 2009, Pearson.

Reference Books:

1. Horowitz - Electronic Devices and Circuits, David A. Bell – 5th Edition, Oxford.
2. Chinmoy Saha, Arindam Halder, Debaati Ganguly - Basic Electronics-Principles and Applications, Cambridge, 2018.

ENGLISH LANGUAGE AND COMMUNICATION SKILLS LAB

I B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS02	HSMC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100
Contact Classes: Nil	Tutorial Classes: Nil	Practical Classes: 36				Total Classes: 36		

Course Objectives:

The course should enable the students to:

1. Facilitate computer-assisted multi-media instruction enabling individualized and independent language learning
2. Sensitize the students to the nuances of English speech sounds, word accent, intonation and rhythm
3. Bring about a consistent accent and intelligibility in students' pronunciation of English by providing an opportunity for practice in speaking
4. Improve the fluency of students in spoken English and neutralize the impact of dialects.
5. Train students to use language appropriately for public speaking, group discussions and interviews.

English Language and Communication Skills Lab (ELCS) shall have two parts:

- a. Computer Assisted Language Learning (CALL) Lab
- b. Interactive Communication Skills (ICS) Lab

Listening Skills

Objectives

1. To enable students develop their listening skills to appreciate its role in the LSRW skills approach to language and improve their pronunciation.
2. To equip students with necessary training in listening so that they can comprehend the speech of people of different backgrounds and regions.

Students should be given practice in listening to the sounds of the language, to be able to recognize them and find the distinction between different sounds, to be able to mark stress and recognize and use the right intonation in sentences.

- Listening for general content
- Listening to fill up information
- Intensive listening
- Listening for specific information

Speaking Skills

Objectives

1. To involve students in speaking activities in various contexts
2. To enable students express themselves fluently and appropriately in social and professional contexts
 - Oral practice
 - Describing objects/situations/people
 - Role play – Individual/Group activities
 - Just A Minute (JAM) Sessions

Exercise-I

CALL Lab: <i>Understand:</i> Listening Skill- Its importance – Purpose- Process- Types- Barriers- Effective Listening. <i>Practice:</i> Introduction to Phonetics – Speech Sounds – Vowels and Consonants – Minimal Pairs- Consonant Clusters- Past Tense Marker and Plural Marker- <i>Testing Exercises</i> ICS Lab: <i>Understand:</i> Spoken vs. Written language- Formal and Informal English. <i>Practice:</i> Ice-Breaking Activity and JAM Session- Situational Dialogues – Greetings – Taking Leave – Introducing Oneself and Others.	
Exercise-II	
CALL Lab: <i>Understand:</i> Structure of Syllables – Word Stress– Weak Forms and Strong Forms – Stress pattern in sentences – Intonation. <i>Practice:</i> Basic Rules of Word Accent - Stress Shift - Weak Forms and Strong Forms- Stress pattern in sentences – Intonation - <i>Testing Exercises</i> ICS Lab: <i>Understand:</i> Features of Good Conversation – Strategies for Effective Communication. <i>Practice:</i> Situational Dialogues – Role Play- Expressions in Various Situations –Making Requests and Seeking Permissions - Telephone Etiquette.	
Exercise-III	
CALL Lab: <i>Understand:</i> Errors in Pronunciation-Neutralising Mother Tongue Interference (MTI). <i>Practice:</i> Common Indian Variants in Pronunciation – Differences between British and American Pronunciation -<i>Testing Exercises</i> ICS Lab: <i>Understand:</i> Descriptions- Narrations- Giving Directions and Guidelines – Blog Writing <i>Practice:</i> Giving Instructions – Seeking Clarifications – Asking for and Giving Directions – Thanking and Responding – Agreeing and Disagreeing – Seeking and Giving Advice – Making Suggestions.	
Exercise-IV	
CALL Lab: <i>Understand:</i> Listening for General Details. <i>Practice:</i> Listening Comprehension Tests - <i>Testing Exercises</i> ICS Lab: <i>Understand:</i> Public Speaking – Exposure to Structured Talks - Non-verbal Communication Presentation Skills. <i>Practice:</i> Making a Short Speech – Extempore- Making a Presentation.	
Exercise-V	
CALL Lab: <i>Understand:</i> Listening for Specific Details. <i>Practice:</i> Listening Comprehension Tests -<i>Testing Exercises</i> ICS Lab: <i>Understand:</i> Group Discussion <i>Practice:</i> Group Discussion	

Reference Books:

- (2022). English Language Communication Skills – Lab Manual cum Workbook. Cengage Learning India Pvt. Ltd.
- Shobha, KN & Rayen, J. Lourdes. (2019). Communicative English – A workbook. Cambridge University Press
- Kumar, Sanjay & Lata, Pushp. (2019). Communication Skills: A Workbook. Oxford University Press
- Board of Editors. (2016). ELCS Lab Manual: A Workbook for CALL and ICS Lab Activities. Orient Black Swan Pvt. Ltd.
1. Mishra, Veerendra et al. (2020). *English Language Skills: A Practical Approach*. Cambridge University Press
 - 2.

Websites:

<https://www.britishcouncil.org>
<https://www.bbc.co.uk>
<https://www.grammarly.com>
<https://www.fluentu.com>
<https://www.cambridgeenglish.org/exams-and-tests/business-preliminary>
<https://www.cambridgeenglish.org/exams-and-tests/business-vantage>

PYTHON PROGRAMMING LAB

I B. TECH-II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS04	ESC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100
COURSE OBJECTIVES The course should enable the students: - To understand the problem-solving approaches. - To learn the basic programming constructs in Python. - To practice various computing strategies for Python-based solutions to real world problems. - To use Python data structures lists, tuples,dictionaries. - To do input/output with files in Python								
COURSE OUTCOMES At the end of the course, student will be able to - Write, test,and debug simple Python programs. - Implement Python programs with conditions and loops. - Develop Python programs stepwise by defining functions and calling them. - Use Python lists, tuples, dictionaries for representing compound data. - Read and write data from/to files in Python								
SYLLABUS								
LIST OF PROGRAMS								
WEEK-1								
- Write a program to perform different Arithmetic Operation so n numbers in Python - Write a Python program which accepts the radius of a circle from the user and compute the area. - Write a Python program to get the Python version you are using. - Write a Python program that accepts an integer(n) and computes the value of $n+n$ $n+n$ n n								
WEEK-2								
- Write a Python program to convert temperatures to and from Celsius,Fahrenheit. [Formula:$c/5=f-32/9$] - Write a python script to print the current date in the following format "Sun May 29 02:26:23 IST 2017" - A library charges a fine for every book returned late.For first 6 days the fine is 50 paise,for 10-15 days fine is one rupee and above15 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled. - Write a python program to accept the number of days the member is late to return the book and display the fine or the appropriate message.								

WEEK-3	
a) Write a python function to find the largest of three numbers. b) Write a Python function that prints prime numbers in between 50 and 100. c) Write a python program to find the factorial of a number using Recursion. d) Write a function that receives marks received by a student in 6 subjects and returns the average and percentage of these marks. Call this function from main() and print the result in main	
WEEK-4	
Write a program to demonstrate working with Strings and string operations.	
WEEK-5	
Write a program to demonstrate working with dictionaries in python	
WEEK-6	
Write a program to demonstrate working with tuples and List in python.	
WEEK-7	
a) Write a script named hello.py. This script should prompt the user for the names of two text files. The contents of the first file should be input and written to the second file. b) Write a program that inputs a text file. The program should print all of the unique words in the file in alphabetical order	
WEEK-8	
Write python programs to demonstrate class object, static and instance method implementation.	
WEEK-9	
Write python programs to demonstrate Inheritance and Polymorphism	
WEEK-10	
Write python programs to demonstrate Exception Handling in python	
WEEK-11	
Write python programs to demonstrate NumPy library and supporting functions.	
WEEK-12	
a) Draw an Olympic Symbol in Python using Turtle b) Develop an simple login page using GUIT kniter	

TEXTBOOKS

1. Allen B. Downey, Think Python: How to Think like a Computer Scientist, 2nd Edition, Publishers, 2016.
2. Karl Beecher, Computational Thinking: A Beginners Guide to Problem Solving and Programming, 1st Edition, BCS Learning and Development Limited, 2017.

REFERENCE REFERENCES

1. Paul Deitel and Harvey Deitel, Python for Programmers, Pearson Education, 1st Edition, 2021.
2. G Venkatesh and Madhavan Mukund, Computational Thinking: A Primer for Programmers and Data Scientists, 1st Edition, Notion Press, 2021.
3. John V. Guttag, Introduction to Computation and Programming Using Python: With Applications to Third
4. Eric Matthes, Python Crash Course, A Hands on Project Based Introduction to Programming, 2nd Edition, No Starch Press, 2019. <https://www.python.org/>
5. Martin C. Brown, Python: The Complete Reference, 4th Edition, Mc-Graw Hill, 2018.

SEMINAR

I B. TECH- II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS04	HSMC	L	T	P	C	CIE	SEE	Total
		0	0	2	1	50	0	50

COURSE OBJECTIVES:

The course should enable the students to:

Seminar is an important component of learning in an Engineering College, where the student gets acquainted with preparing a report & presentation on a topic.

PERIODICITY / FREQUENCY OF EVALUATION :Twice

PARAMETERS OF EVALUATION:

1. The seminar shall have topic allotted and approved by the faculty.
2. The seminar is evaluated for 50 marks for internal only.
3. The students shall be required to submit the rough drafts of the seminar outputs within one week of the commencement of the class work.
4. Faculty shall make suggestions for modification in the rough draft. The final draft shall be presented by the student within a week thereafter.
5. Presentation schedules will be prepared by Department in line with the academic calendar.

THE SEMINARS SHALL BE EVALUATED IN TWO STAGES AS FOLLOWS:

A. ROUGH DRAFT

In this stage, the student should collect information from various sources on the topic and collate them in a systematic manner. He/ She may take the help of the concerned faculty.

The report should be typed in "MS-Word" file with "calibri" font, with font size of 16 for main heading, 14 for sub-headings and 11 for the body text. The contents should also be arranged in Power Point Presentation with relevant diagrams, pictures and illustrations. It should normally contain 10 to 15 slides, consisting of the followings:

1.	Topic, name of the student & faculty	1 Slide
2.	List of contents	1 Slide
3.	Introduction	1Slides
4.	Descriptions of the topic (point-wise)	6 - 10 Slides
5.	Conclusion	1 - 2 Slides
6.	References/Bibliography	1 Slide

The soft copy of the rough draft of the seminar presentation in MS Power Point format along with the draft report should be submitted to the concerned faculty, with a copy to the concerned HOD within stipulated time.

The evaluation of the rough draft shall generally be based upon the following.

1	Punctuality in submission of rough draft	4
2	Dress Code	6
3	Resources from which the seminar have been based	4
4	Report , and content of Presentation	10
5	Depth of the students knowledge in the subject	10
6	Reception from Questions	10
7	Time Management, Classroom Dynamic	6
	Total Marks	50

After evaluation of the first draft the supervisor shall suggest further reading, additional work and fine tuning, to improve the quality of the seminar work.

Within 7 days of the submission of the rough draft, the students are to submit the final draft incorporating the suggestions made by the faculty.

ENVIRONMENTAL SCIENCE

I B. TECH- II SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS11	MC	L	T	P	C	CIE	SEE	Total
		3	0	0	0	50	0	50
Contact Classes:30	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 30			

COURSE OUTCOMES:

Upon Successful completion of this course, Students will be able to

1. Identify the consequences of human actions on the web of life, global economy and quality of human life.
2. Evaluate the strategies for scientific, social, economic and legal environmental protection.
3. Study the impact of conservation of biodiversity.
4. Analyze the reasons for environmental pollution.
5. Assess the environmental impact of air, water, biological and socio-economical aspects and risk assessment towards sustainable future.

UNIT-I	ECOSYSTEMS	CLASSES: 7
Ecosystems: Definition, Scope and Importance of ecosystem. Classification, structure and function of an ecosystem, Food chains, food web and ecological pyramids. Flow of energy: Bio geo chemical cycles, Bio accumulation, Bio magnification, ecosystem value, services and carrying capacity.		
UNIT-II	NATURAL RESOURCES	CLASSES: 8
Natural Resources: Classification of Resources: Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems-case studies. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources. Land resources: Forest resources. Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy source, case studies.		
UNIT-III	BIODIVERSITY AND BIOTIC RESOURCES	CLASSES: 5
Biodiversity and Biotic Resources: Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values. India as a mega diversity nation: Hotspots of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; conservation of biodiversity: In-Situ and Ex-situ conservation. National Biodiversity Act-Case studies.		

UNIT-IV	ENVIRONMENTAL POLLUTION AND CONTROL TECHNOLOGIES	CLASSES:5
Environmental Pollution and Control Technologies: Environmental Pollution: Classification of pollution, Air Pollution, Water Pollution drinking water quality standards. Soil Pollution Impacts of modern agriculture, Noise Pollution Health hazards, standards. Concepts of bio remediation. Global Environmental Problems and Global Efforts: Ozone depletion and Ozone depleting substances (ODS) Concepts of Bio remediation International conventions / Protocols: Earth summit, Kyoto protocol and Montreal Protocol; NAPCC-Gol Initiatives, COP 24, COP25.		
UNIT-V	ENVIRONMENTAL POLICY, LEGISLATION & EIA	CLASSES:5
Environmental Policy, Legislation & Environmental Impact Assessment(EIA): Environmental Protection act, Legal aspects Air Act1981, Water Act, Forest Act, Wild life Act, Municipal solid waste management, biomedical waste management hazardous waste management and handling rules. Environmental Impact Assessment: EIA structure, methods of baseline data acquisition. Strategies for risk assessment, Towards Sustainable Future: Concept of Sustainable Development, Urban Sprawl, Concept of Green Building.		
TEXT BOOKS: 1. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission. 2. Environmental Studies by R.Rajagopalan, Oxford University Press.		

II B.TECH I SEMESTER SYLLABUS

DISCRETE MATHEMATICS

II B. TECH- ISEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS08	BSC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
Contact Classes:60	Tutorial Classes: Nil	Practical Classes: Nil				Total Classes: 60		

COURSE OBJECTIVES

The course should enable the students to:

1. Help students understand discrete and continuous mathematical structures
2. Impart basics of relations and functions
3. Facilitate students in applying principles of Recurrence Relations to calculate generating
4. Functions and solve the Recurrence relations
5. Acquire knowledge in graph theory

COURSE OUTCOMES

At the end of the course, student will be able to

1. Analyze and examine the validity of argument by using propositional and predicate calculus
2. Apply basic counting techniques to solve the combinatorial problems
3. Apply sets relations and digraphs to solve applied problems
4. Solve the given recurrence relation using different methods such as substitution, Generating function and characteristics roots equation.
5. Use the basic concepts of graph theory and some related theoretical problems

UNIT - I	MATHEMATICAL LOGIC	CLASSES: 11
Statements and notations, Connectives, Well formed formulas, Truth Tables, Tautology, Equivalence implication, Normal forms, Logical Inference, Rules of inference, Direct Method, Direct Method using CP(Conditional Proof), Consistency, Proof of contradiction, Automatic Theorem Proving.		
UNIT - II	RELATIONS	CLASSES: 16
Introduction to set theory, Relations, Properties of Binary Relations, Equivalence Relation, Transitive closure, Compatibility and Partial ordering relations, Lattices, Hasse diagram. Functions: inverse Function, Composition of functions. Algebraic Systems, Semi groups and Monoids		
UNIT - III	ELEMENTARY COMBINATORICS	CLASSES: 12
Basis of counting, Combinations & Permutations, Enumeration of Combinations and Permutations, Enumeration of Combinations and Permutations With repetitions, Enumerating Permutations with Constrained repetitions, Binomial Coefficients, Binomial and Multinomial theorems, The principles of Inclusion – Exclusion, Pigeon- hole principles and its applications.		

UNIT - IV	RECURRENCE RELATION	CLASSES: 11
Generating Functions, Function of Sequences, Calculating Coefficient of generating function, Recurrence relations, Solving recurrence relation by substitution and Generating functions , The method of Characteristics roots,		
UNIT - V	GRAPHS	CLASSES: 10
Basic Concepts, Isomorphism and Subgraphs, Trees and their properties, Spanning Trees- DFS,BFS, Minimal Spanning Trees- Prim's, Kruskal's Algorithm, Planar Graphs, Euler's Formula, Multi graph and Euler circuits, Hamiltonian Graphs, Chromatic number.		
TEXT BOOKS		
1. Discrete Mathematics for computer scientists & Mathematicians, <i>J.L. Mott, A. Kandel, T.P. Baker</i> PHI 2. Discrete Mathematical Structures With Applications to Computer Science, JP Tremblay, R Manohar		
REFERENCE BOOKS		
1. Logic and Discrete Mathematics, <i>Grass Man & Trembley</i>, Pearson Education.		

SOFTWARE ENGINEERING

II B.TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS07	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes:60	Tutorial Classes:Nil	Practical Classes: Nil			Total Classes: 60			
COURSE OBJECTIVES: The course should enable the students to: 1. Familiarize with basic Software engineering methods and practices, and its applications. 2. Explain layered technology in software engineering 3. Teach software metrics and software risks. 4. Familiarize with software requirements and the SRS documents. 5. Facilitate students in software design								
COURSE OUTCOMES: At the end of the course, student will be able to: 1. Understand software development life cycle and select appropriate model suited for diverse software application. 2. Analyse the customer's requirements for a project to be developed and formulate the software requirements document. 3. Conceptualize the system through design with emphases on architectural modelling and user interface. 4. Classify software testing strategies and recommend testing techniques during the construction of software. 5. Examine the application of metrics and software tools during software development								
UNIT – I	INTRODUCTION AND PROCESS MODELS						CLASSES: 12	
Introduction To Software Engineering: The Evolving nature of software engineering, Changing nature of software engineering, Software engineering Layers, The Software Processes, Software Myths. Process Models: A Generic Process Model, Waterfall Model, Incremental Process Models, Evolutionary Process Models, Spiral Model, The Unified Process, Personal and Team Process Models, the Capability Maturity Model Integration (CMMI).								
UNIT-II	REQUIREMENTS ENGINEERING AND DESIGN CONCEPTS						CLASSES: 12	
Requirements Engineering: Functional and Non-Functional Requirements, The Software requirements Document, Requirements Specification, requirements Engineering, Requirements Elicitation and Analysis, Requirement Validation, Requirement Management, System Modelling: Context Models, Interaction Models, Structural Models, Behavioural Model, Model-Driven Engineering. Design Concepts: The Design Process, Design Concepts, The Design Models, Architectural Design: Software Architecture, Architectural Genres, Architectural Styles.								

UNIT-III	DESIGN, IMPLEMENTATION AND SOFTWARE TESTING STRATEGIES	CLASSES: 12
Design And Implementation: Design Patterns, Implementation Issues, Open Source Development. User Interface Design: The Golden Rules, User Interface Analysis and Design, Interface Analysis, Interface Design Steps, Design Evaluation. Software Testing Strategies: A Strategic approach to Software Testing, Strategic Issues, Test Strategies for Conventional Software, Validation Testing, System Testing, The Art of Debugging, White-Box Testing, Black Box Testing.		
UNIT-IV	PRODUCT, PROCESS AND PROJECT METRICS	CLASSES: 12
Product Metrics: A Frame Work for Product Metrics, Metrics for the Requirements Model, Metrics for Design Model, Metrics for Source Code, Metrics for Testing. Process And Project Metrics: Metrics in the Process and Project Domains, Software Measurements, Metrics for Software Quality, Risk Management: Risk versus Proactive Risk Strategies, Software Risks, Risk Identification, Risk Projection, Risk Refinements, Risk Mitigation Monitoring and Management (RMMM).		
UNIT-V	OVERVIEW OF QUALITY MANAGEMENT, PROCESS IMPROVEMENT AND CASE TOOLS	CLASSES: 12
Overview Of Quality Management and Process Improvement: Overview of SEI -CMM, ISO 9000, CMMI, PCMM, TQM and Six Sigma. Overview of Case Tools: Software tools and environments: Programming environments; Project management tools; Requirement's analysis and design modelling tools;		
TEXT BOOKS		
1. Roger S. Pressman (2011), Software Engineering, A Practitioner's approach, 7th edition, McGraw Hill International Edition, New Delhi. 2. Sommerville (2001), Software Engineering, 9th edition, Pearson education, India.		
REFERENCE BOOKS		
1. K. K. Agarwal, Yogesh Singh (2007), Software Engineering, 3rd edition, New Age International Publishers, India. 2. Lames F. Peters, Witold Pedrycz (2000), Software Engineering an Engineering approach, John Wiely& Sons, New Delhi, India. 3. Shely Cashman Rosenblatt (2006), Systems Analysis and Design, 6th edition, Thomson Publications, India.		

DIGITAL ELECTRONICS AND COMPUTER ORGANIZATION

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS28	ESC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes:47	Tutorial Classes:Nil	Practical Classes: Nil			Total Classes: 47			
COURSE OBJECTIVES The course should enable the students to: 1. Understand different number systems and Boolean algebra. 2. Design of combinational and sequential logic circuits 3. Understand different Computer Instructions 4. Understand concepts of register transfer logic and arithmetic operations. 5. Learn different types of memory hierarchy								
COURSE OUTCOMES At the end of the course, student will be able to: 1. Able to perform the conversion among different number systems. 2. Able to design combinational and sequential logic circuits 3. Able to understand different computer instructions 4. Identify basic components and design of the CPU 5. Compare various types of IO mapping and memory mapping techniques								
UNIT – I	NUMBER THEORY and BOOLEAN ALGEBRA						CLASSES: 10	
Representation of numbers of different radix, conversion of numbers from one radix to another radix, r-1's complement and r's complement. 4-bit codes. Basic Theorems and Properties of Boolean algebra, Canonical and Standard Forms, Digital Logic Gates, Universal Logic Gates. K- Map Method.								
UNIT - II	COMBINATIONAL and SEQUENTIAL LOGIC CIRCUITS						CLASSES: 9	
Design of Half adder, full adder, half subtractor, full subtractor. Decoder, Encoder, Multiplexer, De-multiplexer and comparator. basic flip-flops, truth tables and excitation tables (NAND RS latch, NOR RS latch, RS flip-flop, JK flip-flop, T flip-flop, D flip-flop with preset and clear terminals). Classification of sequential circuits (synchronous and asynchronous);								
UNIT - III	BASIC COMPUTER ORGANIZATION AND DESIG						CLASSES: 10	
COMPUTER ARITHMETIC: Addition and Subtraction, Multiplication and Division algorithms, Floating-point Arithmetic Operations, Decimal arithmetic operations BASIC COMPUTER ORGANIZATION AND DESIGN: Instruction codes, Computer Registers, Computer Instructions and Instruction cycle. Memory-Reference Instructions.								
UNIT – IV	REGISTER TRANSFER AND MICRO-OPERATIONS:						CLASSES: 9	

Central processing unit: Stack organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Complex Instruction Set Computer (CISC) Reduced Instruction Set Computer (RISC), Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro-Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logic shift unit

UNIT – V**MEMORY SYSTEM****CLASSES: 9**

INPUT OUTPUT ORGANIZATION: I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA.

MEMORY ORGANIZATION: Memory Hierarchy, Main memory, Auxiliary memory, Associate memory, Cache memory

TEXT BOOKS

1. Switching and Finite Automata Theory- ZviKohavi & Niraj K. Jha, 3rd Edition, Cambridge.
2. Digital Design- Morris Mano, PHI, 3rd Edition.
3. M. Morris Mano (2006), Computer System Architecture, 3rd edition, Pearson/PHI, India.
4. John P. Hayes (1998), Computer Architecture and Organization, 3rd edition, Tata McGrawHill

REFERENCE BOOKS

1. Introduction to Switching Theory and Logic Design – Fredriac J. Hill, Gerald R. Peterson, 3rd Ed, John Wiley & Sons Inc.
2. William Stallings (2010), Computer Organization and Architecture- designing for performance, 8th edition, Prentice Hall, New Jersey.
3. Anrew S. Tanenbaum (2006), Structured Computer Organization, 5th edition, Pearson Education Inc,

OBJECT ORIENTED PROGRAMMING THROUGH JAVA

II B. TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT02	ESC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes:53	Tutorial Classes:Nil	Practical Classes: Nil			Total Classes: 53			
COURSE OBJECTIVES: The course should enable the students to: 1. Use object oriented programming concepts to solve real world problems. 2. Demonstrate the user defined exceptions by exception handling keywords (try, catch, throw, throws and finally). 3. Use multi threading concepts to develop inter process communication. 4. Develop java application to interact with database by using relevant software component (JDBC Driver). 5. Solve real world problems using Collections								
COURSE OUTCOMES: At the end of the course, student will be able to 1. Use object oriented programming concepts to solve real world problems. 2. Demonstrate the user defined exceptions by exception handling keywords (try, catch, throw, throws and finally). 3. Use multi-threading concepts to develop inter process communication. 4. Develop java application to interact with database by using relevant software component (JDBC Driver). 5. Build the internet-based dynamic applications using the concept of applets								
UNIT-I	JAVA BASICS						Classes: 12	
JAVA BASICS: Review of Object oriented concepts, History of Java, Java buzzwords, JVM architecture, Data types, Variables, Scope and life time of variables, arrays, operators, control statements, type conversion and casting, simple java program, constructors, methods, Static block, Static Data, Static Method ,String and String Buffer Classes, Using Java API Document.								
UNIT-II	INHERITANCE, POLYMORPHISM, PACKAGES AND INTERFACES						Classes: 11	
INHERITANCE AND POLYMORPHISM: Basic concepts, Types of inheritance, Member access rules, Usage of this and Super key word, Method Overloading, Method overriding, Abstract classes, Encapsulation, Need for encapsulation in java, Data hiding vs Encapsulation, getter and setter methods, Dynamic method dispatch, Usage of final keyword. PACKAGES AND INTERFACES: Defining package, Access protection, importing packages, Defining and Implementing interfaces, and Extending interfaces								
UNIT-III	EXCEPTION HANDLING AND FILES						Classes: 10	
I EXCEPTION HANDLING: Exception types, Usage of Try, Catch, Throw, Throws and Finally keywords, Built-in Exceptions, Creating own Exception classes. I / O STREAMS AND FILES: Concepts of streams, Stream classes- Byte and Character stream, Reading console Input and Writing Console output, File Handling								

UNIT-IV	MULTITHREADING AND JDBC	Classes: 10
MULTI THREADING: Concepts of Thread, Thread life cycle, creating threads using Thread class and Runnable interface, Synchronization, Thread priorities, Inter Thread communication. JDBC-Connecting to Database - JDBC Type 1 to 4 drives, connecting to a database, querying a Database and processing the results, updating data with JDBC		
UNIT-V	COLLECTION FRAMEWORK	Classes: 10
COLLECTION FRAMEWORK: Introduction to Java Collections, Overview of Java Collection frame work, Generics, Commonly used Collection classes- Array List, Vector, Hash table, Stack, Enumeration, Iterator, String Tokenizer, Random, Scanner, calender and Properties		
Text Books:		
1. Herbert Schildt and Dale Skrien, Java Fundamentals – A comprehensive Introduction , McGraw Hill, 1st Edition, 2013. 2. Herbert Schildt, —Java the complete referencell, McGraw Hill, Osborne, 7th Edition, 2011. 3. T.Budd, —Understanding Object- Oriented Programming with Javall, Pearson Education, Updated Edition (New Java 2 Coverage), 1999.		
Reference Books:		
1. P.J.Dietel and H.M.Dietel , —Java How to programll, Prentice Hall, 6th Edition, 2005. 2. P.Radha Krishna , —Object Oriented programming through Javall, CRC Press, 1st Edition, 2007. 3. S.Malhotra and S. Choudhary, —Programming in Javall, Oxford University Press, 2nd Edition, 2014.		

DATA STRUCTURES

II B. TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6CS05	PCC	3	0	0	3	40	60	100
Contact Classes:48	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 48			
COURSE OBJECTIVES 1. Understand basic concepts about stacks, queues and their applications. 2. The student should be able to choose appropriate data structures, understand the performance, and use it to design algorithms for a specific problem. 3. Understand basic concepts of singly linked lists and singly linked lists. 4. Write programs for sorting and searching and hashing. 5. Use advanced data structures like Tress and Graphs for efficient problem solving. 6. Introduces pattern matching algorithms.								
COURSE OUTCOMES At the end of the course, student will be able to: 1. Solve problems using data structures such as stacks and Queues. 2. Solve problems using singly linked lists and double linked list. 3. Compare various types of searching and sorting techniques in terms of implementation, operation and performance. 4. Implement Tree data structure and its variants. 5. Identify the importance and application of Graph data Structure with problem solving techniques.								
UNIT-I	Introduction to Data Structures, Stacks & Queues					CLASSES: 14		
Introduction: Introduction to Data Structures, abstract data types, The Stack ADT: Stack operations, implementing the Stack using Array, Stack Applications: Balanced Delimiters, Evaluating Postfix Expressions, The Queue ADT: Queue operations, Implementing the Queue using Array, Circular Queue operations, Implementing the Circular Queue using Array, Priority Queues: Priority Queue operations, The Priority Queue ADT Implementation.								
UNIT-II	Linked Lists					CLASSES:12		
Linked List ADT - Singly Linked Lists (Insertion, Deletion and Traversing), Doubly Linked Lists (Insertion, Deletion and Traversing), Circular Linked Lists (Insertion, Deletion and Traversing), Stacks implementation Using Linked list, Queues Implementation Using Linked list.								
UNIT-III	Searching and Sorting					CLASSES:8		
Searching: Linear Search - Binary search. Sorting: Bubble Sort - Selection Sort - Insertion Sort – Quick Sort - Merge Sort - Heap sort, comparison of sorting methods.								
UNIT-IV	Trees					CLASSES:14		
Trees –Terminology, Representation of Trees ,Binary Tree, Binary Tree traversals, Binary search trees: operations-Insertion, Delete and Searching – AVL Trees: operations- Insertion, Delete and Searching, B-Trees, B-Tree of order m, height of a B-Tree, insertion, deletion and searching Red-Black Trees, Splay Trees, Comparison of Search Trees.								
UNIT-V	Graphs & Text Processing					CLASSES:12		
Graphs: Introduction, Definition, Terminology, Graph ADT, Graph traversals- DFS and BFS, Dijkstra’s Algorithm, Bellman Ford Algorithm, Floyd Warshall Algorithm. Text Processing: Pattern matching algorithms-Brute force, Knuth Morris-Pratt algorithm, Tries-Standard Tries, Compressed Tries, and Suffix tries.								

TEXTBOOKS

1. Fundamentals of Data Structures in C, 2 nd Edition, E. Horowitz, S. Sahni and Susan Anderson Freed, Universities Press.
2. Data Structures using C A. S.Tanenbaum, Y. Langsam, and M.J. Augenstein, PHI/Pearson Education.

REFERENCE BOOKS

1. Data Structures: A Pseudocode Approach with C, 2 nd Edition, R. F. Gilberg and B.A.Forouzan, Cengage Learning.

DATA STRUCTURES LAB

II B.TECH-I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS06	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100
COURSE OBJECTIVES The course should enable the students to: 1. It covers various concepts of C programming language 2. It introduces searching and sorting algorithms 3. It provides an understanding of data structures such as stacks and queues.								
COURSE OUTCOMES The course should enable the students to: 1. Ability to develop C programs for computing and real-life applications using basic elements like control statements, arrays, functions, pointers and strings, and data structures like stacks, queues and linked lists. 2. Ability to Implement searching and sorting algorithms								
LIST OF EXPERIMENTS								
WEEK-1	INTRODUCTION TO STRUCTURES							
1. Write a program that implement structures (its operations) using i) Structures ii)Array of Structures iii)Nested Structures iv)Pointer to Structures								
WEEK-2	STACK OPERATIONS							
2. Write a program that implement Stack (its operations) using array and Linked list								
WEEK-3	QUEUE OPERATIONS							
3. Write a program that implement Queue (its operations) using array and Linked list								
WEEK-4	SINGLY LINKED LIST DOUBLY LINKED LIST							
4. Write a program that uses functions to perform the following operations on singly linked list & doubly linked list Creation ii) Insertion iii) Deletion iv) Traversal								
WEEK-5	CIRCULAR LINKED LIST							
5. Write a program that uses functions to perform the following operations on circular linked list. Creation ii) Insertion iii) Deletion iv) Traversal								
WEEK-6	SORTING TECHNIQUES - I							
6. Write a program that implements the following sorting methods. i) Bubble sort ii) Insertion sort iii) Selection sort								

WEEK-7	SORTING TECHNIQUES - II
7. Write a program that implements the following sorting methods. i) Quick sort ii) Merge sort iii) Heap sort	
WEEK-8	TREE TRAVERSALS
8. Write a program to implement the tree traversal methods. i) Pre-order ii) In order iii) Post order	
WEEK-9	BINARY SEARCH TREES
9. Write a program to implement i) Binary Search tree ii) B Trees	
WEEK-10	BINARY SEARCH TREES
10. Write a program to implement i) B+ Trees ii) AVL trees iii) Red-Black Trees	
WEEK-11	GRAPH TRAVERSAL
11. Write a program to implement the graph traversal methods. i) DFS ii) BFS	
WEEK-12	PATTERN MATCHING
12. Implement a Pattern matching algorithms using Brute force, Knuth-Morris-Pratt	
TEXTBOOKS	
1. Fundamentals of Data Structures in C, 2nd Edition, E. Horowitz, S. Sahni and Susan Anderson Freed, Universities Press. 2. Data Structures using C A.S. Tanenbaum, Y. Langsam, and M.J. Augenstein, PHI/Pearson Education.	

OBJECT ORIENTED PROGRAMMING THROUGH JAVA LAB

II B.TECH I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT03	ESC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100
Contact Classes:Nil	Tutorial Classes:Nil	Practical Classes:36			Total Classes:36			
COURSE OUTCOMES: At the end of the course the students are able to: 1. Implement Object Oriented programming concept using basic syntax's of control Structures, strings and function for developing skills of logic building activity. 2. Understand the use of different exception handling mechanisms and concept of multi-threading for robust and efficient application development. 3. Understand and implement concepts on file streams and operations in java programming for a a given application programs. 4. Develop java application to interact with database by using relevant software component (JDBC Driver).								
LIST OF EXPERIMENTS								
WEEK-1	JAVA BASICS							
a. Write a java program that prints all real solutions to the quadratic equation $ax^2+bx+c=0$. Read in a, b, c and use the quadratic formula. b. The Fibonacci sequence is defined by the following rule. The first two values in the sequence are 1 and 1. Every subsequent value is the sum of the two values preceding it. Write a java program that uses both recursive and non-recursive functions.								
WEEK-2	ARRAYS							
a. Write a java program to sort given list of integers in ascending order. b. Write a java program to multiply two given matrices								
WEEK-3	STRINGS							
a. Write a java program to check whether a given string is palindrome. b. Write a java program for sorting a given list of names in ascending order.								
WEEK-4	OVERLOADING & OVERRIDING							
a. Write a java program to implement method overloading and constructors overloading. b. Write a java program to implement method overriding.								
WEEK-5	INHERITANCE							
a. Write a java program to create an abstract class named Shape that contains two integers and an empty method named print Area (). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method print Area () that prints the area of the given shape.								
WEEK-6	INTERFACES							

a. Write a program to create interface A in this interface we have two method meth1 and meth2. Implements this interface in another class named MyClass. b. Write a program to give example for multiple inheritance in Java.	
WEEK-7	EXCEPTION HANDLING
a. Write a program that reads two numbers Num1 and Num2. If Num1 and Num2 were not integers, the program would throw a Number Format Exception. If Num2 were zero, the program would throw an Arithmetic Exception Display the exception..	
WEEK-8	I/O STREAMS
a. Write a java program that reads a file name from the user, and then displays information about whether the file exists, whether the file is readable, whether the file is writable, the type of file and the length of the file in bytes. b. Write a java program that displays the number of characters, lines and words in a text file.	
WEEK-9	MULTI THREADING
a. Write a java program that implements a multi-thread application that has three threads. First thread generates random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number	
WEEK-10	GENERIC
a. Write a Java program to swap two different types of data using Generics. b. Write a Java program to find maximum and minimum of two different types of data using Generics	
WEEK-11	COLLECTIONS
Create a linked list of elements. a. Delete a given element from the above list. b. Display the contents of the list after deletion	
WEEK-12	CONNECTING TO DATABASE
a. Write a java program that connects to a database using JDBC and does add, delete, modify and retrieve operations.	
TEXT BOOKS:	
1. P. J. Deitel, H. M. Deitel, "Java for Programmers", Pearson Education, PHI, 4 th Edition, 2007. 2. P. Radha Krishna, "Object Oriented Programming through Java", Universities Press, 2 nd Edition, 2007 3. Bruce Eckel, "Thinking in Java", Pearson Education, 4 th Edition, 2006. 4. Sachin Malhotra, Saurabh Chaudhary, "Programming in Java", Oxford University Press, 5 th Edition, 2010.	

SKILL DEVELOPMENT (ANDROID)

II B.TECH I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AI04	PCC	L	T	P	C	CIE	SEE	Total
		0	0	4	2	0	100	100
Contact Classes:Nil	Tutorial Classes:Nil	Practical Classes:36			Total Classes:36			
Course Objectives: The course should enable the students to: 1. Build programming logic and thereby developing skills in Programming. 2. Help students in learning the basics of android studio. 3. Aid students in understanding how to use the android studio and use the databases.								
Course Outcomes: □ At the end of the course the students are able to: 1.Build programming logic and thereby developing skills in Programming. 2. Develop their own applications.								
LIST OF EXPERIMENTS								
WEEK-1								
1. Create an android application to demonstrate any five types of text boxes.								
WEEK-2								
1.Create an android application to demonstrate Check boxes.								
2.Create an android application to demonstrate radio buttons.								
WEEK-3								
1. Create an android application to demonstrate Image View.								
2. Create an android application to demonstrate Scroll View.								
WEEK-4								
1. Create an android application to demonstrate List View.								
2. Create an android application to demonstrate Grid View.								
WEEK-5								
1. Create an android application to demonstrate page navigation.								
2. Create an android application to demonstrate any three types of layouts.								

WEEK-6	
1. Create a simple calculator application. 2. Create an android application to demonstrate Map View.	
WEEK-7	
1. Create an android application to demonstrate registration form. 2. Create an android application to demonstrate login form by connecting to the database. 3. Create an android application to retrieve a data from the database and display it.	

CONSTITUTION OF INDIA

II B.Tech I SEMESTER:								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS06	MC	L	T	P	C	CIE	SEE	Total
		2	0	0	0	50	0	50
Contact Classes-Nil	Tutorial Classes-Nil	Practical Classes-36			Total Classes-36			
COURSEOBJECTIVES: The course should enable the students to: 1. Understand the need for constitution 2. Appreciate the fundamental duties and rights of the citizens of India. 3. Explain the role and amendments of the constitution in a democratic society. 4. Describe the directive principles of state policy and their significance. 5. List the key features of the constitution, union government and state government.								
COURSE OUTCOMES: At the end of this course students will be able to: Students will be able to: 1. Create awareness about the constitutional values and objectives written in the Indian constitution. 2. List fundamental rights and fundamental duties of Indian citizens. 3. Identify the division of legislative, executive and financial powers between the union and state governments. 4. Understand the working of Indian democracy, its institutions and processes at the local, state and union levels. 5. Explain the functions and responsibilities of election commission of India and union public service commission.								
UNIT-I	HISTORY OF MAKING OF THE INDIAN CONSTITUTION					Classes:12		
Introduction to the constitution of India, the making of the constitution and salient features of the constitution.								
UNIT-II	PHILOSOPHY OF THE INDIAN CONSTITUTION					Classes:12		
Preamble Salient Features, Contours of Constitutional Rights & Duties: Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties, Amendment of the constitutional powers and procedures.								
UNIT-III	UNION GOVERNMENT					Classes:12		
Union Government, Union Legislature (Parliament), Lok Sabha and Rajya Sabha (with powers and functions), president of India (with powers and functions), Prime minister of India (With powers and Functions), Union judiciary (Supreme court), Jurisdiction of the supreme court.								

UNIT-IV	STATE GOVERNMENT	Classes:12
State Government, State legislature (Legislative Assembly/ Vidhan Sabha, Legislative council/ Vidhan parishad), powers and functions of the state legislature, State executive, Governor of the state (with powers and functions), The chief Minister of the state (with powers and functions), State Judiciary (High Courts)		
UNIT-V	ELECTION COMMISSION	Classes:12
Election Commission: Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.		
TEXTBOOKS:		
1. M.V.Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice–Hall of India Pvt. Ltd. New Delhi 2. Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Rights Guarantees in India, Oxford University Press, 2012 3. P.M.Bakshi, The constitution of India, Universal Law Publishing Co., 4. The Constitution of India, 1950 (Bare Act), Government Publication. 5. Dr.S.N.Busi, Dr.B.R.Ambedkar framing of Indian Constitution, 1st Edition, 2015.		
REFERENCE BOOKS:		
1. M.P.Jain, Indian Constitution Law, 7th Edn. Lexis Nexis, 2014. 2. D.D.Basu, Introduction to the Constitution of India, Lexis Nexis, 2015. 3. Indian constitution at work, NCERT 4. Subash Kashyap, Indian Constitution, National Book Trust 5. J.A.Siwach, Dynamics of Indian Government & Politics 6. D.C.Gupta, Indian Government and Politics 7. H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes (Universal Law Publication) 8. J.C.Johari, Indian Government and Politics Hans 9. J.Raj Indian Government and Politics		
Web References:		
1. www.hss.iitb.ac.in/en/lecture-details 2. www.iitb.ac.in/en/event/2nd-lecture-institute-lecture-series-indian-constitution		
MOOC Course		
1. nptel.ac.in/courses/109104074/8 2. nptel.ac.in/courses/109104045/ 3. nptel.ac.in/courses/101104065/		

II B.TECH II SEMESTER SYLLABUS

COMPUTER ORIENTED STATISTICAL METHODS

II B. TECH- II SEMESTER:								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS03	BSC	L	T	P	C	CIA	SEE	Total
		3	1	0	4	40	60	100
Contact Classes-42	Tutorial Classes- Nil	Practical Classes-Nil				Total Classes-42		
COURSE OBJECTIVES 1. Evaluation of Probability distribution of Discrete and Continuous random variables and their moments. 2. The concept of correlation and regression, covariance and sampling distribution 3. Evaluation of the given data for appropriate test of hypothesis for large samples. 4. Evaluation of the given data for appropriate test of hypothesis for small samples and one way ANOVA 5. To learn the concept of Markov chain, transition probabilities in discrete & continuous time and Stochastic simulation techniques.								
COURSE OUTCOMES At the end of the course, student will be able to: 1. Evaluation of Probability distribution of Discrete and Continuous random variables and their moments. 2. Apply the concept of correlation and regression to find covariance and Sampling distribution 3. Evaluate the given data for appropriate test of hypothesis for Large samples. 4. Evaluate the given data for appropriate test of hypothesis for small samples and one way ANOVA 5. Recognize if a given stochastic system with finite number of states is a Markov chain or not and also identify classes of states in Markov chains and Characterize the states.								
UNIT-I	RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS					Classes :8		
Random Variables – Discrete and Continuous. Probability distributions, mass function/ density function of a probability distribution, mathematical expectation. Binomial, Poisson, Normal distributions -their Properties. Moments about origin, central moments, skewness, Kurtosis and find the mean and variance.								
UNIT-II	CORRELATION & REGRESSION AND SAMPLING DISTRIBUTIONS					Classes-9		
Coefficient of correlation, the rank correlation, Covariance of two random variables. Regression- Regression Coefficient, The lines of regression. Sampling: Definitions of population, sampling, statistic, parameter. Types of sampling, Expected values of Sample mean and variance, sampling distribution, Standard error, Sampling distribution of means and sampling distribution of variance. Parameter estimation- Point estimation and interval estimation								
UNIT-III	TESTING OF HYPOTHESIS - I					Classes:9		
Testing of hypothesis: Null hypothesis, Alternate hypothesis, Type I& Type II errors – critical region, confidence interval, Level of significance. One sided test, Two sided test. Large sample tests: (i) Test of Equality of means of two samples, equality of sample mean and population mean (cases of known variance & unknown variance, equal and unequal variances) (ii) Tests of significance of difference between sample S.D and population S.D. (iii) Tests of significance difference between sample proportion and population proportion, difference between two sample proportions.								

UNIT-IV	TESTING OF HYPOTHESIS-II	Classes:8
Student t-distribution, its properties; Test of significance sample mean and population mean, difference between means of two small samples. Snedecor's F- distribution and its properties. Test of equality of two population variances. Chi-square distribution, its properties, Chi-square test of goodness of fit. One way ANOVA		
UNIT-V	STOCHASTIC PROCESSES AND MARKOV CHAINS	Classes:8
Introduction to Stochastic processes- Markov process. Transition Probability, Transition Probability Matrix, First order and Higher order Markov process, n-step transition probabilities, Markov chain, Steady state condition, Markov analysis.		
Text Books:		
1. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010. 2. Probability and Statistics for Engineers by Richard Arnold Johnson, Irwin Miller and John E. Freund, New Delhi, Prentice Hall. 3. Probability and Statistics for Engineers and Sciences by Jay L. Devore, Cengage Learning		
Reference Books:		
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. Fundamentals of Mathematical Statistics by S.C. Guptha&V.K. Kapoor, S. Chand 3. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross, Academic Press		
Web references:		
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/		
E -Text Books:		
1. https://www.e-booksdirectory.com/details.php?ebook=10166		
MOOCS Course:		
1. https://swayam.gov.in/ 2. https://onlinecourses.nptel.ac.in/		

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS

II B. TECH- II SEMESTER								
Course Code	Category	Hours /Week			Credits	Maximum Marks		
A6HS08	HSC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes-60	Tutorial Classes-Nil	Practical Classes-Nil			Total Classes-60			
COURSE OBJECTIVES: The course should enable the students to: To enable the student to understand and appreciate, with a particular insight, the importance of certain basic issues governing the business operations namely; demand and supply, production function, cost analysis,markets,forms of business organizations,capital budgeting and financial accounting and financial analysis.								
COURSE OUTCOMES: At the end of the course, the student will: 1. Understand the market dynamics namely, demand and supply, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures. 2. Gain an insight into how production function is carried out to achieve least cost combination of inputs and cost analysis. 3. Develop an understanding of 4. Analyze how capital budgeting decisions are carried out. 5. Understanding the framework for both manual and computerized accounting process 6. Know how to analyze and interpret the financial statements through ratio analysis.								
UNIT-I	Introduction & Demand Analysis					CLASSES: 10		
Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.								
UNIT-II	Production & Cost Analysis: Production Function					CLASSES: 10		
Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems) - Managerial Significance								
UNIT-III	Markets & New Economic Environment					CLASSES: 12		
Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing: Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of different forms of Business Organisation: Sole Proprietorship, Partnership, Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario								
UNIT-IV	Capital Budgeting					CLASSES: 14		

Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements, Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capital Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method, Accounting Rate of return (ARR) and Net Present Value Method (simple problems).		
UNIT-V	Introduction to Financial Accounting & Financial Analysis	CLASSES: 14
Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, and Trial Balance - Final Accounts (Trasing Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du Pont Chart.		
TEXT BOOKS		
1. Varshney & Maheswari: Managerial Economics, Sultan Chand, 2009. 2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013. 3. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.		
REFERENCE BOOKS		
1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012. 2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012. 3. Lipsey & Chrystel, Economics, Oxford University Press, 2012. 4. Domnick Salvatore: Managerial Economics In a Global Economy, Thomson, 2012. 5. Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012. 6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012. 7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012. 8. Dwivedi: Managerial Economics, Vikas, 2012. 9. Shailaja & Usha: MEFA, University Press, 2012. 10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012. 11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011. 12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.		

ARTIFICIAL INTELLIGENCE

II B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AI02	PCC	L	T	P	C	CIE	SEE	Total
		2	0	0	2	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. The difference between optimal reasoning vs human like reasoning
2. Understand the notions of state space representation, exhaustive search, heuristic search along with the time and space complexities
3. Different knowledge representation techniques
4. Understand the applications of AI: namely Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Apply AI techniques to solve problems of Game Playing, Expert Systems, Machine Learning.
2. Formulate an efficient problem space for problem solving and Represent knowledge using the appropriate technique
3. Design and develop expert systems using Prolog and solve uncertainty problems.
4. Analyze real world problems and implement using ANN, CNN and AI techniques.
5. Implement the concepts of AI in Healthcare, Robotics and Agriculture

UNIT - I	INTRODUCTION & PROBLEM SOLVING	CLASSES: 14
Introduction: History, Intelligent Systems, Foundations of AI, Sub areas of AI, Applications. Problem Solving – State-Space Search and Control Strategies: Introduction, General Problem Solving, Characteristics of Problem, Exhaustive Searches, Heuristic Search Techniques, Iterative-Deepening A*, Constraint Satisfaction, Game Playing, Alpha-Beta Pruning		
UNIT - II	LOGIC PROGRAMMING & KNOWLEDGE REPRESENTATION	CLASSES: 12
Logic Concepts and Logic Programming: Introduction, Propositional Logic, Natural Deduction System, Axiomatic System, Semantic Tableau System in Propositional Logic, Resolution Refutation in Propositional Logic, Predicate Logic, Logic Programming. Knowledge Representation: Introduction, Approaches to Knowledge Representation, Knowledge Representation using Semantic Network, Extended Semantic Networks for KR, Knowledge Representation using Frames.		
UNIT - III	EXPERT SYSTEMS & PROLOG	CLASSES: 16
Expert System and Applications: Introduction, Phases in Building Expert Systems, Expert System Architecture, Expert Systems Vs Traditional Systems, Application of Expert Systems, Uncertainty Measure: Bayesian Belief Networks, Certainty Factor Theory, Dempster - Shafer Theory. Prolog: Basics of Prolog, Applications of Prolog, Representation, Structure, Backtracking, Type of Prolog, variables in Prolog, Operators in Prolog, Input and Output Terms in Prolog.		
UNIT - IV	ANN, CNN	CLASSES: 14
Artificial Neural Networks: Introduction, Artificial Neural Networks, Single-Layer Feed Forward Networks, Multi-Layer Feed-Forward Networks, Radial-Basis Function Networks, Design Issues of Artificial Neural Networks, Recurrent Networks. Convolutional Neural Networks: Pooling, Padding, Layers used to build ConvNets, Working of CNN, Python Libraries used for AI: Tensor flow, Keras, Theano, Pytorch, Scikit Learn, Pandas, Image Classification using CNN with Python		

UNIT - V	APPLICATIONS AND CASE STUDIES OF AI	CLASSES: 12
Applications of AI in the real world, Future of AI, Benefits and Risks of AI. Case Studies: <i>AI in Healthcare</i>: Medical Imaging and Diagnostic, <i>AI in Agriculture</i>: Autonomous Tractors, Controlling Pest Infestations, Soil and Crops Health Monitoring, <i>AI in Robotics</i>: Use of Artificial Intelligence in Robotics		
TEXT BOOKS		
1. Saroj Kaushik. Artificial Intelligence. Cengage Learning. 2011 2. Russell, Norvig: Artificial intelligence, A Modern Approach, Pearson Education, Second Edition. 2004		
REFERENCE BOOKS		
1. Rich, Knight, Nair: Artificial intelligence, Tata McGraw Hill, Third Edition 2009. 2. Introduction to Artificial Intelligence by Eugene Charniak, Pearson. 3. Introduction to Artificial Intelligence and expert systems Dan W. Patterson. PHI. 4. Artificial Intelligence by George Fluger Pearson fifth edition.		

DATABASE MANAGEMENT SYSTEMS

II B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS09	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes-54	Tutorial Classes-Nil		Practical Classes-Nil			Total Classes-54		

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the concepts of database management systems and ER Diagrams.
2. Gain knowledge on relational models and integrity constraints.
3. Understand SQL queries and Normalization of relations.
4. Understand the basics of transaction processing and concurrency control.
5. Understand the database storage structures and access techniques

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Recognize the basic concepts and the applications of database systems.
2. Design ER-models to represent simple database application scenarios and convert ER-Model to Relational Model.
3. Demonstrate SQL queries and apply Normalization techniques.
4. Be acquainted with the basics of transaction processing and concurrency control.
5. Familiarity with database storage structures and access techniques

UNIT-I	INTRODUCTION TO DATABASE AND DATABASE DESIGN	Classes: 11
INTRODUCTION: Introduction and applications of DBMS, Purpose of data base, Data Independence, Database System architecture- Levels, Database users and DBA. DATABASE DESIGN: Database Design Process, ER Diagrams - Entities, Attributes, Relationships, Constraints, keys, extended ER features, Generalization, Specialization, Aggregation, Conceptual design with the E-R model		
UNIT-II	RELATIONALMODEL , RELATIONAL ALGEBRA AND CALCULUS	Classes: 11
INTRODUCTION TO THE RELATIONAL MODEL: Integrity constraint over relations, enforcing integrity constraints, querying relational data, logical database design: E-R to relational, introduction to views, destroying/altering tables and views Relational Algebra, Relational Calculus: Tuple relational Calculus, Domain relational calculus		
UNIT-III	INTRODUCTION TO SQL & SCHEMA REFINEMENT	Classes: 10
INTRODUCTION TO SQL: Basics of SQL:DDL, DML, structure – creation, alteration, defining constraints – Primary key, foreign key, unique, not null, check, IN operator, Functions - aggregate functions, Built-in functions – numeric, date, string functions, set operations, sub-queries, correlated sub-queries, Use of group by, having, order by, join and its types, Exist, Any, All CONCEPT OF JOINS : Inner Join, Outer Join, Left Outer Join, Right Outer Join, Self-Join SCHEMA REFINEMENT: Problems caused by redundancy, decompositions, problems related to decomposition, reasoning about functional dependencies, FIRST, SECOND, THIRD normal forms, BCNF, lossless join decomposition, multi-valued dependencies, FOURTH normal form, FIFTH normal form		

UNIT-IV	TRANSACTION MANAGEMENT AND CONCURRENCY CONTROL	Classes: 11
Introduction to Transactions: Transaction Concept, Transaction State, ACID Properties, Implementation of Atomicity and Durability, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, testing for serializability, Lock Based Protocols, Timestamp Based Protocols, Validation- Based Protocols, Multiple Granularity, Recovery and Atomicity, Log-Based Recovery, Recovery with Concurrent Transactions. TCL Commands – Save point, Commit and Rollback		
UNIT-V	OVERVIEW OF STORAGE AND INDEXING. PL/SQL.	Classes: 11
Data on External Storage, File Organization and Indexing, Cluster Indexes, Primary and Secondary Indexes. Tree based indexing: Intuitions for tree Indexes, Indexed Sequential Access Methods (ISAM), B+ Trees : A Dynamic Index Structure. PL/SQL : Overview of Triggers, Stored Procedures: triggers-Row level table level and active databases, Stored Procedures IN, OUT parameters. DBA – Introduction to DBA, Creating Users, Grant/Revoke Permissions on tables using DCL Commands		
Text Books:		
1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, "Database System Concepts", Sixth Edition, Tata McGraw Hill, 2011. 2. Raghurama Krishnan, Johannes Gehrke, "Data base Management Systems", TATA McGraw Hill, 3rd Edition, 2007. Michael McLaughlin, Oracle Database 11g PL/SQL Programming, Oracle press. 4. RamezElmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Fourth Edition, Pearson / Addison Wesley, 2007		
Reference Books:		
1. Peter Rob, Carlos Coronel, Database Systems Design Implementation and Management, 7th edition, 2009. 2. Scott Urman, Michael McLaughlin, Ron Hardman, "Oracle database 10g PL/SQL programming ", 6th edition, Tata McGraw Hill, 2010 3. S.K.Singh, "Database Systems Concepts, Design and Applications", First edition, Pearson Education, 2006. 4. R.P. Mahapatra & Govind Verma, Database Management Systems, Khanna Publishing House, 2013.		

Design and Analysis of Algorithms

II B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS15	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes-50	Tutorial Classes-Nil	Practical Classes-Nil			Total Classes-50			

COURSE OUTCOMES:

At the end of this course students will be able to:

1. Identify various Time and Space complexities of various algorithms
2. Understand Tree Traversal method and Greedy Algorithms
3. Apply Dynamic Programming concept to solve various problems
4. Apply Backtracking, Branch and Bound concept to solve various problems
5. Implement different performance analysis methods for non deterministic algorithms

UNIT-I		Classes: 10
INTRODUCTION: Algorithm, pseudo code for expressing algorithms, performance analysis-space complexity, time complexity, asymptotic notation- big (O) notation, omega notation, theta notation and little (o) notation, recurrences, probabilistic analysis, disjoint set operations, union and find algorithms.		
UNIT-II		Classes: 10
DIVIDE AND CONQUER: General method, applications-analysis of binary search, quick sort, merge sort, AND OR Graphs. GREEDY METHOD: General method, Applications-job sequencing with deadlines, Fractional knapsack problem, minimum cost spanning trees, Single source shortest path problem.		
UNIT-III		Classes: 10
GRAPHS (Algorithm and Analysis): Breadth first search and traversal, Depth first search and traversal, Spanning trees, connected components and bi-connected components, Articulation points. DYNAMIC PROGRAMMING: General method, applications - optimal binary search trees, 0/1 knapsack problem, All pairs shortest path problem, Travelling sales person problem, Reliability design.		
UNIT- IV		Classes: 10
BACKTRACKING: General method, Applications- n-queen problem, Sum of subsets problem, Graph coloring and Hamiltonian cycles. BRANCH AND BOUND: General method, applications - travelling sales person problem, 0/1 knapsack problem- LC branch and bound solution, FIFO branch and bound solution.		

UNIT-V	Classes: 10
NP-HARD AND NP-COMPLETE PROBLEMS: Basic concepts, non-deterministic algorithms, NP-hard and NP-complete classes.	
TEXT BOOKS	
1. Ellis Horowitz, Satraj Sahni, Rajasekharam (2007), Fundamentals of Computer Algorithms, 2nd edition, University Press, New Delhi.	
REFERENCE BOOKS	
1. R. C. T. Lee, S. S. Tseng, R.C. Chang and T. Tsai (2006), Introduction to Design and Analysis of Algorithms A strategic approach, McGraw Hill, India. 2. Allen Weiss (2009), Data structures and Algorithm Analysis in C++, 2nd edition, Pearson education, New Delhi. 3. Aho, Ullman, Hopcroft (2009), Design and Analysis of algorithms, 2nd edition, Pearson education, New Delhi	

ARTIFICIAL INTELLIGENCE LAB

II B. TECH- II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AI03	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100
Contact Classes-Nil	Tutorial Classes-Nil	Practical Classes-36			Total Classes-36			
Course Objectives: 1. Study the concepts of Artificial Intelligence. 2. Learn the methods of solving problems using Artificial Intelligence 3. Learn the knowledge representation techniques, reasoning techniques and planning 4. Identify and Apply Artificial Intelligence concepts to solve real world problems. 5. Introduce the concepts of Expert Systems and machine learning.								
Course Outcomes: At the end of this course students will be able to: 1. Apply the good programming skills to formulate the solutions for computational problems 2. Design and develop solutions for informed and uninformed search problems in AI. 3. Utilize advanced packages like NLTK for implementing artificial intelligence. 4. Demonstrate and enrich knowledge to select and apply AI tools to synthesize information and develop models within constraints of application area 5. Develop a minor project in multidisciplinary areas to demonstrate team work through reports and presentation.								
LIST OF EXPERIMENTS								
Week-1								
Write a Program to Implement Breadth First Search using Python.								
Week-2								
Write a Program to Implement Depth First Search using Python								
Week-3								
Write a Program to Implement a Tic-Tac-Toe game using Python.								
Week-4								
Write a program to implement hill climbing Algorithm								

Week-5	
Write a program to solve Tower of Hanoi	
Week-6	
Write a program to solve 8-Puzzle problem	
Week-7	
Write a Program to Implement Alpha-Beta Pruning using Python.	
Week-8	
Write a program to solve traveling salesman problem.	
Week-9	
Write a program to solve water jug problem	
Week-10	
Write a program to solve the Monkey Banana problem	
Week-11	
Write a Program to Implement 8-Queens Problem using Python	
Week-12	
Write a program to implement the expert system to diagnosis a particular disease	

DATABASE MANAGEMENT SYSTEMS LAB

II B.Tech II SEMESTER:								
Course Code	Category	Hours / Week			credits	Maximum Marks		
A6CS10	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100
Contact Classes-Nil	Tutorial Classes-Nil	Practical Classes-36			Total Classes-36			
COURSE OBJECTIVES: The course should enable the students to: 1. Apply the basic concepts of Database Systems and Applications. 2. Use the basics of SQL and construct queries using SQL in database creation and interaction 3. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system. 4. Analyze and Select storage and recovery techniques of database system.								
COURSE OUTCOMES: At the end of this course students will be able to: 1. Apply the basic concepts of Database Systems and Applications. 2. Develop an ER model for a given database. 3. Use the basics of SQL and construct queries using SQL in database creation and interaction. 4. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system. 5. Analyze and Select storage and recovery techniques of database system. 6. Develop Procedures, Cursors, and Triggers in database system.								
LIST OF EXPERIMENTS								
Week-1	DDL Commands							
a. Creation of Tables using SQL- Overview of using SQL tool and Data types in SQL b. Altering Tables and c. Dropping Tables								
Week-2	Create Table with Primary key and Foreign Key& DML Commands							

- Creating Tables (along with Primary and Foreign keys), Practicing DML commands-
 - a. Insert,
 - b. Update
 - c. Delete.

Week-3	Selection Queries
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- Practicing Select command using following operations
 - a. AND, OR
 - b. ORDER BY
 - c. BETWEEN
 - d. LIKE
 - e. Apply CHECK constraint

Week-4	AGGREGATE FUNCTIONS and Views
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- Practice Queries using following functions
 - a. COUNT,
 - b. SUM,
 - c. AVG,
 - d. MAX,
 - e. MIN,
- Apply constraint on aggregation using
 - a. GROUP BY,
 - b. HAVING,
- VIEWS Create , Modify and Drop

Week-5	Nested QUERIES
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- Practicing Nested Queries using
 - a. UNION,
 - b. INTERSECT,
 - c. CONSTRAINTS
 - d. IN

Week-6	CO- RELATED NESTED QUERIES
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- Practicing Co – Related Nested Queries using
 - a. EXISTS,
 - b. NOT EXISTS. ANY, ALL

Week-7	JOIN QUERIES
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- Practicing Join Queries using - Inner join - Outer join - Equi join - Natural join	
Week-8	TRIGGERS
Practicing on Triggers - creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger.	
Week-9	PROCEDURES
Procedures- Creation of Stored Procedures, Execution of Procedure, and Modification of Procedure	
Week-10	CURSORS
Cursors- Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor.	
Week-11	PL/SQL Part 1
- Practice PL/SQL – - block structure, - variables, - data types,	
Week-12	PL/SQL Part 2
- Practice PL/SQL – - operators, - control structures; - aseca	
Reference Books:	
- Database System Concepts, by Silberschatz, Sudarshan, and Korth, 6th edition. - Database management System by RaghuRamaKrishna, 3rd edition	
Web References:	
http://www.learnadb.com/databases/how-to-convert-er-diagram-to-relational-database https://www.w3schools.com/sql/sql_create_table.asp http://www.edugrabs.com/conversion-of-er-model-to-relational-model/?upm_export=print http://ssyu.im.ncnu.edu.tw/course/CSDB/chap14.pdf http://web.cs.ucdavis.edu/~green/courses/ecs165a-w11/8-query.pdf	

REAL-TIME/FIELD-BASED PROJECT

II B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AI05	PWC	L	T	P	C	CIE	SEE	Total
		-	-	4	2	50	-	50

Evaluation Procedures:

Real-Time(or)Field-based Research Project course: The Internal Evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two Mid-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e. 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation.

The student is deemed to have failed, if he

1. Does not submit a report on the Project, or
2. Does not make a presentation of the same before the internal committee as per schedule, or
3. Secures less than 40% marks in this course.

COURSE OUTCOMES:

1. Demonstrate a sound technical knowledge of selected project topic.
2. Apply algorithmic, mathematical and scientific reasoning to a variety of computational problems.
3. Make use of latest tools/software associated with programming languages to generate effective results
4. Implement software systems that meet specified design and performance requirements
5. Communicate effectively, both orally and in writing.

GENDER SENSITIZATION

II B. TECH - II SEMESTER:

Course Code	Category	Hours /Week			Credits	Maximum Marks		
A6HS05	MC	L	T	P	C	CIE	SEE	Total
		0	0	2	0	50	00	50
Contact Classes:19	Tutorial Classes:Nil	Practical Classes:Nil			Total Classes:19			

COURSE OBJECTIVES

The course should enable the students :

1. To provide a critical perspective on the socialization of men and women.
2. To introduce students to information about some key biological aspects of genders.
3. To expose the students to debates on the politics and economics of work.
4. To help students reflect critically on gender violence.
5. To expose students to more egalitarian interactions between men and women.

COURSE OUTCOMES

At the end of the course the students are able to:

6. Develop a better understanding of important issues related to gender in contemporary India.
7. Sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender
8. Attain a finer grasp of how gender discrimination works in our society and how to counter it
9. Men and Women students and professionals will be better equipped to work and live together as equals.
10. Create a sense of appreciation of women in all walks of life.

UNIT-I	UNDERSTANDING GENDER	CLASSES: 03
Introduction: Introduction to Gender, What is Gender, Why should we study it. Socialization: Making Women, Making Men-Preparing for Woman hood. Growing up Male. First lessons in Caste: Different Masculinities.		
UNIT-II	GENDER ROLES AND RELATIONS	CLASSES: 03
Two or Many-Struggles with Discrimination-Missing Women-Sex Selection and Its Consequences Declining Sex Ratio. Demographic Consequences-Gender Spectrum: Beyond the Binary		
UNIT-III	GENDER AND LABOUR	CLASSES: 03
Di Valuation of Labour-Housework: The Invisible Labor- "My Mother doesn't Work." "Share the Load."-Work: Its Politics and Economics-Fact and Fiction. Unrecognized and Unaccounted work. Additional Reading: Wages and Conditions of Work.		
UNIT-IV	GENDER-BASED VIOLENCE	CLASSES: 04
Sexual Harassment: Say No! -Sexual Harassment, not Eve-teasing- Coping with Everyday Harassment-Further Reading: "Chupulu". Domestic Violence: Speaking Out Is Home a Safe Place? -When Women Unite [Film].Rebuilding Lives. Thinking about Sexual Violence Blaming the Victim-"I Fought for my Life...."Additional Reading: The Caste Face of Violence.		

UNIT-V	GENDER AND COEXISTENCE	CLASSES: 03
Gender Issues- Just Relationships: Being Together as Equals Mary Kom and Onler. Love and Acid just do not Mix. Love Letters. Mothers and Fathers. Rosa Parks The Brave Heart.		
REFERENCE BOOKS		
1. .Menon, Nivedita. Seeing like a Feminist. New Delhi: Zubaan-Penguin Books, 2012 2. AbdulaliSohaila. "I Fought For My Life...and Won."Available online at: http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdual/		
WEB REFERENCES		
1. http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdual/		
E-TEXT BOOKS		
1. Abdul ali Sohaila. "I Fought For My Life...and Won." Available online at: http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdual/		

III B.TECH I SEMESTER SYLLABUS

AUTOMATA AND COMPILER DESIGN

III B. TECH - I SEMESTER:

Course Code	Category	Hours /Week			Credits	Maximum Marks		
A6IT11	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Analyze the major concepts of language translation and phases of compiler design.
2. Enumerating top down and bottom up parsing techniques used in compilation process.
3. Apply the effectiveness of optimization.
4. Introduction the syntax directed translation and type checking.
5. Illustrate the various optimization techniques for designing various optimizing compilers.

COURSE OUTCOMES

The course should enable the students to:

1. Applying the compiler construction tools and Describes the Functionality of each stage of compilation process
2. Construct parsing tables for different types of parsing techniques and syntax directed translations.
3. Apply the different representations of intermediate code generation.
4. Apply code optimization techniques to different programming languages.
5. Generate object code for natural language representations.

UNIT-I	FORMAL LANGUAGE AND INTRODUCTION TO COMPILERS	Classes:10
FORMAL LANGUAGES AND REGULAR EXPRESSION: Languages, Definition: regular expressions, Finite Automata -DFA, NFA. Conversion of regular expression to NFA, Conversion of NFA to DFA, Equivalence of NFA and DFA. INTRODUCTION TO COMPILERS: Definition of compiler, interpreter and its differences, the phases of a compiler, pass and phases of translation, bootstrapping, LEX-lexical analyzer generator.		
UNIT-II	PARSERS	Classes: 14
PARSING: Parsing, role of parser, context free grammar, derivations, parse trees, ambiguity, elimination of left recursion, left factoring, classes of parsing, top down parsing - LL(1) grammars. BOTTOM UP PARSING: Definition of bottom up parsing, handles, handle pruning, stack implementation of shift-reduce parsing, conflicts during shift-reduce parsing, LR grammars, LR parsers-simple LR, canonical LR (CLR) and Look Ahead LR (LALR) parsers, YACC-automatic parser generator.		
UNIT-III	TYPE CHECKING	Classes: 12
TYPE CHECKING: Definition of type checking, type expressions, type systems, static and dynamic checking of types, specification of a simple type checker, equivalence of type expressions, type conversions. SYNTAX DIRECTED TRANSLATION: Syntax directed definition, construction of syntax trees, S-Attributed and L-Attributed definitions.		

UNIT– IV	INTERMEDIATE CODE GENERATION	Classes: 12
RUN TIME ENVIRONMENTS: Source language issues, Storage organization, storage-allocation Strategies, parameter passing. INTERMEDIATE CODE GENERATION: Intermediate forms of source programs– abstract syntax tree, polish notation and three address code, types of three address statements and its implementation, syntax directed translation into three-address code, translation of simple statements.		
UNIT– V	CODE OPTIMIZATION & GENERATION	Classes: 12
CODE OPTIMIZATION: basic blocks and flow graphs, optimization of basic blocks, principal sources of optimization, directed a cyclic graph (DAG) representation of basic block. CODE GENERATION: Machine dependent code generation, object code forms, peephole optimization.		
Text Books:		
I. K.L.P Mishra,N.Chandra sekaran(2003),Theory of computer science- Automata Languages and computation, 2nd edition, Prentice Hall Of India,NewDelhi, India. II. Compilers Principles,Techniques and Tools Aho, Ullman, Ravisethi,Pearson Education.		
Reference Books:		
Introduction to Theory of computation.Sipser,2nd Edition,Thomson.,2009. Modern Compiler Construction in C,Andrew W.Appel Cambridge University Press. Kenneth C.Louden(1997),Compiler Construction–Principles and Practice,1st edition,PWS Publishing. Elements of Compiler Design,A.Meduna,Auerbach Publications,Taylor and Francis Group. Principles of Compiler Design, V.Raghavan, TMH.		

MACHINE LEARNING

III B. TECH - I SEMESTER:

Course Code	Category	Hours /Week			Credits	Maximum Marks		
A6AI06	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes:62	Tutorial Classes:Nil	Practical Classes:Nil				Total Classes:62		

Course Objectives

The course should enable the students to:

1. Ability to comprehend the concept of supervised and unsupervised learning techniques
2. Differentiate regression, classification and clustering techniques and to implement their algorithms.
3. Analyze the performance of various machine learning techniques and to select appropriate features for training machine learning algorithms.
4. Be familiar with basic machine learning algorithms with classification and clustering.
5. Gain experience of doing independent study and research

Course Outcomes

Upon successful completion of the course, the student is able to

1. Recognize the characteristics of machine learning that makes it useful to solve real-world problems.
2. Evaluate exploratory data analysis and Data preparation and preprocessing on different datasets.
3. Design and implement machine learning solutions of classification, regression problems.
4. Choose an appropriate clustering technique to solve real world problems.
5. Choose a suitable machine learning model, implement and examine the performance of the chosen model for a given real world problems.

UNIT-I	INTRODUCTION	Classes: 12
Machine Learning Foundations: Introduction to machine learning, learning problems and scenarios, need for machine learning, types of learning, standard learning tasks, the Statistical Learning Framework, Probably Approximately Correct (PAC) learning.		
UNIT-II	DESCRIPTIVE STATISTICS	Classes: 12
Data representation, types of data: Quantitative data-nominal, ordinal, Qualitative data- interval and continuous, central tendency- calculating mean mode median, mean vs median, variability, variance, standard deviation, Mean Absolute Deviation using sample data set, finding the percentile, inter quartile range, Box Plot, Outlier, whisker, calculating correlation, covariance, causation. Exploratory data analysis, Data preparation, Data pre processing and Data visualization		
UNIT-III	SUPERVISED LEARNING	Classes: 14
Learning a Class from Examples, types of Classification- Linear, Non-linear, Multi-class and Multi-label classification.		

Perceptron: – multi layer neural networks – back propagation - learning neural networks structures – support vector machines: – soft margin SVM and hard margin – going beyond linearity – generalization and over fitting – regularization – validation.

Decision Trees: ID3, Classification and Regression Trees (CART), Regression: Linear Regression, Multiple Linear Regression, Logistic Regression.

UNIT-IV

UNSUPERVISED LEARNING

Classes: 12

Introduction to clustering, Hierarchical: AGNES, DIANA, Partitional: K-means clustering, K-Mode Clustering, Self-Organizing Map, Expectation Maximization, Gaussian Mixture Models, Principal Component Analysis (PCA), Locally Linear Embedding (LLE), Factor Analysis, Neural Networks: Introduction, Perceptron, Multi layer Perceptron, Support vector machines: Linear and Non Linear, Kernel Functions, K-Nearest Neighbors.

UNIT-V

ENSEMBLE AND PROBABILISTIC LEARNING

Classes: 12

Ensemble Learning Model Combination Schemes, Voting, Error-Correcting Output Codes, Bagging: Random Forest Trees, Boosting: Adaboost, Gradient Boosting, Xg boost, Stacking, Bayesian Learning, Bayes Optimal Classifier, Naïve Bayes Classifier, Bayesian Belief Networks, Mining Frequent Patterns.

Text Books:

1. Giuseppe Bonaccorso, —Machine Learning AlgorithmsII, 2nd Edition, Packt, 2018
2. Tom Mitchel —Machine LearningII, Tata McGraW Hill, 2017.

Reference Books:

1. Ethem Alpaydin, —Introduction to Machine LearningII, PHI, 2004
2. Stephen Marshland, —Machine Learning: An Algorithmic Perspectivell,CRC PressTaylor & Francis, 2nd Edition, 2015
3. Abhishek Vijavargia —Machine Learning using PythonII, BPB Publications, 1 st Edition, 2018
INTRODUCTION TO MACHINE LEARNING
4. Introduction to Machine Learning The Wikipedia Guide
5. Ethem Alpaydin, Introduction to machine learning, second edition, MIT press.

MOOCS Course:

- 1 <https://swayam.gov.in/>
- 2 <https://onlinecourses.nptel.ac.in/>

UNIT-IV	JQUERY	Classes: 16
JQUERY: Introduction to jQuery – Selectors – Elements: Manipulations, Changing and Setting elements – Event Models: Event handlers – Animations & Effects – Functions – Plugins		
UNIT-V	FLASK	Classes: 10
FLASK Basics: FLASK installation, Basic Structure of application, Routing, variable rules, URL building, HTTP methods, Template, static files. FLASK Advance: Request object, Response object, sending form data to template, Redirect errors, message flashing, file uploading, define and access database.		
Text Books:		
1. Web Technologies, Uttam K Roy, Oxford University Press 2. Beginning Angular JS, Grant, Andrew— A Press 2014 3. Pro Mean Stack Development, 1st Edition, ELadElrom, Apress O'Reilly 4. Bear Bibeault and Yehuda Katz, jQuery in Action, 2008. 5. MongoDB – The Definitive Guide, 2nd Edition, Kristina Chodorow, O'Reilly 6. Flask Web Development, 2nd Edition, Miguel Grinberg, March 2018, O'Reilly Media, Inc.,.		
Reference Books:		
1. Harvey & Paul Deitel & Associates, Harve Deitel and Abbey Deitel, "Internet and World Wid Web - How To Program", Fifth Edition, Pearson Education, 2011. 2. Gopalan N.P. and Akilandeswari J., Web Technology, Prentice Hall of India, 2011.		
Web references:		
1. https://www.w3schools.com 2. https://www.tutorialspoint.com 3. https://www.javatpoint.com/		

CLOUD COMPUTING (PROFESSIONAL ELECTIVE - I)

III B. TECH I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6DS29	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students:

1.Inculcate the concept of Virtualization

2.Familiarize the concepts of cloud computing and services

3.Explain cloud mechanisms and architecture

4.Explain cloud security and disaster recovery

5.Describe the cloud real time platforms

COURSE OUTCOMES

At the end of the course, students will be able to:

1.Implement the concept of Virtualization to meet the needs of cloud computing

2. Illustrate the concepts of cloud and demonstrate their use in storage systems, cloud models.

3. Analyze various cloud models and apply them to solve problems on the cloud.

4. Make use of Cloud security and disaster recovery mechanisms to resolve the issues in cloud.

5.Examine the real world problems and select an appropriate cloud mechanism to provide an effective solution

UNIT - I	INTRODUCTION TO COMPUTING PARADIGMS & CONCEPTS	CLASSES: 10
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Computing Paradigms:

High-Performance Computing, Parallel Computing, Distributed Computing, Cluster Computing, Grid Computing, Cloud Computing, Bio computing, Mobile Computing, Quantum Computing, Optical Computing, Nano computing.

UNIT - II	HISTORY OF CLOUD COMPUTING AND MODELS	CLASSES: 12
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HISTORY OF CLOUD COMPUTING AND MODELS:

Cloud Computing: Origin and influences, Basic concepts and terminology, Goals and benefits, Risks and challenges. Cloud Models, roles and boundaries, Cloud characteristics, Cloud delivery models, Cloud deployment models.

UNIT - III	CLOUD COMPUTING TECHNIQUES, ARCHITECTURE AND SERVICES	CLASSES: 12
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CLOUD COMPUTING MECHANISMS AND ARCHITECTURE:

Cloud-Enabling Technology: Broadband networks and internet architecture, Data center technology, Virtualization technology, Web technology, Multitenant technology, Service technology. Cloud Architectures: Architecture - Workload distribution, Resource pooling, Dynamic scalability, Elastic resource capacity, Service load balancing, Cloud bursting, Elastic disk provisioning, Redundant storage.

UNIT - IV	CLOUD SECURITY, PLANNING AND DISASTER RECOVERY	CLASSES: 12
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CLOUD SECURITY, PLANNING DISASTER RECOVERY:

Cloud Security: Data, Network and host security, Cloud security services and cloud security possible solutions. Cloud Disaster Recovery: Disaster recovery planning, Disasters in the cloud, Disaster management, Capacity planning and cloud scale.

UNIT - V	CLOUD SERVICE PROVIDERS- USING CASE STUDIES	CLASSES: 10
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CLOUD SERVICES USING CASE STUDIES: Case Studies: Software-as-a-Service (SaaS) - Salesforce.com, Facebook, Platform-as-a-Service (PaaS) - Google App Engine, MS-Azure and IBM Blue mix; Infrastructure-as-a- Service (IaaS) - Amazon EC2, Amazon S3 and Netflix.

TEXT BOOKS

- 1) Thomas Erl and Ricardo Puttini Cloud Computing Concepts, Technology and Architecture, Pearson, 2013.
- 2) Essentials of cloud Computing: K. Chandrasekhran, CRC press, 2014
- 3) Ivanka Menken and Gerard Blokdijs, Cloud Computing Virtualization Specialist Complete Certification Kit-Study Guide Book, Lightning Source, 2009

REFERENCE BOOKS

- 1) Barrie Sosinsky, Cloud Computing Bible, Wiley India Pvt Ltd, 2011.
- 2) Rajkumar Buyya, James Broberg and Andrzej Goscinski, Cloud Computing Principles and Paradigms, John Wiley and Sons, 2011.
- 3) John W. Ritting house and James F. Ransome, Cloud Computing Implementation, Management and Security, CRC Press, Taylor & Francis Group, 2010.

DATA MINING TECHNIQUES (PROFESSIONAL ELECTIVE – I)

III B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6AI11	PEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 60	Tutorial Classes: 00	Practical Classes: NIL			Total Classes: 60			
COURSE OBJECTIVES The course should enable the students to: 1. Learn data mining concepts and understand association rules mining. 2. Discuss classification algorithms learn how data is grouped using clustering techniques. 3. Develop the abilities of critical analysis to data mining systems and applications. 4. Implement practical and theoretical understanding of the technologies for data mining 5. Understand the strengths and limitations of various data mining models COURSE OUTCOMES At the end of the course, student will be able to: 1. Perform the pre-processing of data and apply mining techniques on it. 2. Identify the association rules, classification and clusters in large data sets. 3. Apply supervised learning techniques on given data sets 4. Apply unsupervised learning techniques on given data sets 5. Classify web pages, extracting knowledge from the web.								
UNIT - I	INTRODUCTION TO DATA MINING						CLASSES: 14	
Introduction to Data Mining: Introduction, What is Data Mining, Definition, Knowledge Discovery from Data (KDD), What Kinds of Data can be Mined, Data Mining Tasks, Integration of Data Mining System With A Database or Data Warehouse System, Types of Data Sets and Attribute Values Prepossessing: Data Quality, Major Tasks in Data Pre-processing , Data Cleaning and Data Integration, Data Reduction, Data Transformation and Data Discretization , Measures of Similarity and Dissimilarity- Basics								
UNIT - II	ASSOCIATION RULES						CLASSES: 12	
Association Rules: Problem Definition, Frequent Item Set Generation, The APRIORI Principle, Support and Confidence Measures, Association Rule Generation; APRIORI Algorithm, The Partition Algorithms, FP-Growth Algorithms, Compact Representation of Frequent Item Set- Maximal Frequent Item Set, Closed Frequent Item Set.								
UNIT - III	CLASSIFICATION						CLASSES: 14	

Classification: Problem Definition, General Approaches to solving a classification problem ,Evaluation of Classifiers , Classification techniques, Decision Trees-Decision tree Construction , Methods for Expressing attribute test conditions, Measures for Selecting the Best Split, Algorithm for Decision tree Induction ; Naive-Bayes Classifier, Bayesian Belief Networks; K- Nearest neighbor classification-Algorithm and Characteristics.

UNIT - IV**CLUSTERING****CLASSES: 10**

Clustering: Problem Definition, Clustering Overview, Evaluation of Clustering Algorithms, Partitioning Clustering-K-Means Algorithm, K-Means Additional issues, PAM Algorithm; Hierarchical Clustering-Agglomerative Methods and divisive methods, Basic Agglomerative Hierarchical Clustering Algorithm, Specific techniques, Key Issues in Hierarchical Clustering, Strengths and Weakness; Outlier Detection.

UNIT - V**WEB AND TEXT MINING****CLASSES: 10**

Web And Text Mining: Introduction, web mining, web content mining, web structure mining, we usage mining, Text mining –unstructured text, episode rule discovery for texts, hierarchy of categories, text clustering.

TEXT BOOKS

1. Data Mining: Concepts and Techniques - Jiawei Han, Micheline Kamber, Jian Pei (2012), 3rd edition, Elsevier, United States of America.
2. Introduction to Data Mining, Pang-Ning Tan, Vipin Kumar, Michael Steinbach, Pearson Education
3. Data mining Techniques and Applications, Hongbo Du Cengage India Publishing

Web references:

1. <https://www.geeksforgeeks.org/data-mining-techniques/>
2. [https://www.javatpoint.com/data-mining-techniques.](https://www.javatpoint.com/data-mining-techniques)
3. <https://www.talend.com/resources/data-mining-techniques/>
4. <https://www.simplilearn.com/types-of-data-mining-techniques-article>

E -Text Books:

1. https://books.google.co.in/books?id=pQws07tdpjoC&printsec=copyright&redir_esc=y#v=onepage&q&f=false

MOOCS Course:

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

IMAGE PROCESSING (PROFESSIONAL ELECTIVE – I)

III B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6AI12	PEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 54	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 54			

Course Objectives

The course should enable the students to:

1. Study the image fundamentals and mathematical transforms necessary for image processing.
2. Study Image enhancement techniques.
3. Study image restoration procedures.
4. Study the image compression procedures.

Course Outcomes

At the end of the course, student will be able to:

1. Review the fundamental concepts of a digital image processing system.
2. Perform image enhancement techniques in spatial and frequency domain.
3. Apply the concept of image restoration.
4. Apply the concept of image segmentation.
5. Elucidate the mathematical modeling of image compression.

UNIT-I	Digital Image Fundamentals	Classes: 12
Digital Image through Scanner, Digital Camera. Concept of Gray Levels. Gray Level to Binary Image Conversion. Sampling and Quantization. Relationship between Pixels. Imaging Geometry. 2D Transformations-DFT, DCT, KLT and SVD.		
UNIT-II	Image Enhancement	Classes: 10
Image Enhancement in Spatial Domain Point Processing, Histogram Processing, Spatial Filtering, Enhancement in Frequency Domain, Image Smoothing, Image Sharpening.		
UNIT-III	Image Restoration	Classes: 10
Image Restoration Degradation Model, Algebraic Approach to Restoration, Inverse Filtering, Least Mean Square Filters, Constrained Least Squares Restoration, Interactive Restoration.		
UNIT-IV	Image Segmentation	Classes: 10

Image Segmentation Detection of Discontinuities, Edge Linking and Boundary Detection, Thresholding, Region Oriented Segmentation.

UNIT-V**Image Compression****Classes: 12**

Image Compression Redundancies and their Removal Methods, Fidelity Criteria, Image Compression Models, Source Encoder and Decoder, Error Free Compression, Lossy Compression.

Text Books:

Digital Image Processing: R.C. Gonzalez & R. E. Woods, Addison Wesley/ Pearson Education, 2 nd Ed, 2004.

Reference Books:

- 1.Fundamentals of Digital Image Processing: A. K. Jain, PHI.
- 2.Digital Image Processing using MATLAB: Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddins: Pearson Education India, 2004.
- 3.Digital Image Processing: William K. Pratt, John Wiley, 3rd Edition, 2004.

SOFTWARE TESTING METHODOLOGIES (PROFESSIONAL ELECTIVE – I)

III B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6AI13	PEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 54	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 54			

Course Objectives

The course should enable the students to:

1. Study the fundamental concepts of software testing which includes objectives, process, criteria, strategies, and methods.
2. Discuss various software testing types and levels of testing like black and white box testing along with levels unit test, integration, regression, and system testing.
3. Provide knowledge on transaction flow testing and data flow testing techniques so that the flow of the program is tested as well.
4. Learn the domain testing, path testing and logic based testing to explore the testing process easier.

Course Outcomes

At the end of the course, student will be able to:

1. Able to identify the various bugs and correcting them after knowing the consequences of the bug.
2. Apply Transaction Flow Testing & Data flow testing methods for a software process.
3. Apply domain testing methods for a software process.
4. Apply paths and logic based testing methods for a software process.
5. Apply states and graph matrices testing methods for a software process.

UNIT-I	Introduction & Flow graphs and Path testing	Classes: 12
Introduction: Purpose of testing, Dichotomies, model for testing, consequences of bugs, taxonomy of bugs. Flow graphs and Path testing: Basics concepts of path testing, predicates, path predicates and achievable paths, path sensitizing, path instrumentation, application of path testing.		
UNIT-II	Transaction Flow Testing & Data flow testing	Classes: 10
Transaction Flow Testing: Transaction flows, transaction flow testing techniques. Data Flow testing: Basics of data flow testing, strategies in data flow testing, application of data flow testing.		
UNIT-III	Domain Testing	Classes: 10
Domain Testing: Domains and paths, Nice & ugly domains, domain testing, domains and interfaces testing, domain and interface testing, domains and testability.		

UNIT-IV	Paths & Logic Based Testing	Classes: 10
Paths, Path products and Regular expressions: Path products & path expression, reduction procedure, applications, regular expressions & flow anomaly detection. Logic Based Testing: Overview, decision tables, path expressions, kv charts, specifications.		
UNIT-V	States & Graph Matrices	Classes: 12
State, State Graphs and Transition testing: State graphs, good & bad state graphs, state testing, Testability tips. Graph Matrices and Application: Motivational overview, matrix of graph, relations, power of a matrix, node reduction algorithm, building tools		
Text Books:		
1. Software Testing techniques – Boris Beizer, Dreamtech, second edition. 2. Software Testing Tools – Dr.K.V.K.K.Prasad, Dreamtech.		
Reference Books:		
1. The craft of software testing – Brian Marick, Pearson Education. 2. Software Testing Techniques – SPD(Oreille). 3. Software Testing in the Real World – Edward Kit, Pearson. 4. Effective methods of Software Testing, Perry, John Wiley. 5. Art of Software Testing – Meyers, John Wiley.		

OPERATING SYSTEMS

III B.TECH- I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS11	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

1. To explain main components of OS and their working
2. To familiarize the operations performed by OS as are source Manager
3. To familiarize with Dead Locks
4. To impart various scheduling policies of OS
5. To teach the different memory management techniques.

COURSE OUTCOMES

1. Analyze the different structures and services of operating system.
2. Analyze various algorithms used for OS services with respect to defined/chosen criteria.
3. Solve the resource allocation and sharing problems.
4. Assess different methods to solve OS problems.
5. Analyze the memory management approaches of operating systems.

UNIT-I	OPERATING SYSTEMS OVERVIEW	CLASSES:14
OPERATING SYSTEM OVERVIEW: Introduction, operating system operations, process management, memory management, storage management, protection and security, distributed systems. OPERATING SYSTEMS STRUCTURES: Operating system services and systems calls, system programs, Operating system structure, operating systems generations		
UNIT -II	PROCESS MANAGEMENT	CLASSES:16
PROCESS MANAGEMENT: Process concepts, process state, process control block, scheduling queues, process scheduling, Scheduling algorithms, multi-threaded programming. CONCURRENCY AND SYNCHRONIZATION: Process synchronization, critical section problem, Peterson's solution, synchronization hardware, semaphores, classic problems of synchronization, reader sand writers problem, dining philosophers problem, monitors, synchronization examples(Solaris).		
UNIT -III	DEADLOCKS	CLASSES:14
DEADLOCKS: System model, deadlock characterization, deadlock prevention, detection and avoidance, recovery from deadlock banker's algorithm. MEMORY MANAGEMENT: Swapping, contiguous memory allocation, paging, structure of the page table, Segmentation, virtual memory, demand paging, page-replacement algorithms, allocation of frames, thrashing.		
UNIT -IV	FILESYSTEM	CLASSES:12

FILESYSTEM: Concept of a file, access methods, directory structure, file system mounting, file sharing, protection.

File system implementation: file system structure, file system implementation, directory implementation, Allocation methods, free-space management, efficiency and performance.

UNIT-V**MASS STORAGE STRUCTURE****CLASSES:14**

MASS STORAGE STRUCTURE: Over view of mass storage structure, disk structure, disk attachment, disk scheduling algorithms, swap space management, stable storage implementation, tertiary storage structure.

I/O: Hardware, application I/O interface, kernel I/O subsystem, transforming I/O request to hardware operations, streams, performance

TEXTBOOKS

1. Operating System Principles-Abraham Silberchatz, PeterB.Galvin,Greg Gagne 7th Edition, John Wiley.
2. Advanced programming in the UNIX environment,W.R. Stevens,Pearson education.

REFERENCE BOOKS

1. OperatingSystems-InternalsandDesignPrinciples,WilliamStallings,FifthEdition– 2005,PearsonEducation/PHI
2. Operating System A Design Approach- Crowley,TMH.
3. Modern Operating Systems, AndrewS.Tanenbaum2ndedition,Pearson/PHI
4. UNIX programming environment, Kernighan and Pike,PHI/Pearson Education
5. UNIX Internals-The New Frontiers, U.Vahalia, Pearson Education.

WEB PROGRAMMING LAB

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AI08	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand basic concepts of HTML and CSS to create static web pages

2. Facilitate students to build dynamic, responsive web pages using Angular JS

3. Familiarize basics of Node.js and MongoDB

4. Understand the concepts of jQuery to design client-side web page

5. Familiarize various concepts Flask framework to implement web applications.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Apply the concepts of JavaScript to perform client-side validation and create dynamic web pages.

2. Create web application using AngularJS that depend on MVC architecture.

3. Create applications using Node.js and MongoDB

4. Design and implement client-side web page using jQuery.

5. Design and implement web application using Flask.

WEEK-1	Design a home page which will display your information, i.e. Bio data, using Image Link and File Link to upload images and necessary documents.
WEEK-2	Create a web page with four frames (Picture, table, list, and hyperlink).
WEEK-3	Write a program to demonstrate Event Handling -Validation of registration form -Open a Window from the current window -Change color of background at each click of button or refresh of a page -Display calendar for the month and year selected from combo box On Mouse over event.
WEEK-4	Write an HTML page including any required Java script that takes a number from one text field in the range of 0 to 999 and shows it in range of 0 to 999 and shows it in another text field in words. If the number is out of range, it should show “out of range” and if it is not a number, it should show “not a number” message in the result box.
WEEK-5	Implement the following web applications using AngularJS: A simple calculator web application that takes two numbers and an operator (+, -, /, * and %) from an HTML page and returns the result page with the operation performed on the operands
WEEK-6	Implement the following web applications using AngularJS: A user validation web application, where the user submits the login name and password to the server. The name and password are checked if the data matches, a successful login page is returned. Otherwise a failure message is shown to the user.

WEEK-7	Write a program to show the workflow of JavaScript code executable by creating web server in Node.js.
WEEK-8	Create a text file src.txt and add the following data to it using Nodejs. Mongo Express Angular Node.
WEEK-9	Write a program to perform various CRUD (Create-Read-Update-Delete) operations using Mongoose library functions.
WEEK-10	Create a div using jQuery with style tag
WEEK-11	Create a Zebra Stripes table effect using jQuery
WEEK-12	Develop web application using Flask.
TEXT BOOKS	
1. Web Technologies, Uttam K Roy, Oxford University Press 2. Beginning Angular JS, Grant, Andrew— A Press 2014 3. Pro Mean Stack Development, 1st Edition, ELadElrom, Apress O'Reilly 4. Bear Bibeault and Yehuda Katz, jQuery in Action, 2008. 5. MongoDB – The Definitive Guide, 2nd Edition, Kristina Chodorow, O'Reilly 6. Flask Web Development, 2nd Edition, Miguel Grinberg, March 2018, O'Reilly Media, Inc.,.	
REFERENCE BOOKS	
1. Harvey & Paul Deitel & Associates, Harvey Deitel and Abbey Deitel, "Internet and World Wide Web – How To Program", Fifth Edition, Pearson Education, 2011. 2. Gopalan N.P. and Akilandeswari J., Web Technology, Prentice Hall of India, 2011.	
Web references:	
1. https://www.w3schools.com 2. https://www.tutorialspoint.com 3. https://www.javatpoint.com/	

MACHINE LEARNING LAB

III B.TECH-I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AI09	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

Contact Classes: Nil

Tutorial Classes: Nil

Practical Classes: 36

Total Classes: 36

COURSE OBJECTIVES

1. Make use of Data sets in implementing the machine learning algorithms
2. Implement the machine learning concepts and algorithms in any suitable language of choice.
3. To impart knowledge on the basic concepts underlying machine learning.
4. To acquaint with the process of selecting features for model construction.
5. To familiarize different types of machine learning techniques.

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Design and implement machine learning solutions to classification, regression problems.
2. Analyze the complexity of Machine Learning algorithms and their limitations.
3. Apply appropriate data sets to the Machine Learning algorithms.
4. Identify and apply Machine Learning algorithms to solve real world problems
5. Apply supervised and unsupervised techniques on various data sets.

LIST OF EXPERIMENTS

WEEK-1	BASICS Write a program to demonstrate the following a) Operation of data types in Python. b) Different Arithmetic Operations on numbers in Python. c) Create, concatenate and print a string and access substring from a given string. d) Append, and remove lists in python. e) Demonstrate working with tuples in python. f) Demonstrate working with dictionaries in python.
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WEEK-2	STATISTICAL OPERATIONS Using python write a NumPy program to compute the a) Expected Value b) Mean c) Standard deviation d) Variance e) Covariance f) Covariance Matrix of two given arrays.
WEEK-3	DATA PREPROCESSING – CONTINUOUS / DISCRETE DATA For a given set of training data examples stored in a .CSV file, demonstrate Data Preprocessing in Machine learning with the following steps a) Getting the dataset. b) Importing libraries. c) Importing datasets. d) Finding Missing Data. e) Finding Outliers f) Splitting dataset into training and test set. g) Feature scaling.
WEEK-4	DATA PREPROCESSING – CATEGORICAL DATA For a given set of training data examples stored in a .CSV file, demonstrate Data Preprocessing In Machine learning with the following steps a) Getting the dataset. b) Importing libraries. c) Importing datasets. d) Finding Missing Data. e) Encoding Categorical Data. f) Splitting dataset into training and test set. g) Feature scaling.
WEEK-5	DECISION TREE Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.
WEEK-6	LINEAR REGRESSION Build a linear regression model using python for a particular data set by a) Splitting Training data and Test data. b) Evaluate the model (intercept and slope). c) Visualize the training set and testing set d) predicting the test set result e) compare actual output values with predicted values.
WEEK-7	MULTIPLE LINEAR REGRESSION Build a multiple linear regression model using python for a particular data set by a) Splitting Training data and Test data. b) Evaluate the model (intercept and slope). c) Visualize the training set and testing set d) predicting the test set result e) compare actual output values with predicted values.

WEEK-8	LOGISTIC REGRESSION The dataset contains information of users from a company's database. It contains information about UserID, Gender, Age, EstimatedSalary, and Purchased. Use this dataset for predicting that a user will purchase the company's newly launched product or not by Logistic Regression model. <table><tr><th>User ID</th><th>Gender</th><th>Age</th><th>EstimatedSalary</th><th>Purchased</th></tr><tr><td>15624510</td><td>Male</td><td>19</td><td>19000</td><td>0</td></tr><tr><td>15810944</td><td>Male</td><td>35</td><td>20000</td><td>0</td></tr><tr><td>15668575</td><td>Female</td><td>26</td><td>43000</td><td>0</td></tr><tr><td>15603246</td><td>Female</td><td>27</td><td>57000</td><td>0</td></tr><tr><td>15804002</td><td>Male</td><td>19</td><td>76000</td><td>0</td></tr><tr><td>15728773</td><td>Male</td><td>27</td><td>58000</td><td>0</td></tr><tr><td>15598044</td><td>Female</td><td>27</td><td>84000</td><td>0</td></tr><tr><td>15694829</td><td>Female</td><td>32</td><td>150000</td><td>1</td></tr><tr><td>15600575</td><td>Male</td><td>25</td><td>33000</td><td>0</td></tr><tr><td>15727311</td><td>Female</td><td>35</td><td>65000</td><td>0</td></tr><tr><td>15570769</td><td>Female</td><td>26</td><td>80000</td><td>0</td></tr><tr><td>15606274</td><td>Female</td><td>26</td><td>52000</td><td>0</td></tr><tr><td>15746139</td><td>Male</td><td>20</td><td>86000</td><td>0</td></tr><tr><td>15704987</td><td>Male</td><td>32</td><td>18000</td><td>0</td></tr><tr><td>15628972</td><td>Male</td><td>18</td><td>82000</td><td>0</td></tr><tr><td>15697686</td><td>Male</td><td>29</td><td>80000</td><td>0</td></tr><tr><td>15733883</td><td>Male</td><td>47</td><td>25000</td><td>1</td></tr><tr><td>15617482</td><td>Male</td><td>45</td><td>26000</td><td>1</td></tr><tr><td>15704583</td><td>Male</td><td>46</td><td>28000</td><td>1</td></tr></table>	User ID	Gender	Age	EstimatedSalary	Purchased	15624510	Male	19	19000	0	15810944	Male	35	20000	0	15668575	Female	26	43000	0	15603246	Female	27	57000	0	15804002	Male	19	76000	0	15728773	Male	27	58000	0	15598044	Female	27	84000	0	15694829	Female	32	150000	1	15600575	Male	25	33000	0	15727311	Female	35	65000	0	15570769	Female	26	80000	0	15606274	Female	26	52000	0	15746139	Male	20	86000	0	15704987	Male	32	18000	0	15628972	Male	18	82000	0	15697686	Male	29	80000	0	15733883	Male	47	25000	1	15617482	Male	45	26000	1	15704583	Male	46	28000	1
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WEEK-9	CLUSTERING A python program to implement K-Means, Hierarchical Clustering and PCA.																																																																																																				
WEEK-10	KNN & SVM Write a Python program to implement KNN and SVM.																																																																																																				
WEEK-11	NAIVE BAYES Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.																																																																																																				
WEEK-12	RANDOM FOREST ALGORITHM Implement Random Forest Algorithm using Python.																																																																																																				
TEXTBOOKS																																																																																																					
1. Aurelien Géron - Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition. September 21019, O'Reilly Media, Inc., ISBN: 9781492032649. 2. Tom Mitchel "Machine Learning", Tata McGraW Hill, 2017.																																																																																																					

LINUX PROGRAMMING LAB

III Year I Semester:

Course Code	Category		Hours / Week			Credits	Maximum Marks		
A6IT15	PCC	L	T	P	C	CIE	SEE	Total	
			0	0	2	1	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Apply the basic commands to handle the Linux Environment
2. Use the Shell/C scripting constructs to modify the file system content.
3. Implement the process management concepts using C language.
4. Design Client-Server Applications using Sockets and TCP/UDP protocols.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Write and use the Shell Scripts in managing Linux Environment
2. Construct C Scripts to handle the File system in Linux.
3. Implement the IPC Mechanisms in Linux operating system using C language
4. Build and Analyze the Client-Server Environment using various protocols.

LIST OF EXPERIMENTS

Week 1:

Study and Practice on various commands like man, passwd, tty, script, clear, date, cal, cp, mv, ln, rm, unlink, mkdir, rmdir, du, df, mount, umount, find, unmask, ulimit, ps, who, w.

Week 2:

Study and Practice on various commands like cat, tail, head, sort, nl, uniq, grep, egrep, fgrep, cut, paste, join, tee, pg, comm, cmp, diff, tr, awk, tar, cpio.

Week 3:

- A. Write a shell script that accepts a filename, starting and ending line numbers as arguments and displays all the lines between the given line numbers.
- B. Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.

Week 4:

- A. Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
- B. Write a shell script that receives any number of file names as arguments, checks if every argument supplied is a file or a directory and reports accordingly. Whenever the argument is a file, the number of lines on it is also reported.

Week 5:

- A. Write a shell script that accepts a list of file names as its arguments, counts and reports the occurrence of each word that is present in the first argument file on other argument files.
- B. Write a shell script to list all of the directory files in a directory.

Week 6:

- A. Practice awk commands
- B. Write an awk script to count the number of lines in a file that do not contain vowels.
- C. Write an awk script to find the number of characters, words and lines in a file.

Week 7:

- A. Write a c program that makes a copy of a file using standard I/O and system calls.
- B. Implement in C the following UNIX commands using System calls.
 - A. cat B. mv
- C. Implement in C the following UNIX commands using System calls.
 - A. ls B. ls -l C. ls -a

Week8:

- A. Write a C program to create a child process and allow the parent to display “parent” and the child to display “child” on the screen.
- B. Write a C program to create a Zombie process.
- C. Write a C program that illustrates how an orphan is created.

Week 9:

- A. Write a C program that illustrates suspending and resuming processes using signals.
- B. Write a program to handle the signals like **SIGINT, SIGQUIT, SIGFPE**.

Week 10:

- A. Write C programs that illustrate communication between two unrelated processes using named pipe (FIFO).
- B. Write a C program in which a parent writes a message to a pipe and the child reads the message.

Week11:

- A. Implement **message queue** form of IPC.
- B. Implement a shared **memory** form of IPC.

Week 12:

- A. Write Client server program in C for connection oriented communication between server and client process using unix domain sockets to perform the following
- B. Write a Client server program in C for connection oriented communication between server and client process using internet domain sockets.

TEXT BOOKS:

1. Unix System Programming using C++, T.Chan, PHI.
2. Unix Concepts and Applications, 4th Edition, Sumitabha Das, TMH,2006.

REFERENCES:

1. Beginning Linux Programming, 4th Edition, N.Matthew, R.Stones,Wrox, Wiley India Edition,rp-2008
2. Linux System Programming, Robert Love, O'Reilly, SPD.
3. Advanced Programming in the Unix environment, 2nd Edition, W.R.Stevens, Pearson Education.
4. Unix Network Programming ,W.R.Stevens,PHI.
5. Unix for programmers and users, 3rd Edition, Graham Glass, King Ables, Pearson Education.

MOOC'S / INDEPENDENT STUDY

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AI16	PWC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	-	100	100

Evaluation Procedures:

There shall be an Industry training (or) Internship (or) Industry oriented Mini-project (or) Skill Development Courses (or) Paper Presentation Reputed Journal (or) Industry Oriented Mini Project in collaboration with an industry of their specialization: Students shall register for this immediately after-Year II Semester Examinations and pursue it during summer vacation/semester break & during III Year without effecting regular coursework. Internship at reputed organization (or) Skill development courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in III-year II semester before end semester examination. It shall be evaluated for 100 external marks. The Committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented MiniProject (or) Internship etc, Internal Supervisor and a Senior Faculty Member of the Department. There shall be **NO internal marks** for Industry Training (or) Internship (or) Mini-Project (or) Skill Development Courses (or) Paper Presentation in reputed journal (or) Industry Oriented Mini Project.

COURSE OUTCOMES:

1. Demonstrate a sound technical knowledge of selected project topic.
2. Apply algorithmic, mathematical and scientific reasoning to a variety of computational problems.
3. Make use of latest tools/software associated with programming languages to generate effective results
4. Implement software systems that meet specified design and performance requirements
5. Communicate effectively, both orally and in writing.

HUMAN VALUES AND PROFESSIONAL ETHICS

III Year I Semester:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS10	MC	L	T	P	C	CIE	SEE	Total
		2	0	0	0	50	-	50

COURSE OBJECTIVES:

The course should enable the students :

1. To help the students appreciate the essential complementarity between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.

2. To facilitate the development of a Holistic perspective among students towards life, profession and happiness, based on a correct understanding of the Human reality and the rest of Existence. Such a holistic perspective forms the basis of Value based living in a natural way.

3. To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually satisfying human behaviour and mutually enriching interaction with Nature.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

It ensures students sustained happiness through identifying the essentials of human values and skills.

1. It facilitates a correct understanding between profession and happiness

2. It helps students understand practically the importance of trust, mutually satisfying human behaviour and enriching interaction with nature.

3. Ability to develop appropriate technologies and management patterns to create harmony in professional and personal life.

UNIT-I	Introduction	Classes: 12
Need, basic Guidelines, Content and Process for Value Education: Understanding the need, basic guidelines, content and process for Value Education. Self Exploration - what is it? - its content and process; 'Natural Acceptance' and Experiential Validation - as the mechanism for self exploration. Continuous Happiness and Prosperity - A look at basic Human Aspirations. Right understanding, Relationship and Physical Facilities - the basic requirements for fulfillment of aspirations of every human being with their correct priority. Understanding Happiness and Prosperity correctly - A critical appraisal of the current scenario. Method to fulfill the above human aspirations: understanding and living in harmony at various levels.		
UNIT-II	Understanding Harmony in the Human Being - Harmony in Myself!	Classes: 12
Understanding human being as a co-existence of the sentient 'I' and the material 'Body'. Understanding the needs of Self ('I') and 'Body' - Sukh and Suvidha. Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer). Understanding the harmony of I with the Body: Sanyam and Swasthya; correct appraisal of Physical needs, meaning of Prosperity in detail. Programs to ensure Sanyam and Swasthya.		
UNIT-III	Understanding Harmony in the Family and Society - Harmony in Human - Human Relationship	Classes: 12
Understanding harmony in the Family the basic unit of human interaction. Understanding values in human - human relationship; meaning of Nyaya and program for its fulfilment to ensure Ubhay-tripti; Trust (Vishwas) and Respect (Samman) as the foundational values of relationship. Understanding the meaning of Vishwas; Difference between intention and competence. Understanding the meaning of Samman, Difference between respect and differentiation; the other salient values in relationship. Understanding the harmony in the society (society being an extension of family): Samadhan, Samridhi, Abhay, Sah-astiva as comprehensive Human Goals. Visualizing a universal harmonious order in society - Undivided Society (AkhandSamaj), Universal Order (SarvabhaumVyawastha) - from family to world family!		
UNIT-IV	Understanding Harmony in the nature and Existence - Whole existence as Co-existence	Classes: 12

Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature - recyclability and self-regulation in nature. Understanding Existence as Coexistence (Sah-astitva) of mutually interacting units in all-pervasive space. Holistic perception of harmony at all levels of existence.		
UNIT-V	Implications of the above Holistic Understanding of Harmony on Professional Ethics	Classes: 12
Natural acceptance of human values, Definitiveness of Ethical Human Conduct, Basic for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics: A. Ability to utilize the professional competence for augmenting universal human order, B. Ability to identify the scope and characteristics of people-friendly and eco-friendly production systems, C. Ability to identify and develop appropriate technologies and management patterns for above production systems.D. Case studies of typical holistic technologies, management models and production systems. Strategy for transition from the present state to Universal Human Order. E. At the level of individual: as socially and ecologically responsible engineers, technologists and managers F. At the level of society: as mutually enriching institutions and organizations.		
TEXT BOOKS:		
1. R. R. Gaur, R Sangal, G P Bagaria, 2009, A Foundation Course in Human Values and Professional Ethics. 2. Prof. K. V. Subba Raju, 2013, Success Secrets for Engineering Students, Smart Student Publications, 3rd Edition.		
REFERENCE BOOKS:		
1. Ivan Illich, 1974, Energy & Equity, The Trinity Press, Worcester, and HarperCollins, USA 2. E. F. Schumaner, 1973, Small is Beautiful: a study of economics as if people mattered. Blond & Briggs, Britain. 3. A Nagraj, 1998 Jeevan Vidya ekParichay, Divya Path Sansthan, Amarkantak. 4. Sussan George, 1976, How the Other Half Dies, Penguin Press, Reprinted 1986, 1991. 5. P. L. Dhar, R. R. Gaur, 1990, Science and Humanism, Commonwealth Publishers. 6. A. N. Tripathy, 2003, Human Values, New Age International Publishers. 7. Subhas Palekar, 2000, How to practice Natural Farming, Pracheen(Vaidik) Krishi Tantra Shodh, Amravati. 8. Donella H. Meadows, Dennis L. Meadows, Jorgen Randers, William W. Behrens III, 1972, Limits to Growth - Club of Rome's report, Universe Books. 9. E G Seebauer& Robert L.Berry, 2000, Fundamentals of Ethics for Scientists & Engineers, Oxford University Press. 10. M Govindrajan, S Natrajan & V. S Senthil kumar, Engineering Ethics (including Humna Values), Eastern Economy Edition, Prentice Hall of India Ltd.		
WEB REFERENCES:		
1. Value Education website, http://www.uptu.ac.in 2. Story of Stuff, http://www.storyofstuff.com 3. Al Gore, An Inconvenient Truth, Paramount Classics, USA 4. Charle Chaplin, Modern Times, United Artists, USA 5. IIT Delhi, Modern Technology - the Untold Story		

III B.TECH II SEMESTER SYLLABUS

COMPUTER NETWORKS

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS18	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES: Student will be able								
1. To introduce the fundamental s of various types of computer networks								
2. To demonstrate the TCP/IP and OSI models with merits and demerits								
3. To explore the various layers of OSI model								
4. To introduce UDP and TCP models								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to								
1. Identify computer networks and its components.								
2. Identify the different types of network topologies and protocols.								
3. Enumerate the layers of the OSI model and TCP/IP. Explain the function(s) of each layer.								
4. Select and use various subnetting and routing mechanisms.								
5. Design a network diagram for a given scenario.								
UNIT-I	INTRODUCTION					CLASSES: 14		
INTRODUCTION: Network applications, network hardware, network software, reference models: OSI, TCP/IP,Internet, Connection oriented network - X.25, frame relay. THE PHYSICAL LAYER: Theoretical basis for communication, guided transmission media, wireless transmission, the public switched telephone networks, mobile telephone system.								
UNIT-II	DATA LINK LAYER					CLASSES: 14		
THE DATA LINK LAYER: Design issues, error detection and correction, elementary data link protocols, sliding window protocols. THE MEDIUM ACCESS SUBLAYER: Channel allocations problem, multiple access protocols, Ethernet, Data Link Layer Switching, Wireless LAN, Broadband Wireless, Bluetooth								
UNIT-III	NETWORK LAYER					CLASSES: 12		
THE NETWORK LAYER: Network layer design issues, routing algorithms, Congestion control algorithms,Internetworking, the network layer in the internet (IPv4 and IPv6), Quality of Service.								
UNIT-IV	TRANSPORT LAYER					CLASSES: 12		
THE TRANSPORT LAYER: Transport service, elements of transport protocol, Simple Transport Protocol, Internet transport layer protocols: UDP and TCP.								
UNIT-V	APPLICATION LAYER					CLASSES: 12		
THE APPLICATION LAYER: Domain name system, electronic mail, World Wide Web: architectural overview, dynamicweb document and http. APPLICATION LAYER PROTOCOLS: Simple Network Management Protocol, File Transfer Protocol, Simple MailTransfer Protocol, Telnet.								

TEXT BOOKS:

1. Computer networks-Andrew S Tanenbaum, 4th edition, pearson education Data communication and networking-Behrouz. A. Forouzan , fifth edition, TMH, 2013

REFERENCE BOOKS:

1. Behrouz A. Forouzan (2006), Data communication and Networking, 4th Edition, Mc Graw-Hill, India.
2. Kurose, Ross (2010), Computer Networking: A top down approach, Pearson Education, India.

NATURAL LANGUAGE PROCESSING

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AI14	PCC	L	T	P	C	CIE	SEE	Total
		3		-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Provide theoretical concepts of language processing that shows how to explore interesting bodies of text.
2. Familiarize with fundamental topics in language processing that include tagging, classification, and information extraction using tiny Python programs.
3. Facilitate understanding of formal grammar to describe the structure of an unlimited set of sentences.
4. Acquaint with methods to parse a sentence, recognize its syntactic structure and construct - representations of meaning.
5. Familiarize with design of existing corpora, the typical workflow for creating a corpus and the life cycle of a corpus

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Discuss basic concepts and applications of Natural Language processing
2. Analyze words using prefix, suffix, grammar, PoS.
3. Analyze structures of Semantic and Discourse to determine what words and phrases mean in relationship
4. Analyze Natural language Generation and apply machine translation
5. Implement levels of NLP system using the Components or lexical resources to demonstrate Morphology syntax of a language.

UNIT – I	OVERVIEW AND LANGUAGE MODELING	CLASSES: 10
Overview: Origins and challenges of NLP Language and Grammar-Processing Indian Languages-NLP Applications-Information Retrieval. Language Modeling: Introduction-Variou Grammar-based Language Models-Statistical Language Model.		
UNIT – II	WORD LEVEL AND SYNTACTIC ANALYSIS	CLASSES: 12
Word Level Analysis: Introduction Morphological Parsing-Spelling Error Detection and correction-Words and Word classes-Part-of Speech Tagging. Parsing: Constituency Parsing - Probabilistic Parsing.		
UNIT – III	SEMANTIC ANALYSIS AND DISCOURSE PROCESSING	CLASSES: 10
Semantic Analysis: Introduction- Meaning Representation-Lexical Semantics Ambiguity-Word Sense Disambiguation. Discourse Processing: Introduction- cohesion-Reference Resolution Discourse Coherence and Structure.		

UNIT – IV	NATURAL LANGUAGE GENERATION AND MACHINE TRANSLATION	CLASSES: 12
Natural Language Generation: Introduction-Architecture of NLG Systems Generation Tasks and Representations-Application of NLG. Problems in Machine Translation, Characteristics of Indian Languages- Machine Translation Approaches Translation involving Indian Languages.		
UNIT – V	APPLICATIONS AND LEXICAL RESOURCES	CLASSES: 10
Information Extraction, Automatic Text Categorization and Text Summarization, Question-Answering System. LEXICAL RESOURCES: Introduction - WordNet- FrameNet – Stemmers - POS Tagger, Research Corpora, NLTK.		
TEXT BOOKS		
Tanveer Siddiqui, U.S. Tiwary, “Natural Language Processing and Information Retrieval”, Oxford University Press, 2008.		
REFERENCE BOOKS		
1. Daniel Jurafsky and James H Martin, “Speech and Language Processing: An introduction to Natural Language Processing, Computational Linguistics and Speech Recognition”, Prentice Hall, 2nd Edition, 2008. 2. James Allen, Benjamin/cummings, “Natural Language Understanding”, 2nd edition, 1995.		

DATA ANALYTICS USING R (PROFESSIONAL ELECTIVE – II)

III B.TECH-II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6DS02	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Contact Classes: 56 Tutorial Classes: Nil Practical Classes: Nil Total Classes: 56

COURSE OBJECTIVES:

The course should enable the students to

1. Use object oriented programming concepts to solve real world problems.
2. Demonstrate The User Defined Exceptions by exception handling keywords (try,catch,throw, throws and finally).
3. Use multi threading concepts to develop interprocess communication.
4. Develop a java application to interact with database by using relevant software component (JDBC Driver).
5. Solve Real World Problems Using Collections

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Use object oriented programming concepts to solve real world problems.
2. Demonstrate The User Defined Exceptions by exception handling keywords (try,catch,throw, throws and finally).
3. Use multi threading concepts to develop interprocess communication.
4. Develop java application to interact with database by using relevant software component (JDBC Driver).

Build the internet-based dynamic applications using the concept applets

UNIT-I	Introduction to R	Classes:12
Introduction to R-Features of R, Installation of R, Getting Started, Variables in R, Mathematical Operators and Vectors, Assigning Variables, Special Numbers, classes, different types of numbers, changing classes, examining variables, the workplace, library of package, getting to know a package. Input of Data, Output in R, In-Built Functions in R, Packages in R		
UNIT-II	Data Types & Functions	Classes:11
Data Types of R-Vectors, Matrices, Arrays, Lists, Factors, Data Frame-Basic Expressions Vectors – sequences, lengths, names, indexing vectors, vector recycling and repetition, Matrices and Arrays – creating arrays and matrices, rows, columns, dimensions, indexing arrays, Arithmetic. Decision-Making Structures – Loops, User- Defined Functions, User-Defined Package, Reports using R markdown, Conditional – if and else, vectorized if, multiple selection, Loops – repeat loops, while loops, for loops, Advanced looping – replication, looping over lists, looping over arrays, Multiple – input apply, instant vectorization.		
UNIT-III	Lists, Strings Data Frames	Classes:10
Lists, Functions, Strings and Factors Lists – creating lists, automatic and recursive variables, list dimensions and arithmetic, indexing lists, conversion between vectors and lists, Combining lists, NULL, Pairlists, Data Frames – Creating Data Frames, Indexing Data Frames, Basic Data Frame Manipulation, Functions – Creating and Calling Functions, Passing functions, variable scope, Strings – Constructing and printing strings, Formatting numbers, Special characters, Changing case, Extracting Substrings.		

UNIT-IV	Regression and Clustering	Classes:10
Statistics with R Summarizing data, calculating relative frequencies, tabulating factors and creating contingency tables, testing categorical variables for independence, calculating quantiles of a dataset, converting data into z-scores, t-test, testing sample proportions, testing normality, comparing means of two samples, testing correlation for significance, linear regression in R, logistic regression in R, clustering with R.		
UNIT-V	Visualization	Classes:10
R-Visualization Techniques Basic Visualization, PieChart, BarChart, Histograms, LineChart, KernelDensityPlots, Quantile-Quantile (Q-Q) Plot, Box-and-Whisker Plot, Violin Plot, Dot Chart, Bubble Plot, Image Plot, Mosaic Plot.		
TEXTBOOKS		
1. Data Analytics with R, Dr. Bharti Motwani, ISBN: 9788126576463, Wiley India		
REFERENCE BOOKS		
1. Teetor, P., R Cookbook, Mumbai: O'Reilly India/ Shroff Publishers, 2014. 2. Cotton, R. Learning R, Mumbai: O'Reilly India/ Shroff Publishers, 2014. 3. Nathan Metzler, R Programming for Beginners, 2019 4. Sandip Rakshit, R Programming for Beginners, 2017.		

OBJECT ORIENTED ANALYSIS AND DESIGN (OOAD) (PROFESSIONAL ELECTIVE-II)

III B. TECH- II SEMESTER:								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6DS25	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes:60	Tutorial Classes:NIL	Practical Classes:			Total Classes:60			
COURSE OBJECTIVES The course should enable the students to: 1. Introduce various designing techniques and methods for object oriented 2. Performance analysis with real time system 3. Demonstrate a familiarity with object-oriented data and system. 4. Give clear idea on implementing design with UML diagram like state diagram, activity diagram, use case diagram etc 5. Inculcate necessary skills to handle complexity in software design COURSE OUTCOMES At the end of the course, student will be able to: 1. Describe Object Oriented Analysis and Design concepts and apply them to solve problems 2. Prepare Object Oriented Analysis and Design documents for a given problem using Unified Modelling Language 3. Ability to analyse and model software specifications. 4. Ability to abstract object-based views for generic software systems. 5. Ability to deliver robust software components.								
UNIT-I						Classes :8		
What is modelling? Object Oriented Thinking, History of UML Building Blocks of UML, Introduction to OMG standards MDA, XMI, UML 2.0. RUP emphasizing Inception, Elaboration, Construction, Transition Phases. 4+1 Architecture, UML Meta model. Abstraction, the three models, Object and class concepts Basic Class diagrams, Need, purpose & application of UML diagrams, Link and association concepts, Generalization & Inheritance, Navigation of class models.								
UNIT-II						Classes-9		
State Modelling: Events, States, Transitions and Conditions, State diagrams, State diagram behaviour. Need, purpose Advanced object and class concepts Use Case Modelling: Actor Identification, Actor Classification, Actor Generalization, Use Cases Identification, Communication, Uses/Include and Extend Associations, writing a Formal Use Cases, Use Case realizations. Need, purpose								
UNIT-III						Classes:8		
Activity Diagram: Activity and Actions, Initial and Final Activity, Activity Edge, Decision and Merge Points, Fork and Join, Input and Output Pins, Activity Group, Activity Partitions, Constraints on Action, Swim Lanes. Sequence Diagram: Context, Objects and Roles, Links, Object Life Line, Message or stimulus, Activation/Focus of Control, Modelling Interactions.								

UNIT-IV		Classes:8
Collaboration Diagram: Objects and Links, Messages and stimuli, Active Objects, Communication Diagram, Iteration Expression, Parallel Execution, Guard Expression, Timing Diagram. Design Using UML Activity Diagram Introduction to Patterns General Responsibility Assignment Software Patterns (GRASP) : Introduction, Creator , Information Expert, Low coupling, Controller, High Cohesion, Polymorphism , Pure fabrication, Indirection, Protected Variations. Gang of Four (GoF): Introduction, Categories of Patterns (Creational, Structural and Behavioral Patterns), Singleton, Adapter, State, and Strategy.		
UNIT-V		Classes:8
Component Diagram, Interfaces and ports, Deployment diagrams, Need, purpose & application of above diagrams two, three tier architecture, Concept of Forward Engineering and Reverse Engineering of UML Diagrams Development stages, Development life cycle, devising a system concepts, Elaborating a concept. Preparing problem statements, Overview of analysis, Domain class models, Domain state model, Domain Interaction model.		
Text Books:		
1. Blaha ,Rumbaugh:"Object Oriented Modeling and Design with UML"(2/e) Pearson Education.		
Reference Books:		
1. Dathan, Ramnath: "Object Oriented Analysis, Design &Implementation,"OUP. 2. McRobb& Farmer: "Object Oriented System Analysis & Design"McGraw Hill. 3. Booch, Rumbaugh& Jacobson: "The UML User guide"Pearson Education. 4. Whitten & Bentley: "System Analysis & Design Methods"TataMcGraw Hill. 5. Booch: "Object Oriented Analysis & Design with Applications",Pearson Education. 6. Visual modelling with Rational Rose and UML by Terry Quatrani, by Addison-Wesley Professional		

INTERNET OF THINGS (IOT)

(PROFESSIONAL ELECTIVE - II)

III B.Tech - II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6DS23	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 44	Tutorial Classes: 08	Practical Classes: Nil			Total Classes: 52			
COURSE OBJECTIVES The course should enable the students - To introduce the terminology, technology and its applications - To introduce the Raspberry PI platform, that is widely used in IoT applications - To introduce the implementation of web-based services on IoT devices - Compare different Application protocols for IoT. Infer the role of Data Analytics and Security in IoT. - Identify sensor technologies for sensing real world entities and understand the role of IoT in various domains of Industry.								
COURSE OUTCOMES At the end of the course, student will be able to: - Interpret the impact and challenges posed by IoT networks leading to new architectural models. - Compare and contrast the deployment of smart objects and the technologies to connect them to network. - Appraise the role of IoT protocols for efficient network communication. - Elaborate the need for Data Analytics and Security in IoT. - Illustrate different sensor technologies for sensing real world entities and identify the applications of IoT in Industry								
UNIT - I	Introduction to IoT					CLASSES: 08		
What is IoT, Genesis of IoT, IoT and Digitization, IoT Impact, Convergence of IT and IoT, IoT Challenges, IoT Network Architecture and Design, Drivers Behind New Network Architectures, Comparing IoT Architectures, A Simplified IoT Architecture, The Core IoT Functional Stack, IoT Data Management and Compute Stack.								
UNIT - II	Smart Objects					CLASSES: 08		
Smart Objects: The “things” in IoT, Sensors, Actuators, and Smart Objects, Sensor Networks, Connecting Smart Objects, Communications Criteria, IoT Access Technologies.								
UNIT - III	IP as the IoT Network Layer					CLASSES: 10		
IP as the IoT Network Layer, The Business Case for IP, The need for Optimization, Optimizing IP for IoT, Profiles and Compliances, Application Protocols for IoT, The Transport Layer, IoT Application Transport Methods.								
UNIT - IV	Data and Analytics for IoT					CLASSES: 10		
Data and Analytics for IoT, An Introduction to Data Analytics for IoT, Machine Learning, Big Data Analytics Tools and Technology, Edge Streaming Analytics, Network Analytics, Securing IoT, A Brief History of OT Security, Common Challenges in OT Security, How IT and OT Security Practices and Systems Vary, Formal Risk Analysis Structures: OCTAVE and FAIR, The Phased Application of Security in an Operational Environment								

UNIT - V	IoT Physical Devices and Endpoints	CLASSES: 08
IoT Physical Devices and Endpoints - Arduino UNO: Introduction to Arduino, Arduino UNO, Installing the Software, Fundamentals of Arduino Programming. IoT Physical Devices and Endpoints - RaspberryPi: Introduction to RaspberryPi, About the RaspberryPi Board: Hardware Layout, Operating Systems on RaspberryPi, Configuring RaspberryPi, Programming RaspberryPi with Python, Wireless Temperature Monitoring System Using Pi, DS18B20 Temperature Sensor, Connecting Raspberry Pi via SSH, Accessing Temperature from DS18B20 sensors, Remote access to RaspberryPi, Smart and Connected Cities, An IoT Strategy for Smarter Cities, Smart City IoT Architecture, Smart City Security Architecture, 08 Smart City Use-Case Examples.		
TEXT BOOKS		
1. David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry, "IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of 1st Edition, Pearson Education (Cisco Press Indian Reprint). (ISBN: 978-9386873743) 2. Srinivasa K G, CENGAGE Learning India, 2017		
REFERENCE BOOKS		
1. Vijay Madiseti and ArshdeepBahga, "Internet of Things(A Hands-on-Approach)", 1st Edition, VPT, 2014. (ISBN: 978-8173719547) 2. Raj Kamal, "Internet of Things: Architecture and Design Principles", 1st Edition, McGraw Hill Education, 2017. (ISBN: 978-9352605224)		

FULL STACK DEVELOPMENT (PROFESSIONAL ELECTIVE – II)

III B.Tech - II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT10	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Become knowledgeable about the most recent web development technologies
2. Idea for creating two tier and three tier architectural web applications
3. Design and Analyze real time web applications.
4. Constructing suitable client and server side applications.
5. learn core concept of both front end and back end programming.

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Develop a fully functioning website and deploy on a web server.
2. Gain Knowledge about the front end and back end tools
3. Find and use code packages based on their documentation to produce working results in a project
4. Create web pages that function using external data
5. Implementation of web application employing efficient database access

UNIT I

Classes: 10

Web Development Basics: Web development Basics - HTML & Web servers Shell - UNIX CLI Version control - Git & Github HTML, CSS

UNIT II

Classes: 12

Frontend Development: Javascript basics OOPS Aspects of JavaScript Memory usage and Functions in JS AJAX for exchange with server jQuery Framework jQuery events, UI components etc. JSON data format.

UNIT III

Classes: 13

REACT JS: Introduction to React React Router and Single Page Applications React Forms, Flow Architecture and Introduction to Redux More Redux and Client-Server Communication

UNIT IV

Classes: 10

Java Web Development: JAVA PROGRAMMING BASICS, Model View Controller (MVC) Pattern MVC Architecture using Spring RESTful API using Spring Framework Building an application using Maven

UNIT V

Classes: 10

Databases & Deployment: Relational schemas and normalization Structured Query Language (SQL) Data persistence using Spring JDBC Agile development principles and deploying application in Cloud.

TEXTBOOKS:

1. Web Design with HTML, CSS, JavaScript and JQuery Set Book by Jon Duckett Professional JavaScript for Web Developers Book by Nicholas C. Zakas
2. Learning PHP, MySQL, JavaScript, CSS & HTML5: A Step-by- Step Guide to Creating Dynamic Websites by Robin Nixon
3. Full Stack JavaScript: Learn Backbone.js, Node.js and MongoDB. Copyright © 2015 BYAZAT MARDAN

REFERENCES:

1. Full-Stack JavaScript Development by Eric Bush.
2. Mastering Full Stack React Web Development Paperback – April 28, 2017 by TomaszDyl , Kamil Przeorski , Maciej Czarnecki

SCRIPTING LANGUAGES (Professional Elective-III)

III B.Tech - II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CT05	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Contact Classes: 60	Tutorial Classes: Nil	Practical Classes: Nil				Total Classes: 60		

Course Objectives

The course should enable the students

1. To study the basics of scripting languages like Java script, Perl, PHP and Ruby
2. To understand the requirements of Scripting Languages
3. to identify the uses of Scripting Languages
4. To introduce in-depth knowledge of programming features of Perl and PHP.
5. To state the implementation and applications of Scripting.

Course Objectives

At the end of the course, student will be able:

1. To Comprehend the differences between typical scripting languages and typical system and application programming languages.
2. To implement the design of programs for simple applications.
3. To write and apply Perl & PHP scripts.
4. Gain knowledge of the strengths and weakness of Perl, and Ruby.
5. To create software systems using scripting languages such as Perl, PHP, and Ruby.

UNIT-I	INTRODUCTION TO SCRIPTING LANGUAGES	Classes: 11
Introduction to Scripts and Scripting Languages —Scripts and Programs, Uses for Scripting Languages, Web Scripting. Java Script: Variables, Data Types, Operators, Conditional statements , Loops, Arrays, Functions, Objects - Predefined objects, Accessing objects, Object Methods		
UNIT-II	JAVASCRIPT PROGRAMMING	Classes: 12
JavaScript programming of reactive webpages elements: Java Script Events-Mouse events, Keyboard events, Form events, window events, Event handlers, Frames, Form object , Java Script Form Validation.		
UNIT-III	PERL	Classes: 11
PERL : Data Types, Variables, Scalars, Operators, Conditional statements ,Loops, Arrays ,Strings, Hashes, Lists, Built-in Functions, Pattern matching and regular expression operators		

UNIT-IV	HYPERTEXT PREPROCESSOR	Classes: 12
PHP: Data Types, Variables, Operators, Conditional statements, Loops, Arrays- Indexed Array, Associative Array, String Functions, Functions-Parameterized Function, Call By Value, Call all By Reference, File Handling, PHP Form handling.		
UNIT-V	RUBY	Classes: 14
Ruby: Data types, Variables, Operators, Conditional statements, Loops, Methods, Blocks, Modules, Arrays , Strings, Hashes, File I/O, Ruby Form handling.		
Text Books:		
1. The World of Scripting Languages, David Barron, Wiley Publications. 2. Learning PHP, MySQL, JavaScript, CSS&HTML5: A Step-by- Step Guide to Creating Dynamic Websites 3rd Edition, O'Reilly Publications		
Reference Books:		
1. The Ruby Programming Language, David Flanagan and Yukihiro Matsumoto, O'Reilly Publications. 2. Beginning Java Script with Dom scripting and AJAX, Russ Ferguson, Christian Heilmann ,Apress. 3. Programming Perl, Larry Wall, T. Christiansen and J.Orwant, O'Reilly, SPD. 4. Open Source Web Development with LAMP using Linux Apache, MySQL, Perl and PHP, J. Lee and B. Ware (Addison Wesley) Pearson Education.		

PATTERN RECOGNITION (PROFESSIONAL ELECTIVE – III)

III B. TECH- II SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6AI28	PEC	L	T	P	C	CIE	SEE	Total
		3	-	0	3	40	60	100
Contact Classes: 48	Tutorial Classes: 00	Practical Classes: Nil				Total Classes: 48		
Prerequisites 1.Students are expected to have knowledge basic linear algebra, basic probability theory and basic programming techniques 2.A course on “Computational Mathematics” 3 A course on “Computer Oriented Statistical Methods” Course Objectives The course should enable the students to: 1. This course introduces fundamental concepts, theories, and algorithms for pattern recognition and machine learning. 2. Understand Pattern Representation, Nearest Neighbor Based Classifier, Bayes,Classifier. 3. Get knowledge on Hidden Markov Models, Decision Trees. 4. Acquire Support Vector Machines 5. Understand Clustering and an application of hand-written digit recognition. Course Outcomes At the end of the course, student will be able to: 1.Understand the theory, benefits, inadequacies and possible applications of various machine learning and pattern recognition algorithms 2. Understanding, identify and employ suitable machine learning techniques in classification of pattern recognition 3. Understand hidden markov models,decision tree. 4. Analyze support vector machines, combination of classifiers 5. Acquire partitional clustering, clustering and an application of hand-written digit recognition								
UNIT-I	INTRODUCTION TO PATTERN RECOGNITION						Classes: 09	
Introduction: What is Pattern Recognition, Datasets for Pattern Recognition, Different Paradigms for Pattern Recognition. Representation: Data Structures for Pattern Representation, Representation of Clusters, Proximity Measures, Size of Patterns, Abstractions of the Data Set, Feature Extraction, Feature Selection, Evaluation of Classifier, Evaluation of Clustering.								
UNIT-II	CLASSIFIERS BASED ON BAYES DECISION THEORY						Classes: 10	
Nearest Neighbor Based Classifier: Nearest Neighbor Algorithm, Variants of the NN Algorithm use of the Nearest Neighbor Algorithm for Transaction Databases, Efficient Algorithms, Data Reduction, Prototype Selection. Bayes Classifier: Bayes Theorem, Minimum Error Rate Classifier, Estimation of Probabilities, Comparison with the NNC, Naïve Bayes Classifier, Bayesian Belief Network.								
UNIT-III	HIDDEN MARKOV MODELS,DECISION TREE						Classes: 10	
Hidden Markov Models: Markov Models for Classification, Hidden Morkov Models, Classification using HMMs. Decision Trees: Introduction, Decision Tree for Pattern Classification, Construction of Decision Trees, Splitting at the Nodes, Overfitting and Pruning, Examples of Decision Tree Induction.								
UNIT-IV	SUPPORT VECTOR MACHINES,COMBINATION OF CLASSIFIERS						Classes: 08	

Support Vector Machines: Introduction, Learning the Linear Discriminant Functions, Neural Networks, SVM for Classification. Combination of Classifiers: Introduction, Methods for Constructing Ensembles of Classifiers, Methods for Combining Classifiers.

UNIT-V**PARTITIONAL CLUSTERING, CLUSTERING****Classes: 11**

Clustering: Why is Clustering Important, Hierarchical Algorithms, Partitional Clustering, and Clustering Large Data Sets. An Application-HandWritten Digit Recognition: Description of the Digit Data, Preprocessing of Data, Classification Algorithms, Selection of Representative Patterns, Results.

Text Books:

1. Pattern Recognition: An Algorithmic Approach: Murty, M. Narasimha, Devi, V. Susheela, Springer Pub, 1 st Ed.

Reference Books:

1. Machine Learning - Mc Graw Hill, Tom M. Mitchell. 2. Fundamentals Of Speech Recognition: Lawrence Rabiner and Biing- Hwang Juang. PrenticeHall Pub.

Web references:

1. <https://www.coursera.org/learn/ibm-ai-workflow-feature-engineering-bias-detection>
2. <https://www.coursera.org/learn/3d-modeling-rhinoscript>
3. <https://www.coursera.org/learn/text-retrieval>
4. https://onlinecourses.nptel.ac.in/noc23_ee119/preview
5. <https://www.coursera.org/learn/deteccion-objetos>

E -Text Books:

[https://darmanto.akakom.ac.id/pengenalanpola/Pattern%20Recognition%204th%20Ed.%20\(2009\).pdf](https://darmanto.akakom.ac.id/pengenalanpola/Pattern%20Recognition%204th%20Ed.%20(2009).pdf)

MOOCS Course:

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

NEURAL NETWORKS

(PROFESSIONAL ELECTIVE – III)

III B.TECH-II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AI26	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

Contact Classes: 56 Tutorial Classes: Nil Practical Classes: Nil Total Classes: 56

COURSE OBJECTIVES

The course should enable the students to:

1. Understand the biological neural network and to model equivalent neuron models.

2. Understand the architecture, learning algorithm and issues of various feed forward and feedback neural networks.

3. Understand the data needs of deep learning.

4. Understand the context of neural networks and deep learning

5. Understand Back Propagation Algorithm and Unconstrained Organization Techniques

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Create different neural networks of various architectures both feed forward and feed backward.

2. Perform the training of neural networks using various learning rules.

3. Implement Simple perception, Perception learning algorithm, Modified Perception learning algorithm, and Adaptive linear combiner, Continuous perception, learning in continuous perception.

4. Perform the testing of neural networks and do the perform analysis of these networks for various pattern recognition applications.

5. Analyze the limitation of Single Layer Perceptron and Develop MLP with 2 hidden layers, Develop Delta learning rule of the output layer and Multilayer feed forward neural network with continuous perceptions.

UNIT-I	LEARNING PROCESS	CLASSES:10
Introduction: A Neural Network, Human Brain, Models of a Neuron, Neural Networks viewed as Directed Graphs, Network Architectures, Knowledge Representation, Artificial Intelligence and Neural Networks Learning Process: Error Correction Learning, Memory Based Learning, Hebbian Learning, Competitive, Boltzmann Learning, Credit Assignment Problem, Memory, Adaption, Statistical Nature of the Learning Process		
UNIT-II	PERCEPTRONS	CLASSES:12
Single Layer Perceptrons: Adaptive Filtering Problem, Unconstrained Organization Techniques, Linear Least Square Filters, Least Mean Square Algorithm, Learning Curves, Learning Rate Annealing Techniques, Perceptron –Convergence Theorem, Relation Between Perceptron and Bayes Classifier for a Gaussian Environment Multilayer Perceptron: Back Propagation Algorithm XOR Problem, Heuristics, Output Representation and Decision Rule, Computer Experiment, Feature Detection		

UNIT-III	BACK PROPAGATION	CLASSES:12
Back Propagation: Back Propagation and Differentiation, Hessian Matrix, Generalization, Cross Validation, Network Pruning Techniques, Virtues and Limitations of Back Propagation Learning, Accelerated Convergence, Supervised Learning.		
UNIT-IV	SELF ORGANIZATION MAPS	CLASSES:10
Self-Organizing Maps (SOM): Two Basic Feature Mapping Models, Self-Organization Map, SOM Algorithm, Properties of Feature Map, Computer Simulations, Learning Vector Quantization, Adaptive Pattern Classification		
UNIT-V	NEURO DYNAMICS	CLASSES:12
Neuro Dynamics: Dynamical Systems, Stability of Equilibrium States, Attractors, Neuro Dynamical Models, Manipulation of Attractors as a Recurrent Network Paradigm Hopfield Models – Hopfield Models, Computer Experiment		
TEXTBOOKS		
1. Simon Haykin, Neural Networks a Comprehensive Foundations, PHI edition.		
REFERENCE BOOKS		
1. Artificial Neural Networks - B. Vegnanarayana Prentice Hall of India P Ltd 2005 2. Neural Networks in Computer Inteligance, Li Min Fu MC GRAW HILL EDUCATION 2003 3. Neural Networks -James A Freeman David M S Kapura Pearson Education 2004		
WEB REFERENCES		
1. https://www.tutorialspoint.com/artificial_neural_network/index.htm 2. https://www.geeksforgeeks.org/neural-networks-a-beginners-guide/		

INFORMATION RETRIEVAL SYSTEMS (PROFESSIONAL ELECTIVE – III)

III B. TECH- II SEMESTER								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6IT26	PEC	L	T	P	C	CIE	SEE	Total
		3	-	0	3	40	60	100
Pre-requisites: Relational Database Management Systems, Data Mining Course Objectives: The course should enable the students to: 1. To use different information retrieval techniques in various application areas 2. To apply IR principles to locate relevant information large collections of data 3. To analyze performance of retrieval systems when dealing with unmanaged data sources 4. To Implement Retrieval Systems For Web Search Tasks. Course Outcomes: After the successful completion of the course, the students would be able to: 1. Understand the basics of web search 2. Identify the data structures like Inverted Indices used in Information retrieval systems 3. Apply the different techniques on compression of index including the dictionary and its posting list 4. Understand and Apply various text search algorithms 5. Identify the evaluation metrics of information system								
UNIT-I	INTRODUCTION						Classes: 09	
Definition, Objectives, Functional Overview, Relationship to DBMS, Digital libraries and Data Warehouses. Information Retrieval System Capabilities - Search, Browse, Miscellaneous.								
Unit II:	Cataloging and Indexing						Classes:	
Objectives, Indexing Process, Automatic Indexing, Information Extraction. Data Structures: Introduction, Stemming Algorithms, Inverted file structures, N - gram data structure, PAT data structure, Signature file structure, Hypertext data structure. Automatic Indexing: Classes of automatic indexing, Statistical indexing, Natural language, Concept indexing, Hypertext linkages								
Unit III:	Document and Term Clustering						Classes:	
Introduction, Thesaurus generation, Item clustering, Hierarchy of clusters. User Search Techniques: Search statements and binding, Similarity measures and ranking, Relevance feedback, Selective dissemination of information search, Weighted searches of Boolean systems, Multimedia Searching.								
UNIT-IV	Information presentation and Text Search Algorithms						Classes: 10	
Information presentation Introduction, Presentation of the hits, Display of the Item, Collaborative Filtering, Multimedia presentation, Human Perception and Presentation. Text Search Algorithms Introduction, Software text search algorithms, Hardware text search systems.								
UNIT-V	Information System Evaluation						Classes: 08	

Introduction, Measures used in system evaluations, Multimedia Information retrieval Evaluation, Measurement example – TREC Evolution.

Text Books:

1. Gerald Kowalski, Information Retrieval Architecture and Algorithms, Springer Science, 2011.
2. Ricardo Baeza-Yates, Modern Information Retrieval, Pearson Education, 2007.
3. David A Grossman and Ophir Frieder, Information Retrieval: Algorithms and Heuristics, 2nd Edition, Springer, 2004

Reference Books:

1. William B Frakes and Ricardo Baeza - Yates, Information Retrieval Data Structures and Algorithms, Pearson Education, 1992.
2. Robert Korfhage, John Wiley & Sons, Information Storage and Retrieval.
3. Christopher D. Manning and Prabhakar Raghavan, Introduction to Information Retrieval, Cambridge University Press, 2008

DATA ANALYTICS USING R LAB (PROFESSIONAL ELECTIVE – II)

III B. TECH- II SEMESTER:

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6DS03	PEC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Install and set up R and RStudio.
2. Understand R data types, R functions
3. Understand the R Programming Language
4. Exposure on solving of data science problems.
5. Perform operations on Correlation and regression using R.

COURSE OUTCOMES:

At the end of this course students will be able to:

1. Demonstrate how to install and configure RStudio
2. Perform basic operations using R Studio.
3. Import, review, manipulate and summarize data-sets in R
4. Make use of R studio to analyze data by working on datasets.
5. Apply various concepts to write programs in R
6. Analyze data and generate reports based on the data.

LIST OF EXPERIMENTS

WEEK-1	
	Download and install R-Programming environment and install= basic packages using install.packages() command in R.
WEEK-2	
	Learn all the basics of R-Programming (Data types, Variables, Operators etc.)
WEEK-3	
	R AS CALCULATOR APPLICATION a. Using with and without R objects on console b. Using mathematical functions on console c. Write an R script, to create R objects for calculator application and save in Specific edition in disk.

WEEK-4	
DESCRIPTIVE STATISTICS IN R A. Write a R script to find basic descriptive statistics using summary, str, quartile function on mtcars & cars datasets. B. Write a R script to find subset of dataset by using subset (), aggregate () functions on iris dataset.	
WEEK-5	
Learn the basics of functions in R and implement with examples.	
WEEK-6	
Implement data frames in R. Write a program to join columns and rows in a data frame using cbind() and rbind() in R	
WEEK-7	
Implement different String Manipulation functions in R.	
WEEK-8	
READING AND WRITING DIFFERENT TYPES OF DATASETS Reading different types of data sets (.txt, .csv) from web and disk and writing in file in specific disk location. A. Reading Excel data sheet in R B. Reading XML dataset in R	
WEEK-9	
VISUALIZATIONS A. Find the data distributions using box and scatter plot. B. Find the outliers using plot.	
WEEK-10	
CORRELATION AND COVARIANCE a. Find the correlation matrix. b. Plot the correlation plot on dataset and visualize giving an overview of relationships among data on iris data Analysis of covariance: variance (ANOVA), if data have categorical variables on iris data	
WEEK-11	COLLECTIONS

REGRESSION MODEL

Import a data from web storage. Name the dataset and now do Logistic Regression to find out relation between variables that are affecting the admission of a student in a institute based on his or her GRE score, GPA obtained and rank of the student. Also check the model is fit or not. require (foreign), require(MASS).

WEEK-12**CONNECTING TO DATABASE****a. MULTIPLE REGRESSION MODEL**

Apply multiple regressions, if data have a continuous independent variable. Apply on above dataset.

b. REGRESSION MODEL FOR PREDICTION

Apply regression Model techniques to predict the data on above dataset.

TEXT BOOKS

1. Data Analytics Using R, Seema Acharya. McGraw Hill Education, First Edition
2. R For Beginners by Sandip Rakshit, MC GRAW HILL INDIA.

OBJECT ORIENTED ANALYSIS AND DESIGN (OOAD) LAB (PROFESSIONAL ELECTIVE - II)

III B. TECH- II SEMESTER:								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6DS26	PEC	L	T	P	C	CIA	SEE	Total
		0	0	3	1.5	40	60	100
Contact Classes:	Tutorial Classes:	Practical Classes:			Total Classes:			
COURSE OBJECTIVES The course should enable the students to: 1. To capture the requirements specification for an intended software system 2. To draw the UML diagrams for the given specification 3. To map the design properly to code 4. To test the software system thoroughly for all scenarios 5. To improve the design by applying appropriate design patterns.								
COURSE OUTCOMES At the end of the course, student will be able to: 1. Develop a working understanding of formal object-oriented analysis and design processes 2. Develop an appreciation for and understanding of the risks inherent to large-scale software development 3. Learn (through experience!) techniques, processes, and artifacts that can mitigate these risks 4. Develop the skills to determine which processes and OOAD techniques should be applied to a given project, and 5. Develop an understanding of the application of OOAD practices from a software project management perspective								
LIST OF EXPERIMENTS								
WEEK-1								
Write Problem Statement for System / Project								
WEEK-2								
Prepare a use case model, from the given description using UML 2 notations.								

WEEK-3		Classes:
Prepare Activity Model from the given description using UML 2 notations.		
WEEK-4		Classes:
Prepare a class diagram from the given problem description using UML2.0and Implement the class diagram with a suitable object oriented language.		
WEEK-5		
Prepare a Design Model from an Analysis Model.		
WEEK-6		
Prepare a Sequence diagram from the given problem description using UML2.0 and Implement the Sequence diagram with a suitable object oriented language.		
WEEK-7		
Prepare a state model from the given problem description, draw a state diagram using UML2 notations and Implement the state model with a suitable object oriented language.		
WEEK-8		
Identification and Implementation of GRASP pattern.		
WEEK-9		
Identification and Implementation of GOF pattern.		
WEEK-10		
Draw Component Diagram for Online Examination System. Draw Component Diagram for Order Processing Application. Draw Deployment Diagram for Online Student Admission System.		
WEEK-11		
State Chart Diagram for Online Examination System. State Chart Diagram for Online Fund Transfer through Netbanking Banking System.		
WEEK-12		
Activity Diagram Online Banking System. Draw Activity Diagram for Online Examination System.		

FULL STACK DEVELOPMENT LAB

III B.Tech - II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT14	PEC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

The course should enable the students to:

1. Usage of various front and back end Tools
2. They can understand and create applications on their own
3. Demonstrate and Designing of Websites can be carried out.
4. Develop web based application using suitable client side and server side code.
5. Implement web based application using effective database access

LIST OF EXPERIMENTS

Week 1:

Write a program to create a simple web page using HTML.

Week 2:

Write a program to create a website using HTML CSS and JavaScript?.

Week 3:

Write a program to build a Chat module using HTML CSS and JavaScript?

Week 4:

Write a program to create a simple calculator Application using React JS.

Week 5:

Write a program to create a voting application using React JS.

Week 6:

Write a program to create and Build a Password Strength Check using JQuery.

Week 7:

Write a program to create and Build a star rating system using JQuery

Week8:

Create a project on Project Management application. This is going to be the toughest of all with a lot of features. They should have a framework of a basic social media site with users being able to communicate with each other. Users should be able to create projects and tasks within projects. Users must be able to assign tasks to other users and must be able to comment on it just like on a social media post. To start with, each user can have a calendar view and a kanban style board. Users must be able to close and archive tasks as well as projects when completed.

Week 9:

Create a project on Content Management System for a blog Using the CMS users must be able to design a web page using the drag and drop method. Users should be able to add textual or media content into placeholders that are attached to locations on the web page using drag and drop method.

Week 10:

Create a project on Grocery delivery application Assume this project is for a huge online departmental store. Assume that they have a myriad of grocery items at their godown. All items must be listed on the website, along with their quantities and prices. Users must be able to sign up and purchase groceries. The system should present

him with delivery slot options, and the user must be able to choose his preferred slot. Users must then be taken to the payment page where he makes the payment with his favourite method.

Week11:

Create a project e-commerce portal for used furniture sales Assume this project is for a startup that is acquiring used furniture from users at a price, refurbishing them and selling them off at a margin. The website must display second-hand furniture that is currently being sold.

TEXTBOOKS:

1. Web Design with HTML, CSS, JavaScript and JQuery Set Book by Jon Duckett Professional JavaScript for Web Developers Book by Nicholas C. Zakas

2. Learning PHP, MySQL, JavaScript, CSS & HTML5: A Step-by- Step Guide to Creating Dynamic Websites by Robin Nixon

3. Full Stack JavaScript: Learn Backbone.js, Node.js and MongoDB. Copyright © 2015 BYAZAT MARDAN

REFERENCES:

1. Full-Stack JavaScript Development by Eric Bush.

2. Mastering Full Stack React Web Development Paperback – April 28, 2017 by Tomasz Dyl , Kamil Przeorski , Maciej Czarnecki

IOT LAB								
III B.Tech - II SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6DS24	PEC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100
Contact Classes-Nil	Tutorial Classes-Nil	Practical Classes-36			Total Classes-36			
COURSE OBJECTIVES The course should enable the students to: 1. Introduce the raspberry PI platform that is widely used in IoT applications. 2. Introduce the implementation of distance sensors on IoT devices. 3. Apply the IoT concept in simple real life applications. 4. Implement an IoT based home automation system. 5. Apply IoT concepts in advanced applications. COURSE OUTCOMES At the end of the course, student will be able to: 1. Ability to introduce the concept of M2M (machine to machine) with necessary protocols and get awareness in implementation of distance sensors. 2. Describe what IoT is and how it works today. 3. Able to understand building blocks of Internet of Things and characteristics. 4. Design and program IoT devices. 5. Recognize the factors that contributed to the emergence of IoT								
LIST OF EXPERIMENTS								
WEEK -1								
1. Using raspberry pi a. Calculate the distance using a distance sensor. b. Basic LED functionality.								
WEEK -2								
2. Using Arduino a. Calculate the distance using a distance sensor. b. Basic LED functionality. c. Calculate temperature using a temperature sensor.								
WEEK -3								
3. Using Node MCU a. Calculate the distance using a distance sensor. b. Basic LED functionality. c. Calculate temperature using a temperature sensor.								

WEEK -4	
4. Connection of an Arduino board with ESP8266 wifi module.	
WEEK -5	
5. IoT based control of an LED using Arduino.	
WEEK -6	
6. IoT and cloud based data logger using LM35 and Arduino	
WEEK -7	
7. IoT based home automation using Arduino.	
WEEK -8	
8 .IoT based street light control using Arduino.	
WEEK -9	
9. IoT based DC motor speed control using Arduino.	

NATURAL LANGUAGE PROCESSING LAB

III B.TECH-II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AI15	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100
Contact Classes: Nil		Tutorial Classes: Nil		Practical Classes: 36		Total Classes: 36		

COURSE OBJECTIVES

The course should enable the students to:

1. To provide theoretical concepts of language processing that shows how to explore interesting bodies of text.
2. To familiarize with fundamental topics in language processing that include tagging, classification, and information extraction using tiny Python programs.
3. To facilitate understanding of formal grammar to describe the structure of an unlimited set of sentences.
4. To acquaint with methods to parse a sentence, recognize its syntactic structure and construct representations of meaning.
5. To familiarize with design of existing corpora, the typical workflow for creating a corpus and the life cycle of a corpus

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Apply the concept of natural language processing (NLP) using Natural Language Toolkit (NLTK).
2. Build text corpora with tokenization, Stemming, Lemmatization and apply visualization techniques.
3. Evaluate the classifiers and choose the best classifier.
4. Access WordNet and Treebank and apply regular expression pattern recognition methods.
5. Create Artificial Intelligence applications for text data.

LIST OF EXPERIMENTS

WEEK-1	Choose an English word and see how it is used in the different example texts by making concordances.
WEEK-2	Counting Vocabulary a) How many words (tokens) are there in the given text. b) How many different words (types) are there in the given text c) How many times does the word occur in the text d) What is this as a percentage of all the words in the text?
WEEK-3	Implement a basic spelling corrector in Python.
WEEK-4	Perform Tokenization, Stemming, and Lemmatization to carry out the analysis with text corpora.
WEEK-5	Write a code to remove stop words with NLTK in Python.
WEEK-6	Analyze text data using Constituency Parsing and Probabilistic Parsing.
WEEK-7	Build Bag of Words model (BoW) in NLP.
WEEK-8	Find most similar sentence in the file to the input sentence.
WEEK-9	Implement Wordnet Lemmatizer with appropriate POS tag
WEEK-10	Creating a basic chatbot using Python

WEEK-11	Build language translator using Python
WEEK-12	Implement Machine Learning based Text Classification in Python.
TEXTBOOKS	
1. Multilingual natural Language Processing Applications: From Theory to Practice – Daniel M. Bikeland ImedZitouni, Pearson Publication 2. Natural Language Processing and Information Retrieval: Tanvier Siddiqui, U.S. Tiwary.	

MINI PROJECT

III B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AI10	PCC	L	T	P	C	CIE	SEE	Total
		-	-	4	2	-	100	100

Evaluation Procedures:

There shall be an Industry training (or) Internship (or) Industry oriented Mini-project (or) Skill Development Courses (or) Paper Presentation Reputed Journal (or) Industry Oriented Mini Project in collaboration with an industry of their specialization: Students Shall register for this immediately after II-Year II Semester Examinations and pursue it during summer vacation/semester break & during III Year without effecting regular course work. Internship at reputed organization (or) Skill development courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in III-year II semester before end semester examination. It shall be evaluated for 100 external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project (or) Internship etc, Internal Supervisor and a Senior Faculty Member of the Department. There shall be **NO internal marks** for Industry Training (or) Internship (or) Mini-Project (or) Skill Development Courses (or) Paper Presentation in reputed journal (or) Industry Oriented Mini Project.

COURSE OUTCOMES:

1. Demonstrate a sound technical knowledge of selected project topic.
2. Apply algorithmic, mathematical and scientific reasoning to a variety of computational problems.
3. Make use of latest tools/software associated with programming languages to generate effective results
4. Implement software systems that meet specified design and performance requirements
5. Communicate effectively, both orally and in writing.

IV B.TECH I SEMESTER SYLLABUS

DEEP LEARNING

IV B. TECH- I SEMESTER								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6AI17	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 60	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 60			
Course Objectives The course should enable the students to: 1. Learn Deep learning techniques and their applications. 2. Acquire the knowledge of neural network architectures, Deep learning methods and algorithms. 3. Understand CNN and RNN algorithms and their applications. Course Outcomes At the end of the course, student will be able to: 1. Understand various learning models. 2. Design and develop various Neural Network Architectures. 3. Understand approximate reasoning using Convolution Neural Networks. 4. Analyze and design Deep learning algorithms in different applications. 5. Ability to apply CNN and RNN techniques to solve different applications.								
UNIT-I	BASICS					Classes: 12		
Historical Trends in Deep Learning, McCulloch Pitts Neuron, Thresholding Logic, Perceptron, Single Layer Percetron, Multiple Layer Perceptron, Representation Power of MLPs, Maximum likelihood estimation, Sigmoid Neurons, Gradient Descent, Feed forward Neural Networks, Curse of Dimensionality.								
UNIT-II	INTRODUCTION TO DEEP LEARNING					Classes: 12		
Learning Algorithms and motivation of Deep Learning, Gradient-Based Learning, Multi-layer perceptron, Back-propagation, Vanishing Gradient Problem, Capacity, Overfitting and Underfitting, Activation Functions: RELU, LRELU, ERELU, Regularization-dropout, drop connect, optimization methods for neural networks- Adagrad, adadelta, rmsprop, adam, NAG.								
UNIT-III	AUTO ENCODERS & REGULARIZATION					Classes: 12		
Auto encoders : Autoencoders, Regularized Autoencoders, Denoising Autoencoders, Representational Power, Layer size and Depth of Autoencoders, Stochastic Encoders and Decoders, Contractive Encoders. Regularization: Bias Variance Tradeoff, L2 regularization, Early stopping, Dataset augmentation, Parameter sharing and tying, Injecting noise at input, Ensemble methods, Dropout, Greedy Layer wise Pre- training, Better activation functions, Better weight initialization methods, Batch Normalization								
UNIT-IV	CONVOLUTION NEURAL NETWORKS					Classes: 12		

Overview of Convolutional Neural Networks Architecture-Motivation, Layers, Kernels, Convolution operation, Padding, Stride, Pooling, Non-linear layer, Stacking Layers, Popular CNN Architectures: LeNet, AlexNet, ZFNet, VggNet.		
UNIT-V	RECURRENT NEURAL NETWORKS	Classes: 12
Recurrent Neural Networks, Bidirectional RNNs, Encoder-decoder sequence to sequence architectures, Deep Recurrent Networks, Recursive Neural Networks, Long Short Term Memory Networks.		
Text Books:		
1. Goodfellow. I., Bengio. Y. and Courville. A., “ Deep Learning”, MIT Press, 2016.		
Reference Books:		
1. Tom M. Mitchell, “Machine Learning”, MacGraw Hill, 1997. 2. Stephen Marsland, “Machine Learning - An Algorithmic Perspective “, CRC Press, 2009. 3. LiMin Fu, “Neural Networks in Computer Intelligence”, McGraw-Hill edition, 1994.		

Knowledge Representation and Reasoning

IV B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6AI18	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Contact Classes: 52	Tutorial Classes: NIL	Practical Classes: NIL			Total Classes: 52			
Course Objectives The course should enable the students to: 1. Investigate the key concepts of Knowledge Representation (KR) techniques and different notations. 2. Integrate the KR view as a knowledge engineering approach to model organizational knowledge. 3. Introduce the study of ontologies as a KR paradigm and applications of ontologies. 4. Understand various KR techniques and process, knowledge acquisition and sharing of ontology.								
Course Outcomes At the end of the course, student will be able to: 1. Analyze and design knowledge based systems intended for computer implementation. 2. Acquire theoretical knowledge about principles for logic-based representation and reasoning. 3. Ability to understand knowledge-engineering process 4. Ability to implement production systems, frames, inheritance systems and approaches to handle uncertain or incomplete knowledge.								
UNIT-I	INTRODUCTION TO KNOWLEDGE REPRESENTATION						Classes: 10	
The Key Concepts: Knowledge, Representation, Reasoning, Why knowledge representation and reasoning, Role of logic Logic: Historical background, Representing knowledge in logic, Varieties of logic, Name, Type, Measures, Unity Amidst diversity								
UNIT-II	Ontology						Classes: 10	
Ontological categories, Philosophical background, Top-level categories, Describing physical entities, Defining abstractions, Sets, Collections, Types and Categories, Space and Time								
UNIT-III	Knowledge Representations						Classes: 10	
Knowledge Engineering, Representing structure in frames, Rules and data, Object-oriented systems, Natural language Semantics, Levels of representation								
UNIT-IV	Processes						Classes: 10	
Times, Events and Situations, Classification of processes, Procedures, Processes and Histories, Concurrent processes, Computation, Constraint satisfaction, Change Contexts: Syntax of contexts, Semantics of contexts, First-order reasoning in contexts, Modal reasoning in contexts, Encapsulating objects in contexts.								

UNIT-V	Knowledge Soup	Classes: 12
Vagueness, Uncertainty, Randomness and Ignorance, Limitations of logic, Fuzzy logic, Nonmonotonic Logic, Theories, Models and the world, Semiotics Knowledge Acquisition and Sharing: Sharing Ontologies, Conceptual schema, Accommodating multiple paradigms, Relating different knowledge representations, Language patterns, Tools for knowledge acquisition		
Text Books:		
1. Knowledge Representation logical, Philosophical, and Computational Foundations by John F. Sowa, Thomson Learning. 2. Knowledge Representation and Reasoning by Ronald J. Brachman, Hector J. Levesque, Elsevier.		

MOBILE COMPUTING (PROFESSIONAL ELECTIVE - IV)

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CY22	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students:

1. Make the student understand the concept of mobile computing paradigm, its novel applications and limitations.
2. Understand the typical mobile networking infrastructure through a popular GSM protocol
3. The issues and solutions of various layers of mobile networks, namely MAC layer, Network Layer & Transport Layer
4. Knowledge of mobile environments & data delivery models.
5. The ad hoc networks and related concepts.

COURSE OUTCOMES:

At the end of the course, students will be able to:

1. Able to think and develop new mobile application.
2. Able to take any new technical issue related to this new paradigm and come up with a solution(s).
3. Able to develop new ad hoc network applications and/or algorithms/protocols.
4. Able to understand & develop any existing or new protocol related to mobile environment
5. Understand the platforms and protocols used in a mobile environment.

UNIT - I	INTRODUCTION TO MOBILE COMMUNICATIONS, MOBILE COMPUTING	CLASSES: 10
Introduction: Mobile Communications, Mobile Computing – Paradigm, Promises/Novel Applications and Impediments and Architecture; Mobile and Handheld Devices, Limitations of Mobile and Handheld Devices. GSM – Services, System Architecture, Radio Interfaces, Protocols, Localization, Calling, Handover, Security, New Data Services, GPRS.		
UNIT - II	INTRODUCTION TO MEDIUM ACCESS CONTROL (MAC)	CLASSES: 12
(Wireless) Medium Access Control (MAC) :Motivation for a specialized MAC (Hidden and exposed terminals, Near and far terminals), SDMA, FDMA, TDMA, CDMA, Wireless LAN/(IEEE 802.11)		
UNIT - III	UNDERSTANDING MEDIUM ACCESS CONTROL (MAC) AND OTHER PROTOCOLS	CLASSES: 12
Mobile Network Layer: IP and Mobile IP Network Layers, Packet Delivery and Handover Management, Location Management, Registration, Tunneling and Encapsulation, Route Optimization, DHCP.		
UNIT - IV	BASICS OF MOBILE TRANSPORT LAYER AND QOS	CLASSES: 12
Mobile Transport Layer: Conventional TCP/IP Protocols, Indirect TCP, Snooping TCP, Mobile TCP, Other Transport Layer Protocols for Mobile Networks. Database Issues: Database Hoarding & Caching Techniques, Client-Server Computing & Adaptation, Transactional Models, Query processing, Data Recovery Process &QoS Issues.		

UNIT - V	DATA DISSEMINATION AND SYNCHRONIZATION	CLASSES: 10
Data Dissemination and Synchronization : Communications Asymmetry, Classification of Data Delivery Mechanisms, Data Dissemination, Broadcast Models, Selective Tuning and Indexing Methods, Data Synchronization – Introduction, Software, and Protocols.		
TEXT BOOKS		
1. Jochen Schiller, "Mobile Communications", Addison-Wesley, Second Edition, 2009. 2. Raj Kamal, "Mobile Computing", Oxford University Press, 2007, ISBN: 0195686772		
REFERENCE BOOKS		
1. ASOKE K TALUKDER, HASAN AHMED, ROOPA R YAVAGAL, "Mobile Computing, Technology Applications and Service Creation" Second Edition, Mc Graw Hill. 2. UWE Hansmann, LotharMerk, Martin S. Nocklous, Thomas Stober, "Principles of Mobile Computing," Second Edition, Springer.		

ROBOTIC PROCESS AUTOMATION (PROFESSIONAL ELECTIVE - IV)

III B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6CT06	PEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 54	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 54			

Course Objectives

The course should enable the students to:

1. Understand the RPA and the ability to differentiate it from other types of automation.
2. Explain about using data tables to store and easily manipulate data in memory.
3. Model The Workflow Of Different scrapping methodologies.
4. Understand to handle the exceptions and Interpret the events that can be used to trigger actions.
5. Select the packages and to release them for the update of the bots.

Course Outcomes

At the end of the course, student will be able to:

1. Understand the RPA and the ability to differentiate it from other types of automation.
2. Explain about using data tables to store and easily manipulate data in memory.
3. Model the workflow of different scrapping methodologies.
4. Understand to handle the exceptions and Interpret the events that can be used to trigger actions.
5. Select the packages and to release them for the update the bots.

UNIT-I	INTRODUCTION TO ROBOTIC PROCESS AUTOMATION	Classes: 12
History of Automation - What is RPA - RPA vs Automation - Processes & Flowcharts - Programming Constructs in RPA - What Processes can be Automated - Types of Bots - Workloads which can be automated - RPA Advanced Concepts - Standardization of processes - RPA Development methodologies - Difference from SDLC - Robotic control flow architecture		
UNIT-II	RPA TOOL INTRODUCTION AND BASICS	Classes: 10
Introduction to RPA Tool - The User Interface - Variables - Managing Variables - Naming Best Practices - The Variables Panel - Generic Value Variables - Text Variables - True or False Variables - Number Variables - Array Variables - Date and Time Variables - Data Table Variables - Managing Arguments - Naming Best Practices - The Arguments Panel - Using Arguments - About Imported Name spaces - Importing New Name spaces - Control Flow - Control Flow Introduction - If Else Statements - Loops - Advanced Control Flow - Sequences - Flowcharts - About Control Flow		
UNIT-III	ADVANCED AUTOMATION CONCEPTS & TECHNIQUES	Classes: 10
Recording Introduction Basic and Desktop Recording Web Recording Input/Output Methods Screen Scraping Data Scraping Scraping advanced techniques Selectors Defining and Assessing Selectors Customization Debugging Dynamic Selectors Partial Selectors RPA Challenge Image, Text & Advanced Citrix Automation Introduction to Image & Text Automation Image based automation Keyboard based automation Information Retrieval Advanced Citrix Automation challenges Best Practices Using tab for Images Starting Apps Excel DataTables & PDF DataTables in RPA-Excel and DataTable basics Data Manipulation in excel Extracting Data from PDF Extracting a single piece of data Anchors Using anchors in PDF.		

UNIT-IV	HANDLING USER EVENTS & ASSISTANT BOTS, EXCEPTION HANDLING	Classes: 10
Monitoring system event triggers Hotkey trigger Mouse trigger System trigger Monitoring image and element triggers Launch in gang assistant bot on keyboard event. EXCEPTION HANDLING: Debugging and Exception Handling-Debugging Tools-Strategies for solving issues-Catching errors.		
UNIT-V	DEPLOYING AND MAINTAINING THE BOT	Classes: 12
Publishing using publish utility - Creation of Server - Using Server to control the bots - Creating a provision Robot From the Server - Connecting a Robot to Server - Deploy the Robot to Server - Publishing and managing updates -Managing packages-Uploading packages-Deleting packages		
Text Books:		
Alok Mani Tripathi, "Learning Robotic Process Automation", Packt Publishing, 2018.		
Reference Books:		
1. Richard Murdoch, Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant", Independently Published, 1st Edition 2018. 2. Srikanth Merianda, "Robotic Process Automation Tools, Process Automation and their benefits: Understanding RPA and Intelligent Automation", Consulting Opportunity Holdings LLC, 1st Edition 2018. 3. Lim Mei Ying, "Robotic Process Automation with Blue Prism Quick Start Guide: Create software robots and automate business processes", Packt Publishing, 1st Edition 2018.		
WEB REFERENCES:		
1. https://www.uipath.com/rpa/robotic-process-automation 2. https://www.academy.uipath.com		

COGNITIVE COMPUTING (PROFESSIONAL ELECTIVE - IV)

IV B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6AI21	PEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 60	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 60			

Course Objectives

The course should enable the students to:

1. Use the Innovation Canvas to justify potentially successful products.
2. Discuss about natural language technologies to business problems.
3. Understand the hadoop and analytical database
4. Explain various descriptive models.
5. Learning from patterns in healthcare data.

Course Outcomes

At the end of the course, student will be able to:

1. Understand applications in Cognitive Computing.
2. Understand Natural language processor role in Cognitive computing
3. Apply various types of data bases
4. Evaluate the process of taking a product to market.
5. Use smarter transportation infrastructure using analytics

UNIT-I	FOUNDATION OF COGNITIVE COMPUTING	Classes: 13
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Foundation of Cognitive Computing: cognitive computing as a new generation, the uses of cognitive systems, system cognitive, gaining insights from data, Artificial Intelligence as the foundation of cognitive computing, understanding cognition

Design Principles for Cognitive Systems: Components of a cognitive system, building the corpus, bringing data into cognitive system, machine learning, hypotheses generation and scoring, presentation and visualization services

UNIT-II	NATURAL LANGUAGE PROCESSING IN SUPPORT OF A COGNITIVE SYSTEM	Classes: 13
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Natural Language Processing in support of a Cognitive System: Role of NLP in a cognitive system, semantic web, Applying Natural language technologies to Business problems Representing knowledge in Taxonomies and Ontologies: Representing knowledge, Defining Taxonomies and Ontologies, knowledge representation, models for knowledge representation, implementation considerations

UNIT-III	RELATIONSHIP BETWEEN BIG DATA AND COGNITIVE COMPUTING	Classes: 12
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Relationship between Big Data and Cognitive Computing: Dealing with human-generated data, defining big data, architectural foundation, analytical data warehouses, Hadoop, data in motion and streaming data, integration of big data with traditional data

Applying Advanced Analytics to cognitive computing: Advanced analytics is on a path to cognitive computing, Key capabilities in advanced analytics, Using advanced analytics to create value, Impact of open source tools on advanced analytics

UNIT-IV	THE BUSINESS IMPLICATIONS OF COGNITIVE COMPUTING	Classes: 12
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The Business Implications of Cognitive Computing: Preparing for change, advantages of new disruptive models, knowledge meaning to business, difference with a cognitive systems approach, meshing data together differently, using business knowledge to plan for the future answering business questions in new ways, building business specific solutions, making cognitive computing a reality, cognitive application changing the market.

UNIT-V	BUILDING A COGNITIVE HEALTH CARE APPLICATION	Classes: 10
Building a cognitive health care application: Foundations of cognitive computing for healthcare, constituents in healthcare ecosystem, learning from patterns in healthcare Data, using cognitive applications to improve health and wellness. Smarter cities-Cognitive Computing in Government: Characteristics of smart city, rise of open data movement with fuel cognitive cities, internet of everything and smarter cities, understanding the ownership and value of data, building a smarter transportation infrastructure using analytics to close workforce skills gap.		
Text Books:		
1. Judith H Hurwitz, Marcia Kaufman, Adrian Bowles , “Cognitive computing and Big Data Analytics” , Wiley		
Reference Books:		
1. Vijay V Raghavan, Venkat N. Gudivada, Venu Govindaraju, Cognitive Computing: Theory and Applications: Volume 35 (Handbook of Statistics), North Holland.		

AUGMENTED REALITY AND VIRTUAL REALITY (PROFESSIONAL ELECTIVE - IV)

IV B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6AI22	PEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 54	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 54			

Course Objectives

The course should enable the students to:

1. Familiarize the students with the fundamentals of Virtual Reality.
2. Impart the knowledge of 3D orientation for understanding the behaviour of VR system with the environment.
3. Deal with the Development Tools and Frameworks in Virtual Reality.
4. Introduce the applications of Virtual Reality Systems.
5. Introduce technology and features of augmented reality.

Course Outcomes

At the end of the course, student will be able to:

1. Describe the basic concepts of Virtual Reality and 3D Computer Graphics.
2. Apply 3D manipulation techniques in Virtual Reality.
3. Analyze Development Tools and Frameworks in Virtual Reality.
4. Develop a Virtual Reality application.
5. Evaluate Augmented Reality Systems.

UNIT-I	INTRODUCTION TO VR AND AR	Classes: 10
Introduction to VR and AR: History of VR and AR, Technology and Features of Augmented Reality, Comparison of AR and VR, Challenges with AR, AR Systems and Functionality, Human factors, Human visual system, Perception of depth, color, contrast, resolution, Stereo Rendering, VR Hardware: Head-coupled displays etc. VR Software, Geometric Modelling: From 2D to 3D, 3D space curves, 3D boundary representation. The Graphics Pipeline and OpenGL, Overview and Transformations, Rotation, translation, scaling, mode view matrix, projection matrix, Lighting and Shading, OpenGL Shading Language (GLSL), GLSL vertex and fragment shaders.		
UNIT-II	VISUAL COMPUTATION IN VIRTUAL REALITY	Classes: 12
Visual computation in virtual reality: 3D Interaction Techniques: 3D Manipulation Techniques and Input Devices, 3D Travel Tasks, Travel Techniques, Theoretical Foundations of Wayfinding, Types of Centred Wayfinding Support, Evaluating Wayfinding Aids, System Control, Classification, Graphical Menus, Voice Commands, Gestural Commands, Tools, Multi-modal System Control Techniques, Case Study: Mixing System Control Methods, Symbolic Input Tasks.		
UNIT-III	FRAMING USING 3D VIRTUAL REALITY	Classes: 10
Framing using 3D virtual reality: Development Tools and Frameworks in Virtual Reality: VR. X3D Standard; Vega, MultiGen, Virtools etc., World Space, World Coordinate, World Environment, Objects - Geometry, Position / Orientation, Hierarchy, Bounding Volume, Scripts and other attributes, VR Environment - VR Database, Tessellated Data, LODs, Graphical User Interface, Control Panel, 2D Controls.		
UNIT-IV	VR APPLICATIONS	Classes: 12
VR Applications: Pose Tracking I, Tracking with light house, Pose Tracking II, Advanced positional tracking, Panoramic Imaging and Cinematic, VR Spatial Sound and the Vestibular System, VR Engines and Other Aspects of VR, Latency, eye tracking, post-rendering warp. The Future: Virtual environment, modes of interaction Application of VR in Digital Entertainment: VR Technology in Film & TV Production. VR Technology in Physical Exercises and Games, Demonstration of Digital Entertainment by VR.		

UNIT-V	MIXED REALITY APPLICATION DEVELOPMENT	Classes: 10
Augmented and Mixed Reality: Taxonomy, technology and features of augmented reality, difference between AR and VR, Challenges with AR, AR systems and functionality, Augmented reality methods, visualization techniques for augmented reality, wireless displays in educational augmented reality applications, mobile projection interfaces, marker-less tracking for augmented reality, enhancing interactivity in AR environments, evaluating AR systems.		
Text Books:		
1. LaValle "Virtual Reality", Cambridge University Press, 2016. 2. John Vince, —Virtual Reality Systems —, Pearson Education Asia, 2007.		
Reference Books:		
1. Alan B Craig, William R Sherman and Jeffrey D Will, —Developing Virtual Reality Applications: Foundations of Effective Designll, Morgan Kaufmann, 2009. 2. Alan B. Craig, Understanding Augmented Reality, Concepts and Applications, Morgan Kaufmann, 2013.		

COMPUTER VISION (PROFESSIONAL ELECTIVE - V)

IV B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6AI23	PEC	L	T	P	C	CIA	SEE	Total
		3	-	--	3	40	60	100
Contact Classes: 46	Tutorial Classes: NIL	Practical Classes: NIL			Total Classes: 46			

Course Objectives

The course should enable the students to:

1. To understand the basic concepts of computer vision and segmentation.
2. To gain knowledge in foundation of image formation and image analysis.
3. To understand the Basic concepts of Recognition.
4. To learn the various concepts of Computer Vision in other application areas

Course Outcomes

At the end of the course, student will be able to:

1. Understand the fundamental problems of computer vision.
2. Implement various techniques and algorithms used in computer vision.
3. Acquire knowledge and understanding of Feature detection and matching.
4. Demonstrate awareness of the current key research issues in computer vision.
5. Exhibit knowledge in Image stitching and Recognition.

UNIT-I	Introduction	Classes: 10
Image formation - Geometric primitives and transformations - Geometric primitives - 2D transformations - 3D transformations - 3D rotations - 3D to 2D projections - Lens distortions – Photometric image formation - Lighting - Reflectance and shading – Optics - The digital camera - Sampling and aliasing – Color – Compression		
UNIT-II	Feature Detection and Matching	Classes: 08
Points and patches - Feature detectors - Feature descriptors - Feature matching - Feature tracking - Application: Performance driven animation - Edges - Edge detection - Edge linking - Application: Edge editing and enhancement Lines - Successive approximation - Hough transforms - Vanishing points -Application: Rectangle detection.		
UNIT-III	Segmentation	Classes: 10
active contours - Snakes - Dynamic snakes and CONDENSATION – Scissors - Level Sets - Application: Contour tracking and rotoscoping – Split and merge - Watershed - Region splitting - Region merging - Graph-based segmentation - Probabilistic aggregation – Mean shift and mode finding - K-means and mixtures of Gaussians - Mean shift – Normalized cuts - Graph cuts and energy-based methods - Application: Medical image segmentation.		
UNIT-IV	Structure from Motion	Classes: 10
Triangulation - Two-frame structure from motion – Factorization - Bundle adjustment - Constrained structure and motion. Dense motion estimation - Translational alignment - Parametric motion - Spline-based motion - Optical flow - Layered motion.		

UNIT-V	Image Stitching and Recognition	Classes: 08
Motion models - Global alignment – Compositing - Recognition - Object detection - Face detection - Pedestrian detection - Face recognition – Eigenfaces - Active appearance and 3D shape models - Instance recognition - Geometric alignment - Large databases - Category recognition - Bag of words - Part-based models - Recognition with segmentation - Context and scene understanding - Learning and large image collections - Recognition databases and test sets.		
Text Books:		
1. Forsyth, A., D. and Ponce, J., Computer Vision: A Modern Approach, Pearson Education, 2nd Edition, 2012. 2. Szeliski, R., Computer Vision: Algorithms and Applications, Springer-Verlag London Limited, 1st Edition, 2011.		
Reference Books:		
1. Gonzalez C. R., and Woods E. R., Digital Image Processing, Addison-Wesley, 4 edition, 2018. 2. Hartley, R. and Zisserman, A., Multiple View Geometry in Computer Vision, Cambridge University Press, 2 nd Edition, 2003. 3. Fukunaga, K., Introduction to Statistical Pattern Recognition, Academic Press, Morgan Kaufmann, 2 nd Edition, 1990. 4. Trucco and Verri, Introductory Techniques for 3D Computer Vision, Prentice Hall, 1998.		
Web references:		
1. https://www.mltut.com/best-resources-to-learn-computer-vision/ 2. https://deeplearning.lipingyang.org/computer-vision-resources/		
E -Text Books:		
1. Szeliski, R. (2010). Computer vision: algorithms and applications. Springer Science & Business Media. (book website; PDF link) . [MIT, UC Berkeley, Princeton CV course primary textbook]. 2. Prince, S. J. (2012). Computer vision: models, learning, and inference (book website; PDF link). Cambridge University Press. 3. Hartley, R., & Zisserman, A. (2003). <i>Multiple view geometry in computer vision</i> . Cambridge university press. 4. Marr, D. (1982). <i>Vision: A computational investigation into the human representation and processing of visual information</i> . MIT Press. Cambridge, Massachusetts. (book website)		
MOOCS Course:		
https://onlinecourses.nptel.ac.in/noc21_cs101/preview		

PROMPT ENGINEERING (PROFESSIONAL ELECTIVE - V)

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6AI24	PEC	3	0	0	3	40	60	100
Contact Classes-60	Tutorial Classes-Nil	Practical Classes-Nil			Total Classes-60			
COURSEOBJECTIVES: The course should enable the students to: 1. Gain a comprehensive understanding of generative AI concepts and applications. 2. Explore various types of generative models and their functionalities. 3. Develop expertise in prompt engineering techniques and their application in various tasks. 4. Design and build real-world generative AI projects with effective prompts. 5. Understand the ethical considerations and responsible use of generative AI. COURSE OUTCOMES: At the end of the course, the student will: 1. Explain and differentiate between diverse generative AI models (GANs, VAEs, Transformers) and their applications. 2. Master core and advanced prompt engineering techniques for effective and creative output across domains. 3. Build generative AI applications using popular tools and libraries, including chatbots, with specific customizations. 4. Apply ethical considerations and best practices to the development and deployment of generative AI. 5. Stay up-to-date with emerging research and trends in generative AI for future applications.								
UNIT - I	Foundations of Generative AI (10 hours)					CLASSES: 12		
History and Evolution: a. Early AI research and the emergence of machine learning b. Development of neural networks and deep learning c. Rise of generative AI models and their impact Core Concepts: d. Definition and types of generative AI models: GANs, VAEs, Transformer-based models, etc. e. Algorithmic principles and learning processes f. Strengths and limitations of different models Applications: g. Text generation: creative writing, poetry, scripts, news articles, code generation h. Image generation: photorealistic images, art, design, fashion i. Music composition: diverse styles, moods, instruments, and genres j. Chatbots: conversational AI, customer service, education, healthcare, entertainment k. Other creative applications: game design, product design, scientific research								
UNIT - II	Prompt Engineering for Generative AI (15 hours)					CLASSES: 8		

Fundamentals:

- a. Importance and impact of prompts in influencing model outputs
- b. Core techniques: context, creativity, specificity, examples, multi-step prompting, conditional prompting
- c. Advanced techniques: chain-of-thought prompting, temperature control, human-in-the-loop approaches

Best Practices:

- d. Avoiding bias and promoting ethical considerations
- e. Troubleshooting and optimizing prompts for improved results
- f. Evaluating outputs and iterating for continuous improvement
- g. Prompt engineering for specific tasks:
 - i. Crafting engaging and informative conversations for chatbots
 - ii. Tailoring prompts for different conversational purposes (e.g., customer service, personal assistant)
 - iii. Utilizing advanced techniques for multi-turn dialogues and personalized responses

UNIT - III Generative AI Tools and Technologies (10 hours)**CLASSES: 14**

1. Popular Tools and Libraries:

- a. Google AI Platform Vertex
- b. Hugging Face Transformers
- c. Specialized tools: ChatGPT, Bard, Bard-Chat, LaMDA

UNIT - IV Real-World Applications and Case Studies (10 hours)**CLASSES: 14**

3. Identifying Real-World Problems:

- a. Analyzing potential applications of generative AI in diverse domains
- b. Exploring existing case studies and successful implementations
- c. Brainstorming innovative solutions for specific challenges

4. Case Studies:

- a. Customer service chatbots for e-commerce platforms
- b. Educational chatbots for personalized learning experiences
- c. Healthcare chatbots for providing information and support
- d. Entertainment chatbots for interactive storytelling and gaming
- e. Other case studies across various industries and domain

UNIT - V Responsible AI and the Future (5 hours)**CLASSES: 10**

3. Ethical Considerations:

- a. Bias and fairness in generative AI models and chatbots
- b. Explainability and interpretability of model outputs
- c. Risks and challenges of generative AI technology
- d. Privacy and security concerns related to chatbots and personal data

4. Best Practices for Responsible AI:

- a. Promoting diversity and inclusion in model development
- b. Mitigating bias and ensuring fairness in outputs
- c. Implementing explainability techniques to understand model decisions
- d. Fostering transparency and responsible use of generative AI

5. The Future of Generative AI:

- a. Emerging trends and advancements in generative AI technology
- b. Potential future applications and societal implications
- c. Exploring the future of chatbots and their role in human-machine interaction

TEXT BOOKS:

1. Prompt Engineering for Large Language Models by Brian Lester, Rami Al-Rfou, Noah Constant, Michael Collins, and Alexander M. Rush (2022) - This book offers a detailed analysis of prompt engineering

techniques, including best practices, advanced techniques, and their applications in various generative tasks.

2. The Art of Prompt Engineering: Exploring the Power of Language Models by Jason Brownlee (2023) - This book provides a practical guide to prompt engineering, covering various tools and techniques for effective prompt design.

Reference Books:

1. Generative Deep Learning: Teaching Machines to Paint, Write, and Dream by David Foster, Andrej Karpathy, Ian Goodfellow (2nd Edition, 2019) - This textbook provides a comprehensive introduction to the fundamental concepts and algorithms of generative AI, covering various models like GANs, VAEs, and Transformer-based models.
2. Hugging Face Transformers: A Practical Guide by Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Manoury, Thomas Hernandez, and Pierric Cistac (2020) - This book serves as a practical guide to using the Hugging Face Transformers library for various NLP tasks, relevant to text generation and chatbot development.

QUANTUM COMPUTING (PROFESSIONAL ELECTIVE - V)

IV B. TECH- I SEMESTER								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6AI25	PEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 60	Tutorial Classes: 00	Practical Classes: NIL			Total Classes: 60			
Course Objectives The course should enable the students : 1. To translate fluently between the major mathematical representations and its quantum operations. 2. To implement' basic quantum algorithms. 3. To explain quantum decoherence in systems for computation. 4. To discuss the physical basis of uniquely quantum phenomena.								
Course Outcomes At the end of the course, student will be able to: 1. Identify the working of a Quantum Computing Program, its architecture and program model. 2. Identify the implications of quantum computation in the field of mathematics and associated fields 3. Explore building blocks and programming methodologies for quantum computing. 4. Explore programming models for quantum computing 5. Apply quantum computation algorithms and use them.								
UNIT-I	INTRODUCTION TO QUANTUM COMPUTING						Classes: 12	
Motivation for Studying Quantum Computing, Major players in the industry (IBM, Microsoft, Rigetti, D-Wave etc), Origin of Quantum Computing, Overview of major concepts in Quantum Computing (Qubits and multi-qubits states, Bra-ket notation, Bloch Sphere representation, Quantum Superposition, Quantum Entanglement.								
UNIT-II	MATH FOUNDATION FOR QUANTUM COMPUTING						Classes: 12	
Matrix Algebra: Basis vectors and orthogonality, inner product and Hilbert spaces, matrices and tensors, unitary operators and projectors, Dirac notation, Eigenvalues and EigenVectors.								
UNIT-III	BUILDING BLOCKS FOR QUANTUM PROGRAM						Classes: 12	
Architecture of a Quantum Computing Platform, Details of q-bit system of information representation (Block Sphere, Multi-qubits States, Quantum superposition of qubits (valid and invalid superposition), Quantum Entanglement, Useful states from Quantum algorithmic perspective e.g.Bell State.Operation on qubits: Measuring and transforming using gates, Quantum Logic gates and Circuit: Pauli, Hadamard, phase shift.								
UNIT-IV	PROGRAMMING MODEL FOR A QUANTUM COMPUTING						Classes: 12	
Programming model for a Quantum Computing Program, Steps performed on classical computer, Steps performed on Quantum Computer, Moving data between bits and qubits.								

UNIT-V	QUANTUM ALGORITHMS	Classes: 12
Major Algorithms: Shor's Algorithm, Grover's Algorithm, Deutsch's Algorithm, Deutsch -Jozsa Algorithm.		
Text Books:		
1. Quantum Computation and Quantum Information, Michael A. Nielsen, Cambridge University Press. 2. Quantum Computing Explained, David McMahon, Wiley		
Reference Books:		
1. IBM Experience: https://quantumexperience.ng.bluemix.net 2. Microsoft Quantum Development Kit https://www.microsoft.com/enus/quantum/development-kit 3. Forest SDK PyQuil: https://pyquil.readthedocs.io/en/stable/		
Web references:		
1. https://www.qi.damtp.cam.ac.uk/files/PartIIIQC/Part%202%20QIC%20lecturenotes.pdf 2. https://homes.cs.washington.edu/~oskin/quantum-notes.pdf 3. https://homepages.cwi.nl/~rdewolf/qcnotes.pdf 4. https://www.worldscientific.com/worldscibooks/10.1142/4261#t=aboutBook		
E -Text Books:		
1. https://profmcruz.files.wordpress.com/2017/08/quantum-computation-and-quantum-information-nielsen-chuang.pdf		
MOOCS Course:		
1. https://swayam.gov.in/ 2. https://onlinecourses.nptel.ac.in/		

CRYPTOGRAPHY AND NETWORK SECURITY (PROFESSIONAL ELECTIVE - V)

IV B. TECH- I SEMESTER:								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6CY06	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes:54	Tutorial Classes:	Practical Classes:				Total Classes:54		
COURSE OBJECTIVES The course should enable the students : 1. To understand the objectives of information security. 2. To inculcate confidentiality, integrity, authentication and availability. 3. To Introduce various cryptographic algorithms and public-key cryptosystem 4. Demonstrate a PGP key pair and encrypted e- mail message. 5. To give a clear idea on fundamental ideas of public-key cryptography, web security and Firewalls COURSE OUTCOMES At the end of the course, student will be able to: 1. Ability to understand basic cryptographic algorithms. 2. Ability to understand message and web authentication and security issues.. 3. Ability to identify information system requirements for both of them such as client and server. 4. Understand the current legal issues towards information security. 5. Understand Intrusions and intrusion detection.								
UNIT-I	Cryptography Concepts					Classes :12		
What is Introduction, The need for security, Security approaches, Principles of security, Types of Security attacks, Security services, Security Mechanisms, A model for Network Security. Cryptography Concepts and Techniques: Introduction, plain text and cipher text, substitution techniques, transposition techniques, encryption and decryption, symmetric and asymmetric key cryptography, steganography, key range and key size, possible types of attacks.								
UNIT-II	Symmetric key Ciphers and Asymmetric key Ciphers					Classes:12		
Symmetric key Ciphers: Block Cipher principles, DES, AES, Blowfish, RC5, IDEA, Block cipher operation, Stream ciphers, RC4. Asymmetric key Ciphers: Principles of public key cryptosystems, RSA algorithm, Elgamal Cryptography, Diffie-Hellman Key Exchange, Knapsack Algorithm.								
UNIT-III	Cryptographic Hash Functions					Classes:14		
Message Authentication, Secure Hash Algorithm (SHA-512), Message authentication codes: Authentication requirements, HMAC, CMAC, Digital signatures, Elgamal Digital Signature Scheme. Key Management and Distribution: Symmetric Key Distribution Using Symmetric & Asymmetric Encryption, Distribution of Public Keys, Kerberos, X.509 Authentication Service, Public – Key Infrastructure.								

UNIT-IV	Transport-level Security	Classes:10
Web security considerations, Secure Socket Layer and Transport Layer Security, HTTPS, Secure Shell (SSH). Wireless Network Security: Wireless Security, Mobile Device Security, IEEE 802.11 Wireless LAN, IEEE 802.11i Wireless LAN Security.		
UNIT-V	E-Mail Security	Classes:10
Pretty Good Privacy, S/MIME IP Security: IP Security overview, IP Security architecture, Authentication Header, Encapsulating security payload, Combining security associations, Internet Key Exchange.		
Text Books:		
1. Cryptography and Network Security - Principles and Practice: William Stallings, Pearson Education, 6 th Edition. 2. Cryptography and Network Security: Atul Kahate, McGraw Hill, 3 rd Edition.		
Reference Books:		
1. Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan, Wiley India, 1 st Edition. 2. Cryptography and Network Security: Forouzan Mukhopadhyay, McGraw Hill, 3 rd Edition. 3. Information Security, Principles, and Practice: Mark Stamp, Wiley India. 4. Principles of Computer Security: WM. Arthur Conklin, Greg White, TMH. 5. Introduction to Network Security: Neal Krawetz, CENGAGE Learning. 6. Network Security and Cryptography: Bernard Menezes, CENGAGE Learning.		

DEEP LEARNING LAB

IV B. TECH-I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AI19	PCC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	40	60	100

COURSE OBJECTIVES

The course should enable the students :

1. The objective of this lab is to get an overview of the various Deep learning techniques and can able to demonstrate those using python.
2. Implement classification of linearly separable Data using Perceptron
3. Implement Back Propagation methods.
4. Implement a Recurrent Neural Network for IMDB movie review classification problem

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Implement deep neural networks to solve real world problems
2. Make use of Activation functions for better results in real time applications
3. Choose appropriate pre-trained model to solve real time problem
4. Interpret the results of two different deep learning models
5. Implement LSTM model and apply the concepts on real world examples

WEEK-I	SINGLE LAYER PERCEPTRON	
Implement Single Layer Perceptron in Python		
WEEK-II	MULTI LAYER PERCEPTRON	
Implement Multi Layer Perceptron in Python		
WEEK-III	ACTIONATION FUNCTIONS	
Demonstrate normalization of input data, basic activation functions such as the softmax, sigmoid,dsigmoid, etc		
WEEK-IV	NEURAL NETWORK	
Build a neural network for logistic regression to minimize the cost function and update the parameters.		
WEEK-V	BACKWORD PROPAGATION	
Implement backward propagation neural network for a two class classification with a single hidden layer, non-linear activation function like tanh and compute the cross entropy loss.		
WEEK-VI	DEEP NEURAL NETWORK-ReLU	
Build a deep neural network with more than one hidden layer, non-linear functions like ReLU.		

WEEK-VII	DEEP NEURAL NETWORK	
Build deep neural network to any classification problem and compare its accuracy to logistic regression.		
WEEK-VIII	REGULARIZATION	
Apply Regularization techniques in deep learning model with backward propagation		
UNIT-IX	MINIBATCH	
Implement minibatch optimization techniques to improve the performance of deep learning models.		
UNIT-X	CONVOLUTION NEURAL NETWORKS	
Implement Convolution Neural Networks in Python		
UNIT-XI	RECURRENT NEURAL NETWORKS	
Implement a Recurrent Neural Network for IMDB movie review classification problem		
UNIT-XII	LONG SHORT TERM MEMORY NETWORKS	
Implement LSTM in Python		
Software Packages required: • Keras • Tensorflow • PyTorch		
TEXT BOOKS		
1. Reza Zadeh and Bharath Ramsundar, "Tensorflow for Deep Learning", O'Reilly publishers, 2018.		
REFERENCES		
1. https://github.com/fchollet/deep-learning-with-python-notebooks		

RESEARCH PROJECT PHASE-I

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
	PCC	L	T	P	C	CIE	SEE	Total
		-	-	8	4	100	-	100

Evaluation Procedures:

The UG research project shall be initiated at the end of the IV Year I Semester and the duration of the project work is one semester. The student must present Project Stage-I during IV Year I Semester before II Mid examinations, in consultation with his Supervisor, the title, objective and plan of action of his Project work to the departmental committee for approval before commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can start his research project work.

UG research project work shall be carried out in two stages: Research Project Stage–I for approval of project before Mid-II examinations in IV Year I Semester and Research Project Stage–II during IV Year II Semester. Student has to submit research project work report at the end of IV Year II Semester. The research project shall be evaluated for 100 marks before commencement of SEE Theory examinations.

For Research Project Stage–I, the departmental committee consisting of the Head of the Department, project supervisor and a senior faculty member shall approve the project work to begin before II Mid-Term examination of IV Year I Semester. The student is deemed to be not eligible to register for the Project work, if he doesn't submit report Research Project Stage - I or does not make a presentation of the same before the evaluation committee as per schedule.

A student who has failed may reappear once for the above evaluation, when it is scheduled again; if he fails in such 'one reappearance' evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

For Research Project Stage – II, the external examiner shall evaluate the project work for 60 marks and the internal project committee shall evaluate it for 40 marks. Out of 40 internal marks, the departmental committee consisting of Head of the Department, Project Supervisor and a Senior Faculty Member shall evaluate the project work for 20 marks and Project Supervisor shall evaluate for 20 marks. The topics for Industry Oriented Mini Project/Internship/SDC etc. and the research Project shall be different from the topic already taken. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together

COURSE OUTCOMES:

1. Demonstrate a sound technical knowledge of selected project topics.
2. Acquire the skills required for developing projects by working in teams, enhancing the research capabilities.
3. Apply the skills acquired from subjects into implementation by working on real world case

studies

4. Work effectively in teams to design and implement solutions to computational problems
5. Communicate effectively, both orally and in writing.

IV B.TECH II SEMESTER SYLLABUS

ORGANIZATIONAL BEHAVIOR

IV B. TECH- II SEMESTER:								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6HS15	HSMC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES: The course should enable the students : 1. To provide understanding of the basic concepts of organizational behavior. 2. To highlight the importance of individual behaviors in an organizational setting. 3. To describe the various aspects of group behaviors in an organization. 4. To learn and appreciate different cultures and innovatively engage in solving organizational challenges. 5. To elaborate on the impact of work–life balance and importance of managing stress and emotions in the workplace.								
COURSE OUTCOMES: At the end of the course, student will be able to 1. Gain the conceptual framework and the theories underlying Organizational Behavior. 2. Describe the different theories related to Individual behavior in the Organization. 3. Analyze Group Behavioral influence in the Organization. 4. Demonstrate skills required for working in groups and explain the essential concepts of organizational culture, climate, change and organizational development. 5. Learn the concepts of work life balance and managing stress and emotions in the work place.								
UNIT-I	INTRODUCTION						Classes :12	
The concept of Organizational Behavior - Importance of OB to the field of Management – Hawthorne Studies - Approaches to Organizational Theory – Organizational Behavior Models - Opportunities and Challenges for OB – Disciplines that contribute to OB.								
UNIT-II	INDIVIDUAL BEHAVIOUR						Classes:12	
Learning – Theories, Principles and Process – Perception – Importance - Perception Model - Factors influencing Perception – Personality – Determinants and Theories – Attitudes and Values – Motivation – Nature, Techniques and Theories.								
UNIT-III	GROUP BEHAVIOUR						Classes:12	
Group Dynamics – Group Formation – Group Norms – Group Decision making – Group Cohesiveness – Team Building – Leadership – Styles and Theories – Power and Politics – Conflict Management – Types and Sources – Conflict Resolution								

UNIT-IV	ORGANIZATIONAL DYNAMICS	Classes:12
Organizational Culture – Organizational Climate - Organizational Change: Approaches and Resistance to change– Organizational Development – Organizational Structure - Communication: Meaning, Types, Process and Barriers- Power – Sources of Power – Leadership – Theories and Styles		
UNIT-V	CONTEMPORARY ISSUES IN ORGANIZATIONAL BEHAVIOUR	Classes:12
Quality of Work Life - Work life balance – Stress Management – Importance and techniques – Emotional Intelligence.		
Suggested Readings:		
1. Stephen Robbins, Timothy A. Judge, Organizational Behaviour, PHI Pvt. Ltd. 2. Nelson, Quick, Khandelwal, Organizational Behaviour, Cengage Learning 3. Williams, Tripathy, MGMT-Principles of Management, Cengage Learning. 4. L.M.Prasad, Principles and Practice of Management, Sultan and Chand. 5. Udai Pareek and Sushama Khanna, Organization Behavior, Oxford Publishing 6. S.S. Khanka, , Organization Behavior, Oxford Publishing S Chand Publishing.		

ETHICAL HACKING

(PROFESSIONAL ELECTIVE - VI)

IV B.TECH-II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CY10	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
Conducted classes:60	Tutorial Classes:Nil	Practical Classes: Nil			Total Classes:63			
COURSE OBJECTIVES								
1.The course should enable the students :								
2.Introduce the methodologies and framework of hacking for enhancing security.								
3.Understand the impacts of hacking and types of hackers.								
4.Learn about information security models.								
5.Develop an understanding of information security programs, business perspectives, and controlled attack planning.								
6.Explore the framework of hacking steps, including reconnaissance, enumeration, vulnerability analysis, exploitation, deliverables, and integration.								
COURSE OUTCOMES								
At the end of the course, student will be able to:								
1.Gain knowledge about ethical hacking tools.								
2.Interpret the results of a controlled attack.								
3.Understand the role of politics and limitations in ethical hacking.								
4.Learn how to use metrics for test planning.								
5.Recognize the dangers associated with penetration testing.								
UNIT-I	Introduction to Hacking						CLASSES:15	
Hacking impacts and the role of hackers.								
Planning a security test and sound operations.								
Framework: reconnaissance, enumeration, vulnerability analysis, exploitation, final analysis, deliverable, and integration.								
Information Security Models: Computer Security, Network Security, Service Security, Application Security, Security Architecture.								
Information Security Programs: The process of security, risk analysis, and ethical hacking.								
UNIT-II	The Business Perspective						CLASSES:14	
Understanding business objectives and security policies. Reviewing previous test results and business challenges. Planning controlled attacks: limitations, attack types, sources, knowledge requirements, multi-phased attacks. Teaming, attack structures, security consultants, logistics, and law enforcement considerations.								
UNIT-III	Preparing for a Hack						CLASSES:10	
Technical preparation and engagement management.								
Reconnaissance: Social engineering, physical security, and internet reconnaissance.								
UNIT-IV	Enumeration and Exploitation						CLASSES:14	
Enumeration techniques, attack strategies, soft objectives, and preparation for the next phase.								
Exploitation techniques: Intuitive testing, evasion tactics, operating systems, password crackers, rootkits, applications, wardialing, network vulnerabilities.								
UNIT-V	Deliverable and Integration						CLASSES:10	

Deliverable documentation, aligning findings, and presenting results.

Integration: Summarizing results, mitigation strategies, defense planning, incident management, security policies.

TEXT BOOKS

James S. Tiller, *The Ethical Hack: A Framework for Business Value Penetration Testing*, Auerbach Publications, CRC Press.

REFERENCE BOOKS

- EC-Council, *Ethical Hacking and Countermeasures Attack Phases*, Cengage Learning.
- Michael Simpson, Kent Backman, James Corley, *Hands-On Ethical Hacking and Network Defense*, Cengage Learning.

BUSINESS INTELLIGENCE AND ANALYTICS

(PROFESSIONAL ELECTIVE-VI)

IV B.TECH-II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6DS22	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
Conducted classes:60	Tutorial Classes:Nil	Practical Classes: Nil			Total Classes:60			
COURSE OBJECTIVES								
The course should enable the students :								
1. To introduce the Business Intelligence, Analytics and Decision Support system 2. To introduce the list of technologies for Decision making, Automated decision systems 3. Explain sentiment analysis techniques 4. Illustrate Multi-criteria Decision making systems, predictive modeling techniques 5. Understand the emergence of business analytics as a competitive strategy.								
COURSE OUTCOMES								
At the end of the course, student will be able to:								
7. Discuss the impact of Business Intelligence (BI) theories, architectures, and methodologies on the organizational decision making process. 8. Explain how different data can be integrated for querying and reporting to improve the performance of marketing and sales strategies. 9. Conduct enterprise-wide data requirements analysis to create a BI solution. 10. Develop neural network based systems. 11. Develop Ad hoc queries, reports, spreadsheets, dashboards and mobile BI applications using business analytics to answer complex business questions using data from a variety of sources, such as data files and relational/NoSQL databases.								
UNIT-I	An Overview of Business Intelligence					CLASSES:12		
Analytics, and Decision Support, Information Systems Support for Decision Making, An Early Framework for Computerized Decision Support, The Concept of Decision Support Systems, A Framework for Business Intelligence, Business Analytics Overview, Brief Introduction to Big Data Analytics								
UNIT-II	Decision Making					CLASSES:12		
Introduction and Definitions, Phases of the Decision, Making Process, The Intelligence Phase, Design Phase, Choice Phase, Implementation Phase, Decision Support Systems Capabilities, Decision Support Systems Classification, Decision Support Systems Components.								
UNIT-III	Neural Networks and Sentiment Analysis					CLASSES:12		
Basic Concepts of Neural Networks, Developing Neural Network-Based Systems, Illuminating the Black Box of ANN with Sensitivity, Support Vector Machines, A Process Based Approach to the Use of SVM, Nearest Neighbor Method for Prediction, Sentiment Analysis Overview, Sentiment Analysis Applications, Sentiment Analysis Process,, Sentiment Analysis, Speech Analytics								
UNIT-IV	Model-Based Decision Making					CLASSES:12		
Decision Support Systems modeling, Structure of mathematical models for decision support, Certainty, Uncertainty, and Risk, Decision modeling with spreadsheets, Mathematical programming optimization, Decision Analysis with Decision Tables and Decision Trees, Multi- Criteria Decision Making With Pair wise Comparisons.								
UNIT-V	Automated Decision Systems and Expert Systems					CLASSES:12		

Automated Decision Systems, The Artificial Intelligence field, Basic concepts of Expert Systems, Applications of Expert Systems, Structure of Expert Systems, Knowledge Engineering, Development of Expert Systems.

TEXT BOOKS

1. Ramesh Sharda, Dursun Delen, Efraim Turban, J.E. Aronson, Ting-Peng Liang, David King, "Business Intelligence and Analytics: System for Decision Support", 10th Edition, Pearson Global Edition, 2013

REFERENCE BOOKS

1. Business Intelligence Guidebook by Rick Sherman
2. Data Science for Business : What you need to know about Data Mining and Data Analytic Thinking by Foster Provost and Tom Fawcett

E-COMMERCE (PROFESSIONAL ELECTIVE - VI)

IV B.TECH-II SEMESTER

Course Code	Category		Hours / Week			Credits		Maximum Marks	
A6IT33	PEC		L	T	P	C		CIE	SEE
	3	0	0	3	40	60	100		Total

Course Objectives:

The course should enable the students to:

1. Become familiar with state of the art Electronic Model, Payment Mechanisms.
2. Understand the basic principle of E-Government, Securities, and Supply Chain Mechanisms.
3. Evaluate and observe various online businesses Management.
4. Provide guiding principles behind the design and strategy of the customer web interface.
5. Understand the traditional and new communication/marketing approaches that create competitive advantage in the New Economy.

Course Outcomes:

Upon successful completion of the course, the student is able to

1. Analyze the principles of E-commerce and basics of World Wide Web.
2. Understand the concept of electronic data interchange and its legal, social and technical aspects.
3. Analyze the security issues over the web, the available solutions and future aspects of e-commerce security.
4. Develop the concept of E-banking, electronic payment system
5. Create different types of payment modes.

Unit I

Classes: 12

Introduction Overview of Electronic Commerce - Definition of Electronic Commerce - E Business - Potential Benefits of E Commerce - Impact of E Commerce on Business Models - E Commerce Applications - Market Forces influencing highway - The global information distribution networks - The regulatory environment for E Commerce.

Unit II

Classes: 12

Electronic Data Interchange (EDI), Electronic Commerce and the Internet Introduction - traditional EDI systems - Benefits - Data Transfer and Standards - Financial EDI – EDI Systems and the Internet - Legal, Security and Private Concerns - Authentication - Internet Trading Relationships Consumer to Business (B2C) Business to Business (B2C), Consumer to Consumer (C2C) Government to Citizen - Features and Benefits - Portal Vs. Website. Impact. - Intra Organizational E Commerce - Supply Chain Management.

Unit III

Classes: 12

Cryptography and Authentication Introduction - Messaging Security Issues - Confidentiality - Integrity - Authentication. Encryption Techniques - Integrity Check Values and Digital Signatures - Good Encryption Practices – Key Management - key management tasks - Additional Authentication Methods. Firewalls -Definition - component - Functionality - securing the firewall - factors considered in securing the firewall - Limitations.

Unit IV

Classes: 12

Electronic Payment Mechanisms Introduction - The SET protocol - SET vs. SSL - Payment Gateway - Certificate Issuance – Certificate Trust Chain - Cryptography Methods - Dual Signatures - SET Logo - Compliance Testing - Status of Software - Magnetic Strip Cards - Smart Cards - Electronic Cheques - Electronic Cash - Third party Processors and Credit Cards - Risk and electronic payment system - Designing Electronic payment systems.

Unit V

Classes: 12

E-Commerce Strategy and Implementation Strategic Planning for E-Commerce, Strategy in Action, Competitive Intelligence on the Internet, Implementation; Plans and Execution, Project and Strategy assessment - Managerial Issues – Global Electronic Commerce - Future of Electronic Commerce.

Text Books:

1. Electronic Commerce - Security, Risk Management and Control, Greenstein and Feinman, Irwin Mcgraw-Hill, 2000.
2. Frontiers of electronic commerce – Kalakata, Whinston, Pearson.
3. E-Commerce fundamentals and applications Henry Chan, Raymond Lee, Tharam Dillon, Elizabeth Chang, John Wiley

Reference Books:

1. E-Commerce, S.Jaiswal – Galgotia.
2. E-Commerce, Efrain Turban, Jae Lee, David King, H.Michael Chang.
3. Electronic Commerce – Gary P.Schneider – Thomson. 4. E-Commerce – Business, Technology, Society, Kenneth C.Taudon, Carol

BLOCKCHAIN TECHNOLOGY (PROFESSIONAL ELECTIVE - VI)

IV B.TECH-II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CY25	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
ContactClasses:64	Tutorial Classes:Nil	Practical Classes:Nil			Total Classes:64			
COURSE OBJECTIVES The Course Should Enable The Students to: 1. Understand How Blockchain systems(mainly Bitcoin and Ethereum)work 2. Knowledge on cryptocurrency 3. Design Smart Contracts And Distributed Applications 4. Build smart Contracts And distributed applications 5. Integrate Ideas From Blockchain Technology Into Their own projects								
COURSE OUTCOMES At The End Of the course, student will be able to: 1. Explain design principles of Bitcoin and Ethereum. 2. Design,build,and deploy a distributed application. 3. Analyze the working of Smart Contracts 4. Analyze and apply the working of smart contracts 5. Create-centralized apps on Ethereum								
UNIT-I	UNDERSTANDING BLOCKCHAIN						CLASSES:10	
Introduction to Blockchain, network and mechanism, history of block chain, types blockchain, blocks and transactions, Peer to Peer systems, the block structure in block chain, Distributed Ledger Technology (DLT), Digital Signatures, Hashes as addresses, Using a key as identity.								
UNIT-II	BLOCKCHAIN 1.0:CRYPTOCURRENCY						CLASSES:10	
Bitcoin Basics, Tokens and Icons, Distributed Consensus, Consensus in Bitcoin, The Double-Spend and Byzantine Generals' Computing Problems, How a Cryptocurrency Works.								
UNIT-III	BLOCKCHAIN SMART CONTRACTS						CLASSES:10	
Introduction to Ethereum, Smart Contracts, Transactions in Ethereum, Consensus Mechanism in Ethereum, Programming using Solidity, Ethereum Accounts, Ether, Gas, structuring a contract,ERC20 Standards.								

UNIT-IV	BUILDING SMART CONTRACTS	CLASSES:10
Wallet Development Projects, Blockchain Development Platforms and APIs, Best Practices and Auditing, Smart Contract Mini Project.		
UNIT-V	BUILDING APPS	CLASSES:08
Introduction Hyperledger , Key Frameworks and Tools, Programming using Golang, DApp MiniProject.		
TEXT BOOKS		
1. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder, Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction, Princeton University Press		
REFERENCE BOOKS		
1. Antonopoulos, Mastering Bitcoin: Unlocking Digital Cryptocurrencies 2. Satoshi Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System 3. DR. Gavin Wood, "ETHEREUM: A Secure Decentralized Transaction Ledger," Yellow		

RESEARCH PROJECT PHASE-II

IV B. TECH- II SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
	PWC	L	T	P	C	CIE	SEE	Total
		-	-	22	11	40	60	100

Evaluation Procedures:

The UG research project shall be initiated at the end of the IV Year I Semester and the duration of the project work is one semester. The student must present Project Stage – I during IV Year I Semester before II Mid examinations, in consultation with his Supervisor, the title, objective and plan of action of his Project work to the departmental committee for approval before commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can start this research project work.

UG research project work shall be carried out in two stages: Research Project Stage–I for approval of project before Mid-II examinations in IV Year I Semester and Research Project Stage–II during IV Year II Semester. Student has to submit research project work report at the end of IV Year II Semester. The research project shall be evaluated for 100 marks before commencement of SET Theory examinations.

For Research Project Stage–I, the departmental committee consisting of the Head of the Department, project supervisor and a senior faculty member shall approve the project work to begin before II Mid-Term examination of IV Year I Semester. The student is deemed to be not eligible to register for the Project work, if he doesn't submit report Research Project Stage - I or does not make a presentation of the same before the evaluation committee as per schedule.

A student who has failed may reappear once for the above evaluation, when it is scheduled again; if he fails in such 'one reappearance' evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

For Research Project Stage – II, the external examiner shall evaluate the project work for 60 marks and the internal project committee shall evaluate it for 40 marks. Out of 40 internal marks, the departmental committee consisting of Head of the Department, Project Supervisor and a Senior Faculty Member shall evaluate the project work for 20 marks and Project Supervisor shall evaluate for 20 marks. The topics for Industry Oriented Mini Project/Internship/SDC etc., and the research Project shall be different from the topic already taken. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together

COURSE OUTCOMES:

1. Demonstrate a sound technical knowledge of selected project topics.
2. Acquire the skills required for developing projects by working in teams, enhance the research capabilities.
3. Apply the skills acquired from subjects into implementation by working on real world case studies

4. Work effectively in teams to design and implement solutions to computational problems
5. Communicate effectively, both orally and in writing.

OPEN ELECTIVES OFFERED BY AERONAUTICAL DEPARTMENT

OE1		OE2	
A6AE19	Fundamentals of Avionics	A6AE25	Introduction to jets and rockets
A6AE20	Introduction to Aerospace Technology	A6AE26	Non-Destructive Testing Methods
OE3			
A6AE40	Introduction to Aircraft Industry		
A6AE41	Unmanned Aerial Vehicles		

FUNDAMENTALS OF AVIONICS

V SEMESTER - OPEN ELECTIVE - I								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AE19	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The purpose of this subject is to provide the students with the theoretical background and engineering applications. 1. Overview on Aviation using Electronics 2. Basic understanding about major electronics systems used for communication 3. Basic understanding about major devices, display and flight controls used in aircraft								
COURSE OUTCOMES At the end of the course the students are able to: 1. To explain the instrumentation used in avionics. 2. To classify various ranges of the communication techniques used in aircraft. 3. To distinguish between network systems, controlling parts & surfaces 4. To compare various principles of navigation systems 5. To build phenomena of auto pilot control system								
UNIT-I	BASICS & FLIGHT DECK AND DISPLAY SYSTEMS							
BASICS: Basic principles of Avionics, Typical avionics sub system in civil/military aircraft and space vehicles. FLIGHT DECK AND DISPLAY SYSTEMS: Flight deck display technologies, CRT, LED, LCD, Touch screen, Head up display, electronic instrumentation systems.								
UNIT-II	COMMUNICATION SYSTEMS							
AUDIO AND COMMUNICATION SYSTEMS: Aircraft audio systems, basic audio transmitter and receiver principles, VHF communication system, UHF communication systems.								
UNIT-III	FREQUENCY RANGING SYSTEM							
RANGING AND LANDING SYSTEMS: VHF Omnitrange, VOR receiver principles, distance maturity equipment, principles of operation, Instrument landing system, and localizer and glide slope. POSITIONING SYSTEM: Global positioning system principles, triangulation, position accuracy, applications in aviation								
UNIT-IV	NAVIGATION SYSTEM							
INERTIAL NAVIGATION SYSTEM: Principle of Operation of INS, navigation over earth, components of inertial Navigation systems, accelerometers, gyros and stabilized platform. SURVELLIENCE SYSTEM: ATC surveillance systems principles and operation interrogation and replay standards, Collision avoidance system, ground proximity warning system								
UNIT-V	AUTO FLIGHT SYSTEM							
AUTO FLIGHT SYSTEM: Automatic flight control systems fly by wire and fly by light technologies, flight director systems, and flight management systems.								

TEXT BOOKS
1. N. S. Nagaraja (1996), Elements of electronic navigation, 2nd edition, Tata McGraw Hill, New Delhi. 2. Janes W. Wasson, Jeppesen Sandersen (1994), Avionic systems Operation and maintenance,
REFERENCE BOOKS
1. Albert Hel Frick (2010), Principle of Avionics, 6th edition, Avionics Communications Inc, India. 2. H. J. Pallet (2010), Aircraft Instrumentation and Integrated systems, Pearson Education, New Delhi. 3. J. Powell (1998), Aircraft Radio Systems, Pitman publishers, London

INTRODUCTION TO AEROSPACE TECHNOLOGY

V SEMESTER: OPEN ELECTIVE - I								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AE20	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: - To introduce the basic concepts of Aerospace technology and the current developments in the field. - To provide knowledge on the basic principles on which the development of aerodynamics, Structures, propulsion and satellite systems.								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: - Compare the atmosphere conditions of different altitudes for spacecraft system - Analyze how lift, drag and thrust are generated and understand which components constitute them - Analyze the flight performance parameters with respective stability condition - Distinguish the different systems used in a satellite mission - Design lighter than air vehicle using Catia software								
UNIT-I	HISTORY OF FLIGHT- THE AEROSPACE ENVIRONMENT							
Balloons and dirigibles, heavier than air aircraft, commercial air transport, introduction of jet aircraft, helicopters, missiles, conquest of space, commercial use of space, exploring solar system and beyond. Earth's atmosphere, the temperature extremes of space, laws of gravitation, low earth orbit, microgravity, benefits of microgravity. The near earth radioactive environment. The magnetosphere. Environmental impact on spacecraft. Meteoroids and micrometeoroids, space debris. Planetary environments								
UNIT-II	AERODYNAMICS AND FLIGHT VEHICLE PROPULSION							
Anatomy of the airplane, helicopter, launch vehicles and missiles, space vehicles. Static forces and moments on the vehicle. Understanding engineering models. Aerodynamics of wings and bodies. Generation of lift. Sources of drag. Force and moment coefficients, center of pressure. Thrust for flight, the propeller, the jet engine, rocket engines- description, principles of operation. Governing equations.								
UNIT-III	FLIGHT VEHICLE PERFORMANCE AND STABILITY							
Performance parameters. Performance in steady flight, cruise, climb, range, endurance; accelerated Flight- symmetric maneuvers, turns, sideslips, takeoff and landing. Flight vehicle stability- longitudinal, lateral and directional- static, dynamic; trim, control. Handling qualities of airplanes								
UNIT-IV	SATELLITE SYSTEMS ENGINEERING- HUMAN SPACE EXPLORATION							
Satellite missions, an operational satellite system, elements of satellite, satellite subsystems. Satellite structures, mechanisms and materials. Power systems. Communication and telemetry. Thermal control. Attitude determination and control. Propulsion and station keeping. Space missions. Mission objectives. Case studies. Human space flight missions- goals, historical background. The Soviet and US missions. The Mercury, Gemini, Apollo (manned flight to the moon), Skylab, Apollo-Soyuz, Space Shuttle. International Space Station, extravehicular activity								

UNIT-V	INTRODUCTION TO ENGINEERING DESIGN, AIR TRANSPORTATION
Design as a critical component of engineering education- as a skill- the process, design thinking, design drawing. Design for mission, performance and safety requirements. Concurrent engineering. Computer aided engineering, design project. Example: the lighter-than – air vehicle student design project at MIT. Air Transportation Systems- civil, military- objectives- principal constituents- the vehicle, the ground facilities, the organization- role. Regulation- national and international. Indian effort- civil and military- in the field of Aerospace Engineering.	
TEXT BOOKS	
1. Newman, D., Interactive Aerospace Engineering and Design, (with software and reference material on CD), McGraw-Hill, 2002, ISBN 0-07-112254-0. 2. Anderson, J.D., Introduction to Flight, fifth edition, Tata McGraw-Hill, 2007, ISBN: 0-07-006082	
REFERENCE BOOKS	
1. Russell Mikel, Aerospace and Aeronautical Engineering, Willford press, 2017. 2. Ajoykumar Kundu, Mark A Price and David Riordan, Conceptual Design: An Industrial Approach, Wiley-Blackwell, 2019.	

INTRODUCTION TO JETS AND ROCKETS

VI Semester: OPEN ELECTIVE -II								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AE25	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The course is intended to serve as an introduction to air breathing propulsion systems and Rocket Propulsion Systems. 1. Illustrate an overview of aerospace propulsion system. 2. Identify the foundation in fundamentals of thermodynamics. 3. Compare the ideal components and characteristics of jet engine 4. Interpret the performance of nozzles 5. Simplify the ideal performance analysis of rocket engines. 6. Select appropriate fuel for aerospace application.								
COURSE OUTCOMES At the end of the course the students are able to: 1. Explain the complexity in working of various engines 2. Interpret the elementary principles of thermodynamic cycles as applied to propulsion analysis 3. Analyze the process involved in individual components 4. Compare the nozzles with various operating conditions. 5. Determine Equations of motion in free space, Tsiokovsky's equation. 6. Classify the types of fuel in aviation and aerospace engineering.								
UNIT-I	INTRODUCTION TO AEROSPACE PROPULSION							
Propulsion system, Propulsive Systems – Evolution, Development, Growth and Challenges. Fundamentals of Thermodynamics – Variables, Thermodynamic Process, Introduction to IC Engines and Reciprocating Engines, Propellers and Working of Propellers.								
UNIT-II	PRINCIPLES OF JET PROPULSION							
Fundamentals of jet propulsion, Working Principle, Analysis of Ideal Jet Engine cycle, Engine components- merit-significance- ideal component characteristics, Classification – turbo jet, turbo fan, turbo prop and Ramjet engines. Basic Problems based on Engine Cycle.								
UNIT-III	RAMJET, SCRAMJET ENGINES AND NOZZLES							
Speed limitations of gas turbines, Basics of Ramjets, Combustors for liquid fuel ramjet engines, Combustion Instability and its Suppression, Solid fuel Ramjet Engines, SCRAM jet engines, Applications of RAM Jet and SCRAM Jet Engines to Missiles with Examples, Nozzles- Types of Nozzles, Converging-Diverging Nozzle, Variable Nozzle and Effects of Pressure Ratios on Engine Performance.								
UNIT-IV	ROCKET THEORY							
Applications of Rockets, Types of Rockets, Basics of Thermal Rocket Engine-Thermodynamics and Ideal Performance Analysis, Equations of motion-Rocket Motion in free space, Tsiokovsky's equation, Rocket Parameters, Burnout range, Burnout Velocity. Practical Problems								

UNIT-V	PROPELLANT ROCKETS
Solid Propulsion-Solid Propellant Rockets, Basic Configuration and Performance, Propellant Grain and Configuration, Propellant Characteristics Combustion Chamber, Ignition Process Liquid Propulsion - Design consideration of liquid rocket combustion chamber, injector, and propellant feed lines, valves, propellant tank outlet and helium pressurized and turbine feed systems-.BIO Fuels and Impact on the Atmosphere, Aviation turbine fuels - Requirements of aviation fuels of kerosene type.	
TEXT BOOKS	
1. Mechanics and Thermodynamics of Propulsion – Philip G Hill & Carl R Peterson , Pearson Publication – 2ndEdt 2. Rocket Propulsion Elements, Sutton, G.P., John Wiley, 1993.	
REFERENCE BOOKS	
1. The Jet Engine – Rolls Royce 2. Gas Turbines and Jet and Rocket Propulsion, M. L. Mathur, R. P. Sharma, Standard Publishers Distributors.	

NON-DESTRUCTIVE TESTING METHODS

VI SEMESTER - OPEN ELECTIVE - II								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AE26	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. To impart knowledge about the non-destructive testing methods. 2. To provide knowledge on the selection of NDT methods for application in engineering industries. 3. Classify the various NDT methods for detecting defects in the structural components. 4. Judge defects basing on data representation of testing								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: 1. Recognize various non-destructive techniques for engineering industries. 2. Select appropriate non-destructive technique for defects detection in manufactured/operating parts. 3. Perform inspection using major non-destructive testing methods. 4. Understand the importance and application of NDT in Aerospace structural analysis 5. Determine the defects basing on the principal of radiography								
UNIT-I	SURFACE TECHNIQUES							
Introduction to non-destructive testing (NDT) - importance of NDT techniques - types of NDT techniques - visual testing (direct and remote visual inspection) - principle and types of liquid penetrant tests (LPT) - advantages and limitations of LPT - applications of LPT.								
UNIT-II	MAGNETIC PARTICLE TESTING							
Introduction to magnetic particle testing (MPT) – principle, magnetization methods - demagnetization – advantages, limitations and applications of MPT, eddy current testing (ECT) method- principle advantages, limitations and applications								
UNIT-III	ULTRASONIC TESTING							
Introduction to ultrasonic testing (UT) -- principle of UT – UTprobes - UT inspection methods (pulse echo, transmission and phased array techniques) - advantages, limitations and applications								
UNIT-IV	RADIOGRAPHY TESTING							
Introduction to radiography testing (RT) - sources of X-rays and Gamma rays - characteristics of X-rays and Gamma rays (absorption, scattering) - filters and screens - film radiography and digital radiography (shadow formation, exposure factors, film handling and storage) --exposure charts - penetrameters - safety issues.								
UNIT-V	SPECIAL TECHNIQUES							
Acoustic emission testing (AET) principle, advantages, limitations - instrumentation and application of AET - infra red thermography (IRT) - contact and non-contact inspection methods - acoustic holography.								

TEXT BOOKS
1. Baldev Raj, T. Jayakumar, M. Thavasimuthu, "Practical Non-Destructive Testing", Narosa Publishing, London, 2012. 2. Paul E. Mix, "Introduction to Non Destructive Testing", A Training Guide, Wiley- Interscience, New Jersey, USA, June 2005.
REFERENCE BOOKS
1. ASM Metals Handbook, V-17, "Non-Destructive Evaluation and Quality Control", American Society of Metals, Metals Park, Ohio, USA, 2001 2. W.T. McGonnagle, "Non-Destructive Testing", McGraw Hill Book Co., USA, 2013.

INTRODUCTION TO AIRCRAFT INDUSTRY

VII SEMESTER: OPEN ELECTIVE -III								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AE40	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The aim is to introduce students the overview of the Aircraft Industry. The course covers basic design process, Aircraft Stability & control and different mechanical systems. 1. Familiarize students with major types of aircraft industry. 2. Illustrate the role of regulatory bodies and its business context. 3. Exemplify the organisational structure of airlines and airports 4. Illustrate the importance of air navigation services and its types. 5. Familiarize students with air safety and security as a means for ensuring passenger safety.								
COURSE OUTCOMES At the end of the course the students are able to: 1. Discuss the types of aerospace industry and evolution of flying machine to airline alliances. 2. Explain standardisation and the importance of regulatory bodies. 3. Outline the organisational structure of airlines and airports. 4. Demonstrate major types of air navigation services and air traffic control services. 5. Illustrate the major processes involved in air safety and security.								
UNIT-I	AIRCRAFT INDUSTRY OVERVIEW							
Flying machine to airline alliances, Types of Aerospace Industry, Introduction to ages of engineering, Aerospace Manufacturing, Introduction to the space environment & human space exploration.								
UNIT-II	REGULATORY AND BUSINESS CONTEXT							
Regulatory bodies, Introduction to standardization, Standards and recommended practices (SARPS), Freedoms of the air, Role of governments, Major agreements and treaties, Business side of the industry – Key performance indicators (revenue passenger kilometer, yield etc.), Industry characteristics of passenger airlines.								
UNIT-III	AIRLINES AND AIRPORTS							
Introduction to airlines, Organizational structure, Types of airline personnel, Flight crew and cabin crew, Role of training with airlines, Organizational culture of an airline Introduction to airports – Airport personnel, Processing passengers and freight, Security								
UNIT-IV	AIR NAVIGATION SERVICES							
Introduction to air navigation services, Types of navigation services, Air traffic control (ATC) services, Structure of controlled airspace, Air traffic and control services, Classifications of airspace, Flight information services and alerting services, Air navigation facilities and air navigation service (ANS) providers								

UNIT-V	AIR SAFETY AND SECURITY
Introduction to air safety and security, Manufacturer responsibilities, Maintenance procedures, Airside safety, Safety in the air, Issues in air safety, Accident and incident investigation, Security – Response of International Civil Aviation Organization (ICAO).	
TEXT BOOKS	
1. Peter Belobaba, Amedeo Odoni and Cynthia Barnhart (2015), The Global Airline Industry (Aerospace Series), John Willey & Sons, Ltd, United Kingdom. 2. Connor R Walsh, Airline Industry – Strategies, Operations and Safety, Nova Science Publishers, New York.	
REFERENCE BOOKS	
1. Andrew R Thomas (2011), Soft Landing: Airline Industry Strategy, Service and Safety, A press, New York 2. Ahmed Abdelghany, Khaled Abdelghany (2009), Modelling Applications in the Airline Industry, Ashgate Publishing Ltd, England.	

UNMANNED AERIAL VEHICLES

VII SEMESTER - OPEN ELECTIVE - III								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AE41	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: - To understand the evolution and applications of unmanned aerial vehicles. - To illustrate the subsystems of UAVs. - To explain the process involved in design of UAVs.								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: - Classify the Unmanned Aerial Vehicles. - Calculate the basic performance parameters for aircraft. - Identify and illustrate various payloads and propulsive systems. - Explain the functioning of subsystems in UAVs. - Illustrate the design process for a UAV.								
UNIT-I	INTRODUCTION TO UAV							
Historical Development, Overview of UAV Systems and sub-systems, Classification, Informal Categories, The Tier System, Classification Change.								
UNIT-II	BASIC AERODYNAMICS AND PERFORMANCE of UAV							
Basic Aerodynamic Equations, Air foils, lift, drag, moments, Aircraft Polar, The Real Wing and Airplane, Induced Drag, Total Air-Vehicle Drag, Flapping Wings, Rotary wings. Climbing Flight, Range, Endurance, Gliding Flight								
UNIT-III	PROPULSIVE SYSTEMS, STRUCTURES AND LOADS, PAYLOAD							
Thrust Generation and basic thrust equation, Sources of Power, Loads, types of loads, Materials, construction techniques. Payloads-Reconnaissance/Surveillance Payloads, Weapon Payloads, Other payloads								
UNIT-IV	UAV SUBSYSTEMS							
Mission Planning and Control Station- Types, Physical Configuration, Planning and Navigation, MPCS Interfaces, Modes of control, piloting and controlling mission, Autopilot system Launch Systems- Basic Considerations, launch Methods for Fixed, rotary wing UAV Recovery Systems								
UNIT-V	BASICS DESIGN AND CASE							
Introduction to Design and Selection of the System - Conceptual Phase, Preliminary Design, Detail Design, Selection of the System Case study on Indian UAVs (Rustom, Lakshya, AURA) Case study on Israeli-Heron, US-MQ9 Reaper								
TEXT BOOKS - Paul Fahlstrom, Thomas Gleason - Introduction to UAV Systems-Wiley (2012) - Reg Austin, "Unmanned Air Systems: UAV Design, Development and Deployment", First Edition, Wiley Publishers, 2015.								
REFERENCE BOOKS - Mirosaw Adamski, "Power units and power supply systems in UAV", New Edition, Taylor and Francis Group publishers, 2014. - Skafidas, "Microcontroller Systems for a UAV", KTH, TRITA-FYS 2002:51 ISSN 0280-316X. 34, 2002.								

OPEN ELECTIVES OFFERED BY COMPUTER SCIENCE AND ENGINEERING

OE1		OE2	
A6CS48	Software Project Management Fundamentals	A6CS09	Database Management System
A6CS37	Semantic Web	A6CS43	Data Warehousing and Business Intelligence
OE3			
A6CS51	Artificial Intelligence and Applications		
A6CS36	Cyber Security & Cyber laws		

SOFTWARE PROJECT MANAGEMENT FUNDAMENTALS

III B.TECH - II SEM - (OPEN ELECTIVE-I)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS48	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVE The course will enable the students to: 1. Understanding the specific roles within a software organization as related to project and process management. 2. Understanding the basic infrastructures competences (e.g., process modelling and measurements.). 3. Describe the principles, techniques, methods & tools for model based management of software projects, assurance of product quality and process adherence (quality assurance), as well as experienced based creation and improvements of models (process management). 4. Understanding the basic steps of project planning, project management, quality assurance, and process management and their relationships. 5. To provide basic project management skills with a strong emphasis on issues and problems associated with delivering successful IT projects.								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: 1. To describe and determine the purpose and importance of project management from the perspectives of planning, tracking and completion of project. 2. To compare and differentiate organization structures and project structures. 3. To implement a project to manage project schedule, expenses and resources with the application of suitable project management tools. 4. Understand and practice the process of project management and its application in delivering successful IT projects. 5. Conduct Case Study on “The Command Center Processing and Display System-Replacement” (CCPDS-R)								
UNIT - I	CONVENTIONAL SOFTWARE MANAGEMENT						CLASSES: 12	
The waterfall model, conventional software Management performance. Evolution of Software Economics: Software Economics, pragmatic software cost estimation.								
UNIT - II	IMPROVING SOFTWARE ECONOMICS						CLASSES: 12	
Reducing Software product size, improving software processes, improving team effectiveness, improving automation, Achieving required quality, peer inspections. The old way and the new: The principles of conventional software engineering, principles of modern software management, transitioning to an iterative process.								
UNIT - III	LIFE CYCLE PHASES						CLASSES: 12	
Engineering and production stages, inception, Elaboration, construction, transition phases. Artefacts of the process: The artefact sets, Management artefacts, Engineering artefacts, programmatic artefacts. Model based software architectures: A management perspective and technical perspective.								

UNIT - IV	WORK FLOWS OF THE PROCESS	CLASSES: 14
Software Process Workflows, Inter Trans Workflows. Checkpoints Of The Process: Major Mile Stones, Minor Milestones, Periodic Status Assessments. Iterative Process Planning: Work Break Down Structures, Planning Guidelines, Cost And Schedule Estimating, Interaction Planning Process, Pragmatic Planning. Project Organizations And Responsibilities: Line-Of-Business Organizations, Project Organizations, Evolution Of Organizations. Process Automation: Automation Building Blocks, The Project Environment.		
UNIT - V	PROJECT CONTROL AND PROCESS INSTRUMENTATION	CLASSES: 15
The Server Core Metrics, Management Indicators, Quality Indicators, Life Cycle Expectations Pragmatic Software Metrics, Metrics Automation. Tailoring The Process: Process Discriminates, Example. Future Software Project Management: Modern Project Profiles, Next Generation Software Economics, Modern Process Transitions.		
TEXT BOOKS		
1. Software Project Management, Walker Royce, Pearson Education, 2012 2. Software Project Management, Bob Hughes & Mike Cotterell, Fifth edition, TataMc-GrawHill, 2006		
REFERENCE BOOKS		
1. Applied Software Project Management, Andrew Stellman & Jennifer Greene, O'Reilly, 2006 2. Head First PMP, Jennifer Greene & Andrew Stellman, O'Reilly, 2007 3. Software Engineering Project Management, Richard H. Thayer & Edward Yourdon, second edition, Wiley India, 2004. 4. Agile Project Management, Jim Highsmith, Pearson education, 2004 5. The art of Project management, Scott Berkun, O'Reilly, 2005. 6. Software Project Management in Practice, Pankaj Jalote, Pearson Education, 2002		
WEB LINKS		
1. https://www.tutorialspoint.com/software_engineering/software_project_management.htm 2. https://ebooks.lpude.in/management/mba/term_4/DCAP304_DCAP515_SOFTWARE_PROJECT_MANAGEMENT.pdf		
MOOCS : https://onlinecourses.nptel.ac.in/noc19_cs70/preview		
EBOOK : https://www.edulearn.net.in/wp-content/uploads/2019/09/MG6088-Software-Project-Management.pdf		

SEMANTIC WEB

III B.TECH - II SEMESTER - (OPEN ELECTIVE - I)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS37	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course should enable the students to: 1. Understand the sematic web and its importance. 2. Understand the attribute languages and how to use in semantic web 3. Understand the Rule based languages 4. Understand the different methods to development in internet 5. Learn different types of semantic web tools.								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: 1. Classify the ontologies towards to the internet. 2. Apply the attribute languages on internet. 3. Apply the rule based languages for improving the usage of semantic web. 4. Analyze the methods to development of the internet. 5. Use semantic web tools for efficient development and usage of web.								
UNIT - I	THE FUTURE OF THE INTERNET						CLASSES: 12	
Introduction, The Syntactic Web, The Semantic Web, How the Semantic Web Will Work. Ontology in Computer Science: Defining the Term Ontology, Differences Among Taxonomies, Thesauri, and Ontologies, Classifying Ontologies, Web Ontologies, Web Ontology Description Languages, Ontology, Categories, and Intelligence.								
UNIT - II	KNOWLEDGE REPRESENTATION IN DESCRIPTION LOGIC						CLASSES: 12	
Introduction, An Informal Example, The Family of Attributive Languages, Inference Problems. RDF and RDF Schema: Introduction, XML Essentials, RDF, RDF Schema, A Summary of the RDF/RDF Schema Vocabulary. OWL: Introduction, Requirements for Web Ontology Description Languages, Header Information, Versioning, and Annotation Properties, Properties, Classes, Individuals, Data types, A Summary of the OWL Vocabulary.								
UNIT - III	RULE LANGUAGES						CLASSES: 14	
Introduction, Usage Scenarios for Rule Languages, Datalog, RuleML, SWRL, TRIPLE. Semantic Web Services: Introduction, Web Service Essentials, OWL-S Service Ontology, An OWL-S Example.								
UNIT - IV	METHODS FOR ONTOLOGY DEVELOPMENT						CLASSES: 15	
Introduction, Uschold and King Ontology Development Method, Toronto Virtual Enterprise Method, Methontology, KACTUS Project Ontology Development Method, Lexicon-Based Ontology Development Method, Simplified Methods. Ontology Sources: Introduction, Metadata, Upper Ontologies, Other Ontologic of Interest, Ontology Libraries.								
UNIT - V	SEMANTIC WEB SOFTWARE TOOLS & SOFTWAREAGENTS						CLASSES: 12	
Introduction, Metadata and Ontology Editors, Reasoners, Other tools. Introduction, Agent Forms, Agent Architecture, Agents in the Semantic web Context. Semantic Desktop: Introduction, Semantic Desktop Metadata, Semantic Desktop Ontologies, Semantic Desktop Architecture, Semantic Desktop Related Applications. Ontology Application in Art: Introduction, Ontologies for the Description of Works of Art, Metadata Schemas for The Description of Works of Art, Semantic Annotation of Art Images.								

TEXT BOOKS

1. Semantic Web- Concepts, Technologies and applications, 2007th edition (24 January 2007) K.Breitman, Marco Antonio Casanova and Walter Truszowski, Springer..

REFERENCE BOOKS

1. Information Sharing on the Semantic Web, Heiner Stuckenschmidt, Frankvan Harmelen, Springer.
2. Semantic Web Primer, Grigoris Antoniou, FrankVan
3. Semantic Web Services: Concepts, Technologies and Applications, Rudi Studer, Stephan Grimm, Andrees Abeker, Springer
4. Towards the Semantic Web: Ontology Driven Knowledge Management, John Davis, Dieter

WEB LINKS

1. <https://www.techtarget.com/searchcio/definition/Semantic-Web>
2. <https://www.w3.org/RDF/Metalog/docs/sw-easy>
MOOC:1. <https://www.coursera.org/learn/web-data>
EBOOK:1. <https://open.hpi.de/courses/semanticweb>

DATABASE MANAGEMENT SYSTEMS

IV B.TECH - I SEMESTER - (OPEN ELECTIVE-II)									
Course	Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS09	OEC	L	T	P	C	CIE	SEE	Total	
		3	-	-	3	40	60	100	
COURSE OBJECTIVES The course should enable the students to: 1. Discuss the basic database concepts, applications, data models, schemas and instances. 2. Design Entity Relationship model for a database. 3. Demonstrate the use of constraints and relational algebra operations. 4. Describe the basics of SQL and construct queries using SQL 5. Understand the importance of normalization in databases. 6. Demonstrate the basic concepts of transaction processing and concurrency control. 7. Understand the concepts of database storage structures and identify the access techniques.									
COURSE OUTCOMES At the end of the course the students are able to: 1. Use the basic concepts of Database Systems in Database design 2. Design a Database using ER Modeling 3. Apply relational algebra and calculus concepts AND SQL for a wide range of database operations 4. Apply SQL queries and PL/SQL queries to interact with Database 5. Grasp transaction management principles including transaction states and concurrency control techniques									
UNIT - I	INTRODUCTION						CLASSES: 09		
Introduction and applications of DBMS, Purpose of database, Data Independence, Database System architecture-Levels, Database users and DBA. DATABASE DESIGN: Database Design Process, ER Diagrams - Entities, Attributes, Relationships, Constraints, keys, extended ER features, Generalization, Specialization, Aggregation, Conceptual design with the E-R model.									
UNIT - II	RELATIONAL MODEL & SCHEMA REFINEMENT						CLASSES: 09		
RELATIONAL MODEL: Introduction to the relational model, Integrity constraints over relations, Enforcing integrity Constraints, Querying relational data, Logical database design: E-R to relational, Introduction to views, Destroying/altering tables and views. SCHEMAREFINEMENT AND NORMAL FORMS: Introduction to schema refinement, functional dependencies, reasoning about FDs. Normal forms: 1NF, 2NF, 3NF, BCNF, properties of decompositions, normalization, schema refinement in database design.									
UNIT - III	RELATIONAL ALGEBRA CALCULUS & SQL						CLASSES: 11		

RELATIONAL ALGEBRA AND CALCULUS: Relational algebra operators, relational calculus – Tuple and domain relational calculus. SQL: Basics of SQL, DDL,DML,DCL, structure - creation, alteration, defining constraints - Primary key, foreign key, unique, not null, check, IN operator, Functions - aggregate functions, Built-in functions — numeric, date, string functions, set operations, sub-queries, correlated sub-queries		
UNIT - IV	SQL and PL/SQL	CLASSES: 11
SQL: Use of group by, having, order by clauses, join and its types, Exist, Any, All clauses. Transaction control commands - Commit, Rollback, Save point, PL/SOL: Environment, block structure, variables, operators, data types, control structures; cursors, stored procedures, Triggers.		
UNIT - V	TRANSACTION & CONCURRENCY CONTROL	CLASSES: 10
TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, concurrent executions, Serializability, recoverability, testing for serializability. CONCURRENCY CONTROL AND RECOVERY SYSTEM: Concurrency control, lock based protocols, Time-stamp based protocols, validation based protocols, multiple granularity and deadlock handling. Recovery system - failure classification, storage structure, recovery and atomicity, log based recovery.		
TEXT BOOKS		
1. Abraham Silbers Ghatz, Henry F. Korth, S. Sudharshan, "Database System Concepts". Sixth Edition, Tata McGraw Hill, 2011 2. Raghurama Krishnan, Johannes Gehrke, "Database Management Systems", TATA McGraw Hill. 3rd Edition, 2007. 3. R.P. Mahapatra & Govind Verma, Database Management Systems, Khanna Publishing House, 2013. 4. Michael McLaughlin, Oracle Database 11g PL/SOL Programming, Oracle press.		
REFERENCE BOOKS		
1. Peter Rob, Carlos Coronet, Database Systems Design Implementation and Management, 7th edition, 2009. 2. Scott UnTian. Michael McLaughlin, Ron Hardman. "Oracle database 10g PL/SQL programming ". 6th edition, Tata McGraw Hill, 2010 3. S.K Singh, "Database Systems Concepts. Design and Applications", First edition, Pearson Education, 2006. 4. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems". Fourth Edition. Pearson/Addition wesley, 2007		
WEB REFERENCES		
1. https://www.learndb.com/databases/how-to-convert-er-diagram-to-relational-database 2. https://www.w3schools.com/sql/sql_create_table.asp 3. https://www.ibm.com/docs/en/zos-basic-skills?topic=zos-what-is-database-management-system&external_link=true 4. https://www.cs.ucdavis.edu/~green/courses/ecs165a-w11/		
E-TEXT BOOKS		
1. https://www.freebookcentre.net/Database/Free-Database-Systems-Books-Download.html 2. https://www.ddegjust.ac.in/studymaterial/mca-3/ms-11.pdf		

DATA WAREHOUSING AND BUSINESS INTELLIGENCE

IV B.TECH - I SEMESTER - (OPEN ELECTIVE-II)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS43	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. This course is concerned with extracting data from the information systems that deal with the day-to-day operations and transforming it into data that can be used by businesses to drive high-level decision making 2. Students will learn how to design and create a data warehouse, and how to utilize the process of extracting, transforming, and loading (ETL) data into data warehouses.								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: 1. Understand architecture of data warehouse and OLAP operations. 2. Understand Fundamental concepts of BI and Analytics 3. Application of BI Key Performance indicators and Design of Dashboards, Implementation of Web Analytics 4. Understand Utilization of Advanced BI Tools and their Implementation. 5. Implementation of BI Techniques and BI Ethics.								
UNIT - I	DATA WAREHOUSE						CLASSES:12	
DATA WAREHOUSE: Data Warehouse-Data Warehouse Architecture- Multidimensional Data Model-Data cube and OLAP Technology-Data Warehouse Implementation -DBMS schemas for Decision support - Efficient methods for Data cube computation.								
UNIT - II	BUSINESS INTELLIGENCE						CLASSES: 10	
Business Intelligence: Introduction – Definition, Leveraging Data and Knowledge for BI, BI Components, BI Dimensions, Information Hierarchy, Business Intelligence and Business Analytics. BI Life Cycle. Data for BI - Data Issues and Data Quality for BI.								
UNIT - III	BI IMPLEMENTATION						CLASSES: 10	
BI Implementation - Key Drivers, Key Performance Indicators and Performance Metrics, BI Architecture/Framework, Best Practices, Business Decision Making, Styles of BI-vent-Driven alerts-A Cyclic process of Intelligence Creation. The value of Business intelligence -Value driven and Information use.								
UNIT - IV	ADVANCED BI						CLASSES:10	
Advanced BI – Big Data and BI, Social Networks, Mobile BI, emerging trends, Description of different BI-Tools (Pentaho, KNIME)								
UNIT - V	BI IMPLEMENTATION						CLASSES: 12	
Business intelligence implementation-Business Intelligence and integration implementation-connecting in BI systems- Issues of legality- Privacy and ethics- Social networking and BI.								

TEXT BOOKS

1. Data Mining – Concepts and Techniques - JIAWEI HAN & MICHELINE KAMBER, Elsevier.
2. Rajiv Sabherwal “Business Intelligence” Wiley Publications, 2012.

REFERENCE BOOKS

1. Efraim Turban, Ramesh Sharda, Jay Aronson, David King, Decision Support and Business Intelligence Systems, 9th Edition, Pearson Education, 2009.
2. David Loshin, Business Intelligence - The Savy Manager's Guide Getting Onboard with Emerging IT, Morgan Kaufmann Publishers, 2009
3. Philo Janus, Stacia Misner, Building Integrated Business Intelligence Solutions with SQL Server, 2008 R2 & Office 2010, TMH, 2011.
4. Business Intelligence Data Mining and Optimization for decision making [Author: Carlo-Verellis] [Publication: (Wiley)]
5. Data Warehousing, Data Mining & OLAP- Alex Berson and Stephen J. Smith- Tata McGrawHill Edition, Tenth reprint 2007
6. Building the Data Warehouse- W. H. Inmon, Wiley Dreamtech India Pvt. Ltd.
7. Data Mining Introductory and Advanced topics –MARGARET H DUNHAM, PEA.

WEB LINKS

1. <https://www.coursera.org/learn/data-warehousing-business-intelligence>
2. https://www.vssut.ac.in/lecture_notes/lecture1428550844.pdf

ARTIFICIAL INTELLIGENCE AND APPLICATIONS

IV B.TECH - II SEMESTER - (OPEN ELECTIVE-III)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS51	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. The difference between optimal reasoning vs. human like reasoning
2. Understand the notions of state space representation, exhaustive search, heuristic search along with the time and space complexities
3. Apply the Expert and rule based systems in knowledge representation techniques
4. Understand the applications of AI: namely Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Apply AI techniques to solve problems of Game Playing, Expert Systems, Machine Learning.
2. Formulate an efficient problem space for problem solving and Represent knowledge using the appropriate technique
3. Design and develop expert systems using Prolog and solve uncertainty problems.
4. Analyze real world problems and implement using ANN, CNN and AI techniques.
5. Implement the concepts of AI in Healthcare, Robotics and Agriculture.

UNIT - I	INTRODUCTION AND PROBLEM SOLVING	CLASSES: 14
Introduction: History, Intelligent Systems, Foundations of AI, Sub areas of AI, Applications. Problem Solving – State-Space Search and Control Strategies: Introduction, General Problem Solving, Characteristics of Problem, Exhaustive Searches, Heuristic Search Techniques, Iterative-Deepening A*, Constraint Satisfaction, Game Playing, Alpha-Beta Pruning.		
UNIT - II	LOGIC AND KNOWLEDGE REPRESENTATION	CLASSES: 14
Logic Concepts and Logic Programming: Introduction, Propositional Logic, Natural Deduction System, Axiomatic System, Semantic Tableau System in Propositional Logic, Resolution Refutation in Propositional Logic, Predicate Logic, Logic Programming. Knowledge Representation: Introduction, Approaches to Knowledge Representation, Knowledge Representation using Semantic Network, Extended Semantic Networks for KR, Knowledge Representation using Frames.		
UNIT - III	EXPERT SYSTEM AND APPLICATIONS	CLASSES:14
Expert System and Applications: Introduction, Phases in Building Expert Systems, Expert System Architecture, Expert Systems Vs. Traditional Systems, rule based expert systems black board systems truth maintenance systems, Application of Expert Systems.		
UNIT - IV	ANN, CNN	CLASSES:12
Artificial Neural Networks: Introduction, Artificial Neural Networks, Single-Layer Feed Forward Networks, Multi-Layer Feed-Forward Networks, Radial-Basis Function Networks, Design Issues of Artificial Neural Networks, Recurrent Networks. Convolutional Neural Networks: Pooling, Padding, Layers used to build ConvNets, Working of CNN, Python Libraries used for AI: Tensor flow, Keras, Theano, Pytorch, Scikit Learn, Pandas, Image Classification using CNN with Python.		

UNIT - V	APPLICATIONS & CASE STUDIES OF AI	CLASSES: 11
Applications of AI in real world, Future of AI, Benefits and Risks of AI. Case Studies: AI in Healthcare: Medical Imaging and Diagnostic, AI in Agriculture: Autonomous Tractors, Controlling Pest Infestations, Soil and Crops Health Monitoring, AI in Robotics: Use of Artificial Intelligence in Robotics		
TEXT BOOKS		
1. Saroj Kaushik. Artificial Intelligence. Cengage Learning. 2011 2. Russell, Norvig: Artificial intelligence, A Modern Approach, Pearson Education, Second Edition. 2004		
REFERENCE BOOKS		
1. Rich, Knight, Nair: Artificial intelligence, Tata McGraw Hill, Third Edition 2009. 2. Introduction to Artificial Intelligence by Eugene Charniak, Pearson. 3. Introduction to Artificial Intelligence and expert systems Dan W.Patterson. PHI. 4. Artificial Intelligence by George Flugerrearson fifth edition.		
WEB LINKS		
1. https://www.ibm.com/topics/artificial-intelligence#:~:text=At%20its%20simplest%20form%2C%20artificial,in%20conjunction%20with%20artificial%20intelligence2.https://www.techtarget.com/searchenterpriseai/definition/AI-Artificial-Intelligence 2. MOOCS:https://www.mygreatlearning.com/great-lakes-artificial-intelligence-and-machine-learning? 3. E BOOK: https://www.freebookcentre.net/CompuScience/Free-Artificial-Intelligence-Books-Download.html		

CYBER SECURITY AND CYBER LAWS

IV B.TECH - II SEMESTER - (OPEN ELECTIVE-III)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS36	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course should enable the students to: 1. To learn cybercrime cyber security and cyber law. 2. To understand the cyber-attacks. 3. To understand information gathering. 4. To learn how to detect and prevent a cyber-attack and tools for mitigating them. 5. Develop Competencies For Dealing With Frauds And Deceptions (Confidence Tricks, Scams) And Other Cyber Crimes.								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: 1. Explain the basics of cyber security, cybercrime and cyber law. 2. Classify various types of attacks and learn the tools to launch the attacks. 3. Apply various tools to perform information gathering. 4. Understand Cybercrimes and cyber laws. 5. Explore The Legal And Policy Developments In Various Countries To Regulate Cyberspace.								
UNIT - I	INTRODUCTION						CLASSES: 16	
Cyber Security-definition, Need for Cyber Security, Protection, Security, risk, threat, vulnerability, exploit, attack, confidentiality, integrity, availability, non-repudiation, authentication , authorization, codes, plain text, encryption, decryption, cipher text, key, ciphers, Symmetric and asymmetric cryptography, Public key , private key ,Crypt analysis, Cyber forensics.								
UNIT - II	CYBER ATTACKS						CLASSES:14	
CYBER ATTACKS (definitions and examples): Denial-of-service attacks, Man-in-the middle attack, Phishing, spoofing and spam attacks, Drive by attack, Password attack, SQL injection attack, Cross-site scripting attack, Eavesdropping attack, Birthday attack, Malware attacks, Social Engineering attacks								
UNIT - III	HANDLING THE ATTACKS						CLASSES: 11	
Firewalls, logging and intrusion detection systems, e-mail security, security issues in operating systems, ethics of hacking and cracking.								
UNIT - IV	CYBER CRIMES						CLASSES: 12	
Cyber Crime and Information Security – History of Cyber Crime ,classifications of Cyber Crimes – Tools and Methods –Password Cracking, Key loggers, Spywares, SQL Injection – Network Access Control – Cloud Security – Web Security – Wireless Security								
UNIT - V	CYBER SPACE AND CYBER LAWS						CLASSES: 16	
Cyber Space- Fundamental definitions -Interface of Technology and Law – Jurisprudence and-Jurisdiction in Cyber Space - Indian Context of Jurisdiction - Enforcement agencies – Need for IT act - UNCITRAL – E-Commerce basics .Information Technology Act, 2000 - Aims and Objects — Overview of the Act – Jurisdiction Cyber Law, A Global Perspective on Cyber Crimes; Cyber Laws – The Indian IT Act – Cybercrime and Punishment.								

TEXT BOOKS
1. Anand Shinde, "Introduction to Cyber Security Guide to the World of Cyber Security", Notion Press, 2021. 2. Nina Godbole, Sunit Belapure, "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", Wiley Publishers, 2011.
REFERENCE BOOKS
1. David Kim, Michael G. Solomon, "Fundamentals of Information Systems Security", Jones & Bartlett Learning Publishers, 2013 2. Patrick Engebretson, "The Basics of Hacking and Penetration Testing: Ethical Hacking and Penetration Testing Made easy", Elsevier, 2011 3. Justice Yatindra Singh: Cyber Laws, Universal Law Publishing Co., New Delhi 4. Farouq Ahmed, Cyber Law in India, New Era publications, New Delhi

OPEN ELECTIVES OFFERED BY DEPARTMENT OF ECE

OE1		OE2	
A6EC55	Microprocessors and Interfacing	A6EC57	Microcontrollers and Applications
A6EC56	Principles of Communications	A6EC58	Fundamentals of Image processing
OE3			
A6EC59	Introduction to Sensors and Actuators		
A6EC60	Introduction to Computer Vision		

MICRO PROCESSORS AND INTERFACING

OPEN ELECTIVE COURSE - I								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC55	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course should enable the students to: 1. Understand the basic of 8, 16 bit microprocessor architectures and its functionalities. 2. Develop an assembly language programming skills of various processors. 3. Interface different peripheral devices with microprocessors and microcontrollers. 4. Interface memory devices to 8086 processor. 5. Analyze Serial communication schemes of microprocessor based systems.								
COURSE OUTCOMES Upon successful completion of the course, the student is able to 1. Understand the architecture of microprocessor. 2. Understand the programming model of microprocessors. 3. Interface different external peripherals with microprocessors. 4. Analyze a problem and formulate appropriate computing solution for processor based application. 5. Develop an assembly language program for specified application in communication.								
UNIT-I	MICROPROCESSORS ARCHITECTURE						CLASSES: 09	
Overview of 8085, 8086 architecture- functional diagram, Register organization, memory segmentation, Memory addresses, physical memory organization.								
UNIT-II	SIGNAL DESCRIPTION OF 8086						CLASSES: 09	
Signal description of 8086, timing diagrams, Interrupt structure of 8086, Vector interrupt table, Interrupts of 8086 Processor, Interrupt service routine.								
UNIT-III	INSTRUCTION SET AND ASSEMBLY LANGUAGE PROGRAMMING OF 8086						CLASSES: 09	
Programming model, Addressing modes, Instruction set, Assembler directives, Programs involving logical, branch and call instructions, Sorting, evaluating arithmetic expressions, and string manipulations.								
UNIT-IV	I/O INTERFACE						CLASSES: 09	
Introduction to 8255 PPI, various modes of operation of 8255 PPI, interfacing 8255 to 8086, Stepper motor interfacing, D/A &A/D converter, Memory interfacing to 8086.								
UNIT-V	SERIAL COMMUNICATION INTERFACE						CLASSES: 09	
Serial communication standards, serial data transfer schemes, 8251 USART architecture, Interfacing of 8251/USART to 8086 Processor.								

TEXT BOOKS

1. D.V.Hall, Microprocessors and Interfacing. TMGH, 2nd edition 2006.
2. Advanced microprocessors and peripherals-A.K ray and K.M.Bhurchandani, TMH, 2nd edition 2006.
3. R.S.Gaonkar, Microprocessor Architecture: Programming and Applications with the 8085/8080A, Penram International Publishing, 1996.

REFERENCE BOOKS

1. D A Patterson and J H Hennessy, "Computer Organization and Design The hardware and software interface. Morgan Kaufman Publishers.
2. Micro computer system 8086/8088 family architecture, programming and design- By Liu and GA Gibson, PHI, 2nd Ed.,

WEB REFERENCES

1. <http://www.freebookcentre.net/electronics-ebooks-download/Microprocessor-and-Microcontroller.html>
2. <http://coen.boisestate.edu/smluo/smluo-courses/ece-332-microprocessors-fall07/lecture-notes/>
3. <http://www.freebookcentre.net/electronics-ebooks-download/Introduction-to-Microcontrollers-Lecture-Notes.html>

E-TEXT BOOKS

1. <http://gen.lib.rus.ec/book/index.php?md5=67C5AC79DC8180A7F0641609D0C7800C>
2. <http://www.faadooengineers.com/threads/9039-8085-microprocessor-by-RAMESH-GANOKAR-ebook-pdf-download>
3. https://e.edim.co/123389964/The_8051_Microcontroller_Architecture_Programming_And_Applications.pdf
4. https://e.edim.co/123389964/A.K._Ray_and_K.M._BhurchandiAdvanced_Microprocessors_and_Peripherals_3e-Tata_Mcgraw_Hill.pdf

MOOC COURSE

1. <https://www.mooc-list.com/tags/microprocessors>
2. <https://www.coursera.org/courses?query=microprocessor>

PRINCIPLES OF COMMUNICATIONS

OPEN ELECTIVE COURSE-I								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC56	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
PREREQUISITE: Knowledge on Electron devices and Electronic circuits								
COURSE OBJECTIVES The students are able to: 1. Describe the basic concepts of analog communication 2. Significance of digital & analog communication 3. Compare the digital modulation techniques 4. Learn the various multiple access techniques 5. Understanding the Satellite and Optical fiber concepts.								
COURSE OUTCOMES Upon completion of the course, students will be able to: 1. Understand the concepts of analog modulations 2. Illustrate the basic concepts of digital modulations. 3. Analyse the various digital modulation methodologies. 4. Distinguish the concepts of multiple access techniques. 5. Differentiate the various orbits and associated satellite launches								
UNIT-I	FUNDAMENTALS OF ANALOG COMMUNICATION						CLASSES: 09	
Modulation, Principles of Amplitude Modulation, AM modulator and Demodulator, Types of Amplitude modulation (DSBSC, SSBSC, VSB). Angle modulation FM and PM waveforms, phase deviation and modulation index, frequency deviation and percent modulation.								
UNIT-II	DIGITAL TRANSMISSION						CLASSES: 09	
Introduction, Sampling theorem, Types of pulse modulations, PCM, companding, differential pulse code modulation, delta modulation, adaptive delta modulation, Inter symbol interference.								
UNIT-III	DIGITAL MODULATION TECHNIQUES						CLASSES: 09	
Introductions, Shannon limit for information capacity, ASK FSK, BPSK QPSK and DPSK modulators and demodulators, comparison of digital modulation techniques.								
UNIT-IV	SPREAD SPECTRUM AND MULTIPLE ACCESS TECHNIQUES						CLASSES: 09	
Introduction, Pseudo noise sequence, DS spread spectrum with coherent binary PSK, FH spread spectrum, multiple access techniques-FDMA, TDMA and CDMA.								
UNIT-V	SATELLITE AND OPTICAL COMMUNICATION						CLASSES: 09	
Satellite Communication Systems, Keplers Law, LEO, MEO and GEO Orbits, Link model Optical Communication Systems-Fiber losses-scattering, absorption attenuation and bending losses, types of optical Sources and Detectors.								

TEXT BOOKS

1. Wayne Tomasi, —Advanced Electronic Communication SystemsI, 6/e, Pearson Education, 2007.
2. Simon Haykin, —Communication SystemsI, 4th Edition, JohnWiley & Sons., 2001.

REFERENCE BOOKS

1. H.Taub, D L Schilling, G Saha, IPrinciples of Communication I3/e, 2007.
2. B.P.Lathi, IModern Analog And Digital Communication systems I, 3/e, Oxford University Press, 2007
3. Blake — Electronic Communication SystemsI, Thomson Delmar Publications, 2002.
4. Martin S.Roden — Analog and Digital Communication System, 3rd Edition, PHI, 2002.
5. B.Sklar, IDigital Communication Fundamentals and Applications I2/e Pearson Education 2007.

WEB REFERENCES

1. <https://personal.utdallas.edu/~torlak/courses/ee4367/lectures/FIBEROPTICS.pdf>
2. Bricker G (2012) 2-D bar codes, Journal of Computing Sciences in Colleges, **28**:1, (25-32), Online publication date: 1-Oct-2012.

E-TEXT BOOKS

1. https://books.google.co.in/books/about/Principles_Of_Communication.html?id=6Zunu4Acfg8C

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc18_ee26/preview

MICRO CONTROLLER & APPLICATIONS

OPEN ELECTIVE COURSE-II								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC57	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. To introduce the students about architectural features of microcontrollers and its registers 2. To introduce about the instruction set of 8051 3. To know about the I/O Ports and Interrupts of 8051 4. To know about Timers/Counters of 8051 5. To introduce Arduino programming and interfacing of sensors								
COURSE OUTCOMES Upon completion of the course, students will be able to: 1. Understand the architectural features of MCS-51 and select a suitable microcontroller to suit the application. 2. Develop programs for control applications using assembly language and embedded C 3. Use timers and counters for delay generation and event counting and Illustrate the use of interrupts and service routines 4. Write algorithms and develop programs for serial data communication applications. 5. Design microcontroller based-applications for simple real-world applications								
UNIT-I	8051 ARCHITECTURE						CLASSES: 09	
Introduction to microcontrollers, RISC, CISC, Harvard and Von Neumann architectures. Architecture of 8051, Signal descriptions of 8051, General purpose registers of 8051, register banks, Memory organization.								
UNIT-II	INSTRUCTION SET OF 8051						CLASSES: 09	
Assembly language Instruction format, 8051 Addressing modes, Instruction set of 8051: Classification, syntax and function of instructions, Simple programs.								
UNIT-III	I/O PORT AND INTERRUPT PROGRAMMING						CLASSES: 09	
I/O Ports of 8051,Features of I/O ports, configuring I/O ports, interface I/O devices such as LED, buzzer, push-button switch, example programs with assembly & C. Interrupts of 8051 and its priorities, IE and IP registers, Interrupt enabling/disabling and priority setting, example programs in assembly and C.								
UNIT-IV	TIMERS/COUNTERS AND SERIAL I/O						CLASSES: 09	
Bit structure and function of TMOD and TCON registers, Timer/Counter modes of operations, Timer/Counter programs in assembly and C. Bit structure and function of SCON, PCON registers, SBUF register, Serial Communication modes in 8051, programs on serial communication.								
UNIT-V	INTRODUCTION TO ARDUINO						CLASSES: 09	
Introduction to Arduino-uno board, Analog and Digital pins, programming structure of Arduino, introduction to sensors and actuators , Sensor interfacing, programming to sensors, Motor interfacing, LCD interfacing.								
TEXT BOOKS								

1. The 8051 Microcontroller(3rd edition) -Kenneth J Ayala
2. The 8051 Microcontroller & Embedded systems using assembly and C (2ndEdition) – M.A.Mazidi , J.C. Mazidi & R.D.McKinlay ISBN: 81-317-1026-2

REFERENCE BOOKS

1. The 8051 Microcontroller(4th Edition)- MacKenzie , ISBN:81-317-2018-7
2. The 8051 Microcontroller(1st Edition) – Dr. Uma Rao & Andhe Paallavi, ISBN: 81-317-3252-5
3. Microcontrollers & applications, Ramani Kalpathi, & Ganesh Raja , ISBN: 81-888-4918-9
4. Programming Arduino: Getting Started with Sketches, Second Edition (Tab) 2nd Edition – Simon Monk

WEB REFERENCES

1. <https://www.the8051microcontroller.com/web-references>

E-TEXT BOOKS

1. <https://www.freebookcentre.net/Electronics/Microcontroller-Books.html>
2. <https://www.freebookcentre.net/Electronics/Microcontroller-Application-Books.html>

MOOC COURSE

1. <https://nptel.ac.in/courses/117/104/117104072/>

FUNDAMENTALS OF IMAGE PROCESSING

OPEN ELECTIVE COURSE - II								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC58	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES								
The students are able to:								
1. To become familiar with digital image fundamentals								
2. To get exposed to image enhancement techniques in Spatial and Frequency domain.								
3. To learn concepts of degradation function and restoration techniques.								
4. To study the image segmentation and representation techniques.								
5. To become familiar with image compression and recognition methods								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. Know and understand the basics and fundamentals of digital image processing.								
2. Operate on images using the techniques of smoothing, sharpening and enhancement.								
3. Apply the restoration concepts and filtering techniques on digital images.								
4. Learn the basics of segmentation and features extraction.								
5. Analyze the compression and recognition methods for color images.								
UNIT-I	INTRODUCTION OF IMAGE PROCESSING						CLASSES: 09	
Digital Image Fundamentals-Elements of visual perception, image sensing and acquisition, image sampling and quantization, basic relationships between pixels – neighbourhood, adjacency, connectivity, distance measures								
UNIT-II	IMAGE ENHANCEMENT						CLASSES: 09	
Spatial Domain: Gray level transformations – Histogram processing – Basics of Spatial Filtering–Smoothing and Sharpening Spatial Filtering, Image Enhancement in Frequency Domain: Low Pass and High Pass Filters in Frequency Domain, Homomorphic filtering								
UNIT-III	IMAGE TRANSFORMS						CLASSES: 09	
Image Transforms: 2 D- Discrete Fourier Transform and its properties, Discrete Cosine Transform (DCT), Haar Transform, Hadmard Transform, Hotelling Transform and Slant transform.								
UNIT-IV	IMAGE SEGMENTATION						CLASSES: 09	
Introduction-Edge detection, Edge linking and boundary detection – Thresholding – Region based segmentation – Region growing – Region splitting and merging- Watershed segmentation algorithm.								
UNIT-V	IMAGE COMPRESSION						CLASSES: 09	
Image Compression Models, Huffman and Arithmetic Coding, Error Free Compression, Lossy Compression, Lossy and Lossless Predictive Coding, Transform Based Compression, JPEG 2000 Standards.								
TEXT BOOKS								
1. R.C. Gonzalez and R.E. Woods, Digital Image Processing, Second Edition, Pearson Education 3rd edition 2008								
2. Digital Image Processing- S Jayaraman, S Esakkirajan, T Veerakumar- McGraw Hill Edn., 2010.								
REFERENCE BOOKS								

- | |
|--|
| <ol style="list-style-type: none">1. Anil Kumar Jain, Fundamentals of Digital Image Processing, Prentice Hall of India.2nd edition 20042. MuratTekalp , Digital Video Processing" Prentice Hall, 2nd edition 2015.3. S. Sridhar , Digital Image Processing, Oxford University Press, 2nd Ed, 2016. |
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WEB REFERENCES

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| <ol style="list-style-type: none">1. http://homepages.inf.ed.ac.uk/rbf/BOOKS/VERNON/Chap004.pdf |
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E-TEXT BOOKS

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|--|
| <ol style="list-style-type: none">1. https://books.google.co.in/books/about/Digital_Image_Processing.html?id=a62xQ2r_f8wC |
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MOOC COURSE

- | |
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| <ol style="list-style-type: none">1. https://nptel.ac.in/courses/117105079/ |
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INTRODUCTION TO SENSORS AND ACTUATORS

OPEN ELECTIVE COURSE-III								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC59	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course should enable the students to: 1. Understanding basic laws and phenomena on which operation of sensors and actuators - Transformation of energy 2. Create analytical design and development solutions for sensors and actuators 3. To know the basic laws of behaviour of sensors and actuators 4. To able to know about the Standards for Smart Sensor Interface 5. Analyse the development and application of sensors and actuators.								
COURSE OUTCOMES Upon successful completion of the course, the student is able to 1. Apply the fundamental physical and technical base of sensors and actuators 2. Analyze various premises, approaches, procedures and results related to sensors and actuators 3. Analyze basic laws and phenomena that define behaviour of sensors and actuators 4. Apply the Smart Sensor Interface in various applications 5. Develop the application of sensors and actuators.								
UNIT-I	SENSORS & TRANSDUCERS						CLASSES: 09	
Sensors/Transducers: Principles, Classification, Parameters, Characteristics, Environmental Parameters (EP), Characterization. Mechanical and Electromechanical Sensors: Introduction, Resistive Potentiometer, Resistance Strain Gauge, Semiconductor Strain Gauges, Capacitive Sensors, Electrostatic Transducer, Ultrasonic Sensors.								
UNIT-II	THERMAL SENSORS						CLASSES: 09	
Thermal Sensors: Introduction, Gas thermometric Sensors, Thermal Expansion Type Thermometric Sensors, Acoustic Temperature Sensor, Dielectric Constant and Refractive Index Thermo-sensors, Nuclear Thermometer, Thermo-EMF Sensors, Thermal Radiation Sensors, Quartz Crystal Thermo-electric Sensors, Spectroscopic Thermometry, Noise Thermometry.								
UNIT-III	RADIATION SENSORS						CLASSES: 09	
Introduction – Basic Characteristics – Types of Photosensistors– X-ray and Nuclear Radiation Sensors– Fiber Optic Sensors. Electro Analytical Sensors: Introduction – The Electrochemical Cell – The Cell Potential – Standard Hydrogen Electrode (SHE) – Liquid Junction and Other Potentials – Sensor Electrodes.								
UNIT-IV	SMART SENSORS						CLASSES: 09	
Introduction, Primary Sensors, Excitation, Converters, Compensation, Information Coding/Processing, Data Communication, Standards for Smart Sensor Interface, the Automation. Sensors Applications: Introduction, On-board Automobile Sensors (Automotive Sensors), Home Appliance Sensors.								
UNIT-V	ACTUATORS						CLASSES: 09	
Actuators: Pneumatic and Hydraulic Actuation Systems- Actuation systems, Pneumatic and hydraulic systems, Directional Control valves, Pressure control valves, Cylinders, Servo and proportional control valves.								

TEXT BOOKS
1. D. Patranabis, "Sensors and Transducers", PHI Learning Private Limited. 2. W. Bolton, "Mechatronics", Pearson Education Limited.
REFERENCE BOOKS
1. Patranabis, "Sensors and Actuators", 2nd Edition, PHI, 2013.
WEB REFERENCES
1. https://www.journals.elsevier.com/sensors-and-actuators
E-TEXT BOOKS
1. https://www.sciencedirect.com/handbook/handbook-of-sensors-and-actuators
MOOC COURSE
1. https://www.classcentral.com/course/swayam-sensors-and-actuators-14285

INTRODUCTION TO COMPUTER VISION

OPEN ELECTIVE COURSE-III								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC60	OEC	L	T	P	C	CIA	SEE	Total
		3		-	3	40	60	100
COURSE OBJECTIVES The course should enable the students to: 1. To review image processing techniques for computer vision. 2. To understand the concepts of Image Enhancement. 3. To understand the basics of segmentation and its applications. 4. To know about the feature extraction and Hough Transform. 5. To understand three-dimensional image analysis techniques.								
COURSE OUTCOMES Upon successful completion of the course, the student is able to 1. Implement fundamental image processing techniques required for computer vision. 2. Analyze the various spatial and frequency domain filtering. 3. Understand the idea about segmentation 4. Apply Hough Transform for line, circle, and ellipse detections. 5. Apply 3D vision techniques and motion related techniques.								
UNIT-I	IMAGE PROCESSING FOUNDATIONS						CLASSES: 09	
Fundamental steps in Digital Image Processing, Components of Digital Image Processing Image sensing and acquisition, Image formation model, Pixels, Basic relationship between pixels, Sampling and Quantization in Digital Image Processing								
UNIT-II	IMAGE ENHANCEMENT						CLASSES: 09	
Intensity transformations, contrast stretching, histogram equalization, Spatial filtering: Smoothing filters, sharpening filters, Frequency domain filtering, Homomorphic filtering,								
UNIT-III	BAICS OF SEGMENTATION						CLASSES: 09	
Edge detection, Thresholding, Region growing, Fuzzy clustering, Watershed algorithm, Active contour models, and Texture feature based segmentation, Graph based segmentation, Wavelet based Segmentation - Applications of image segmentation.								
UNIT-IV	FEATURE EXTRACTION						CLASSES: 09	
First and second order edge detection operators, Phase congruency, Localized feature extraction -detecting image curvature, shape features, Introduction to Hough transform, shape skeletonization, Boundary descriptors, Moments, Texture descriptors.								
UNIT-V	3D IMAGE VISUALIZATION						CLASSES: 09	
Sources of 3D Data sets, Slicing the Data set, Arbitrary section planes, Volumetric display, Stereo Viewing, Ray tracing, Reflection, Surfaces, Multiple connected surfaces, Image processing in 3D, Measurements on 3D images.								

BOOKS

1. Rafael C. Gonzalez, Richard E. Woods, Digital Image Processing', Pearson, Education, Inc., Second Edition, 2004.
2. Mark Nixon, Alberto Aguado, "Feature Extraction and Image Processing", Academic Press, 2008.
3. D. L. Baggio et al., "Mastering OpenCV with Practical Computer Vision Projects", Packt Publishing, 2012.

REFERENCE BOOKS

1. E. R. Davies, "Computer & Machine Vision", Fourth Edition, Academic Press, 2012.
2. Jan Erik Solem, "Programming Computer Vision with Python: Tools and algorithms for analyzing images", O'Reilly Media, 2012.
3. R. Szeliski, "Computer Vision: Algorithms and Applications", Springer 2011.
4. Simon J. D. Prince, "Computer Vision: Models, Learning, and Inference", Cambridge University Press, 2012.

WEB REFERENCES

1. <https://machinelearningmastery.com/what-is-computer-vision/>

E-TEXT BOOKS

1. <https://machinelearningmastery.com/computer-vision-books/>

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc21_ee23/preview

INTRODUCTION TO MOBILE COMMUNICATIONS

OPEN ELECTIVE COURSE-IV								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC61	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course should enable the students to: - Understand mobile radio communication principles and to study the recent trends adopted in cellular systems and wireless standards. - To fill the skill gap in the domain of Cellular Technology. - Get hands on practice in the field of cellular technology and related disciplines. - To appreciate the contribution of wireless Communication networks to overall technological growth. - To provide an overview of wireless Communication networks are and its applications in communication engineering.								
COURSE OUTCOMES Upon successful completion of the course, the student is able to - Explain the basics of mobile telecommunication system - Illustrate the generations of telecommunication systems in wireless network - Demonstrate knowledge on : cellular concepts like frequency reuse, fading, equalization, and hand off strategies - Determine the functionality of network layer and Identify a routing protocol for a given Adhoc networks - Implement the functionality of Transport and Application layer								
UNIT-I	INTRODUCTION TO MOBILE COMPUTING						CLASSES: 09	
Applications of Mobile Computing- Generations of Mobile Communication Technologies-MAC Protocols –SDMA-TDMA-FDMA-CDMA								
UNIT-II	MOBILE TELE COMMUNICATION SYSTEM						CLASSES: 09	
GSM–Architecture–Protocols–Connection Establishment–Frequency Allocation –Routing–Mobility Management – Security–GPRS-UMTS-Architecture								
UNIT-III	CELLULAR CONCEPT						CLASSES: 09	
Limitations of conventional mobile system, concept of frequency reuse, cluster size, cellular system Architecture, channel assignment strategies, call hand off strategies-hard hand off and soft hand off, prioritizing hand off.								
UNIT-IV	WIRELESS NETWORKS						CLASSES: 09	
Wireless networks – Advantages and applications of Wireless LAN, WLAN technology – RF and IR wireless LAN, diffuse, IEEE802.11architecture, Physical layer, MAC layer, Introduction to WI-FI, Bluetooth architecture.								
UNIT-V	MOBILE NETWORK AND TRANSPORT LAYER						CLASSES: 09	
Introduction to Mobile IP, requirements, IP packet delivery, Agent discovery, Registration, Tunneling and encapsulation, Optimization, Reverse tunneling; Mobile adhoc networks–Routing, Destination sequence distance vector, Dynamic source routing.								

TEXT BOOKS

1. Theodore S. Rappaport - Wireless Communications Principles and Practice, 2nd Edition, Pearson Education, 2003.
2. Andreas F. Molisch – Wireless Communications, JohnWiley, 2nd Edition, 2006.

REFERENCE BOOKS

1. Kamilo Feher-Wireless Digital Communications, PHI, 2003
2. W.C.Y. Lee - Mobile Cellular Communications, 2nd Edition, McGraw-Hill, 1995.
3. Yi Bing Lin - Wireless and Mobile Network Architectures, 2nd Edition, Wiley, 2008.

WEB REFERENCES

1. https://ptolemy.berkeley.edu/books/leeseshia/releases/LeeSeshia_DigitalV2_2.pdf

E-TEXT BOOKS

1. <https://ptolemy.berkeley.edu/books/leeseshia/>

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc20_ee98/preview

BASICS OF EMBEDDED SYSTEMS DESIGN

OPEN ELECTIVE COURSE-IV								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A5EC62	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course should enable the students to: 1. Explains about the basic functions, structure, concepts and applications of embedded systems. 2. Describes the Core of the Embedded System 3. Explains about the tools used to develop in an embedded environment 4. Gives the knowledge about the development of embedded software using RTOS. 5. Gives the knowledge about the Advance Architectures								
COURSE OUTCOMES Upon successful completion of the course, the student is able to 1. Understand the selection procedure of Processors in the embedded domain. 2. Design Procedure for Embedded Firmware. 3. Implement the Real time Operating Systems applications in Embedded Systems 4. Analyze the Correlation between task synchronization and latency issues 5. Understand the concepts of Advance architecture								
UNIT-I	INTRODUCTION TO EMBEDDED SYSTEMS						CLASSES:09	
Definition of Embedded System, Embedded Systems Vs. General Computing Systems, History of Embedded Systems, Classification, Major Application Areas, Purpose of Embedded Systems, Characteristics and Quality Attributes of Embedded Systems								
UNIT-II	TYPICAL EMBEDDED SYSTEM						CLASSES:09	
Core of the Embedded System: General Purpose and Domain Specific Processors, ASICs, PLDs, Commercial Off-The-Shelf Components (COTS), Sensors and Actuators.								
UNIT-III	EMBEDDEDS OF TWARE DEVELOPMENT TOOLS						CLASSES:09	
Host and target machines, linker/locators for embedded software, getting embedded software in to the target system, debugging techniques: Testing on host machine, using laboratory tools.								
UNIT-IV	RTOS BASED EMBEDDED SYSTEM DESIGN						CLASSES:09	
Operating System Basics, Types of Operating Systems, Tasks, Process and Threads, Multiprocessing and Multitasking, Task Scheduling. Message queues, mailboxes and pipes.								
UNIT-V	INTRODUCTION TO ADVANCED ARCHITECTURES						CLASSES:09	
ARM and SHARC, processor and memory organization and instruction level parallelism; networked embedded systems: bus protocols, I2Cbus and CANbus.								
TEXT BOOKS								
1. Introduction to Embedded Systems-Shibu K.V, McGraw-Hill.								
REFERENCE BOOKS								

1. Embedded Systems-Raj Kamal, TMH.
2. Embedded System Design-Frank Vahid, Tony Givargis, John Wiley.
3. Embedded Systems –Lyla,Pearson,2013
4. An Embedded Software Primer-David E.Simon, Pearson Education.

WEB REFERENCES

1. <https://www.elprocus.com/embedded-system-design/>

E-TEXT BOOKS

1. <https://www.phindia.com/Books/BookDetail/9788120347304/embedded-system-design-chattopadhyay>

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc20_ee98/preview

OPEN ELECTIVES OFFERED BY EEE

OE1		OE2	
A6EE55	Electrical Wiring and Safety Measures	A6EE57	Neural Networks and Fuzzy Logic
A6EE56	Electrical Materials	A6EE58	Energy Storage Systems
OE3			
A6EE59	Solar Energy and Applications		
A6EE65	Instrumentation and Control		

ELECTRICAL WIRING AND SAFETY MEASURES

III YEAR II SEMESTER - (OPEN ELECTIVE – I)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE55	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES								
The course will enable the students to:								
1. To Study the wiring diagram of residential.								
2. To understand the Safety measures of Electrical wiring								
3. Apply the Different types of safety rules in Electrical Systems.								
4. Design the Different types of Residential Electrification Methods.								
5. Identifying Different types of Substations.								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. To Analyze the safety measures & state safety precautions.								
2. To Understand the Different Methods of Earthlings.								
3. Apply the Different types of safety rules in Electrical Systems.								
4. Design the Different types of Residential Electrification Methods.								
5. Identifying Different types of Substations.								
SYLLABUS								
UNIT-I	BASICS OF ELECTRICAL INSTALLATIONS						CLASSES: 12	
Electric Supply System, Three phase four wire distribution system, Protection of Electric Installation against over load, short circuit and Earth fault, General requirements of electrical installations, testing of installations, Types of loads, Systems of wiring, Service connections, Service Mains, Sub-Circuits, Location of Outlets, Location of Control Switches, Location of Main Board and Distribution board, Guide lines for Installation of Fittings, Load Assessment, Permissible voltage drops and sizes of wires, estimating and costing of Electric installations.								
UNIT-II	EARTHING						CLASSES: 08	
Introduction & importance, Factors affecting Earth Resistance, Methods of earthing Substation and Transmission tower earthing, Neutral and Earth wire, Transformer Neutral Earthing.								
UNIT-III	SAFETY & PREVENTION OF ACCIDENTS						CLASSES: 08	
Definition of terminology used in safety, I.E. Act & statutory regulations for safety of persons & equipments working with electrical installation. Dos & don'ts for substation operators as listed in IS. Meaning & causes of electrical accidents factors on which severity of shock depends.								
UNIT-IV	RESIDENTIAL BUILDING ELECTRIFICATION						CLASSES: 10	
General rules guidelines for wiring of Residential Installation and positioning of equipments. Principles of circuit design in lighting and power circuits. Procedures for designing the circuits and deciding the number of sub-circuits. Method of drawing single line diagram & wiring diagram.								

UNIT-V	RULES : SUBSTATION AND METERS	CLASSES: 12
Rule 28: Voltage level definitions. Rule 30: Service lines & apparatus on consumer premises. Rule 31: Cut-out on consumer's premises. Rule 46: Periodical inspection & testing of consumer's installation. Rule 47: Testing of consumer's installation. Rule 54: Declared voltage of supply to consumer. Rule 55: Declared frequency of supply to consumer. Rule 56: Sealing of meters & cut-outs. Rule 77: Clearances above ground of the lowest conductor. Rule 79: Clearances between conductors & trolley wires. Rule 87: Lines crossing or approaching each other. Rule 88: Guarding.		
TEXT BOOKS		
1. K.B. Raina, S.K.Bhattacharya Electrical Design; Estimating and costing New Age International (p) Limited, New Delhi Surjit Singh. 2. Electrical Estimating and costing Dhanpat Rai and company, New Delhi .J.B.Gupta. 3. A course in Electrical Installation, Estimating & costing, S.K.Kataria & sons, S.L. Uappal. 4. Electrical wiring Estimating and costing, Khanna Publication. ,A.K.Sawhney.		
REFERENCE BOOKS		
1. Electrical Machine Design Danpat Rai & co. 2. The Electricity Rule 2005 Universal Law Publishing Co. Pvt. Ltd. N. Alagapan, S. Ekambaram 3. Electrical Estimating and costing Tata Mc Graw Hill Publication, New Delhi ,Surjit Singh 4. TarlokSibgh Installation, Commissioning & Maintenance of Electrical Equipment, S.K.Kataria & Sons 5. B.V.S.Rao Operation & Maintenance of Electrical Machines Vol I & II, Media Promoters & Publisher Ltd. Mumbai		
WEB REFERENCES		
1. https://electrical-engineering-portal.com › Technical Articles 2. https://www.st-andrews.ac.uk/staff/policy/healthandsafety/publications/electricalsafety/ 3. https://www.cpwd.gov.in/Publication/Internal2013.pdf		
E-TEXT BOOKS		
1. https://books.google.co.in/books?isbn=0323170064 2. https://www.jove.com/science.../electrical-safety-precautions-and-basic-equipment		
MOOC COURSE		
1. https://nptel.ac.in/courses/103106071/5 2. https://nptel.ac.in/courses/108108099/28 3. https://nptel.ac.in/courses/124107001/		

ELECTRICAL MATERIALS

III Year II Semester - (OPEN ELECTIVE – I)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE56	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To analyze the various Conducting materials.
2. To explain the various Semi conducting materials.
3. To explain The Dielectrics and Insulators.
4. To explain the Different Magnetic Materials.
5. To describe the Optical properties of Fibers

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Analyze the various Conducting materials.
2. Explain the various Semi conducting materials.
3. Explain The Dielectrics and Insulators.
4. Explain the Different Magnetic Materials.
5. Describe the Optical properties of Fibers.

SYLLABUS

UNIT-I	CONDUCTORS	CLASSES: 10
Classification: High conductivity, high resistivity materials, fundamental requirements of high conductivity materials and high resistivity materials, mobility of electron in metals, commonly used high conducting materials, copper, aluminum, bronze brass, properties, characteristics, constantan, platinum, nichrome, properties, characteristics and applications, materials used for contacts.		
UNIT-II	SEMICONDUCTORS	CLASSES: 08
General concepts, energy bands, types of semiconductors, Fermi Dirac distribution, intrinsic Semi-conductors, extrinsic Semi-conductors, hall effect, drift, mobility, diffusion in Semiconductors, and their applications, superconductors.		
UNIT-III	DIELECTRICS AND INSULATORS	CLASSES: 12
Properties Of Gaseous, Liquid And Solid Dielectric, Dielectric As A Field Medium, Electric Conduction In Gaseous, Liquid And Solid Dielectric, Breakdown In Dielectric Materials, Mechanical And Electrical Properties Of Dielectric Materials, Effect Of Temperature On Dielectric Materials, Polarization, Loss Angle And Dielectric Loss, Petroleum Based Insulating Oils, Transformer Oil, Capacitor Oils, Properties, Solid Electrical Insulating Materials, Fibrous, Paper Boards, Yarns, Cloth Tapes, Sleeving Wood, Impregnation, Plastics, Filling And Bounding Materials, Fibrous, Film, Mica, Rubber, Mica Based Materials, Ceramic Materials, Classification Of Insulation (Solid) And Application In AC And DC Machines.		
UNIT-IV	MAGNETIC MATERIALS	CLASSES: 10

Soft and hard magnetic materials, diamagnetic, paramagnetic and ferromagnetic materials, electric steel, sheet steel, cold rolled grain oriented silicon steel, hot rolled grain oriented silicon steel, hot rolled silicon steel sheet, hysteresis loop, hysteresis loss, magnetic susceptibility, coercive force, curie temperature, magnetostriction.		
UNIT-V	OPTICAL PROPERTIES OF SOLIDS	CLASSES: 10
Photo emission, photo emission materials, electro luminous junction diode, photo emitters, photo transistor, photo resistors, injection lasers, optical properties of semiconductors, application of photo sensitive materials (CRT, Tube light, photo panels etc.).		
TEXT BOOKS		
1. "Electrical Engineering Materials", Dekker, PHI Pbs. 2. "Electrical Engineering Materials", Indulkar, S.Chand		
REFERENCE BOOKS		
1. "Electrical Engineering Materials", Tareev 2. "Electrical Engineering Materials", Yu. Koritsky. 3. "Electrical Engineering Materials", R.K.Rajput, Laxmi Pbs		
WEB REFERENCES		
1. https://physics.info/dielectrics/ 2. https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095631265 3. web.mit.edu/course/6/6.732/www/6.732-pt2.pdf		
E-TEXT BOOKS		
1. https://easyengineering.net/electrical-engineering-materials-by-dekker/ 2. https://www.oreilly.com/library/view/dielectric-materials-for/9781118619780/		
MOOC COURSE		
1. https://nptel.ac.in/courses/108108076/ 2. https://nptel.ac.in/courses/112104203/3 3. https://onlinecourses.nptel.ac.in/noc18_ee14/		

NEURAL NETWORKS AND FUZZY LOGIC

IV YEAR I SEMESTER - (OPEN ELECTIVE – II)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE57	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the fundamental theory and concepts of neural networks, neuro-modeling, several neural network paradigms and its applications.
2. To understand the concepts of fuzzy sets, knowledge representation using fuzzy rules, approximate reasoning, fuzzy inference systems, and fuzzy logic control and other machine intelligence applications of fuzzy logic.
3. To understand the basics of an evolutionary computing paradigm known as genetic algorithms and its application to engineering optimization problems.

COURSE OUTCOMES

The students should be able to

1. Comprehend the concepts of feed forward neural networks
2. Analyze the various feedback networks.
3. Understand the concept of fuzziness involved in various systems and fuzzy set theory.
4. Comprehend the fuzzy logic control and adaptive fuzzy logic and to design the fuzzy control using genetic algorithm.
5. Analyze the application of fuzzy logic control to real time systems.

SYLLABUS

UNIT-I	INTRODUCTION & ESSENTIALS TO NEURAL NETWORKS	CLASSES: 12
Introduction, Humans and Computers, Organization of the Brain, Biological Neuron, Biological and Artificial Neuron Models, Hodgkin-Huxley Neuron Model, Integrate-and-Fire Neuron Model, Spiking Neuron Model, Characteristics of ANN, McCullochPiUs Model, Historical Developments, Potential Applications of ANN. Artificial Neuron Model, Operations of Artificial Neuron, Types of Neuron Activation Function, ANN Architectures, Classification Taxonomy of ANN — Connectivity, Neural Dynamics (Activation and Synaptic), Learning Strategy (Supervised, Unsupervised, Reinforcement), Learning Rules, Types of Application		
UNIT-II	SINGLE & MULTI LAYER FEED FORWARD NEURAL NETWORKS	CLASSES: 10
Introduction, Perceptron Models: Discrete, Continuous and Multi-Category, Training. Algorithms: Discrete and Continuous Perceptron Networks, Perceptron Convergence theorem, Limitations of the Perceptron Model, Applications. Credit Assignment Problem, Generalized Delta Rule, and Derivation of Back-propagation (BP) Training, Summary of Back-propagation Algorithm, Kolmogorov Theorem, Learning Difficulties and Improvements.		
UNIT-III	ASSOCIATIVE MEMORIES-I	CLASSES: 08
Paradigms of Associative Memory, Pattern Mathematics, Hebbian Learning, General Concepts of Associative Memory (Associative Matrix, Association Rules, Hamming Distance, The Linear Associator, Matrix Memories, Content Addressable Memory).		
UNIT-IV	ASSOCIATIVE MEMORIES-II	CLASSES: 10

Bidirectional Associative Memory (BAM) Architecture, BAM Training Algorithms: Storage and Recall Algorithm, BAM Energy Function, Proof of BAM Stability Theorem. Architecture of Hopfield Network: Discrete and Continuous versions, Storage and Recall Algorithm, Stability Analysis, Capacity of the Hopfield Network Summary and Discussion of Instance/Memory Based Learning Algorithms, Applications		
UNIT-V	FUZZY LOGIC	CLASSES: 10
Classical & Fuzzy Sets: Introduction to classical sets – properties, Operations and relations; Fuzzy sets, Membership, Uncertainty, Operations, properties, fuzzy relations, cardinalities, membership functions. Fuzzy Logic System Components: Fuzzification, Membership value assignment, development of rule base and decision making system, De-fuzzification to crisp sets, De-fuzzification methods.		
TEXT BOOKS		
1. Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications, Rajasekharan and Pal, PHI. 2. Neural Networks and Fuzzy Logic, C. Naga Bhaskar, G. Vijay Kumar, BS Publications.		
REFERENCE BOOKS		
1. Artificial Neural Networks, B. Yegnanarayana, PHI. 2. Artificial Neural Networks, Zaruda, PHI. 3. Neural Networks and Fuzzy Logic System, Bail Kosko, PHI. 4. Fuzzy Logic and Neural Networks, M. Amirthavalli, Scitech Publications India Pvt. Ltd. 5. Neural Networks, James A Freeman and Davis Skapura, Pearson Education. 6. Neural networks by Satish Kumar, TIVIH, 2004 7. Neural Networks, Simon Hakens, Pearson Education. 8. Neural Engineering, C.Eliasmith and CH.Anderson, PHI.		
WEB REFERENCES		
1. users.monash.edu/~app/CSE5301/Lnts/LaD.pdf 2. https://engineering.purdue.edu/~tsoukala/rational.html 3. https://pdfs.semanticscholar.org/5e31/c55a00eb3945e3e483caa2e146a95c12f5aa.pdf		
E-TEXT BOOKS		
1. https://www.mheducation.co.in/computer.../neural-networks-fuzzy-systems/text-book 2. www.crectirupati.com/sites/default/files/lecture_notes/NNFL.pdf 3. www.vssut.ac.in/lecture_notes/lecture1423723637.pdf		
MOOCCOURSE		
1. https://nptel.ac.in/noc/individual_course.php?id=noc19-ge07 2. https://nptel.ac.in/courses/108104049/16 3. https://nptel.ac.in/courses/117105084/ 4. https://nptel.ac.in/syllabus/127105006/		

ENERGY STORAGE SYSTEMS

IV Year I Semester - (OPEN ELECTIVE – II)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE58	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. To enable the student to understand the need for energy storage, devices and technologies available and their applications. 2. To Analyze the characteristics of energy from various sources and need for storage 3. To Classify various types of energy storage and various devices used for the purpose 4. To Explain the Pumped hydro Storage System.								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: 1. Analyze the characteristics of energy from various sources and need for storage 2. Classify various types of energy storage and various devices used for the purpose 3. Explain the Pumped hydro Storage System. 4. Identify various real time applications.								
SYLLABUS								
UNIT-I	ELECTRICAL ENERGY STORAGE TECHNOLOGIES						CLASSES: 12	
Characteristics of electricity, Electricity and the roles of EES, High generation cost during peak-demand periods, Need for continuous and flexible supply, Long distance between generation and consumption, Congestion in power grids, Transmission by cable.								
UNIT-II	NEEDS FOR ELECTRICAL ENERGY STORAGE						CLASSES: 08	
Emerging needs for EES, More renewable energy, less fossil fuel, Smart Grid uses, The roles of electrical energy storage technologies, The roles from the viewpoint of a utility, The roles from the viewpoint of consumers, The roles from the viewpoint of generators of renewable energy.								
UNIT-III	FEATURES OF ENERGY STORAGE SYSTEMS						CLASSES: 08	
Classification of EES systems , Mechanical storage systems, Pumped hydro storage (PHS), Compressed air energy storage (CAES), Flywheel energy storage (FES),Electrochemical storage systems, Secondary batteries, Flow batteries, Chemical energy storage, Hydrogen (H2),Synthetic natural gas (SNG).								
UNIT-IV	TYPES OF ELECTRICAL ENERGY STORAGE SYSTEMS						CLASSES: 10	
Electrical storage systems, Double-layer capacitors (DLC), Superconducting magnetic energy storage (SMES), Thermal storage systems, Standards for EES, Technical comparison of EES technologies.								

UNIT-V	APPLICATIONS	CLASSES: 12
Present status of applications, Utility use (conventional power generation, grid operation & service) , Consumer use (uninterruptable power supply for large consumers), New trends in applications, Renewable energy generation, Smart Grid, Smart Micro grid, Smart House, Electric vehicles, Management and control hierarchy of storage systems, Internal configuration of battery storage systems, External connection of EES systems, Aggregating EES systems and distributed generation (Virtual Power Plant), Battery SCADA–aggregation of many dispersed batteries.		
TEXT BOOKS		
1. James M. Eyer, Joseph J. Iannucci and Garth P. Corey, “Energy Storage Benefits and Market Analysis”, Sandia National Laboratories, 2004. 2. The Electrical Energy Storage by IEC Market Strategy Board.		
REFERENCE BOOKS		
1. Jim Eyer, Garth Corey, Energy Storage for the Electricity Grid: Benefits and Market Potential Assessment Guide, Report, Sandia National Laboratories, Feb 2010.		
WEB REFERENCES		
1. https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118991978.hces212		
E-TEXT BOOKS		
1. https://www.pewtrusts.org/~media/.../energy_storage-backs_up_power_supply.pdf 2. https://energy.mit.edu/wp-content/uploads/2018/04/Energy-Storage-for-the-Grid.pdf 3. https://www.adb.org/sites/default/files/.../handbook-battery-energy-storage-system.pdf		
MOOC COURSE		
1. nptel.ac.in/courses/112105221/56 2. nptel.ac.in/courses/108108036/9 3. https://nptel.ac.in/courses/108102047/7		

SOLAR ENERGY AND APPLICATIONS

IV YEAR II SEMESTER - (OPEN ELECTIVE – III)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE59	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES								
The course will enable the students to:								
1. To Study about solar modules and PV system design and their applications 2. To Deal with grid connected PV systems 3. To Discuss about different energy storage systems								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. Understanding of principles and technologies for solar thermal energy collection, conversion and utilization 2. Understanding of solar heating systems, liquid based solar heating systems for buildings. 3. Identify, formulate and solve simple to complex problems of solar thermal energy conversion and storage. 4. Identify and understand solar thermal systems ‘components and their function. 5. Analyze hot water load and solar resource data and use this information to properly size a solar thermal system								
SYLLABUS								
UNIT-I	INTRODUCTION						CLASSES: 10	
Characteristics of sunlight – semiconductors and P-N junctions –behavior of solar cells – cell properties – PV cell interconnection								
UNIT-II	STAND ALONE PV SYSTEM						CLASSES: 10	
Solar modules – storage systems – power conditioning and regulation - MPPT- protection – standalone PV systems design – sizing								
UNIT-III	GRID CONNECTED PV SYSTEMS						CLASSES: 10	
PV systems in buildings – design issues for central power stations – safety – Economic aspect – Efficiency and performance - International PV programs								
UNIT-IV	ENERGY STORAGE SYSTEMS						CLASSES: 10	
Impact of intermittent generation – Battery energy storage – solar thermal energy storage – pumped hydroelectric energy storage								
UNIT-V	APPLICATIONS						CLASSES: 10	
Water pumping – battery chargers – solar car – direct-drive applications –Space – Telecommunications.								
TEXT BOOKS								

1. Solanki C.S., "Solar Photovoltaics: Fundamentals, Technologies And Applications", PHI Learning Pvt. Ltd., 2015.
2. Stuart R. Wenham, Martin A. Green, Muriel E. Watt and Richard Corkish, "Applied Photovoltaics", 2007, Earthscan, UK. Eduardo Lorenzo G. Araujo, "Solar electricity engineering of photovoltaic systems", Progensa, 1994

REFERENCE BOOKS

1. Frank S. Barnes & Jonah G. Levine, "Large Energy storage Systems Handbook", CRC Press, 2011.
2. McNeils, Frenkel, Desai, "Solar & Wind Energy Technologies", Wiley Eastern, 1990 S.P. Sukhatme, "Solar Energy", Tata McGraw Hill, 1987.

WEB REFERENCES

1. <https://www.loc.gov/rr/scitech/tracer-bullets/solar-updatetb.html>
2. <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803100516798>
3. <https://link.springer.com/journal/11949>

E-TEXT BOOKS

1. https://courses.edx.org/c4x/DelftX/ET.3034TU/asset/solar_energy_v1.1.pdf
2. bookstore.teri.res.in/books/9788179935736

MOOC COURSE

1. <https://nptel.ac.in/courses/112105051/>
2. <https://nptel.ac.in/courses/121106014/18>
3. <https://nptel.ac.in/courses/112105050/>
4. <https://nptel.ac.in/syllabus/112105051/>

INSTRUMENTATION AND CONTROL

IV Year II Semester - (OPEN ELECTIVE – III)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE65	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To introduce the basic principles of all measuring instruments
2. To deal with the measurement of voltage, current measurements
3. To understand students how different types of meters work and their construction
4. To understand the different ways of system representations such as Transfer function representation and state space representations and to assess the system dynamic response
5. To assess the system performance using time domain analysis and methods for improving it

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Identify the instruments suitable for typical measurements.
2. Apply the knowledge about transducers and instrument transformers to use them effectively.
3. Improve the system performance by selecting a suitable controller for a specific application
4. Apply various time domain techniques to assess the system performance

SYLLABUS

UNIT-I	INTRODUCTION TO MEASURING INSTRUMENTS	CLASSES: 10
Classification —deflecting, control and damping torques — Ammeters and Voltmeters — PMMC, moving iron type instruments — expression for the deflecting torque and control torque — Errors and compensations, extension of range using shunts and series resistance. Electrostatic Voltmeters-electrometer type and attracted disc type — Extension of range of E.S. Voltmeters.		
UNIT-II	POTENTIOMETERS & INSTRUMENT TRANSFORMERS	CLASSES: 08
Principle and operation of D.C. Crompton's potentiometer—standardization — Measurement of unknown resistance, current, voltage. A.C. Potentiometers: polar and coordinate types standardization — applications. CT and PT — Ratio and phase angle errors.		
UNIT-III	TRANSDUCERS & OSCILLOSCOPES	CLASSES: 12
Definition of transducers, Classification of transducers, Advantages of Electrical transducers, Characteristics and choice of transducers; Principle operation of LVDT and capacitor transducers; LVDT Applications, Strain gauge and its principle of operation, gauge factor, Thermistors, Thermo couples, Piezo electric transducers, photovoltaic, photo conductive cells, photo diodes. CRO: Cathode ray oscilloscope-Cathode ray tube-time base generator- horizontal and vertical amplifiers-CRO probes-applications of CRO - Measurement of phase and frequency-lissajous patterns.		
UNIT-IV	CLASSIFICATION OF CONTROL SYSTEMS	CLASSES: 10
Concepts of Control Systems- Open Loop and closed loop control systems and their differences- Different examples of control systems- Classification of control systems Feed-Back Characteristics, Effects of feedback. Mathematical models – Differential equations – Impulse Response and transfer functions. Block diagram algebra – Representation by Signal flow graph – Reduction using mason's gain formula.		

UNIT-V	TIME RESPONSE ANALYSIS	CLASSES: 10
Standard test signals – Time response of first order systems –Characteristic Equation of Feedback control systems, Transient response of second order systems – Time domain specifications .Effects of proportional derivative, proportional integral systems.		
TEXT BOOKS		
1. Electrical and Electronic Measurements and Instrumentation, R. K. Rajput, S. Chand & Company Ltd. 2. Electrical Measuring Instruments and Measurements, S. C. Bhargava, BS Publications. 3. I. J. Nagrath and M. Gopal", "Control Systems Engineering", New Age International (P) Limited, Publishers, 5th edition, 2009 4. "B. C. Kuo", "Automatic Control Systems", John wiley and sons, 8th edition, 2003.		
REFERENCE BOOKS		
1. Electrical & Electronic Measurement & Instruments, A.K.Sawhney Dhanpat Rai & Co. Publications. 2. Electrical and Electronic Measurements, G. K. Banerjee, PHI Learning Pvt. Ltd. 3. Electrical Measurements and Measuring Instruments, Golding and Widdis, Reem Publications. 4. N. K. Sinha", "Control Systems", New Age International (P) Limited Publishers, 3rdEdition, 1998. 5. "NISE", "Control Systems Engineering", John wiley, 6th Edition, 2011. 6. "Katsuhiko Ogata", "Modern Control Engineering", Prentice Hall of India Pvt. Ltd., 3rd edition, 1998.		
WEB REFERENCES		
1. home.mit.bme.hu/~virosztek/docs/mt.../Principles_of_electrical_measurement.pdf 2. https://www.mccdaq.com/handbook/chapt_4.aspx		
E-TEXT BOOKS		
1. www.vssut.ac.in/lecture_notes/lecture1423813026.pdf 2. www.vssut.ac.in/lecture_notes/lecture1423904331.pdf 3. www.ent.mrt.ac.lk/~rohan/teaching/EN5001/Reading/DORFCH1.pdf		
MOOC COURSE		
1. https://nptel.ac.in/syllabus/108106070/ 2. https://nptel.ac.in/courses/108105064/ 3. https://nptel.ac.in/courses/108101037/		

OPEN ELECTIVES OFFERED BY INFORMATION TECHNOLOGY

OE-1		OE-2	
A6IT28	Adhoc & Sensor Networks	A6IT30	Security Testing
A6IT35	Object Oriented Programming	A6IT36	Human Computer Interaction
OE-3			
A6IT37	Introduction to Computer Networks		
A6IT31	High Performance Computing		

AD-HOC & SENSOR NETWORKS

III Year II Semester - (OPEN ELECTIVE – I)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT28	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the concepts of sensor networks
2. To understand the MAC and transport protocols for ad hoc networks
3. To understand the security of sensor networks
4. To understand the applications of adhoc and sensor networks

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Know the principles and characteristics of wireless sensor networks.
2. Use proactive routing protocols function and their implications on data transmission delay and bandwidth consumption.
3. Find issues in MAC Layer Protocols, transport layer protocols in MANETS and WSNs.
4. Find security issues and familiar with security key management mechanisms MANETS and WSNs.
5. Practice network simulations on various simulators.

SYLLABUS

UNIT-I	INTRODUCTIONS TO AD HOC NETWORKS	CLASSES: 10
Introduction to Ad Hoc Networks - Characteristics of MANETs, Applications of MANETs and Challenges of MANETs. Routing in MANETs - Criteria for classification, Taxonomy of MANET routing algorithms, Topology- based routing algorithms- Proactive : DSDV; Reactive : DSR, AODV; Hybrid: ZRP; Position-based routing algorithms- Location Services -DREAM, Quorum-based; Forwarding Strategies : Greedy Packet, Restricted Directional Flooding-DREAM, LAR.		
UNIT-II	BROADCASTING TECHNIQUES	CLASSES: 10
Data Transmission - Broadcast Storm Problem, Rebroadcasting Schemes -Simple-flooding, Probability-based Methods, Area-based Methods, Neighbor Knowledge-based: SBA, Multipoint Relaying, AHBP. Multicasting : Tree-based : AMRIS, MAODV; Mesh-based : ODMRP, CAMP; Hybrid : AMRoute, MCEDAR.		
UNIT-III	GEOCASTING AND PROTOCOLS	CLASSES: 10
Geocasting : Data-transmission Oriented-LBM; Route Creation Oriented-GeoTORA, MGR. TCP over Ad Hoc TCP protocol overview, TCP and MANETs, Solutions for TCP over Ad hoc		
UNIT-IV	BASICS OF WIRELESS SENSOR NETWORKS	CLASSES: 10
Basics of Wireless, Sensors and Lower Layer Issues : Applications, Classification of sensornetworks, Architecture of sensor network, Physical layer, MAC layer, Link layer, Routing Layer.		
UNIT-V	ISSUES OF WIRELESS SENSOR NETWORKS	CLASSES: 10

Upper Layer Issues of WSN: Transport layer, High-level application layer support, Adapting to the inherent dynamic nature of WSNs, Sensor Networks and mobile robots.

TEXT BOOKS

1. Ad Hoc and Sensor Networks – Theory and Applications, Carlos Corderio Dharma P. Aggarwal, World Scientific Publications, March 2006, ISBN – 981–256–681–3.
2. Wireless Sensor Networks: An Information Processing Approach, Feng Zhao, Leonidas Guibas, Elsevier Science, ISBN – 978-1-55860-914-3 (Morgan Kaufman).

REFERENCE BOOKS

1. Advanced Selenium Web Accessibility Testing, Software Automation Testing Secrets Revealed, Narayanan Palani, 2019
2. Automated Software Testing A Complete Guide - 2020 Edition-by Gerardus Blokdyk (Author).
3. Software Automation Testing Secrets Revealed Part 2 Selenium Webdriver
4. Mastering Web Automated Testing -Use of Selenium with Practical Answers to Testing Interview Questions Narayanan Palani, 2016.
5. J F DiMarzio, "Beginning Android Programming with Android Studio", 4th Edition, Wiley India Pvt Ltd, 2016. ISBN-13: 978-8126565580

WEB LINKS

1. <https://www.guru99.com/selenium-tutorial.html>.
2. <https://www.browserstack.com/guide/automation-testing-tutorial>

OBJECT ORIENTED PROGRAMMING

III Year II Semester - (OPEN ELECTIVE – I)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT35	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand the basic object oriented programming concepts and apply them in problem solving.
2. Illustrate inheritance and polymorphism concepts for reusing the program.
3. Demonstrate on the exception handling mechanism
4. Use multithreading concepts to develop inter process communication
5. Understand the basics of java collection framework

COURSE OUTCOMES

At the end of the course, student will be able to

1. Understand the basic programming concepts and apply them in problem solving
2. Implement object-oriented principles, such as encapsulation, inheritance, polymorphism, and abstraction.
3. Demonstrate the user defined exceptions by exception handling keywords (try, catch, throw, throws and finally).
4. Develop multithreaded applications with synchronization.
5. Build real world applications using Collection framework

SYLLABUS

UNIT-I	JAVA BASICS	CLASSES: 12
JAVA BASICS: Review of Object oriented concepts, History of Java, Java buzzwords, JVM architecture, Data types, Variables, arrays, operators, control statements, type conversion and casting, constructors, methods, Static block, Static Method ,String Handling, Scanner class		
UNIT-II	INHERITANCE, POLYMORPHISM, PACKAGES AND INTERFACES	CLASSES: 11
INHERITANCE AND POLYMORPHISM: Basic concepts, Types of inheritance, Member access rules, Usage of this and Super key word, Polymorphism, Abstract classes, Encapsulation, getter and setter methods, Dynamic method dispatch, Usage of final keyword. PACKAGES AND INTERFACES: Defining package, Access protection, importing packages, Defining and Implementing interfaces, and Extending interfaces		
UNIT-III	EXCEPTION HANDLING AND FILES	CLASSES: 10
EXCEPTION HANDLING: Exception types, Usage of Try, Catch, Throw, Throws and Finally keywords, Built-in Exceptions, Creating own Exception classes. I/O STREAMS AND FILES: Concepts of streams, Stream classes- Byte and Character stream, Reading console Input and Writing Console output, IO/Serialization, File Handling.		
UNIT-IV	MULTITHREADING	CLASSES: 10
MULTI THREADING: Concepts of Thread, Thread life cycle, creating threads using Thread class and Runnable interface, Synchronization, Thread priorities, Inter Thread communication, Concurrency.		

UNIT-V	COLLECTION FRAMEWORK	CLASSES: 10
COLLECTION FRAMEWORK: Introduction to Java Collections, Overview of Java Collection frame work, Generics, Commonly used Collection classes- Array List, Vector, Hash table, Stack, Enumeration, Iterator.		
TEXT BOOKS		
1. Herbert Schildt and Dale Skrien, <i>Java Fundamentals – A comprehensive Introduction</i>, McGraw Hill, 1st Edition, 2013. 2. Herbert Schildt, — <i>Java the complete reference</i>, McGraw Hill, Osborne, 7th Edition, 2011. 3. T.Budd, — <i>Understanding Object- Oriented Programming with Javall</i>, Pearson Education, Updated Edition (New Java 2 Coverage), 1999.		
REFERENCE BOOKS		
1. P.J.Dietel and H.M.Dietel, — <i>Java How to program</i>, Prentice Hall, 6th Edition, 2005. 2. P.Radha Krishna, — <i>Object Oriented programming through Javall</i>, CRC Press, 1st Edition, 2007. 3. S.Malhotra and S.Choudhary, — <i>Programming in Javall</i>, Oxford University Press, 2nd Edition, 2014.		

SECURITY TESTING

IV Year I Semester - (OPEN ELECTIVE – II)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT30	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
COURSE OUTCOMES The course will enable the students to: 1. Understand the basic principles and techniques of how attackers can enter computer systems. 2. Able to analyse the data breaches and audits of information technology security. 3. Able to understand the nitti-gritties of pen tests for educational purposes. 4. Evaluate the societal role of hacking from a social, ethical and economic standpoint								
COURSE OUTCOMES At the end of the course, student will be able to: 1. Explain the basic principles and techniques of how attackers can enter computer systems. 2. Perform analyses of data breaches and audits of information technology security. 3. Evaluate the strengths and weaknesses of various information technology solutions regarding data security. 4. Present and perform demonstrations of pen tests for educational purposes. 5. evaluate the societal role of hacking from a social, ethical and economic standpoint								
SYLLABUS								
UNIT-I	INTRODUCTION TO OWASP						CLASSES: 10	
OWASP: Introduction to web applications security, threats and OWASP principles, introduction to secure design, web server: introduction a secure setup of apache, firewalling a server Browser: general concepts, functionalities, browsers war, configuration (HTTP-cookies, contents, scripting etc. attack to browsers, and users tracking/profiling (third party cookies, super cookies, XSS, CSFR, Command Injection), browser security (add-ons, plugins, same-origin policy etc.) & secure browsing								
UNIT-II	OWASP Privacy preserving						CLASSES: 08	
OWASP Privacy preserving: attacks to privacy, (spyware & backdoors, browser, email etc.) Tracking techniques: (HTTP cookies, third party cookies, browser fingerprinting, CSP) Advanced browser configuration, anonymity and onion routing (Tor). Internet E-mail: Architecture and infrastructure, functions, agents and standards, MIME & PGP, phishing, spamming & spoofing, DKIM, SPF, introduction to email forensics.								
UNIT-III	Introduction to Ethical Hacking Terminology						CLASSES: 12	
Introduction to Ethical Hacking Terminology-Five stages of hacking-Vulnerability Research-Legal implication of hacking Impact of hacking. Foot printing & Social engineering.								
UNIT-IV	Information gathering methodologies						CLASSES: 10	
Information gathering methodologies - Competitive Intelligence- DNS Enumerations- Social Engineering attacks. Scanning & Enumeration Port Scanning-Network Scanning- Vulnerability Scanning- NMAP scanning tool- OS Finger printing Enumeration. System Hacking Password.								

UNIT-V	Sniffers & SQL Injection	CLASSES: 10
Sniffers & SQL Injection Active and passive sniffing - ARP Poisoning- Session Hijacking- DNS Spoofing- Conduct SQL Injection attack – Counter measures. Cracking techniques- Key loggers- Escalating privileges- Hiding Files-Steganography technologies- Countermeasures.		
TEXT BOOKS		
1. Kimberly Graves, “CEH: Official Certified Ethical Hacker Review Guide”, Wiley Publishing Inc., 2007.ISBN: 978-0-7821-4437-6.		
REFERENCE BOOKS		
1. Shakeel Ali & Tedi Heriyanto, “Backtrack -4: Assuring security by penetration testing”, PACKT Publishing., 2011. ISBN: 978-1-849513-94-4.		

HUMAN COMPUTER INTERACTION

IV Year I Semester - (OPEN ELECTIVE – II)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT36	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To learn the foundations of Human Computer Interaction
2. To become familiar with the design technologies for individuals and persons with disabilities.
3. To be aware of mobile HCI.
4. To learn the guidelines for user interface
5. To get an understanding of web interface design

COURSE OUTCOMES

Upon successful completion of the course, the student is able to learn

1. Design effective dialog for HCI
2. Design effective HCI for individuals and persons with disabilities.
3. Assess the importance of user feedback.
4. Explain the HCI implications for designing multimedia/ecommerce/e-learning Web sites.
5. Develop meaningful user interface

SYLLABUS

UNIT-I	FOUNDATION OF HCI	CLASSES: 12
The Human: I/O channels, Memory, Reasoning and problem solving; The computer: Devices, Memory, processing and networks; Interaction: Models, frameworks, Ergonomics, styles, elements, interactivity- Paradigms.		
UNIT-II	DESIGN & SOFTWARE PROCESS	CLASSES: 12
Interactive Design: Basics, process, scenarios, navigation, screen design, Iteration and prototyping. HCI in software process: Software life cycle, usability engineering, Prototyping in practice, design rationale. Design rules: principles, standards, guidelines, rules. Evaluation Techniques, Universal Design.		
UNIT-III	MODELS AND THEORIES	CLASSES: 10
HCI Models: Cognitive models: Socio-Organizational issues and stakeholder requirements – Communication and collaboration models-Hypertext, Multimedia and WWW.		
UNIT-IV	MOBILE HCI	CLASSES: 12
Mobile Ecosystem: Platforms, Application frameworks- Types of Mobile Applications: Widgets, Applications, Games- Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools. - Case Studies.		
UNIT-V	WEB INTERFACE DESIGN	CLASSES: 09
Designing Web Interfaces: Drag & Drop, Direct Selection, Contextual Tools, Overlays, Inlays and Virtual Pages, Process Flow - Case Studies		
TEXT BOOKS		

1. Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, —Human Computer InteractionII, 3rd Edition, Pearson Education, 2004 (UNIT I, II & III)
2. Brian Fling Mobile Design and DevelopmentII, First Edition, O'Reilly Media Inc., 2009 (UNIT – IV)

REFERENCE BOOKS

1. Bill Scott and Theresa Neil, —Designing Web InterfacesII, First Edition, O'Reilly, 2009. (UNIT-V)
2. Interaction Design PRECE, ROGERS, SHARPS. Wiley Dreamtech,
3. User Interface Design, Soren Lauesen, Pearson Education.

INTRODUCTION TO COMPUTER NETWORKS

IV Year II Semester - (OPEN ELECTIVE – III)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT37	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: - Understand and apply different network commands. - Analyse different networking functions and features for implementing optimal solutions Apply different networking concepts for implementing network solution - Implement different network protocols.								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: - Apply the basics of Physical layer in real time applications - Apply data link layer concepts, design issues, and protocols - Apply Network layer routing protocols and IP addressing - Implement the functions of Application layer and Presentation layer paradigms and Protocols								
SYLLABUS								
UNIT-I	INTRODUCTION TO COMPUTER NETWORKS						CLASSES: 10	
Introduction: Network Types, LAN, MAN, WAN, Network Topologies Reference models- The OSI Reference Model- the TCP/IP Reference Model - A Comparison of the OSI and TCP/IP Reference Models, OSI Vs. TCP/IP, Lack of OSI models success Physical Layer –Introduction to Guided Media- Twisted-pair cable, Coaxial cable and Fiber optic cable and unguided media: Wireless-Radio waves, microwaves, infrared. Modern Networks – History of Internet, Wi-Fi Networks, Blue-Tooth and Edge Computing								
UNIT-II	DATA LINK LAYER						CLASSES: 10	
Data link layer: Design issues, Framing: fixed size framing, variable size framing, flow control, error control, error detection and correction codes, CRC, Checksum: idea, one's complement internet checksum, services provided to Network Layer Elementary Data Link Layer protocols: simplex protocol, Simplex stop and wait, Simplex protocol for Noisy Channel. Sliding window protocol: One bit, Go back N, Selective repeat-Stop and wait protocol, Data link layer in HDLC: configuration and transfer modes, frames, control field, point to point protocol (PPP): framing transition phase, multiplexing, multi link PPP.								
UNIT-III	MULTIPLE ACCESS TECHNIQUES USED						CLASSES: 10	
Media Access Control: Random Access: ALOHA, Carrier sense multiple access (CSMA), CSMA with Collision Detection, CSMA with Collision Avoidance, Controlled Access: Reservation, Polling, Token Passing, Channelization: frequency division multiple Access (FDMA), time division multiple access (TDMA), code division multiple access (CDMA). Wired LANs: Ethernet, Ethernet Protocol, Standard Ethernet, Fast Ethernet (100 Mbps), Gigabit Ethernet, 10 Gigabit Ethernet.								

UNIT-IV	NETWORK LAYER DESIGN ISSUES	CLASSES: 10
The Network Layer Design Issues – Store and Forward Packet Switching-Services Provided to the Transport layer- Implementation of Connectionless Service-Implementation of Connection Oriented Service- Comparison of Virtual Circuit and Datagram Networks, Routing Algorithms-The Optimality principle-Shortest path, Flooding, Distance vector, Link state, Hierarchical, Congestion Control algorithms-General principles of congestion control, Congestion prevention policies, Approaches to Congestion Control-Traffic Aware Routing- Admission Control-Traffic Throttling-Load Shedding. Traffic Control Algorithm: Leaky bucket & Token bucket. Internet Working: How networks differ- How networks can be connected- Tunneling, internetwork routing-, Fragmentation, network layer in the internet – IP protocols-IP Version 4 protocol-IPV4 Header Format, IP addresses, Class full Addressing, CIDR, NAT-, Subnets-IP Version 6-The main IPV6 header, Transition from IPV4 to IPV6, Comparison of IPV4 & IPV6- Internet control protocols- ICMP-ARPDHCP		
UNIT-V	TRANSPORT LAYER	CLASSES: 10
The Transport Layer: Transport layer protocols: Introduction-services- port number-User data gram protocol-User datagram-UDP services-UDP applications-Transmission control protocol: TCP services- TCP features-Segment- A TCP connection- windows in TCP- flow control-Error control, Congestion control in TCP. Application Layer: World Wide Web: HTTP, Electronic mail-Architecture- web based mail- email security-TELENET-local versus remote Logging-Domain Name System: Name Space, DNS in Internet, Resolution-Caching- Resource Records- DNS messages- Registrars-security of DNS Name Servers, SNMP		
TEXT BOOKS		
1. Computer Networks — Andrew S Tanenbaum, Fifth Edition. Pearson Education/PHI 2. Data Communications and Networks – Behrouz A. Forouzan, Fifth Edition TMH.		
REFERENCE BOOKS		
1. Data Communications and Networks- Achut S Godbole, Atul Kahate 2. Computer Networks, Mayank Dave, CENGAGE		

HIGH PERFORMANCE COMPUTING

IV Year II Semester - (OPEN ELECTIVE – III)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT31	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To learn about Modern Processors and concepts
2. To understand the concepts of Optimizations
3. To learn about Parallel Computers and Programming
4. To study about Memory Parallel Programming using Open MP and MPI

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Transform algorithms in the computational area to efficient programming code for modern computer architectures
2. Write, organise and handle programs for scientific computations.
3. Use tools for performance optimisation and debugging and Analyse code with respect to performance.
4. Suggest and implement performance improvements
5. Write, organise and handle programs for MIP

SYLLABUS

UNIT-I	MODERN PROCESSORS	CLASSES: 09
Stored-program computer architecture, General-purpose cache-based microprocessor architecture, Memory hierarchies, Multicore processors, Multi-threaded processors, Vector processors.		
UNIT-II	BASIC OPTIMIZATION TECHNIQUES FOR SERIAL CODE	CLASSES: 09
Scalar profiling, Common sense optimizations, Simple measures, large impact, The role of compilers, C++ optimizations		
UNIT-III	PARALLEL COMPUTERS	CLASSES: 09
Taxonomy of parallel computing paradigms, Shared-memory computers, Distributed-memory computers, Hierarchical (hybrid) systems, Networks, Need for parallelization, Parallelism, Parallel scalability		
UNIT-IV	SHARED MEMORY PARALLEL PROGRAMMING WITH OPENMP	CLASSES: 08
introduction to Open MP, Parallel execution, Data scoping, Open MP work sharing for loops, Synchronization, Reductions, Loop scheduling, Tasking, Miscellaneous, Open MP-parallel Jacobi algorithm, Open MP: Wave front parallelization.		
UNIT-V	DISTRIBUTED-MEMORY PARALLEL PROGRAMMING WITH MPI	CLASSES: 09
Message passing, introduction to MPI, A simple example, Messages and point-to-point communication, Collective communication, Nonblocking point-to-point communication, Virtual topologies, MPI parallelization of a Jacobi solver, MPI implementation, Performance properties		

TEXT BOOKS

1. Georg Hager, Gerhard Welle in, "Introduction to High Performance Computing for Scientists and Engineers", Chapman & Hall/CRC Computational Science series, 2011.

REFERENCE BOOKS

1. Charles Severance, Kevin Dowd, "High Performance Computing", O'Reilly Media, 2nd Edition, 1998.
2. Kai Hwang, Faye Alaye Briggs, "Computer Architecture and Parallel Processing", McGraw Hill, 1984

OPEN ELECTIVES OFFERED BY MECH

OE1		OE2	
A6ME55	Basics of Mechanical Engineering	A6ME65	Fundamentals of Operations Research
A6ME60	Fundamentals of Engineering Materials	A6ME70	Introduction to Industrial Engineering
OE3			
A6ME75	Renewable Energy Sources		
A6ME80	Elements of Electric and Hybrid Vehicles		

BASICS OF MECHANICAL ENGINEERING

III Year II Semester - (OPEN ELECTIVE – I)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME55	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the basic concepts of mechanics of rigid and deformable bodies.
2. To know the fundamentals of metallurgy and material science.
3. To be able to conceptualize the principles of thermodynamics.
4. To know the basic elements of fabrication processes and equipment.
5. To know the basic concepts of CAD/CAM.

COURSE OUTCOMES

At the end of the course, the student should be able to:

1. Demonstrate basic concepts of mechanics of rigid and deformable bodies.
2. Explain fundamentals of metallurgy and material science.
3. Discuss basic concepts and principles of thermodynamics.
4. Demonstrate the elements of fabrication processes and equipment.
5. Explain basic concepts of CAD/CAM.

UNIT - I	MECHANICS	CLASSES: 10
MECHANICS OF RIGID BODIES: Force, Force system, Classification of force systems, triangle law, parallelogram law–derivation and problems, polygon law, resolution of forces in coplanar concurrent force system–problems; Definition of friction, laws of friction, coefficient of friction, types of friction, angle of friction, angle of repose, Definition of centroid and center of gravity, problems on centroid of plane areas; Definition of area moment of inertia, parallel axis theorem, perpendicular axis theorem, simple problems on area moment of inertia. MECHANICS OF DEFORMABLE BODIES: Stress, Strain, Types of stresses and strains, Poisson's ratio, stress-strain curve of mild steel, Hooke's Law, factor of safety, mechanical properties of materials, bulk modulus, shear modulus, bars of uniform and varying sections-simple problems.		
UNIT - II	METALLURGY & MATERIAL SCIENCE	CLASSES: 08
Bonds in Solids, crystallization of metals, Classification of steels, Necessity of alloying, types of solid solutions, lever rule, phase rule, cooling curve of pure iron, Iron-Iron carbide equilibrium diagram. Heat treatment–Annealing, quenching, normalizing, Hardening, tempering; Ceramic materials-classification, properties and applications.		
UNIT - III	THERMO DYNAMICS	CLASSES: 12
System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic view points, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Zeroth Law of Thermodynamics – Concept of Temperature – Principles of Thermometry. First law of Thermodynamics – Corollaries, First law applied to a Closed System, applied to a flow system, Steady Flow Energy Equation and Limitations of the First Law, Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements, Carnot cycle and its specialties.		

UNIT - IV	MANUFACTURING	CLASSES: 10
Casting-Types of casting, advantage of casting and its applications, defects in castings; Patterns - Pattern making, Types, Materials used for patterns, pattern allowances; Welding-classification, Heat affected zone in welding, welding defects-causes and remedies. Basic extrusion process and its characteristics, Forging operations and principles; Basic machining operations- drilling, milling, turning, step turning, threading, grinding and shaping; Limits, fits and tolerances- Unilateral and bilateral tolerance system, hole and shaft basis system.		
UNIT - V	CAD/CAM	CLASSES: 10
Fundamentals of CAD/CAM – Benefits of CAD, Computer configuration for CAD applications, Computer peripherals for CAD; Wire frame modeling: Definition, advantages, disadvantages, wire frame entities– analytic entities and synthetic entities. Surface modeling: Definition, advantages, disadvantages, surface entities–analytic entities and synthetic entities; Solid Modeling: Definition, constructive solid geometry, advantages, modeling entities.		
TEXT BOOKS		
1. K. Vijay Kumar Reddy and J. Suresh Kumar, Singer's Engineering Mechanics, B.S Publications 2. Avner, Introduction to physical metallurgy, Tata McGraw-Hill Edition. 3. P.K Nag, Thermodynamics, McGraw Hill Education. 4. R.K Jain, Production Technology, Khanna Publishers. 5. Ibrahim Zeid & R.Sivasubramanian, CAD/CAM Theory & Practice, McGraw Hill Education.		
REFERENCE BOOKS		
1. R.K. Bansal, Engineering Mechanics, Laxmi Publications 2. O.P. Khanna, Material science and metallurgy, Dhanpat Ray Publications 3. Pakirappa, V.Naveen Kumar, V.Naresh, Engineering Thermodynamics, Durga Publishing House 4. P.C Sharma, Production Technology, S.CHAND. 5. P. Radha Krishnan, S. Subramanyan and V. Raju, CAD/CAM, New Age International Publishers. 6. Sadhu Singh, Strength of Materials, Khanna Publishers.		

FUNDAMENTALS OF ENGINEERING MATERIALS

III B. TECH.II SEM - (OPEN ELECTIVE – I)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME60	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. To understand basic structure and crystal arrangement of materials 2. To understand the phase diagrams 3. To understand the advantages of heat treatment processes 4. To understand the application of composite materials.								
COURSE OUTCOMES At the end of the course, the student should be able to: 1. Explain basic concepts of crystal structure such as unit cells, crystal systems of metals etc. 2. Demonstrate the concept of alloying and formation of different types of phases in alloys. 3. Differentiate ferrous and non-ferrous alloys. 4. Explain various heat treatment processes. 5. Classify and explain polymers, ceramics and composites.								
UNIT-I	CRYSTAL STRUCTURE					CLASSES: 10		
Unit cells, crystal systems of metals, Imperfection in solids: Point, line, interfacial and volume defects; dislocation strengthening mechanisms and slip systems, determination of grain size, effect of grain size on the properties of alloys.								
UNIT-II	ALLOYS & PHASE DIAGRAMS					CLASSES: 09		
Alloys- substitutional and interstitial solid solutions. Phase diagrams: Interpretation of binary phase diagrams and microstructure development; eutectic, peritectic, peritectoid and monotectic reactions. Iron Iron-carbide phase diagram and microstructure of ledeburite, austenite, ferrite and cementite.								
UNIT-III	FERROUS AND NON FERROUS ALLOYS					CLASSES: 10		
Alloying of steel, properties of stainless steel and tool steels, maraging steels; cast irons - grey, white, malleable and spheroidal cast irons. Copper and copper alloys- brass, bronze and cupro-nickel; Aluminum and Aluminum alloys.								
UNIT-IV	HEAT TREATMENT OF STEEL					CLASSES: 08		
Annealing, tempering, normalizing and spheroidising, austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening.								
UNIT-V	POLYMERS, CERAMICS AND COMPOSITES					CLASSES: 12		

Classification, properties and applications of polymers, ceramics, composites and nano materials.

TEXT BOOKS

1. V. Raghavan, "Material Science and Engineering", Prentice Hall of India Private Limited, 1999.
2. U. C. Jindal, "Engineering Materials and Metallurgy", Pearson, 2011.
3. Sidney H. Avener, Introduction to Physical Metallurgy, 2nd edition, Tata McGraw hill education (P) Ltd, New Delhi, India, 2007.

REFERENCE BOOKS

1. W. D. Callister, "Materials Science and Engineering-An Introduction", 6th Edition, Wiley India, 2006.
2. Kenneth G. Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 4th Indian Reprint, 2002.
3. V. D. Kodgire, Material Science and Metallurgy for engineers, 1st Edition, Everest, Pune, India, 2006.

FUNDAMENTALS OF OPERATIONS RESEARCH

IV B. TECH.I SEM - (OPEN ELECTIVE-II)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME65	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. To introduce the basic concepts of Operational Research 2. Understand about the types of models and linear programming, transportation and assignment problem 3. To understand the sequencing, game theory and inventory to the students.								
COURSE OUTCOMES At the end of the course, the student should be able to: 1. Describe types of models and solve linear programming problem. 2. Solve transportation and assignment problems. 3. Analyze sequencing and replacement models and apply them for optimization. 4. Apply gaming theory for optimal decision making. 5. Analyze inventory models to optimize the cost								
UNIT-I	INTRODUCTION & ALLOCATION					CLASSES: 10		
Development – Definition– Characteristics and Phases – Types of models – Operations Research models – applications. ALLOCATION: Linear Programming Problem – Formulation – Graphical solution – Simplex method –Artificial variables techniques: Two–phase method, Big-M method; Duality Principle.								
UNIT-II	TRANSPORTATION & ASSIGNMENT PROBLEMS					CLASSES: 10		
TRANSPORTATION PROBLEM: Formulation – Optimal solution, unbalanced transportation problem – Degeneracy. ASSIGNMENT PROBLEM: Formulation – Optimal solution – Variants of Assignment Problem; Traveling Salesman problem.								
UNIT-III	SEQUENCING & REPLACEMENT					CLASSES: 12		
SEQUENCING: Introduction – Flow –Shop sequencing – n jobs through two machines – n jobs through three machines. REPLACEMENT: Introduction – Replacement of items that deteriorate with time – when money value is not counted and counted – Replacement of items that fail completely.								
UNIT-IV	THEORY OF GAMES					CLASSES: 10		
THEORY OF GAMES: Introduction –Terminology– Solution of games with saddle points and without saddle points– 2×2 games – $m \times 2$ & $2 \times n$ games – graphical method – $m \times n$ games - dominance principle.								
UNIT-V	INVENTORY					CLASSES: 08		
INVENTORY: Introduction – Single item, Deterministic models – Types – Purchase inventory models with one price break and multiple price breaks –Stochastic models – demand discrete variable or continuous variable – Single Period model with no setup cost.								

TEXT BOOKS

1. J.K. Sharma, Operation Research: Theory and Applications, 5th Edition, MacMillian India Ltd., 2012.
2. A.C.S. Kumar, Operations Research: Quantitative Analysis for Business Decisions, 1st Edition, Yes Dee Publications, 2015.

REFERENCE BOOKS

1. A.M.Natarajan, P.Balasubramaniam, A. Tamilarasi, Operations Research, 2nd Edition, Pearson Education India, 2014.
2. Harvey M. Wagner, Principles of Operations Research, 2nd Edition, Prentice Hall India Learning Pvt. Ltd., 1980.
3. Frederick S. Hillier, Gerald J. Lieberman, Bodhibrata Nag, Preetam Basu, Introduction to Operations Research, 11th Edition, Tata McGraw Hill India, 2021.

INTRODUCTION TO INDUSTRIAL ENGINEERING

IV B. TECH.I SEM - (OPEN ELECTIVE-II)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME70	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES								
The course will enable the students to:								
1. To understand the Industrial engineering principles, tools and concepts. 2. To know concepts like define, design, build, operate and improve production processes that convert resources to high quality products or services effectively, efficiently and safely. 3. To know the various types of plant layouts and their maintenance.								
COURSE OUTCOMES								
At the end of the course, the student should be able to:								
1. Describe the tools and concepts of industrial engineering principles. 2. Differentiate various types of plant layouts and maintenance. 3. Apply the principles of method study and time study to analyze and optimize work processes for increased effectiveness. 4. Apply Statistical Methods for Quality Control and explain Total Quality Management principles. 5. Use the techniques like human resource management, industrial relations, wage incentive plans, Value engineering for the effective enterprise resource planning.								
UNIT-I	INTRODUCTION					CLASSES: 10		
Definition of industrial engineering (I.E), development, applications, role of an industrial engineer, differences between production management and industrial engineering, quantitative tools of IE and productivity measurement. concepts of management, importance, functions of management, scientific management, Taylor's principles, theory X and theory Y, Fayol's principles of management.								
UNIT-II	PLANT LAYOUT					CLASSES: 12		
Factors governing plant location, types of production layouts, advantages and disadvantages of process layout and product layout, applications, quantitative techniques for optimal design of layouts, plant maintenance, preventive and breakdown maintenance.								
UNIT-III	WORK STUDY					CLASSES: 08		
Importance, types of production, applications, work study, method study and time study, work sampling, PMTS. Micro-motion study, rating techniques, MTM, work factor system, principles of Ergonomics, flow process charts, string diagrams and Therbligs.								
UNIT-IV	STATISTICAL QUALITY CONTROL					CLASSES: 10		
Quality control, Queuing assurance and its importance, SQC, attribute sampling inspection with single and double sampling, Control charts – X and R – charts X and S charts and their applications, numerical examples. TOTAL QUALITY MANAGEMENT: zero defect concept, quality circles, implementation, applications, ISO quality systems. Six sigma – definition, basic concepts.								
Concept of human resource management, personnel management and industrial relations, functions of personnel management, Job-evaluation, its importance and types, merit rating, quantitative methods, wage								

incentive plans, types.

VALUE ANALYSIS: Value engineering, implementation procedure, enterprise resource planning and supply chain management.

TEXT BOOKS

1. O.P Khanna, Industrial Engineering and management, 2nd Edition, Dhanpat Rai publications, 2018.
2. Martand Telsang, Industrial Engineering and Production Management, 3rd Edition, S.Chand & Company Ltd. New Delhi, 2013.

REFERENCE BOOKS

1. Bhattacharya D. K., Industrial Management, 2nd Edition, Vikas publishers, 2011.
2. J.G Monks, Operations Management, 3rd Edition, McGraw-Hill Publishers, 2007.
3. T.R. Banga, S.C.Sharma, N. K. Agarwal, Industrial Engineering and Management Science, 12th Edition, Khanna Publishers, 2008.

RENEWABLE ENERGY SOURCES

IV B. TECH.II SEM - (OPEN ELECTIVE-III)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME75	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the various energy sources and their availability.
2. To know the knowledge of solar radiation and its applications.
3. To analyse the performance of wind energy conversion systems.
4. To understand the Biomass conversion techniques and Geothermal sources and resources

COURSE OUTCOMES

At the end of the course, the student should be able to:

1. Demonstrate various energy sources and their availability.
2. Demonstrate the knowledge of solar radiation and its applications.
3. Analyse the performance of wind energy conversion systems.
4. Explain Biomass conversion techniques and Geothermal sources and resources
5. Explain various methods for harvesting the ocean energy sources.

UNIT-I	ENERGY SOURCES & THEIR AVAILABILITY	CLASSES: 10
Energy Sources and their Availability - Importance of Renewable Energy Resources – Present energy scenario – World energy status – Energy scenario in India – Classification of RES. Introduction to Solar, Wind, Geothermal, Bio-mass, Ocean Energy Sources, Comparison of these Energy Sources, Advantages and Limitations of RES.		
UNIT-II	INTRODUCTION TO SOLAR RADIATION & ITS APPLICATIONS	CLASSES: 10
Introduction to Solar radiation - beam and diffuse radiation, solar constant, measurement of solar radiation. Solar Energy Storage – Collectors: Flat plate and Concentrating collectors, Focusing and non-focusing type. Applications of Solar Energy: Solar water heater – Solar dryer – Solar distillation – Solar Photo-Voltaic Cell.		
UNIT-III	PRINCIPLE OF WIND ENERGY CONVERSION	CLASSES: 12
Principle of Wind Energy Conversion – Basic components of Wind Energy Conversion Systems, Wind mill Components, Various Types and their Constructional Features. Effect of Density, Frequency Variances, Angle of attack, and Wind Speed – Design Considerations of Horizontal and Vertical Axis Wind Machines.		
UNIT-IV	BIOMASS CONVERSION TECHNIQUES , GEOTHERMAL SOURCES AND RESOURCES	CLASSES: 10
Biomass conversion techniques – Biogas Generation – Types of biogas plants – Advantages and disadvantages of biogas plants – Types of Bio-digester – Advantages and disadvantages. Geothermal Sources like hydrothermal, geo-pressured hot dry rock, magma – Simple geothermal Power plant.		
UNIT-V	TIDAL ENERGY & OCEAN THERMAL ENERGY CONVERSION	CLASSES: 08
Tidal Energy – Principle of working, performance and limitations. Wave Energy – Principle of working. Ocean Thermal Energy Conversion – OTEC power plants – working principle and Limitations.		

TEXT BOOKS

1. G. D. Rai, Non-Conventional Energy Sources, 4th Edition, Khanna Publishers, 2000.
2. B H Khan, Non-Conventional Energy Resources, 2nd Edition, Tata McGraw Hill Education Pvt. Ltd, 2011

REFERENCE BOOKS

1. S. Hasan Saeed and D.K. Sharma, Non-Conventional Energy Resources, 3rd Edition, S.K. Kataria & Sons, 2012.
2. N.K. Bansal, M. Kleemann, M. Heliss, Renewable energy sources and conversion technology, Tata McGraw Hill, 1990.
3. S. P. Sukhatme, Solar Energy Principles and Application, Tata McGraw Hill Education Pvt. Ltd, 2009.

ELEMENTS OF ELECTRIC AND HYBRID VEHICLES

IV B. TECH. II SEM - (OPEN ELECTIVE-III)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME80	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES								
The course will enable the students to:								
1. To understand the impact of conventional vehicles on the society and different types of drive train topologies 2. To Evaluate the performance and energy consumption in electric vehicles 3. To understand the assessment, configuration and architecture in hybrid vehicles								
COURSE OUTCOMES								
At the end of the course, the student should be able to:								
1. Analyze the impact of conventional vehicles on the society and different types of drive train topologies 2. Evaluate the performance and energy consumption in electric vehicles 3. Assess the configuration and architecture in hybrid vehicles 4. Choose the type of motor used in electric and hybrid electric vehicles 5. Design the drive train suitable for electric and hybrid vehicles								
UNIT-I	INTRODUCTION TO HYBRID ELECTRIC VEHICLES					CLASSES: 10		
History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies								
UNIT-II	ELECTRIC VEHICLES					CLASSES: 08		
Configurations of EVs, Performance of EVs, Tractive Effort in Normal Driving, Energy Consumption.								
UNIT-III	HYBRID ELECTRIC VEHICLES					CLASSES: 10		
Concept of Hybrid Electric Drive Trains. Architectures of Hybrid Electric Drive Trains.								
UNIT-IV	ELECTRIC PROPULSION SYSTEMS					CLASSES: 12		
Introduction, DC Motor Drives, Induction Motor Drives, Permanent Magnetic BLDC Motor Drives, SRM Drives								
UNIT-V	DESIGN PRINCIPLE OF SERIES AND PARALLEL HYBRID ELECTRIC DRIVE TRAIN					CLASSES: 10		
Design Principles of a Series (Electrical Coupling) Hybrid Drive Train: Operation Patterns, Control Strategies, Design Example. Design Principles of Parallel (Mechanically Coupled) Hybrid Electric Drive Train: Drive Train Configuration and Design Objectives, Control Strategies, Parametric Design of a Drive Train.								
TEXT BOOKS								

1. Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press, 2004.
2. Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003.

REFERENCE BOOKS

1. James Larminie, John Lowry, Electric Vehicle Technology Explained, Wiley, 2003.
2. Allen Fuhs, Hybrid Vehicles and the future of personal transportation, CRC Press, 2011.

OPEN ELECTIVES OFFERED BY HSM

OE1		OE2	
A6HS08	Business Economics and Financial Analysis	A6HS11	Advanced Entrepreneurship
A6HS09	Basics of Entrepreneurship		
OE3			
A6HS12	Management Science		

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS

VI SEMESTER: OPEN ELECTIVE-I								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS08	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES								
The course will enable the students to:								
1. To understand and appreciate, with a particular insight, the importance of certain basic issues governing the business operations 2. To understand the demand and supply, production function, cost analysis, markets, forms of business organizations, capital budgeting and financial accounting and financial analysis.								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. Understand the market dynamics namely, demand and supply, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures. 2. Gain an insight into how production function is carried out to achieve least cost combination of inputs and cost analysis. 3. Develop an understanding of Analyse how capital budgeting decisions are carried out. 4. Understanding the framework for both manual and computerised accounting process 5. Know how to analyse and interpret the financial statements through ratio analysis.								
UNIT-I	INTRODUCTION & DEMAND ANALYSIS						CLASSES: 12	
Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.								
UNIT-II	PRODUCTION & COST ANALYSIS						CLASSES: 12	
Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems) - Managerial Significance.								
UNIT-III	MARKETS & NEW ECONOMIC ENVIRONMENT						CLASSES: 12	
Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing: Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of different forms of Business Organisation: Sole Proprietorship, Partnership, Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario.								
UNIT-IV	CAPITAL BUDGETING						Classes: 12	
Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements, Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capital Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method, Accounting Rate of return (ARR) and Net Present Value Method (simple problems).								

UNIT-V	INTRODUCTION TO FINANCIAL ACCOUNTING & FINANCIAL ANALYSIS	CLASSES: 12
Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, Trial Balance - Final Accounts (Tracing Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du Pont Chart.		
TEXT BOOKS		
1. Varshney & Maheswari: Managerial Economics, Sultan Chand, 2009. 2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013. 3. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.		
REFERENCE BOOKS		
1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012. 2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012. 3. Lipsey & Chrystel, Economics, Oxford University Press, 2012. 4. Domnick Salvatore: Managerial Economics In a Global Economy, Thomson, 2012. 5. Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012. 6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012. 7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012. 8. Dwivedi: Managerial Economics, Vikas, 2012. 9. Shailaja & Usha: MEFA, University Press, 2012. 10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012. 11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011. 12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.		

BASIC OF ENTREPRENEURSHIP

VI SEMESTER: OPEN ELECTIVE-I								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS09	Foundation	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 50	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 50			
COURSE OBJECTIVES The course should enable the students to: - Understand and discover entrepreneurship - Build a strong foundation for students to start, build, and grow a viable and sustainable venture - Develop an entrepreneurial outlook and mindset, critical skills and knowledge - Mitigate three types of risks: Customer, Business Model, and Product/Technical								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: - Entrepreneurship and Innovation minors will be able to sell themselves and their ideas. Students - Master oral and visual presentation skills and establish a foundation of confidence in the skills necessary to cause others to act. - The ability to imagine, recognize, and seize opportunities for innovation and new venture creation in the arts and culture sector - The know-how to bring capital to ideas in both commercial and nonprofit domains.								
UNIT-I	DISCOVER YOURSELF AND IDENTIFY PROBLEMS WORTH SOLVING						CLASSES: 10	
Discover Yourself: Find your flow, Effectuation, Case Study: Tristan Walker: The extroverted introvert, Identify your entrepreneurial style, Identify Problems Worth Solving: What is a business opportunity and how to identify it, Find problems around you that are worth solving, Methods for finding and understanding problems - (Observation, Questioning, DT, Jobs to be done (JTBD), How to run problem interviews to understand the customer's worldview, Introduction to Design Thinking – Process and Examples, Generate ideas that are potential solutions to the problem identified – DISRUPT, Class Presentation: Present the problem you "love"								
UNIT-II	CUSTOMER, BUSINESS MODEL, VALIDATION						CLASSES: 10	
CUSTOMER :Identify Your Customer Segments and Early Adopters - The difference between a consumer and a customer (decision maker), Market Types, Segmentation and Targeting, Defining the personas; Understanding Early Adopters and Customer Adoption Patterns, Identify the innovators and early adopters for your startup; Craft Your Value Proposition - Come up with creative solutions for the identified problems, Deep dive into Gains, Pains and “Jobs-To- Be-Done” (using Value Proposition Canvas, or VPC), Identify the UVP of your solution using the Value Proposition section of the VPC, Outcome-Driven Innovation, Class Presentation: Communicating the Value Proposition- 1 min Customer Pitch. BUSINESS MODEL: Get Started with Lean Canvas. VALIDATION: Develop the Solution, Sizing the Opportunity, Building an MVP.								
UNIT-III	MONEY AND TEAM						CLASSES: 12	
MONEY: Revenue Streams - Basics of how companies make money, Understand income, costs, gross and net margins, Identify primary and secondary revenue, streams ; Pricing and Costs - Value, price, and costs; Different pricing, Understand product costs and operations costs; Basics of unit costing strategies; Financing Your New Venture - How to finance business ideas, Various sources of funds available to								

an entrepreneur and pros and cons of each, What investors expect from you, Practice Pitching to Investors and Corporates.

TEAM: Team Building - Shared Leadership, Role of a good team in a venture's success; What to look for in a team; How do you ensure there is a good fit? Defining clear roles and responsibilities, How to pitch to candidates to join your startup, Explore collaboration tools and techniques - Brainstorming, Mind mapping, Kanban Board, Slack.

UNIT-IV**MARKETING & SALES****CLASSES: 10**

MARKETING & SALES: Positioning - Understand the difference between product and brand and the link between them, Define the positioning statement for your product/service and how it should translate into what your customers should see about that brand in the market place.

Channels & Strategy: Building Digital Presence and leveraging Social media, Creating your company profile page, Measuring the effectiveness of selected channels, Budgeting and planning.

UNIT-V**PLANNING & TRACKING****CLASSES: 08**

Sales Planning : Understanding why customers buy and how buying decisions are made; Listening, Sales planning, setting targets, Unique Sales Proposition (USP); Art of the sales pitch (focus on customers needs, not on product features, Follow-up and closing a sale; Asking for the sale

Planning & Tracking: Importance of project management to launch and track progress, Understanding time management, workflow, and delegation of tasks.

Business Regulation: Basics of business regulations of starting and operating a business; Importance of being compliant and keeping proper documentation, How to find help to get started.

TEXT BOOKS

1. S. R. Bhowmik & M. Bhowmik, Entrepreneurship, New Age International, 2007.
2. Steven Fisher, Ja-nae' Duane, The Startup Equation - A Visual Guidebook for Building Your Startup, Indian Edition, Mc Graw Hill Education India Pvt. Ltd, 2016.
3. D F Kuratko and T V Rao —Entrepreneurship-A South-Asian Perspective —Cengage Learning, 2012. (For PPT, Case Solutions Faculty may visit : login.cengage.com)

REFERENCE BOOKS

1. Vasant Desai — Small Scale industries and entrepreneurship|| Himalaya publishing 2012.
2. Rajeev Roy — Entrepreneurship|| 2e, Oxford, 2012.
3. B.Janakiram and M.Rizwana || Entrepreneurship Development: Text & Cases, Excel Books, 2011.
4. Stuart Read, Effectual Entrepreneurship, Routledge, 2013.
5. Nandan H, Fundamentals of Entrepreneurship, PHI, 2013

WEB REFERENCES

1. <http://freevideolectures.com/Course/3641/Entrepreneurship-Through-the-Lens-of-Venture-Capital>
2. <http://www.onlinevideolecture.com/?course=mba-programs&subject=entrepreneurship>
3. http://nptel.ac.in/courses/122106032/Pdf/7_4.pdf
4. <https://www.scribd.com/doc/21516826/Entrepreneurship-Notes>
5. <http://freevideolectures.com/Course/3514/Economics/-/Management-/Entrepreneurship/50>
6. Journal of Entrepreneurship & Organization Management, Vikalpa, IIMA, IIMB Review, Decision, IIMC, Vision, HBR.

ADVANCED ENTREPRENEURSHIP

VII SEMESTER: OPEN ELECTIVE-II								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS11	Foundation	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 50	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 50			
COURSE OBJECTIVES								
The course should enable the students to:								
1. Understand and discover entrepreneurship								
2. Build a strong foundation for students to start, build, and grow a viable and sustainable venture								
3. Develop an entrepreneurial outlook and mindset , critical skills and knowledge								
4. Mitigate three types of risks : Customer, Business Model, and Product/Technical								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. The knowledge and mastery of organizational and management techniques in the arts, including leadership, organizational design, human resources, legal issues, and finance								
2. The knowledge of the legal and ethical environment impacting business organizations and exhibit an understanding and appreciation of the ethical implications of decisions.								
3. An ability to engage in critical thinking by analyzing situations and constructing and selecting viable solutions to solve problems.								
UNIT-I	ORIENTATION TO GROWTH						CLASSES: 10	
ORIENTATION TO GROWTH: Getting Ready for Growth-Why growth stage is different compared to startup phase, Why Product-Market fit is not enough, Case study, To assess readiness for growth, To chart a growth path.								
UNIT-II	CUSTOMER, BUSINESS MODEL, VALIDATION						CLASSES: 10	
CUSTOMERS :Expanding Customer Base-Revisit your business model and develop few variants (more business model types),Identify additional customer segments that your solution can address, Evaluate business models for the new customer segments, Relook at the Problem Statement (can you expand the scope and scalability of your business by repositioning your problem statement?), Explore additional ways to monetize. BUSINESS MODEL: Get Started with Lean Canvas. VALIDATION: Develop the Solution, Sizing the Opportunity, Building an MVP.								
UNIT-III	TRACTION AND TEAM						CLASSES: 12	
TRACTION : Scaling - How to gain traction beyond early customers, Defining traction (in quantifiable terms) and identifying the most important metrics to measure traction, Calculate cost of new customer acquisition, Estimate your customer lifetime value(LTV),Identifying waste in your operations and focusing your team on what is important for traction. Channels and Strategy - The Bullseye framework, Identify Channels using Bulls Eye Frame work, Measuring the effectiveness of selected channels, Budgeting and planning.								
UNIT-IV	MONEY & SALES						CLASSES: 10	
MONEY: Growing Revenues - Stabilizing key revenue streams, Developing additional revenue streams (licensing, franchising), Exploring new channels and partnerships, Sales Planning - Understanding why customers buy and how buying decisions are made; Listening skills, Sales planning, setting targets, Unique Sales Proposition (USP); Art of the sales pitch (focus on customers needs, not on product features, Follow-up and								

closing a sale; Asking for the sale. Strengthening Sales - Follow-up and closing a sale; Asking for the sale, Building a professional sales team, Sales compensation and incentives, Sales planning, setting targets. Improving Margins - Testing price elasticity, Optimizing costs and operational expenses, Advanced concepts of unit costing. Financial Modeling - Financial modeling of your venture's growth, Analyzing competitor and peer's financial models.		
UNIT-V	SUPPORT	CLASSES: 08
SUPPORT: Legal - Overview of legal issues and their impact on entrepreneurs, Importance of getting professional help (legal and accounting), Importance of being compliant and keeping proper documentation, Patents and Intellectual property, Trademarks. Mentors, Advisors, and Experts - The importance of a Mentor and how to find one, Role of business advisors and experts for specific targets in your growth plan		
TEXT BOOKS		
1. Bygrave, W., & Zacharakis, A. (2017) Entrepreneurship, 4th Edition (3rd Edition is ok too) Wiley. 2. Carree, M. A., Thurik, A. R. "The impact of entrepreneurship on economic growth" In: Audretsch, D.B.Acs. Handbook of Entrepreneurship Research. Berlin: Springer Verlag, 2010. 3. Handbook of Entrepreneurship Research. Berlin: Springer Verlag, 2010.		
REFERENCE BOOKS		
1. Eric, Reis (2017) The Startup Way: How Entrepreneurial Management Transforms Culture and Drives Growth 2. Thurik, A. R., Audretsch, D. B., Stam, E. "The rise of the entrepreneurial economy and the future of dynamic capitalism", Technovation. 3. Audretsch, D. B., Grilo, I., Thurik, A. R. (Eds). The Handbook of Research on Entrepreneurship Policy. Cheltenham, UK: Edward Elgar, 2007.		
WEB REFERENCES		
1. https://www.ediindia.org/ediilibrary.aspx 2. https://www.entrepreneurshipsecret.com/sources-of-information-for-entrepreneurship-development-contd		
KEY REFERENCE		
1. https://www.ediindia.org/EDIILibrary.aspx 2. https://www.entrepreneurshipsecret.com/sources-of-information-for-entrepreneurship-development-contd		

MANAGEMENT SCIENCE

VIII SEMESTER: OPEN ELECTIVE-III								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS12	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES								
The course will enable the students to:								
1. To familiarize & obtain Knowledge with the process of management and to provide basic insights into management practices 2. To understand the structure & Designing of an Organization 3. Familiarize with strategies used for analysis of an Organization 4. To understand the Contemporary Management Issues 5. To familiarize with the management skills which can be applied in the Organizational context to achieve Organizational goals.								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. Knowledge on management theories and practices 2. Understanding designing organizational structure 3. Understanding on the methods & charts used in operations management 4. Ability to understand the market and its environment 5. Understand the processes, functions etc. in Human Resources Management 6. Knowledge on Strategic alternatives 7. Familiar with the practices implemented in management 8. Understand the social responsibilities of Management.								
UNIT-I	INTRODUCTION TO MANAGEMENT AND ORGANIZATION						CLASSES: 12	
Concepts of Management and organization- nature, importance and Functions of Management, Systems Approach to Management-Taylor's Scientific Management Theory- Fayal's Principles of Management- Maslow's theory of Hierarchy of Human Needs- Douglas McGregor's Theory X and Theory Y - Hertzberg Two Factor Theory of Motivation - Leadership Styles, Social responsibilities of Management, Designing Organizational Structures: Basic concepts related to Organization - Departmentation and Decentralization, Types and Evaluation of mechanistic and organic structures of organization and suitability.								
UNIT-II	OPERATIONS AND MARKETING MANAGEMENT						CLASSES: 12	
Principles and Types of Plant Layout-Methods of Production(Job, batch and Mass Production), Work Study - Basic procedure involved in Method Study and Work Measurement - Business Process Reengineering(BPR) - Statistical Quality Control: control charts for Variables and Attributes (simple Problems) and Acceptance Sampling, TQM, Six Sigma, Deming's contribution to quality, Objectives of Inventory control, EOQ, ABC Analysis, Purchase Procedure, Stores Management and Store Records - JIT System, Supply Chain Management, Functions of Marketing, Marketing Mix, and Marketing Strategies based on Product Life Cycle, Channels of distribution.								
UNIT-III	HUMAN RESOURCES MANAGEMENT(HRM)						CLASSES: 12	
Concepts of HRM, HRD and Personnel Management and Industrial Relations (PMIR), HRM vs. PMIR, Basic functions of HR Manager: Manpower planning, Recruitment, Selection, Training and Development, Placement, Wage and Salary Administration, Promotion, Transfer, Separation, Performance Appraisal, Grievance Handling and Welfare Administration, Job Evaluation and Merit Rating - Capability Maturity Model (CMM) Levels - Performance Management System.								

UNIT-IV	PROJECT MANAGEMENT (PERT/CPM)	CLASSES: 12
Network Analysis, Program Evaluation and Review Technique (PERT), Critical Path Method (CPM), Identifying critical path, Probability of Completing the project within given time, Project Cost Analysis, Project Crashing (simple problems)		
UNIT-V	STRATEGIC MANAGEMENT AND CONTEMPORARY STRATEGIC ISSUES	CLASSES: 12
Mission, Goals, Objectives, Policy, Strategy, Programs, Elements of Corporate Planning Process, Environmental Scanning, Value Chain Analysis, SWOT Analysis, Steps in Strategy Formulation and Implementation, Generic Strategy alternatives. Bench Marking and Balanced Score Card as Contemporary Business Strategies.		
TEXT BOOKS		
1. A.R.Aryasri : Management Science, TMH, (Latest edition) 2. Stoner, Freeman, Gilbert, Management, 6th Ed, Pearson Education, New Delhi, 2004.		
REFERENCE BOOKS		
1. Kotler Philip and Keller Kevin Lane: Marketing Management, Pearson, 2012. 2. Koontz and Weihrich: Essentials of Management, McGraw Hill, 2012. 3. Thomas N. Duening and John M. Ivancevich Management - Principles and Guidelines, Biztantra, 2012. 4. Kanishka Bedi, Production and Operations Management, Oxford University Press, 2012. 5. Samuel C. Certo: Modern Management, 2012. 6. Schermerhorn, Capling, Poole and Wiesner: Management, Wiley, 2012. 7. Parnell: Strategic Management, Cengage, 2012. 8. Lawrence R Jauch, R. Gupta and William F. Glueck: Business Policy and Strategic Management Science, McGraw Hill, 2012.		
WEB REFERENCES		
1. https://www.smartzworld.com/notes/management-science-pdf-notes-ms-notes-pdf 2. https://www.alljntuworld.in/download/management-science-notes/		